

BLO-1: Flight Principles

- The maximum aerodynamic efficiency of a wing with elliptical lift distribution can be increased by ...
 - Reducing friction.
 - Reducing span.
 - Increasing the chord.
 - Transferring lift production from wing root to wing tip.
- What is TRUE regarding the International Standard Atmosphere (ISA)?
 - It is an atmospheric model providing a common reference for temperature and pressure over a wide range of altitudes.
 - The earth average atmospheric conditions at sea level.
 - The atmospheric conditions that allow international flight.
 - Non-linear increasing Mach number (speed of sound) within the Troposphere.
- What is the effect on an aerofoil lift characteristic curve ($C_L(\alpha) = C_{L_0} + C_{L_\alpha}\alpha$ for $\alpha < \alpha_S$) of a high-lift device that ONLY increases the camber of the aerofoil?
 - It shifts the curve up by increasing C_{L_0} .
 - It increases the slope C_{L_α} .
 - It increases the stall angle α_S .
 - It reduces $C_D(\alpha)$ but does nothing to $C_L(\alpha)$.
- Which of the following parameters is roughly independent of altitude if we neglect the effects of Reynolds and Mach numbers?
 - Stall angle of attack.
 - Stall speed.
 - Maximum bank angle in a coordinated turn.
 - Maximum available thrust.
- How does maximum bank angle compare between an airplane flying high at an altitude $h_h(\Phi_h)$ and the same airplane, at the same speed, flying at sea level at $h_{SL}(\Phi_{SL})$?
 - $\Phi_h < \Phi_{SL}$.
 - $\Phi_h > \Phi_{SL}$.
 - $\Phi_h = \Phi_{SL}$.
 - Which one is bigger depends on the aerodynamic characteristics of the airplane.
- The drag polar of an airplane changes from $C_D = 0.016 + 0.121C_L^2$ in clean configuration to $C_D = 0.027 + 0.081C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - $C_L = 0.524$.
 - $C_L = 0.368$.
 - $C_L = 0.269$.
 - $C_L = 0.19$.
- A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 35^\circ$) and speed $V = 145$ kts. How long (time t) will it take to complete a 120° turn? (consider $g \simeq 9.8 \text{ m/s}^2$, $1 \text{ m/s} = 1.944 \text{ kts}$.)
 - $t = 22.8 \text{ s}$.
 - $t = 20.4 \text{ s}$.
 - $t = 18 \text{ s}$.
 - $t = 15.2 \text{ s}$.
- An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.21 + 0.37\alpha$ with $\alpha_S = 13.3^\circ$ and $C_D = 0.015 + 0.049C_L^2$. With full flaps, it changes to: $C_L^f = 0.22 + 0.36\alpha$ with $\alpha_S = 15.1^\circ$ and $C_D^f = 0.036 + 0.047C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - $\alpha = 8.19^\circ$.
 - $\alpha = 7.46^\circ$.
 - $\alpha = 2.54^\circ$.
 - $\alpha = 14.4^\circ$.
- An aerobatics airplane with lift coefficient $C_L = 0.013 + 0.23\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 155$ kts with $\alpha = 0.7^\circ$. What is the required speed (V_{if}) to perform a balanced inverted flight at the same altitude with an angle of attack $\alpha = -0.8^\circ$? (Consider $1 \text{ m/s} = 1.944 \text{ kts}$.)
 - $V_{if} = 156 \text{ kts}$.
 - $V_{if} = 169 \text{ kts}$.
 - $V_{if} = 126 \text{ kts}$.
 - $V_{if} = 201 \text{ kts}$.
- An airplane of mass $m = 1090 \text{ kg}$ is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 20.3 < f_{max}$. What can we say about the total (D), the induced (D_i) and the parasitic (D_p) drag? (consider $g \simeq 9.8 \text{ m/s}^2$)
 - $D = 526 \text{ N}$ and $D_p \neq D_i$.
 - $D = 526 \text{ N}$ and $D_p = D_i = 263 \text{ N}$.
 - $D = 789 \text{ N}$ and $D_p = D_i = 394 \text{ N}$.
 - $D = 789 \text{ N}$ and $D_p \neq D_i$.

11. What can be NOT said about Titanium Alloys?

- (a) All other options are correct.
- (b) Excellent corrosion resistance.
- (c) High melting point.
- (d) Allotropic transformation behavior.

12. Which of the following is a valid material selection criteria for aeronautical applications?

- (a) All other options are correct.
- (b) Corrosion resistance.
- (c) Compatibility with other materials.
- (d) Surface treatments.

13. The ability to absorbed energy in an impact rupture is called.

- (a) Resilience.
- (b) Fatigue.
- (c) Ductility.
- (d) Toughness.

14. Which of the following materials is usually NOT used as matrix material in composites?

- (a) Glass.
- (b) Metallic.
- (c) Ceramic.
- (d) Polyester.

15. What can be said about steel?

- (a) It is almost three times denser than Al.
- (b) It is never used in parts of airframe structures which require high mechanical performances.
- (c) Its crystallographyc structure is HCP.
- (d) Around 80 per cent of modern airframe structures are made of steel.

16. Which aircraft has the ICAO designation: A332.

- (a) Airbus 330-200.
- (b) Airbus 300-320.
- (c) Antonov 300-320.
- (d) ATR 33-200.

17. Which of the following statements is TRUE regarding the approach speed in landing procedures?

- (a) The approach speed is about 1.3 times the stall speed.
- (b) Helicopters belong to category E.
- (c) Category A includes the planes with highest speed.
- (d) Category H is dedicated to hyper-sonic planes.

18. Identify the following aircraft.



- (a) XB-70 Valkyrie.
- (b) Tupolev Tu-160.
- (c) Rockwell B-1 Lancer.
- (d) RUSSIAN COUNTERPART.

19. Which company constructs the following aircraft?



- (a) Iljuschin.
- (b) Antonov.
- (c) Airbus.
- (d) Boeing.

20. Which of the following is NOT a wide body aircraft?

- (a) Boeing B757.
- (b) Boeing B767.
- (c) Airbus A300.
- (d) Airbus A310.

BLO-6: Aircraft Systems & Helicopters

21. Which one of the following flight control system options does NOT entitle the pilot to a full control recovery in case of a complete electrical blackout?
- (a) An electro-hydraulic flight control system.
 - (b) An hydro-mechanical flight control system.
 - (c) An electro-hydro-mechanical flight control system.
 - (d) A purely mechanical flight control system.
22. Which of the following is NOT an option to power the hydraulic system in an Airbus A320 type aircraft?
- (a) The pneumatic system.
 - (b) Engine-driven pumps.
 - (c) The ram air turbine (RAT).
 - (d) Electrical motors.
23. Which of the following elements CANNOT be found in an electrical system?
- (a) A power transfer unit (PTU).
 - (b) Buses.
 - (c) Batteries.
 - (d) A transformer/rectifier.
24. Which one of the following elements of a flight control system provides the pilot with feedback on what his actions on the commands are producing on the control surfaces?
- (a) Artificial feel system.
 - (b) Hydraulic actuators.
 - (c) The flight controls computer (FCC).
 - (d) Sensors (gyros, accelerometers ...).
25. What can be said, in terms of power generation (how hydraulic power is produced) and distribution (how and where hydraulic power is used), of the several hydraulic circuits that comprise a redundant hydraulic system?
- (a) They are different in both generation and distribution.
 - (b) They are different in generation but equal in distribution.
 - (c) They are equal in both generation and distribution.
 - (d) They are equal in generation but different in distribution.
26. Which of the following statement is TRUE?
- I An autogiro can perform stationary flight.
 - II The rotors of tandem helicopter are counter-rotating.
 - III The rotors of a coaxial helicopter are contra-rotating.
- (a) II and III.
 - (b) I.
 - (c) II.
 - (d) I and III.
27. The Coanda effect ...
- (a) ... is used to create thrust along the tail boom to compensate the torque created by the main rotor in NOTAR helicopters.
 - (b) ... is used to create thrust along the tail boom to compensate the torque created by the main rotor in coaxial main rotor helicopters.
 - (c) ... is the air flux expelled at the end of the tail of a NOTAR helicopter.
 - (d) ... is the flux interaction created between the two rotors of a coaxial main rotor helicopter.
28. What is WRONG for a helicopter performing a lateral flight?
- (a) The helicopter has identical lift distribution along different blades.
 - (b) The helicopter does NOT have a zero specific fuel consumption.
 - (c) The helicopter requires tail compensation.
 - (d) The helicopter can move independently lateral or longitudinal.
29. The maximum tip speed of the main rotor of an helicopter ...
- (a) ... is limited by the noise produced and compressibility effects.
 - (b) ... is limited by the compressibility effects and the risk of flow separation at the blades.
 - (c) ... is limited by the stored kinetic energy of the main rotor and the risk of flow separation at the blades.
 - (d) All other statements are wrong.
30. For a helicopter in horizontal flight, ...
- (a) ... the distribution of the main rotor speeds induces a roll momentum in the helicopter.
 - (b) ... the distribution of the main rotor speeds is proportional to the radius.
 - (c) ... the retreating blade moves up to compensate the reverse flow zone.
 - (d) ... there is no reverse flow in the main rotor.

BLO-8: Navigation & Altimetry

31. Right after take-off the barometric altimeter requires a calibration called ...
- (a) QNH while below the transition altitude (TA).
 - (b) QNH while below the transition level (TL).
 - (c) STD while below the transition altitude (TA).
 - (d) STD while below the transition layer (TL).
32. Why can't GPS be used as a standalone system for navigation?
- (a) Because the system cannot be trusted, as it does not issue a timely alarm in case of malfunction.
 - (b) Because of its low accuracy.
 - (c) Because it is a military system.
 - (d) Because the Selective Availability signal intentionally degrades the GPS accuracy.
33. An example of azimuthal guidance subsystem would be ...
- (a) the Localizer (ILS/LLZ), which provides the horizontal angle.
 - (b) the Glide Slope (ILS/GS), which provides the vertical angle.
 - (c) the primary surveillance radar.
 - (d) the weather radar.
34. Integrity is ...
- (a) the guarantee that the error will not be higher than some threshold and that, in case it is, an alarm will be received on time.
 - (b) the ability of the total system to perform its function without unscheduled interruptions during the intended operation.
 - (c) the ability of the total system to maintain the aircraft position within a given Total System Error (TSE) with a 95% probability.
 - (d) the ability of the total system to provide the required parameters at the start of the intended operation.
35. Which of the following is an existing navigation option?
- (a) All other answers are correct.
 - (b) Esteemed navigation: IAS, Clock and autonomous navigation systems like IRUs.
 - (c) Navigation by external references: Systems defined in the ICAO Annex 10 like NDB, VOR, TACAN, DME, ILS, MLS, GNSS.
 - (d) Radar navigation.
36. The accuracy of a VOR is about:
- (a) $\pm 1.4^\circ$ (about 90 m crosstrack error at 2 nm).
 - (b) ± 185 m for a limit of 100 aircraft.
 - (c) in CAT-I, a threshold of a 10000 ft-long runway, Azimuth = ± 9.1 m and Elevation = ± 4.1 m.
 - (d) Horizontal = ± 13 m and Vertical = ± 22 m.
37. The Transition Level (TL) is situated:
- (a) Above the Transition Altitude (TA).
 - (b) Below the Transition Altitude (TA).
 - (c) Sometimes above the Transition Altitude (TA) and others below it, depending on the pressure gradient.
 - (d) Sometimes above the Transition Altitude (TA) and others below it, depending on the temperature gradient.
38. The Distance Measurement Equipment is:
- (a) 1 NM accurate.
 - (b) 1° accurate.
 - (c) 2° accurate.
 - (d) 1 rad accurate.
39. Which of these statements about Ground-Based Augmentation System (GBAS) is correct?
- (a) All other statements are correct.
 - (b) GBAS provides higher levels of accuracy and integrity than bare GPS.
 - (c) GBAS is based on differential GPS (dGPS).
 - (d) GBAS provides lower coverage than Satellite-based Augmentation System (SBAS).
40. Why is VHF Omnidirectional Range (VOR) used for en-route (ENR) navigation?
- (a) Because it provides accurate lateral guidance.
 - (b) Because it provides accurate vertical guidance.
 - (c) Because they are cheaper than ILS.
 - (d) They are not used in ENR navigation.

BLO-2: Aircraft Stability & Control

1. What is TRUE when the centre of lift is located behind the center of gravity?
 - (a) Both canard and tailplane have to generate an equilibrating pitching moment.
 - (b) A canard must generate a downforce.
 - (c) A tailplane may generate an unstable moment.
 - (d) Stability does not depend on the location of the centre of gravity.
2. Which of the following manoeuvres can be considered a lateral equilibrium?
 - (a) All are lateral equilibria.
 - (b) Crosswind straight flight.
 - (c) Coordinated banked turn.
 - (d) Operational banked turn.
3. What is FALSE about the phugoid motion?
 - (a) The vertical stabiliser plays an important role in making it stable.
 - (b) It corresponds to a periodic exchange of potential and kinetic energy.
 - (c) It is a slow, periodic, poorly damped mode.
 - (d) It is related to longitudinal flight in the vertical plane.
4. Which statement is FALSE for a positive static stability?
 - (a) All other options are false.
 - (b) Implies good maneuverability.
 - (c) Generates instability.
 - (d) Is achieved when Neutral point is ahead of center of gravity.
5. Which oscillations are coupled in a Dutch roll?
 - (a) Yaw and roll.
 - (b) Yaw and pitch.
 - (c) Pitch and roll.
 - (d) Spiral and roll.
6. Which of the following features are NOT characteristic of the pitch oscillation mode?
 - (a) The angle of attack remains constant along the oscillation.
 - (b) It is fast and usually periodic.
 - (c) It can be dangerous if it is unstable.
 - (d) It can be left unstable if appropriately stabilised with automatic control.
7. Having a stable airplane, which is the first longitudinal mode usually to be completely damped?
 - (a) Pitch oscillation.
 - (b) Phugoid.
 - (c) Return to straight and level flight.
 - (d) All of them are damped equally fast.
8. Which parameters have a disengaging (stabilising) effect on the spiral mode?
 - (a) Positive dihedral (wings up, $C_{l_\beta} < 0$).
 - (b) High directional stability (big fin, $C_{l_\beta} > 0$).
 - (c) High roll damping (big wing span, $C_{l_\beta} < 0$).
 - (d) None of the others.
9. The stability of which mode is critically compromised by the relative position of the aerodynamic centre and the centre of gravity?
 - (a) Pitch oscillation.
 - (b) Phugoid.
 - (c) Dutch roll.
 - (d) Spiral.
10. What does Dynamic Stability around an axis requires, at the very least?
 - (a) All options are correct.
 - (b) The appearance of a restoring moment around this axis upon perturbation.
 - (c) The appearance of a damping moment around this axis during restoration after perturbation.
 - (d) Static stability around this axis.

BLO-3: Aircraft Performances

11. Which of the following statements is TRUE about the velocities for best angle ($V_{\gamma_{max}}$) and for maximum rate of climb ($V(R/C)_{max}$)?
- $V_{\gamma_{max}} < V(R/C)_{max}$.
 - There is no way to know which one of $V_{\gamma_{max}}$ and $V(R/C)_{max}$ is greater because it depends on the type of engine.
 - $V_{\gamma_{max}} = V(R/C)_{max}$.
 - $V_{\gamma_{max}} > V(R/C)_{max}$.
12. Which of the following statements is TRUE? In which of the two possible straight and level flight regimes, stall may occur?
- Stall may occur in flight regime R2.
 - Stall may occur in flight regime R1.
 - Stall may occur in both flight regimes R1 and R2.
 - Stall doesn't occur in flight regime R1 nor in R2.
13. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the altitude decreases?
- It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - It occurs at higher speeds (moves right) and increases (moves up).
 - It occurs at lower speeds (moves left) and decreases (moves down).
 - It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
14. How does head wind affect the maximum angle of climb (AOC) and the corresponding aerodynamic speed V_{AOC} of a given airplane?
- V_{AOC} decreases and AOC increases.
 - V_{AOC} increases and AOC increases.
 - V_{AOC} increases and AOC decreases.
 - V_{AOC} decreases and AOC decreases.
15. Which of the following conditions corresponds to a best angle climb of an airplane, independently of how it is powered (jet or propeller)?
- Maximum excess thrust.
 - Maximum excess power.
 - Minimum power.
 - Minimum drag.
16. The required thrust of a jet airplane is given by $T_R = 0.23\bar{\rho}V^2 + 0.035W^2/\bar{\rho}V^2$. If the airplane has mass $m = 7500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = \rho/\rho_0 = 0.53$, what is velocity (V_E) for maximum endurance? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V_E = 233$ m/s.
 - $V_E = 147$ m/s.
 - $V_E = 307$ m/s.
 - $V_E = 198$ m/s.
17. A propeller-powered airplane of mass $m = 1040$ kg has a required power curve $P_R = 0.27\bar{\rho}V^3 + 0.037W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (consider $g \simeq 9.8$ m/s², 1 m/s = 1.944 kts.)
- $V_{(R/C)_{max}} = 90.7$ kts.
 - $V_{(R/C)_{max}} = 119$ kts.
 - $V_{(R/C)_{max}} = 157$ kts.
 - $V_{(R/C)_{max}} = 68.9$ kts.
18. Given is a jet airplane with thrust of $T_R = 0.33\bar{\rho}V^2 + 0.021W^2/\bar{\rho}V^2$. If the airplane has mass $m = 3950$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.23$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D_{V_E} = 6440$ N.
 - $D_{V_E} = 7440$ N.
 - $D_{V_E} = 6770$ N.
 - $D_{V_E} = 7770$ N.
19. A glider whose maximum aerodynamic efficiency is $f_{max} = 32$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 64$ mi of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination? (consider 1 ft = 0.305 m, 1 km = 0.621 mi.)
- $h = 10500$ ft.
 - $h = 13100$ ft.
 - $h = 6710$ ft.
 - $h = 1640$ ft.
20. The required thrust of a jet-engined airplane of mass $m = 4875$ kg, wing area $S = 42$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.42$, is given by $T_R = 0.37\bar{\rho}V^2 + 0.057W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 59000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.24$)? (consider $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³.)
- $V \simeq 253$ m/s.
 - $V \simeq 73.8$ m/s.
 - $V \simeq 27.9$ m/s.
 - $V \simeq 143$ m/s.

BLO-5: Aerospace Propulsion

21. What is TRUE regarding propulsive power?
- It is the thrust of the engines times the aircraft speed.
 - It is constant with the aircraft speed.
 - It is the mass flow times the exhaust gases speed.
 - None of the other statements is true.
22. Which of the following statements is TRUE?
- None of the other statements is true.
 - The efficiency of the propeller is proportional to the number of blades it has.
 - The more blades the propeller has, the minor are the air perturbations created.
 - A turbofan uses an external propeller.
23. How can you obtain the maximum propulsive efficiency?
- The mass flow must be maximized with the minimum possible acceleration.
 - The thermal efficiency must be maximized.
 - The exhaust gases must be accelerated as much as possible and the mass flow reduced.
 - It is needed to get a compromise between what is written in the other statements.
24. An airplane is flying at a speed of 120 m/s at an altitude for which the atmospheric pressure is 0.315 bar. At the exit of the nozzle, with a diameter of 0.75 m, the pressure of the gases is 2 bar and their speed is 1125 m/s. If the fuel consumption is 2.25 kg/s and the air massflow into the engine is 35 kg/s, how much is its thrust-specific fuel consumption (TSFC)?
- TSFC= 0.0689 g/Ns.
 - TSFC= 0.149 g/Ns.
 - TSFC= 0.0345 g/Ns.
 - TSFC= 0.345 g/Ns.
25. A 10kN rocket engine is ejecting a mass flow of 2.05kg/s, so it is a ...
- ... liquid cryogenic propellant rocket engine.
 - ... liquid storable propellant rocket engine.
 - ... solid propellant rocket engine.
 - ... plasma ion electric rocket engine.
26. From the entrance to the exit of the gases, a 2 spool turbojet is composed by
- diffuser, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle.
 - nozzle, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, diffuser.
 - diffuser, high pressure compressor, low pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle.
 - nozzle, high pressure compressor, low pressure compressor, combustion chamber, low pressure turbine, high pressure turbine, diffuser.
27. The by-pass ratio of ...
- ... a turbofan is higher than a turbojet.
 - ... a turboprop is lower than a turbofan.
 - ... a ramjet is higher than a turbofan.
 - ... a turbofan is lower than a rocket engine.
28. Which of the following engines has better thermal efficiency?
- 4 stroke Diesel engine.
 - 4 stroke Otto engine.
 - 2 stroke Otto engine.
 - None of the other statements is true.
29. To increase the thrust of a propeller engine without anticipating compression phenomena, you should ...
- ... increase the number of blades.
 - ... decrease the number of blades.
 - ... increase the length of the blades.
 - ... decrease the length of the blades.
30. Which of the following statement(s) is/are TRUE?
- Turboprops have high by-pass ratios.
 - Ramjets have no compressor.
 - High by-pass ratio turbofans are usually meant for higher speeds than turbojets.
- I and II.
 - I and III.
 - III.
 - II and III.

BLO-7: Space Systems

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31. What is TRUE regarding the Tsiolkovsky equation ($\Delta V = v_e \log \frac{m_0}{m_1}$)?
- The greater ΔV , the more mass will be consumed, for a given effective exhaust velocity (specific impulse).
 - The smaller ΔV , the more mass will be consumed, for a given effective exhaust velocity (specific impulse).
 - The greater ΔV , the more distance will be traveled by the spacecraft
 - The smaller ΔV , the less mass will be consumed, regardless of the effective exhaust velocity (specific impulse).
32. What is NOT an orbital element?
- Altitude.
 - True anomaly.
 - Longitude of Ascending node.
 - Eccentricity.
33. Which of the following statements is TRUE for a rocket engine?
- The ratio thrust vs weight of a solid propellant engine is higher than the ratio T/W of a liquid propellant.
 - The specific impulse of a solid propellant rocket is higher than the specific impulse of a liquid propellant.
 - The required thrust/weight ratio for an interplanetary mission is higher than for a launching mission.
 - All the other statements are wrong.
34. What is TRUE for a sun-synchronous orbit (SSO)?
- It is a polar orbit ($i \approx 90^\circ$) in which the satellite flies a given point of the surface at the same local time.
 - is a polar orbit ($i \approx 0^\circ$) in which the satellite flies a given point of the surface at the same local time.
 - It is widely used in navigation applications.
 - It is widely used in satellite-communications.
35. To launch a satellite into geostationary orbit, the launcher flies to which cardinal direction?
- East.
 - North-East.
 - North-West.
 - West.
36. What application is the highly elliptical earth orbit (HEO) mainly used for?
- Communications for locations situated in extreme latitudes.
 - Communications for locations situated in temperate latitudes.
 - Communications for locations situated in extreme longitudes.
 - Communications for locations situated in temperate longitudes.
37. What is FALSE for Solid-Propellant Rocket Motors?
- They have a smaller thrust-to-volume ratio.
 - Engine is simpler, safer, and more reliable.
 - Solid propellants are more storable and stable.
 - One ignited, it cannot be turned off.
38. What is TRUE for the specific impulse I_{SP} of a rocket?
- None of the other options is correct.
 - It is related to the time required for engine ignition.
 - It is related to the total fuel burning time.
 - It is related to the inertial mass of the rocket payload.
39. Which is the best orbit suggested for an Earth observation mission?
- Polar orbit.
 - Geostationary orbit.
 - Molniya orbit.
 - Medium Earth orbit.
40. Assuming keplerian Orbits, on what kind of orbit is a space vehicle with speed $V > V_{esc}$ flying?
- Hyperbolic orbit.
 - Parabolic orbit.
 - Elliptic orbit.
 - None of the other answers.

BLO-1: Flight Principles

1. Which is ultimately a resulting effect of air viscosity?
 - (a) All options are correct.
 - (b) Friction drag.
 - (c) Lift.
 - (d) Boundary layer separation.
2. Which component of drag decreases when speed decreases?
 - (a) Parasite drag.
 - (b) Parasite drag and induced drag.
 - (c) Induced drag.
 - (d) None.
3. A wing with varying incidence angle along the span is said to have:
 - (a) Geometric twist.
 - (b) Dihedral.
 - (c) Taper.
 - (d) Sweep.
4. How does ISA temperature change for decreasing altitude below 33000 ft?
 - (a) It increases 2°C per 1000 ft.
 - (b) It decreases 2°F per km.
 - (c) It increases 3.5°F per km.
 - (d) It decreases 6.5°F per 1000 ft.
5. What happens when the flaps are deployed?
 - (a) k decreases.
 - (b) C_{D0} decreases.
 - (c) C_L decreases.
 - (d) None of the others.
6. The drag polar of an airplane changes from $C_D = 0.031 + 0.21C_L^2$ in clean configuration to $C_D = 0.043 + 0.108C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.343$.
 - (b) $C_L = 0.434$.
 - (c) $C_L = 0.181$.
 - (d) $C_L = 0.578$.
7. Calculate the stall speed V_S of an airplane of mass $m = 1840$ kg and wing surface $S = 18.7\text{m}^2$ with $C_L = 0.19 + 0.28\alpha$ with $\alpha_S = 13.9^\circ$ at an altitude such that $\bar{\rho} = 0.58$ (take $g = 9.8\text{ m/s}^2$ and $\rho_0 = 1.235\text{ kg/m}^3$).
 - (a) $V_S = 25.7\text{ m/s}$.
 - (b) $V_S = 24\text{ m/s}$.
 - (c) $V_S = 17.4\text{ m/s}$.
 - (d) $V_S = 30\text{ m/s}$.
8. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 21^\circ$) and speed $V = 210$ kts. How long (time t) will it take to complete a 140° turn? (consider $g \simeq 9.8\text{ m/s}^2$, $1\text{ m/s} = 1.944\text{ kts}$.)
 - (a) $t = 70.2\text{ s}$.
 - (b) $t = 62\text{ s}$.
 - (c) $t = 84.5\text{ s}$.
 - (d) $t = 92.3\text{ s}$.
9. An aerobatics airplane with lift coefficient $C_L = 0.017 + 0.27\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 165$ kts with $\alpha = 0.4^\circ$. What is the required speed (V_{if}) to perform a balanced inverted flight at the same altitude with an angle of attack $\alpha = -0.5^\circ$? (Consider $1\text{ m/s} = 1.944\text{ kts}$.)
 - (a) $V_{if} = 170\text{ kts}$.
 - (b) $V_{if} = 157\text{ kts}$.
 - (c) $V_{if} = 195\text{ kts}$.
 - (d) $V_{if} = 219\text{ kts}$.
10. An airplane of mass $m = 1350$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 17.5 < f_{max}$. What can we say about the total (D), the induced (D_i) and the parasitic (D_p) drag? (consider $g \simeq 9.8\text{ m/s}^2$)
 - (a) $D = 756\text{ N}$ and $D_p \neq D_i$.
 - (b) $D = 756\text{ N}$ and $D_p = D_i = 378\text{ N}$.
 - (c) $D = 1130\text{ N}$ and $D_p = D_i = 566\text{ N}$.
 - (d) $D = 1130\text{ N}$ and $D_p \neq D_i$.

BLO-4: Materials & Structures / Aircraft Recognition

11. The ability for a material to absorb energy is described by which of the following quantities?
- (a) Resilience.
 - (b) Fatigue.
 - (c) Yield strength.
 - (d) Resistance.
12. Which of the following statements is TRUE regarding different material types?
- (a) Polymers have a smaller elasticity E than metal alloys.
 - (b) Composites have a larger density ρ than metal alloys.
 - (c) Composites have a larger yield strength σ_y than metal alloys.
 - (d) Metal alloys have a larger yield strength σ_y than composites.
13. Which statement is WRONG regarding Aluminium Alloys?
- (a) Al has a body-centered cubic (bcc) structure.
 - (b) Al has a low melting point.
 - (c) Al has good corrosion resistance.
 - (d) Al has good mechanical properties.
14. Which of the following materials can be used as reinforcement fiber/filament in composites?
- (a) Spectra.
 - (b) Vinylester.
 - (c) Epoxy.
 - (d) Ceramics.
15. Which of the following is a valid material selection criteria for aeronautical applications?
- (a) All options are correct.
 - (b) Fatigue.
 - (c) Thermal and electric properties.
 - (d) Manufacturing.
16. Which aircraft has the ICAO description: A346 L4J.
- (a) Airbus 340-600: Land plane, 4 engines, jet type.
 - (b) Airbus 340-600: Large plane, 4 engines, jet type.
 - (c) Antonov 34-6: Large plane, 4 engines, jet type.
 - (d) Antonov 340-6: Land plane, 4 engines, jet type.

17. According to ICAO, what is the classification in terms of wake turbulence for an A340-400?

- (a) H: Heavy.
- (b) S: Super.
- (c) L: Light.
- (d) M: Medium.

18. Identify the following aircraft.



- (a) Airbus A300.
- (b) Boeing 757.
- (c) Airbus A320.
- (d) Boeing 737.

19. Which company constructs the following aircraft?



- (a) McDonnell Douglas.
 - (b) Embraer.
 - (c) Airbus.
 - (d) Boeing.
20. Which of the following is a wide body aircraft?
- (a) Airbus A310.
 - (b) Airbus A321.
 - (c) Boeing 757.
 - (d) Boeing 737.

BLO-6: Aircraft Systems & Helicopters

21. Which one of the following elements of a flight control system (FCS) are essential to the implementation of flight augmentation capabilities and automatic control?
- (a) All of them.
 - (b) Computers (FCC, SEC, ELAC, FAC).
 - (c) Hydraulic and/or electro-mechanic actuators.
 - (d) Sensors (gyros, accelerometers, transducers, ...).
22. How many hydraulic systems has an A320?
- (a) 3 but not fully redundant.
 - (b) 3 fully redundant.
 - (c) 2 independent + 1 redundant.
 - (d) 2 redundant + 1 independent.
23. The electrical system of an aircraft such as an A320 is integrated by
- (a) AC and DC circuits, linked by inverters and transformers.
 - (b) AC circuits.
 - (c) DC circuits.
 - (d) Independent AC and DC circuits.
24. All engine tasks are managed by which of the following systems?
- (a) FADEC.
 - (b) PTU.
 - (c) ADIRU.
 - (d) FMGC.
25. What does the FADEC do?
- (a) Performs complete engine management.
 - (b) It is the computer in charge of flight augmentation.
 - (c) Provides AC electrical power.
 - (d) Control ailerons and elevators.
26. What is TRUE about tail rotor compensation for a helicopter?
- (a) It uses around 10% of the engine power.
 - (b) It requires the installation of a second dedicated engine.
 - (c) It is not possible with gas turbine engine helicopters.
 - (d) It is not possible on piston-based engine helicopters.
27. What are the main restrictions regarding minimum allowable blade tip speed for a typical helicopter?
- (a) Kinetic energy and stall effects.
 - (b) Compressibility and noise effects.
 - (c) Compressibility and stall effects.
 - (d) None of the others.
28. What is TRUE for a helicopter performing a lateral flight?
- (a) The helicopter has different lift distribution along different blades.
 - (b) The helicopter has a zero specific fuel consumption.
 - (c) The helicopter requires no tail compensation.
 - (d) The helicopter cannot stop moving laterally.
29. Which one of the following does not constitute a control system in helicopters?
- (a) Thrust reverser control system.
 - (b) Tail rotor pitch control system.
 - (c) Cyclic pitch control system.
 - (d) Collective pitch control system.
30. Which of the following is NOT a tail compensator type of a helicopter?
- (a) Coaxial rotor.
 - (b) NOTAR.
 - (c) Fenestron.
 - (d) Tail rotor.

BLO-8: Navigation & Altimetry

31. What is TRUE regarding True Air Speed (TAS)?
- (a) It indicates the physical speed of the aircraft relative to the surrounding air.
 - (b) It is lower than the Indicated Air Speed (IAS).
 - (c) It results from IAS corrected for instrument, position and installation errors.
 - (d) It is the speed of an aircraft relative to the ground.
32. When the airplane fly, the static pressure is measured as 813 *mbar*. What is the altitude of the airplane approximately?
- (a) 1.8 *km*.
 - (b) 0.5 *km*.
 - (c) 4.8 *km*.
 - (d) We cannot measure it.
33. Which of the following radionavigation aids does NOT provide distance information?
- (a) VOR.
 - (b) DME.
 - (c) Radioaltimeter.
 - (d) GNSS.
34. What is TRUE of using the QNH calibration setting in the altimeter?
- (a) Indicated altitude is referred to an estimated pressure at sea level.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter indicates flight levels.
 - (d) All options are correct.
35. How does a pilot at increasing altitude has to shift the calibration of the altimeter at the Transition Level (TL)?
- (a) From QNE to QNH.
 - (b) From QNH to QNE.
 - (c) From QFE to QNE.
 - (d) From QFE to QNH.
36. What does Dead Reckoning stands for?
- (a) The process of calculating the current position from an initial known position and subsequent speed, direction and time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) A air ship based radio navigation aid.
 - (d) A generic name for ground stations.
37. Which is not a ILS (Instrument Landing System)?
- (a) DME.
 - (b) LOC (Locallizer).
 - (c) Gliding Slope.
 - (d) Marker Beacon.
38. Which navigation system provides the angle referred to true north of a line connecting the aircraft and a tuned ground station?
- (a) VOR.
 - (b) NDB.
 - (c) GPS.
 - (d) ADF.
39. In order to get the air speed, which is not necessary?
- (a) INS (Inertail Navigation System).
 - (b) Static Port.
 - (c) Pitot Tube.
 - (d) Air density information.
40. How does the the GNSS find the receiver's position?
- (a) By measuring the pseudoranges between the GNSS sateliite and the receiver.
 - (b) By communicating with GNSS satellite, receiving the position information.
 - (c) By measuring the geomagnetic north, finding the position.
 - (d) By using the record of the flight acceleration and velocity.

BLO-2: Aircraft Stability & Control

1. Which of the following statements is TRUE?
 - (a) For $C_{l_r} > 0$ one observes Roll-Yaw Coupling.
 - (b) For $C_{l_\beta} > 0$ one observes Dihedral effect.
 - (c) For $C_{l_p} > 0$ one observes Roll Damping.
 - (d) For $C_{n_r} > 0$ one observes Yaw Damping.
2. Which statement is TRUE for the roll subsidence mode?
 - (a) Large aspect ratio wings result in better damping.
 - (b) The damping is related to the Dutch Roll.
 - (c) It is a longitudinal mode.
 - (d) It increases maneuverability.
3. In which of the two possible straight and level flight regimes it is easier to fly stably?
 - (a) Regime 1 (R1).
 - (b) Regime 2 (R2).
 - (c) Both, regimes 1 (R1) and regime 2 (R2).
 - (d) Regime 0 (R0).
4. What of the following assertions is TRUE about specific points on an aircraft?
 - (a) Around the centre of lift, X_L , no pitching aerodynamic moment exists.
 - (b) The centre of gravity, X_{cg} , is a function of α .
 - (c) The aerodynamic centre, X_{ac} , strongly depends on α .
 - (d) None of the others is correct.
5. Dutch roll couples oscillations in ...
 - (a) Roll and yaw.
 - (b) Yaw and pitch.
 - (c) Pitch and roll.
 - (d) Spiral and roll.
6. What is TRUE when the centre of lift is located behind the center of gravity?
 - (a) Both canard and tailplane have to generate an equilibrating pitching moment.
 - (b) A canard must generate a downforce.
 - (c) A tailplane may generate an unstable moment.
 - (d) Stability does not depend on the location of the centre of gravity.
7. How can directional stability be tuned?
 - (a) Vertical stabilizer size.
 - (b) Horizontal stabilizer size.
 - (c) Throttle.
 - (d) Ailerons trim tab size.
8. Which of the following hypotheses is NOT a requirement for longitudinal flight?
 - (a) The angle of attack must be small.
 - (b) Propulsion moments must be orthogonal to the symmetry plane of the aircraft.
 - (c) Aerodynamics forces must be parallel to the symmetry plane of the aircraft.
 - (d) Aircraft reflexion-symmetry plane must be vertical.
9. Which of the following statements is TRUE regarding a negative static stability?
 - (a) All other options are true.
 - (b) Implies good maneuverability.
 - (c) Generates instability.
 - (d) Is achieved when the neutral point is ahead of center of gravity.
10. What does Dynamic Stability around an axis requires, at the very least?
 - (a) All options are correct.
 - (b) The appearance of a restoring moment around this axis upon perturbation.
 - (c) The appearance of a damping moment around this axis during restoration after perturbation.
 - (d) Static stability around this axis.

BLO-3: Aircraft Performances

11. Which of the following performance optima are NOT affected by wind?
- Maximum rate of climb and endurance.
 - Maximum rate of climb and range.
 - Maximum climb angle and endurance.
 - Maximum climb angle and range.
12. Which statement is FALSE about climbs?
- Maximum excess thrust corresponds to the best rate of climb.
 - Maximum excess thrust corresponds to the best angle of climb.
 - Minimum excess thrust corresponds to the best angle of climb.
 - At the absolute ceiling, both best angle and best rate of climb are zero.
13. What is essentially TRUE about maximum endurance conditions at cruise?
- It roughly corresponds to minimum required power in case of a propeller-powered airplane.
 - Endurance is increased by tail wind.
 - High altitude is better.
 - None of the others.
14. How does tail wind affect maximum range R_{max} and the corresponding aerodynamic speed $V_{R_{max}}$ of a given airplane?
- $V_{R_{max}}$ decreases and R_{max} improves.
 - $V_{R_{max}}$ increases and R_{max} improves.
 - $V_{R_{max}}$ decreases and R_{max} worsens.
 - It doesn't affect whatsoever.
15. Which statement is TRUE regarding the drag and power curves?
- The aircraft weight impacts D_{min} and P_{min} .
 - The altitude impacts D_{min} and P_{min} .
 - The altitude impacts D_{min} while the aircraft weight impacts P_{min} .
 - None of the other options are correct.
16. The required thrust of a jet airplane is given by $T_R = 0.13\bar{\rho}V^2 + 0.042W^2/\bar{\rho}V^2$. If the airplane has mass $m = 15500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.21$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (consider $g \simeq 9.8$ m/s².)
- $D = 22400$ N.
 - $D = 25900$ N.
 - $D = 21300$ N.
 - $D = 24800$ N.
17. A propeller-powered airplane of mass $m = 1320$ kg has a required power curve $P_R = 0.24\bar{\rho}V^3 + 0.029W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (consider $g \simeq 9.8$ m/s², 1 m/s = 1.944 kts.)
- $V_{(R/C)_{max}} = 99$ kts.
 - $V_{(R/C)_{max}} = 130$ kts.
 - $V_{(R/C)_{max}} = 171$ kts.
 - $V_{(R/C)_{max}} = 226$ kts.
18. The required thrust of a jet-engined airplane of mass $m = 4550$ kg, wing area $S = 39$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.28$, is given by $T_R = 0.36\bar{\rho}V^2 + 0.051W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 39200$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.32$)? (consider $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³.)
- $V \simeq 200$ m/s.
 - $V \simeq 67.5$ m/s.
 - $V \simeq 133$ m/s.
 - $V \simeq 272$ m/s.
19. A propeller-powered airplane with drag polar $C_D = 0.023 + 0.158C_L^2$, wing surface $S = 31$ m² and mass $m = 1070$ kg is flying at a speed $V = 72.3$ m/s. The available engine power at sea level is $(P_A)_{SL} = 260$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.69$ kg/m³? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $\delta_x = 0.788$.
 - $\delta_x = 0.694$.
 - $\delta_x = 0.651$.
 - $\delta_x = 0.444$.
20. A glider whose maximum aerodynamic efficiency is $f_{max} = 37$ finds an isolated thermal column amidst an otherwise still atmosphere. If the glider flies at $h = 4100$ ft, what is the maximum distance he can reach? (take 1 m = 3.28 ft)
- $d = 46.3$ km.
 - $d = 88.6$ km.
 - $d = 53.2$ km.
 - $d = 49.6$ km.

BLO-5: Aerospace Propulsion

21. How does the thrust of a turbojet engine reacts when ambient pressure rises?

- (a) It increases.
- (b) It decreases.
- (c) It remains approximately constant.
- (d) It increases for centrifugal-compressor turbojets and decreases for axial-compressor turbojets.

22. Which of the following statements is TRUE about turbojet engine nozzles?

- (a) All options are correct.
- (b) They increase airflow velocity and reduce pressure.
- (c) They can accomodate afterburning.
- (d) Supersonic nozzles end in a divergent section.

23. Sort the following engine types from maximum to minimum achievable speed.

- (a) Ramjet, turboprop, turboshaft for helicopter application.
- (b) Turboshaft for helicopter application, turbofan, piston-engine-driven propeller.
- (c) Rocket, turboprop, turbofan.
- (d) Turbofan, turbojet, ramjet.

24. An airplane is flying at a speed of 100 m/s at an altitude for which the atmospheric pressure is 0.265 bar. At the exit of the nozzle, with a diameter of 0.5 m, the pressure of the gases is 2 bar and their speed is 1275 m/s. If the fuel consumption is 2.65 kg/s and the air massflow into the engine is 45 kg/s, how much is its thrust-specific fuel consumption (TSFC)? (Note: $T = (\dot{m}_a - \dot{m}_f)v_{EG} - \dot{m}_a v_{Air} + (p_e - p_0)A_e$.)

- (a) TSFC= 0.0535 g/Ns.
- (b) TSFC= 0.115 g/Ns.
- (c) TSFC= 0.0268 g/Ns.
- (d) TSFC= 0.268 g/Ns.

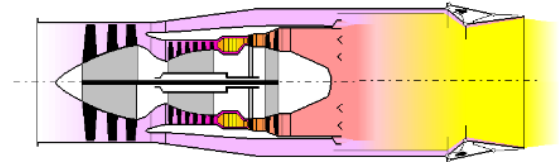
25. Which statement is TRUE for propulsive power?

- (a) It's zero at the beginning of the take-off run.
- (b) It has a peak at the beginning of the take-off run.
- (c) None of the others.
- (d) It depends on thermal efficiency.

26. What does an adapted nozzle means with respect to a turbojet engine?

- (a) Such a final section that makes $P_e = P_0$.
- (b) Temperature adaptation after the combustion chamber.
- (c) Speed adaptation before the compressor.
- (d) Such a final section that makes $V_e = V_0$.

27. What kind of aircraft engine scheme can you see in the following figure?



- (a) Low by-pass ratio turbofan with post-combustion.
- (b) High by-pass ratio turbofan with post-combustion.
- (c) Double spool turbojet with post-combustion.
- (d) Double spool turbojet without post-combustion.

28. A 10kN rocket engine is ejecting a mass flow of 2.05 kg/s. How is this engine type called?

- (a) Liquid cryogenic propellant rocket engine.
- (b) Plasma ion electric rocket engine.
- (c) Liquid storable propellant rocket engine.
- (d) Solid propellant rocket engine.

29. Which of the following statements regarding propeller is TRUE?

- (a) None of the other statements is true.
- (b) A turbofan uses an external propeller.
- (c) The more blades the propeller has, the minor are the air perturbations created.
- (d) The efficiency of the propeller is proportional to the number of blades it has.

30. Which of the following statements are FALSE?

- (I) Turboprops have low by-pass ratios.
- (II) Ramjets have a compressor.
- (III) High by-pass ratio turbofans are usually meant for lower speeds than turbojets.
- (a) I and II.
- (b) I and III.
- (c) III.
- (d) II and III.

BLO-7: Space Systems

31. In which of the following orbits is the satellite always in the vertical of the same fixed point on Earth's surface?
- (a) GEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
32. The Global Positioning System (GPS) provides the users position on the Earth, by using ...
- (a) Distance measuring between GNSS satellites and User receiver.
 - (b) Directly sending the user position.
 - (c) measuring the acceleration of user receiver relative to the GNSS satellites.
 - (d) measuring the angle between the GNSS satellites and the user receiver.
33. When comparing rocket propulsion systems, which of the following statements is TRUE?
- (a) All answers are true.
 - (b) Cold-gas thrusters provide the smallest specific impulse.
 - (c) Chemical propulsion provides higher thrust-to-weight ratios.
 - (d) Electrostatic propulsion provides higher specific impulse than chemical propulsion.
34. What is not the Keplerian orbital elements:
- (a) Orbital height.
 - (b) Inclination.
 - (c) Semimajor axis.
 - (d) True anomaly.
35. What is necessary in order to obtain the maximum propulsive efficiency?
- (a) Maximize the mass flow with the minimum possible acceleration.
 - (b) Maximize thermal efficiency.
 - (c) Maximize the acceleration of the exhaust gases and reduce mass flow.
 - (d) Minimize the mass flow with the maximum possible acceleration.
36. The linear speed of a circular orbit of a satellite is 7.5 km/s. Then, what is this satellite orbit?
- (a) LEO.
 - (b) HEO.
 - (c) MEO.
 - (d) GEO.
37. What is the specific impulse of a rocket engine?
- (a) The thrust per unit mass flow of ejected gases.
 - (b) The time it takes to burn all of the fuel.
 - (c) The energy contained in the fuel divided by the power.
 - (d) The maximum time it takes to reach full speed.
38. What does the Tsiolkovsky's equation deal with?
- (a) The maximum change of speed of the vehicle.
 - (b) The fuel consumption of a rocket.
 - (c) The speed of a rocket.
 - (d) The thrust time of a rocket.
39. Which of the following statements is TRUE for a rocket engine?
- (a) The ratio thrust vs weight of a solid propellant engine is higher than the ratio T/W of a liquid propellant.
 - (b) The required thrust/weight ratio for an interplanetary mission is higher than for a launching mission.
 - (c) The specific impulse of a solid propellant rocket is higher than the specific impulse of a liquid propellant.
 - (d) All other statements are wrong.
40. Which condition is preferred for rocket launch?
- (a) at close to the equator, to the east direction.
 - (b) at close to the poles, to the east direction.
 - (c) at close to the equator, to the west direction.
 - (d) at close to the poles, to the west direction.

BLO-1: Flight Principles

1. What is TRUE for increasing the aspect ratio (λ)?
 - (a) Slope for c_L vs α increases.
 - (b) The drag reduce.
 - (c) Increase effective angle α .
 - (d) Decreasing $C_{L,max}$.
2. Which component of drag increases when speed decreases?
 - (a) Induced drag.
 - (b) Parasite drag and induced drag.
 - (c) Parasite drag.
 - (d) None.
3. Which one of the following manoeuvres entails a higher load factor?
 - (a) Coordinated banked turn.
 - (b) Rectilinear ascent.
 - (c) Rectilinear decent.
 - (d) Straight and level flight.
4. Which of the following is TRUE about the ISA in the Troposphere with increasing altitude?
 - (a) Non-linearly decreasing density.
 - (b) Linearly decreasing pressure.
 - (c) Linearly increasing temperature.
 - (d) Non-linear decreasing Mach number (speed of sound).
5. What is TRUE regarding sweep?
 - (a) It reduces the effective thickness chord ratio.
 - (b) Larger sweep angle result in larger drag coefficient.
 - (c) Smaller sweep angle decreases the lift-to-drag (L/D) ratio.
 - (d) None of the others.
6. The drag polar of an airplane changes from $C_D = 0.037 + 0.24C_L^2$ in clean configuration to $C_D = 0.046 + 0.112C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.265$.
 - (b) $C_L = 0.319$.
 - (c) $C_L = 0.15$.
 - (d) $C_L = 0.096$.
7. An airplane of mass $m = 820 \text{ kg}$ and wing surface $S = 13.1 \text{ m}^2$ and $C_L = 0.16 + 0.26\alpha$ with $\alpha_S = 14.1^\circ$ is performing a bank turn with angle Φ at an altitude such that $\bar{\rho} = 0.66$. Calculate the load factor n for this manoeuvre so that the plane has the same stall speed as for straight-and-level flight under the configuration $C_L = 0.18 + 0.29\alpha$ with $\alpha_S = 14.3^\circ$ at an altitude such that $\bar{\rho} = 0.49$ (take $g = 9.8 \text{ m/s}^2$ and $\rho_0 = 1.225 \text{ kg/m}^3$).
 - (a) $n = 0.916$.
 - (b) $n = 0.798$.
 - (c) $n = 0.905$.
 - (d) $n = 1.22$.
8. An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.23 + 0.38\alpha$ with $\alpha_S = 12.8^\circ$ and $C_D = 0.021 + 0.062C_L^2$. With full flaps, it changes to: $C_L^f = 0.27 + 0.34\alpha$ with $\alpha_S = 14.6^\circ$ and $C_D^f = 0.027 + 0.036C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - (a) $\alpha = 0.659^\circ$.
 - (b) $\alpha = 0.553^\circ$.
 - (c) $\alpha = 0.204^\circ$.
 - (d) $\alpha = 0.915^\circ$.
9. An aerobatics airplane with lift coefficient $C_L = 0.015 + 0.27\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 141 \text{ kts}$ with $\alpha = 0.3^\circ$. What is the required speed (V_{if}) to perform a balanced inverted flight at the same altitude with an angle of attack $\alpha = -0.4^\circ$? (Consider $1 \text{ m/s} = 1.944 \text{ kts}$.)
 - (a) $V_{if} = 73.7 \text{ m/s}$.
 - (b) $V_{if} = 80.9 \text{ m/s}$.
 - (c) $V_{if} = 62.5 \text{ m/s}$.
 - (d) $V_{if} = 55 \text{ m/s}$.
10. An airplane of mass $m = 1220 \text{ kg}$ is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 22.7 < f_{max}$. What can we say about the total (D), the induced (D_i) and the parasitic (D_p) drag? (consider $g \simeq 9.8 \text{ m/s}^2$)
 - (a) $D = 527 \text{ N}$ and $D_p \neq D_i$.
 - (b) $D = 527 \text{ N}$ and $D_p = D_i = 263 \text{ N}$.
 - (c) $D = 790 \text{ N}$ and $D_p = D_i = 394 \text{ N}$.
 - (d) $D = 790 \text{ N}$ and $D_p \neq D_i$.

BLO-4: Materials & Structures / Aircraft Classification & Recognition

11. Which of the following quantities describes the ability for a material to absorb energy?
 - (a) Resilience.
 - (b) Fatigue.
 - (c) Yield strength.
 - (d) Resistance.
12. Which of the following is a valid material selection criteria for aeronautical applications?
 - (a) All options are correct.
 - (b) Fatigue.
 - (c) Thermal and electric properties.
 - (d) Corrosion resistance.
13. Which of the following statements is TRUE regarding different material types?
 - (a) Polymers have a smaller elasticity E than metal alloys.
 - (b) Composites have a larger density ρ than metal alloys.
 - (c) Composites have a larger yield strength σ_y than metal alloys.
 - (d) Metal alloys have a larger yield strength σ_y than composites.
14. Which of the following materials is NOT used as reinforcement fiber/filament in composites?
 - (a) Ceramic.
 - (b) Spectra.
 - (c) Glass.
 - (d) Baron.
15. What is the main advantage of the 2000 aluminum alloy series compared to the 7000 series?
 - (a) It has a better behavior against crack propagation.
 - (b) It has higher toughness.
 - (c) It has a higher Young modulus.
 - (d) All answers are correct.
16. Which aircraft has the ICAO description: B766 L2J.
 - (a) Boeing 767-600: Land plane, 2 engines, jet type.
 - (b) Boeing 767-600: Large plane, 2 engines, jet type.
 - (c) Bombardier 766: Light plane, 2 engines, jet type.
 - (d) Bombardier 76-600: Land plane, 2 engines, jet type.

17. According to ICAO, what is the classification in terms of wake turbulence for an A320-200 (78 tons MTOW)?

- (a) M: Medium.
- (b) S: Super.
- (c) L: Light.
- (d) H: Heavy.

18. Identify the following aircraft.



- (a) Boeing 767.
- (b) Boeing 757.
- (c) Airbus A330.
- (d) Airbus A300.

19. Which company constructs the following aircraft?



- (a) Embraer.
- (b) McDonnell Douglas.
- (c) Airbus.
- (d) Boeing.

20. Which of the following is NOT a wide body aircraft?

- (a) Airbus A321.
- (b) Airbus A310.
- (c) Boeing 767.
- (d) Boeing 787.

BLO-6: Aircraft Systems & Helicopters

21. Which one of the following elements of a flight control system (FCS) are essential to the implementation of flight augmentation capabilities and automatic control?
- (a) All of them.
 - (b) Computers (FCC, SEC, ELAC, FAC).
 - (c) Hydraulic and/or electromechanic actuators.
 - (d) Sensors (gyros, accelerometers, transducers, ...).
22. All engine tasks are managed by which of the following systems?
- (a) FADEC.
 - (b) PTU.
 - (c) ADIRU.
 - (d) FMGC.
23. How many hydraulic systems has an A320?
- (a) 3 but not fully redundant.
 - (b) 3 fully redundant.
 - (c) 2 independent + 1 redundant.
 - (d) 2 redundant + 1 independent.
24. The electrical system of an aircraft such as an A320 is integrated by
- (a) AC and DC circuits, linked by inverters and transformers.
 - (b) AC circuits.
 - (c) DC circuits.
 - (d) Independent AC and DC circuits.
25. Which of the following elements CANNOT power the electrical system of an Airbus A320?
- (a) Pneumatic-driven generator.
 - (b) Hydraulic-driven generator.
 - (c) APU-driven generator.
 - (d) Engine-driven generator.
26. What is the typical main rotor speed for a helicopter?
- (a) 300 rpm.
 - (b) 300 m/s.
 - (c) 300 kts.
 - (d) 300 ft/s.
27. What are the main restrictions regarding minimum allowable blade tip speed for a typical helicopter?
- (a) Kinetic energy and stall effects.
 - (b) Compressibility and noise effects.
 - (c) Compressibility and stall effects.
 - (d) None of the others.
28. Which of the following statements is TRUE for the cyclic control of a helicopter?
- (a) All are correct.
 - (b) Allows control over the X-Y plane.
 - (c) Increases or decreases the angle of attack of some blades.
 - (d) Is actuated by a lever.
29. Which of the following statements is TRUE for a helicopter in horizontal flight?
- (a) The distribution of the main rotor speeds induces a roll momentum in the helicopter.
 - (b) There is no reverse flow in the main rotor.
 - (c) The distribution of the main rotor speeds is proportional to the radius..
 - (d) The retreating blade moves up to compensate the reverse flow zone.
30. What is TRUE for an electro-hydraulic FCS?
- (a) All are correct.
 - (b) It is lighter than a hydro-mechanical FCS.
 - (c) It allows Flight Augmentation Capabilities.
 - (d) It is also known as Fly-by-wire system.

BLO-8: Navigation & Altimetry

31. Which of the following is NOT an ILS (Instrument landing System)?
- (a) DME.
 - (b) Localizer.
 - (c) Marker beacons.
 - (d) Glide Slope.
32. What does accuracy refer to in air navigation?
- (a) To the conformance between calculated and actual position of the aircraft.
 - (b) To the issuing of an alarm whenever the aircraft cannot be guaranteed as being inside the outer tunnel.
 - (c) To the guarantee that the outer tunnel will not disappear during operation.
 - (d) To the guarantee that the tunnel will be in place when the operation starts.
33. Which of the following statements is TRUE for NDB-ADF?
- (a) None of the other answers is correct.
 - (b) The NDB-ADF cannot be used for a non-precision approach.
 - (c) The NDB-ADF provides zenithal guidance.
 - (d) The NDB-ADF provides altitude information.
34. What is TRUE of using the QNH calibration setting in the altimeter?
- (a) Indicated altitude is referred to an estimated sea level's pressure.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter indicates flight levels.
 - (d) All options are correct.
35. If the airplane locates to the North West direction about VOR station, which is the phase difference of received VOR signals?
- (a) 315° .
 - (b) 45° .
 - (c) 135° .
 - (d) 180° .
36. What will the altimeter read, when an aircraft is on a particular runway and the pilot has set the altimeter to the local QFE?
- (a) 0 ft.
 - (b) The altitude of the airport.
 - (c) The altitude of the runway.
 - (d) Further data is required to anticipate the altimeter reading.
37. To estimate the air speed using Pitot Tube, which information is not so relevant?
- (a) Moving direction.
 - (b) Static Pressure.
 - (c) Total Pressure.
 - (d) Air density.
38. What is TRUE for CAS (Calibrated Air Speed)?
- (a) It is the IAS corrected for instrument, position and installation errors based on tabulated data.
 - (b) It is the IAS corrected for compressibility but not for density.
 - (c) It is the actual speed of the aircraft relative to the surrounding air.
 - (d) It is the raw estimation from the pitot-static tube reading.
39. What is TRUE when using the QNE calibration setting in the altimeter?
- (a) All of the other.
 - (b) It is always referred to 1013 hPa.
 - (c) It is only used above the transition altitude.
 - (d) Must switch to QFE/QNH below the transition level.
40. How many GNSS satellites are at least needed to estimate the position on the Earth surface?
- (a) 4.
 - (b) 3.
 - (c) 6.
 - (d) 5.

BLO-2: Aircraft Stability & Control

1. Which of the following statements is TRUE?
 - (a) A crosswind landing implies a zero yaw speed and a non-zero β .
 - (b) A crosswind landing implies a non-zero yaw speed and a non-zero β .
 - (c) A crosswind landing implies a zero yaw speed and a zero β .
 - (d) A crosswind landing implies a non-zero yaw speed and a zero β .
2. Which of the following manoeuvres can be considered a lateral equilibrium?
 - (a) All are lateral equilibria.
 - (b) Crosswind straight flight.
 - (c) Coordinated banked turn.
 - (d) Operational banked turn.
3. What spiral mode is?
 - (a) Slow aperiodic change in bank angle.
 - (b) Slow periodic change in bank angle.
 - (c) Slow periodic change in turn angle.
 - (d) Slow aperiodic change in turn angle.
4. Which statement is FALSE for a positive static stability?
 - (a) All are correct.
 - (b) Implies good maneuverability.
 - (c) Generates instability.
 - (d) Is achieved when the neutral point is ahead of center of gravity.
5. Rudder, elevator and ailerons are meant to generate moments around which axes?
 - (a) Yaw, pitch and roll, respectively.
 - (b) Pitch, roll and yaw, respectively.
 - (c) Roll, yaw and pitch, respectively.
 - (d) Pitch, yaw and roll, respectively.
6. Which of the following hypotheses is NOT a requirement for longitudinal flight?
 - (a) The angle of attack must be small.
 - (b) Propulsion moments must be orthogonal to the symmetry plane of the aircraft.
 - (c) Aerodynamics forces must be parallel to the symmetry plane of the aircraft.
 - (d) Aircraft reflexion-symmetry plane must be vertical.
7. Consider an aircraft that has to be balanced longitudinally. Where do center of gravity needs to be to compensate the pitching moment?
 - (a) Ahead the center of lift, in tailplane configuration.
 - (b) Between the center of lift and the tailplane, in tailplane configuration.
 - (c) Behind the center of lift, in canard configuration.
 - (d) In the center of lift, for any configuration.
8. What is the stability mode that consists in a potential and kinetic energy exchange, that is to say that the plane gains altitude when it loses speed and vice versa, loses altitude when it gains speed?
 - (a) Phugoid mode.
 - (b) Spiral mode.
 - (c) Dutch Roll mode.
 - (d) Roll Subsidence mode.
9. Which of the following statements is TRUE regarding Directional and Lateral Dynamics?
 - (a) The dihedral is the main mechanism for lateral stability.
 - (b) Directional and lateral motions are uncoupled.
 - (c) The vertical stabilizer can be neglected for directional stability.
 - (d) Crosswind flight and banked turn required the command of aileron or rudder.
10. What does Dynamic Stability around an axis requires, at the very least?
 - (a) All options are correct.
 - (b) The appearance of a restoring moment around this axis upon perturbation.
 - (c) The appearance of a damping moment around this axis during restoration after perturbation.
 - (d) Static stability around this axis.

BLO-3: Aircraft Performances

11. Which of the following statements is FALSE about the velocities for best angle ($V_{\gamma_{max}}$) and for maximum rate of climb ($V(R/C)_{max}$)?
- $V_{\gamma_{max}} > V(R/C)_{max}$.
 - There is no way to know which one of $V_{\gamma_{max}}$ and $V(R/C)_{max}$ is greater because it depends on the type of engine.
 - $V_{\gamma_{max}} = V(R/C)_{max}$.
 - $V_{\gamma_{max}} < V(R/C)_{max}$.
12. Which of the following statements is TRUE?
- Stall never occur in flight regime R1.
 - Stall never occur in flight regime R2.
 - Stall may occur in both flight regimes R1 and R2.
 - Stall doesn't occur in flight regime R1 nor in R2.
13. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the altitude decreases?
- It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - It occurs at higher speeds (moves right) and increases (moves up).
 - It occurs at lower speeds (moves left) and decreases (moves down).
 - It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
14. Which of the following statements is TRUE regarding the effect of wind?
- Wind doesn't affect endurance.
 - Wind affects the maximum rate of climb.
 - Wind doesn't affect range.
 - Wind doesn't affect best angle of glide.
15. Consider a gliding flight, what is the condition to achieve *minimum* distance?
- Maximum drag conditions.
 - Maximum descent rate conditions.
 - Maximum required power conditions.
 - Maximum aerodynamic efficiency.
16. The required thrust of a jet airplane is given by $T_R = 0.28\bar{\rho}V^2 + 0.024W^2/\bar{\rho}V^2$. If the airplane has mass $m = 14500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = \rho/\rho_0 = 0.61$, what is velocity (V_E) for maximum endurance? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V_E = 261$ m/s.
 - $V_E = 165$ m/s.
 - $V_E = 345$ m/s.
 - $V_E = 231$ m/s.
17. A propeller-powered airplane of mass $m = 1350$ kg has a required power curve $P_R = 0.19\bar{\rho}V^3 + 0.017W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V(R/C)_{max}$) at sea level ($\bar{\rho} = 1$)? (consider $g \simeq 9.8$ m/s², 1 m/s = 1.944 kts.)
- $V(R/C)_{max} = 92.9$ kts.
 - $V(R/C)_{max} = 122$ kts.
 - $V(R/C)_{max} = 161$ kts.
 - $V(R/C)_{max} = 212$ kts.
18. Given is a jet airplane with thrust of $T_R = 0.41\bar{\rho}V^2 + 0.053W^2/\bar{\rho}V^2$. If the airplane has mass $m = 25700$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.34$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D_{V_E} = 74300$ N.
 - $D_{V_E} = 85700$ N.
 - $D_{V_E} = 78000$ N.
 - $D_{V_E} = 82000$ N.
19. A glider whose maximum aerodynamic efficiency is $f_{max} = 31$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 45$ mi of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination? (consider 1 ft = 0.305 m, 1 km = 0.621 mi.)
- $h = 7620$ ft.
 - $h = 9520$ ft.
 - $h = 4570$ ft.
 - $h = 2260$ ft.
20. The required thrust of a jet-engined airplane of mass $m = 34500$ kg, wing area $S = 132$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.38$, is given by $T_R = 0.36\bar{\rho}V^2 + 0.007W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 117000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.41$)? (consider $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³.)
- $V \simeq 218$ m/s.
 - $V \simeq 86$ m/s.
 - $V \simeq 41.4$ m/s.
 - $V \simeq 301$ m/s.

BLO-5: Aerospace Propulsion

21. Sort the following engine types from maximum to minimum achievable speed.

- (a) Rocket, ramjet, turbojet, turboprop.
- (b) Scramjet, turbofan, turboshaft for helicopter application, piston-engine-driven propeller, rocket.
- (c) scramjet, rocket, piston-engine-driven propeller, turboprop, turbofan.
- (d) Rocket, turbofan, turbojet, ramjet.

22. Why does turbojet engines usually have a convergent nozzle?

- (a) All are correct.
- (b) To accelerate the speed of exhaust gasses.
- (c) To increase thrust.
- (d) After the turbine.

23. A turbofan engine has a typical fuel mass flow $m_f = 22$ t/h and a thrust $T = 310$ kN. Calculate its TSFC.

- (a) TSFC = 19.7 (g/s)/kN.
- (b) TSFC = 11.1 (g/s)/kN.
- (c) TSFC = 44.4 (g/s)/kN.
- (d) TSFC = 28.4 (g/s)/kN.

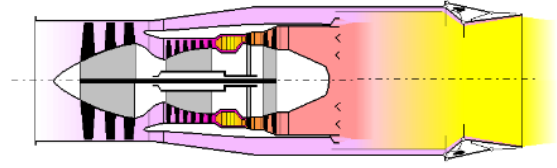
24. Which statement is TRUE for turboprops?

- (a) All other options are true.
- (b) Better power-to-weight ratio and less frontal area than piston engines.
- (c) Limitations of the propeller: Small velocity and low altitude.
- (d) Advantages of the propeller: Excellent performance during take-off, low-speed low-altitude flight.

25. A turbofan engine has a typical fuel mass flow $m_f = 43$ t/h and a thrust $T = 980$ kN. Calculate its TSFC.

- (a) TSFC = 12.2 (g/s)/kN.
- (b) TSFC = 6.86 (g/s)/kN.
- (c) TSFC = 27.4 (g/s)/kN.
- (d) TSFC = 17.6 (g/s)/kN.

26. What kind of aircraft engine scheme can you distinguish in the figure?



- (a) Low by-pass ratio turbofan with post-combustion.
- (b) High by-pass ratio turbofan with post-combustion.
- (c) Double spool turbojet with post-combustion.
- (d) Double spool turbojet without post-combustion.

27. Which of the following turbine engines has best performance at low speed?

- (a) Turboprop.
- (b) Turbojet.
- (c) Scramjet.
- (d) Turbofan.

28. Which of the following is a component of the gas generator?

- (a) All are components of a gas generator.
- (b) Compressor.
- (c) Combustor.
- (d) Turbine.

29. What is true about stoichiometric reaction in heat engines?

- (a) Needs to be constant.
- (b) Requires about 15 units of fuel per 1 unit of air.
- (c) Requires about 15 units of oxygen per 1 unit of fuel.
- (d) Is useless in jet engines.

30. Which statement is FALSE for ramjets?

- (a) Allows for speed about $M = 2$.
- (b) Subsonic combustion.
- (c) Needs another propulsion system for the vehicle to reach supersonic speeds.
- (d) Allows high temperature operation, high thrust-per-weight and high specific thrust.

BLO-7: Space Systems

31. Which orbital element is concerned with covering the whole globe, i.e., polar orbit?
- (a) Inclination should be higher close to 90 degrees.
 - (b) Eccentricity should be 0.
 - (c) The period of the satellite should be large.
 - (d) The speed of the satellite should be fast.
32. Assuming keplerian Orbits, on what kind of orbit is a space vehicle with speed $V > V_{esc}$ flying (V_{esc} escape velocity)?
- (a) Hyperbolic orbit.
 - (b) Parabolic orbit.
 - (c) Elliptic orbit.
 - (d) None of the other answers.
33. Which orbit does give the same local time when the satellite crosses?
- (a) Sun-Synchronous Orbit.
 - (b) Geo Stationary Orbit.
 - (c) Medium Earth Orbit.
 - (d) Molniyar Orbit.
34. What is FALSE for Solid-Propellant Rocket Motors?
- (a) They have a smaller thrust-to-volume ratios.
 - (b) Engine is simpler, safer, and more reliable.
 - (c) Solid propellants are more storable and stable.
 - (d) One ignited, it cannot be turned off.
35. When comparing rocket propulsion systems, which of the following statements is TRUE?
- (a) All answers are true.
 - (b) Cold-gas thrusters provide the smallest specific impulse.
 - (c) Chemical propulsion provides higher thrust-to-weight ratios.
 - (d) Electrostatic propulsion provides higher specific impulse than chemical propulsion.
36. Which of the following propulsive systems is suitable for Earth launches?
- (a) Chemical rockets.
 - (b) Antimatter rockets.
 - (c) Continuous Fusion rockets.
 - (d) Electrostatic rockets.
37. What is TRUE for the specific impulse I_{SP} ?
- (a) All other options are correct.
 - (b) It increases with increasing adiabatic flame temperature.
 - (c) It decreases with increasing molecular weight of the propellant.
 - (d) Higher specific impulse means higher efficiency.
38. The Global Positioning System (GPS) deploys a constellation of ...
- (a) Medium Earth Orbit satellites.
 - (b) Low Earth Orbit satellites.
 - (c) Polar Orbit satellites.
 - (d) Geostationary Orbit satellites.
39. What is the advantage of launching space missions from sites on the Equator?
- (a) ΔV is already large when launching eastwards.
 - (b) ΔV is already small when launching westwards.
 - (c) ΔV is already large when launching northwards to intercept Polar Orbits.
 - (d) All options are correct.
40. What of the following elements are NOT classical orbital elements:
- (a) The speed and the period of satellite.
 - (b) The eccentricity and semimajor axis.
 - (c) The inclination and the longitude of ascending node.
 - (d) The argument of the periapsis and the true anomaly.

BLO-1: Flight Principles

- Which of the following aerodynamic parameters is strongly dependent on altitude?
 - The stall true air speed (stall TAS).
 - The stall equivalent air speed (stall EAS).
 - The stall angle of attack α_s .
 - The slope of the lift characteristic curve $C_{L,\alpha}$.
- Which component of drag increases when speed decreases?
 - Induced drag.
 - Parasite drag and induced drag.
 - Parasite drag.
 - None.
- What physical phenomenon is the source of the quadratic dependence of the drag coefficient on the lift coefficient (the k in $C_D = C_{D_0} + kC_L^2$)?
 - Both wake and wingtip vortices.
 - Only wake.
 - Only wing tip vortices.
 - Viscous friction.
- According to ISA ...
 - the pressure drops non-linearly with altitude in the troposphere.
 - the density drops linearly with altitude in the troposphere.
 - the temperature remains constant at 56.5°C in the altitude range 11 to 50 km.
 - the pressure drops linearly with altitude in the troposphere.
- Which nondimensional parameter allows classifying flight regimes in terms of flow compressibility?
 - The Mach number.
 - The Reynolds number.
 - Viscosity.
 - The lift coefficient.
- The drag polar of an airplane changes from $C_D = 0.017 + 0.14C_L^2$ in clean configuration to $C_D = 0.024 + 0.092C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - $C_L = 0.382$.
 - $C_L = 0.469$.
 - $C_L = 0.557$.
 - $C_L = 0.617$.
- Calculate the stall speed V_S of an airplane of mass $m = 65500\text{ kg}$ and wing surface $S = 62.5\text{ m}^2$ with $C_L = 0.13 + 0.29\alpha$ with $\alpha_S = 14.2^\circ$ at an altitude such that $\bar{\rho} = 0.56$ (take $g = 9.8\text{ m/s}^2$ and $\rho_0 = 1.235\text{ kg/m}^3$).
 - $V_S = 83.6\text{ m/s}$.
 - $V_S = 77.5\text{ m/s}$.
 - $V_S = 56.7\text{ m/s}$.
 - $V_S = 97.8\text{ m/s}$.
- A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 37^\circ$) and speed $V = 160\text{ kts}$. How long (time t) will it take to complete a 200° turn? (consider $g \simeq 9.8\text{ m/s}^2$, $1\text{ m/s} = 1.944\text{ kts}$.)
 - $t = 38.9\text{ s}$.
 - $t = 33.3\text{ s}$.
 - $t = 29.6\text{ s}$.
 - $t = 50.7\text{ s}$.
- An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.22 + 0.34\alpha$ with $\alpha_S = 13.8^\circ$ and $C_D = 0.023 + 0.064C_L^2$. With full flaps, it changes to: $C_L^f = 0.26 + 0.36\alpha$ with $\alpha_S = 14.7^\circ$ and $C_D^f = 0.028 + 0.039C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - $\alpha = 0.668^\circ$.
 - $\alpha = 0.558^\circ$.
 - $\alpha = 0.207^\circ$.
 - $\alpha = 0.851^\circ$.
- An airplane of mass $m = 1180\text{ kg}$ and wing surface $S = 15.2\text{ m}^2$ and $C_L = 0.17 + 0.29\alpha$ with $\alpha_S = 14.3^\circ$ is performing a bank turn with angle Φ at an altitude such that $\bar{\rho} = 0.69$. Calculate the load factor n for this maneuver so that the plane has the same stall speed as for straight-and-level flight under the configuration $C_L = 0.17 + 0.26\alpha$ with $\alpha_S = 14.6^\circ$ at an altitude such that $\bar{\rho} = 0.52$ (take $g = 9.8\text{ m/s}^2$ and $\rho_0 = 1.225\text{ kg/m}^3$).
 - $n = 0.832$.
 - $n = 0.901$.
 - $n = 0.928$.
 - $n = 1.08$.

11. What is WRONG about Titanium Alloys?

- (a) Bad corrosion resistance.
- (b) High melting point.
- (c) They are very expensive.
- (d) Allotropic transformation behavior.

12. Which of the following is a valid material selection criteria for aeronautical applications?

- (a) All other options are correct.
- (b) Corrosion resistance.
- (c) Compatibility with other materials.
- (d) Surface treatments.

13. The ability to absorbed energy in an impact rupture is called.

- (a) Resilience.
- (b) Fatigue.
- (c) Ductility.
- (d) Toughness.

14. Which of the following materials is usually NOT used as matrix material in composites?

- (a) Glass.
- (b) Metallic.
- (c) Ceramic.
- (d) Polyester.

15. What does the designation 7075-T6 refer to?

- (a) An aluminum alloy.
- (b) A titanium jet fighter.
- (c) A hardness scale.
- (d) A type of landing gear fitting.

16. What does the aircraft descriptor H2T mean?

- (a) Helicopter, 2 turboshaft engines.
- (b) Helicopter, twin seater, test-bed.
- (c) Heavy plane, 2 turbojet engines.
- (d) Heavy plane, 2 tiltwing.

17. According to ICAO's aircraft classification regarding approach speed, sort the following categories from slowest to fastest.

- (a) H, A, B, C, D, E.
- (b) A, B, C, D, E, H.
- (c) E, D, C, B, A, H.
- (d) H, E, D, C, B, A.

18. Identify the following aircraft.



- (a) Tupolev Tu-154.
- (b) Boeing B757.
- (c) McDonnell Douglas MD-11.
- (d) Iljuschin Il-62.

19. Which company constructs the following aircraft?



- (a) Boeing.
- (b) Antonov.
- (c) Airbus.
- (d) Comac.

20. Which of the following aircraft can accommodate the highest number of passengers?

- (a) Boeing B777-300.
- (b) Boeing B757-400.
- (c) Airbus A220.
- (d) Airbus A321.

BLO-6: Aircraft Systems & Helicopters

21. Which one of the following flight control system options does NOT entitle the pilot to a full control recovery in case of a complete electrical blackout?
- (a) An electro-hydraulic flight control system.
 - (b) An hydro-mechanical flight control system.
 - (c) An electro-hydro-mechanical flight control system.
 - (d) A purely mechanical flight control system.
22. Which of the following is NOT an option to power the hydraulic system in an Airbus A320 type aircraft?
- (a) The pneumatic system.
 - (b) Engine-driven pumps.
 - (c) The ram air turbine (RAT).
 - (d) Electrical motors.
23. Which of the following components is not a source of electric power?
- (a) Fuel transfer valve.
 - (b) Main Generator.
 - (c) Emergency generator.
 - (d) Battery.
24. A mechanical flight control system ...
- (a) uses a system of rods, levers, cables, pulleys, springs and/or dampers, with no additional power source, to link cockpit controls with control surfaces.
 - (b) is a system with limited self-repair capacity.
 - (c) uses a system of rods, levers, cables, pulleys, springs and/or dampers to link cockpit controls with the hydraulic actuators that operate the control surfaces.
 - (d) uses electronics to link cockpit controls with the hydraulic actuators that operate the control surfaces.
25. What can be said, in terms of power generation (how hydraulic power is produced) and distribution (how and where hydraulic power is used), of the several hydraulic circuits that comprise a redundant hydraulic system?
- (a) They are different in both generation and distribution.
 - (b) They are different in generation but equal in distribution.
 - (c) They are equal in both generation and distribution.
 - (d) They are equal in generation but different in distribution.
26. Which of the following statements (I - III) is/are TRUE?
- I An autogiro can perform stationary flight.
 - II The rotors of tandem helicopter are counter-rotating.
 - III The rotors of a coaxial helicopter are contra-rotating.
- (a) II and III.
 - (b) I.
 - (c) II.
 - (d) I and III.
27. The Coanda effect ...
- (a) ... is used to create thrust along the tail boom to compensate the torque created by the main rotor in NOTAR helicopters.
 - (b) ... is used to create thrust along the tail boom to compensate the torque created by the main rotor in coaxial main rotor helicopters.
 - (c) ... is the air flux expelled at the end of the tail of a NOTAR helicopter.
 - (d) ... is the flux interaction created between the two rotors of a coaxial main rotor helicopter.
28. Helicopters are piloted/controlled...
- (a) by cyclic, collective and anti-torque actuators.
 - (b) longitudinally by an anti-torque system.
 - (c) vertically by a cyclic action.
 - (d) by autorotation.
29. Which of the following phenomena does NOT constitute a limit for maximum blade-tip speed?
- (a) Stored kinetic energy limit.
 - (b) Compressibility limit.
 - (c) Noise limit.
 - (d) Blade stall limit.
30. Engaging in autorotation is ...
- (a) safer the higher and faster the helicopter is flying.
 - (b) safer the lower the helicopter is flying.
 - (c) safer the slower the helicopter is flying.
 - (d) not an option for helicopters.

BLO-8: Navigation & Altimetry

31. An anemometer measures:
- (a) The Indicated Airspeed (IAS).
 - (b) The Ground Speed (GS).
 - (c) The True Airspeed (TAS).
 - (d) The Calibrated/Rectified Airspeed (CAS/RAS).
32. To measure the air speed of the flight, which one is not fundamentally necessary?
- (a) Altitude from static port.
 - (b) Pitot tube for dynamic pressure.
 - (c) Static port for static pressure.
 - (d) Air density information.
33. When the airplane fly, the static pressure is measured as 850 mb (hPa). What is the altitude of the airplane approximately from mean sea level?
- (a) 1.4 km.
 - (b) 0.9 km.
 - (c) 2.5 km.
 - (d) We cannot measure it.
34. When the flight is taking off, the altimeter calibration should be changed from (1) to (2). What is the pair of (1) and (2):
- (a) QNH-QNE.
 - (b) QNE-QFE.
 - (c) QNE-QNH.
 - (d) QFE-QNH.
35. Which one is not ILS (Instrument Landing System)?
- (a) DME.
 - (b) Localizer.
 - (c) Glide slope.
 - (d) Marker beacon.
36. How many GNSS satellites needs for positioning, at least?
- (a) 4.
 - (b) 3.
 - (c) 5.
 - (d) 2.
37. Which navigation system provides the angle referred to true north of a line connecting the aircraft and a tuned ground station?
- (a) VOR.
 - (b) NDB.
 - (c) GPS.
 - (d) ADF.
38. What does the altimeter shows if it has been set to QFE and the aircraft is on ground at the threshold of the runway of an airfield?
- (a) Zero altitude.
 - (b) The altitude of the airport.
 - (c) FL301.
 - (d) ISA+10.
39. If a VOR signals are received with 45° phase difference, then which direction does the airplane located from VOR station?
- (a) North-East.
 - (b) North.
 - (c) South.
 - (d) South-East.
40. Which device is not used essentially for INS (Inertial Navigation System)?
- (a) Magnetometers.
 - (b) Computers.
 - (c) Accelerometers.
 - (d) Gyroscopes.

BLO-1: Flight Principles

1. What is FALSE regarding flap deployment?
 - (a) k increases.
 - (b) C_{D_0} increases.
 - (c) Deploy flaps when high C_L is required.
 - (d) Camber of the airfoil increases.
2. Which one of the following manoeuvres entails a higher load factor?
 - (a) Coordinated banked turn.
 - (b) Rectilinear ascent.
 - (c) Rectilinear decent.
 - (d) Straight and level flight.
3. Consider a plane with positive dihedral. What is the position of the wing root with respect to the wing tip?
 - (a) Lower down.
 - (b) Further ahead.
 - (c) Further back.
 - (d) None of the others.
4. Two identical airplanes flying at the same altitude. One airplane flying at fast speed $V_f(\phi_f)$ and the other one flying at slow speed $V_s(\Phi_s)$. Which statement regarding the maximum bank angle is correct?
 - (a) $\Phi_f > \Phi_s$.
 - (b) $\Phi_f = \Phi_s$.
 - (c) $\Phi_f < \Phi_s$.
 - (d) It depends on the aerodynamic characteristics of the airplane.
5. Which of the following parameters is roughly independent of altitude if we neglect the effects of Reynolds and Mach numbers?
 - (a) Maximum lift coefficient.
 - (b) Stall speed.
 - (c) Maximum lift force.
 - (d) Maximum bank angle in a coordinated turn.
6. An aerobatics airplane with lift coefficient $C_L = 0.013 + 0.17\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 145$ kts (1 m/s = 1.944 kts) with $\alpha = 0.7^\circ$. What should the speed (V_{if}) be to perform a balanced inverted flight at the same altitude with an angle of attack $\alpha = -0.6^\circ$?
 - (a) $V_{if} = 177$ kts.
 - (b) $V_{if} = 190$ kts.
 - (c) $V_{if} = 205$ kts.
 - (d) $V_{if} = 227$ kts.
7. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 37^\circ$) and speed $V = 155$ kts. How long will it take to complete a 60° turn (t)? (consider $g \simeq 9.8$ m/s², 1 m/s = 1.944 kts.)
 - (a) $t = 11.3$ s.
 - (b) $t = 9.87$ s.
 - (c) $t = 13.9$ s.
 - (d) $t = 15.2$ s.
8. An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.28 + 0.39\alpha$ with $\alpha_S = 11^\circ$ and $C_D = 0.011 + 0.06C_L^2$. With full flaps, it changes to: $C_L^f = 0.26 + 0.34\alpha$ with $\alpha_S = 12^\circ$ and $C_D^f = 0.028 + 0.04C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - (a) $\alpha = 1.65^\circ$.
 - (b) $\alpha = 1.45^\circ$.
 - (c) $\alpha = 0.51^\circ$.
 - (d) $\alpha = 2.36^\circ$.
9. The drag polar of an airplane changes from $C_{D,\text{clean}} = 0.028 + 0.079C_L^2$ in clean configuration to $C_{D,\text{flaps}} = 0.044 + 0.071C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 1.41$.
 - (b) $C_L = 1.82$.
 - (c) $C_L = 0.771$.
 - (d) $C_L = 0.489$.
10. What is the drag coefficient ($C_{D_{opt}}$) for the optimal condition that maximizes aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.25 + 0.027C_L^2$?
 - (a) $C_{D_{opt}} = 0.5$.
 - (b) $C_{D_{opt}} = 0.438$.
 - (c) $C_{D_{opt}} = 3.04$.
 - (d) $C_{D_{opt}} = 0.375$.

BLO-2: Aircraft Stability & Control

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11. What is TRUE when the centre of lift is located behind the center of gravity?
- Both canard and tailplane have to generate an equilibrating pitching moment.
 - A canard must generate a downforce.
 - A tailplane may generate an unstable moment.
 - Stability does not depend on the location of the centre of gravity.
12. Lateral stability is influenced by which of the following wing parameters?
- Dihedral and sweep.
 - Aspect ratio and camber.
 - Taper and twist.
 - Span and chord.
13. Order the longitudinal modes from slower to faster.
- Return to straight and level flight, phugoid and pitch oscillation.
 - Return to straight and level flight, pitch oscillation and phugoid.
 - Phugoid, return to straight and level flight and pitch oscillation.
 - Pitch oscillation, return to straight and level flight and phugoid.
14. Which parameters have a disengaging (stabilizing) effect on the spiral mode?
- Positive dihedral (wings up, $C_{l\beta} < 0$).
 - High roll damping (big wing span, $C_{lp} < 0$).
 - High directional stability (big fin, $C_{n\beta} < 0$).
 - None of the others.
15. What of the following assertions is TRUE about specific points on an aircraft?
- Around the centre of lift, X_L , no pitching aerodynamic moment exists.
 - The centre of gravity, X_{cg} , is a function of α .
 - The aerodynamic centre, X_{ac} , strongly depends on α .
 - None of the others is correct.
16. Which of the following statements is TRUE about the response of an airplane to a perturbation in yaw?
- It depends on static directional stability, since there automatically appears a restoring yaw moment.
 - It depends on both directional and lateral stability, since there is no restoring moment and the return to straight and level flight is mostly dynamical.
 - It depends mainly on static longitudinal stability, since there automatically appears a restoring pitch moment.
 - It depends mainly on static lateral stability, since there automatically appears a restoring roll moment.
17. Which of the following hypotheses is NOT a requirement for longitudinal flight?
- The angle of attack must be small.
 - Propulsion moments must be orthogonal to the symmetry plane of the aircraft.
 - Aerodynamics forces must be parallel to the symmetry plane of the aircraft.
 - Aircraft reflexion-symmetry plane must be vertical.
18. What is TRUE for dynamic stability?
- All other options are true.
 - Roll subsidence is stable.
 - Dutch roll is stable.
 - Spiral mode is unstable.
19. Which of the following statements is TRUE regarding static stability?
- $C_{m,\alpha} < 0$ has to be negative.
 - $C_{m,\alpha} = 0$ has to be negative
 - $C_{m,\alpha} > 0$ has to be negative.
 - Center of gravity must be behind of the neutral point.
20. What is TRUE for the elevator trim angle?
- All other options are true.
 - Depends on α_e .
 - Depends on the centering $x_{cg} - x_{np}$.
 - It is independent of V , h , W , γ .

BLO-3: Aircraft Performances

21. Which statement is NOT correct about climbs?
- The best angle of climb corresponds to minimum drag regardless of engine type.
 - The best rate of climb corresponds to maximum excess power.
 - The best angle of climb corresponds to maximum excess thrust.
 - Both best angle and best rate of climb are zero at the absolute ceiling.
22. In which of the two possible straight-and-level-flight regimes, stall may occur?
- R2.
 - R1.
 - In both R1 and R2.
 - Neither in R1 nor in R2.
23. What is TRUE in a constant-altitude coordinated banked turn?
- Weight is outweighed by lift.
 - Weight equals lift.
 - Lift is outweighed by weight.
 - Whether lift or weight is higher is determined by the sense of rotation (left or right).
24. How does tail wind affect maximum range R_{max} and the corresponding aerodynamic speed $V_{R_{max}}$ of a given airplane?
- $V_{R_{max}}$ decreases and R_{max} improves.
 - $V_{R_{max}}$ increases and R_{max} improves.
 - $V_{R_{max}}$ decreases and R_{max} worsens.
 - It doesn't affect whatsoever.
25. Which statement is TRUE regarding the drag and power curves?
- The aircraft weight impacts D_{min} and P_{min} .
 - The altitude impacts D_{min} and P_{min} .
 - The altitude impacts D_{min} while the aircraft weight impacts P_{min} .
 - None of the other options are correct.
26. The required thrust of a jet airplane is given by $T_R = 0.19\bar{\rho}V^2 + 0.043W^2/\bar{\rho}V^2$. If the airplane has mass $m = 18500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.43$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (consider $g \simeq 9.8$ m/s².)
- $D = 32800$ N.
 - $D = 37800$ N.
 - $D = 31100$ N.
 - $D = 39500$ N.
27. A propeller-powered airplane of mass $m = 1590$ kg has a required power curve $P_R = 0.25\bar{\rho}V^3 + 0.029W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (consider $g \simeq 9.8$ m/s², 1 m/s = 1.944 kts.)
- $V_{(R/C)_{max}} = 108$ kts.
 - $V_{(R/C)_{max}} = 142$ kts.
 - $V_{(R/C)_{max}} = 186$ kts.
 - $V_{(R/C)_{max}} = 81.7$ kts.
28. A propeller-powered airplane with drag polar $C_D = 0.025 + 0.121C_L^2$, wing surface $S = 31$ m² and mass $m = 1530$ kg is flying at a speed $V = 138$ kts. The available engine power at sea level is $(P_A)_{SL} = 605$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.51$ kg/m³? (consider $g \simeq 9.8$ m/s², $\rho_0 = 1.225$ kg/m³, 1 m/s = 1.944 kts.)
- $\delta_x = 0.473$.
 - $\delta_x = 0.511$.
 - $\delta_x = 0.39$.
 - $\delta_x = 0.197$.
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 38$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 51$ mi of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination? (consider 1 ft = 0.305 m, 1 km = 0.621 mi.)
- $h = 7040$ ft.
 - $h = 8800$ ft.
 - $h = 6350$ ft.
 - $h = 2440$ ft.
30. The required thrust of a jet-engined airplane of mass $m = 8750$ kg, wing area $S = 71$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.34$, is given by $T_R = 0.17\bar{\rho}V^2 + 0.023W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 32500$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.52$)? (consider $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³.)
- $V \simeq 147$ m/s.
 - $V \simeq 53.2$ m/s.
 - $V \simeq 118$ m/s.
 - $V \simeq 93.9$ m/s.

BLO-4: Aircraft Classification & Recognition / Materials & Structures

31. The S-N curve is related to ...

- (a) fatigue.
- (b) plasticity.
- (c) elasticity.
- (d) hardness.

32. Which of the following is a valid material selection criteria for aeronautical applications?

- (a) All options are correct.
- (b) Fatigue.
- (c) Thermal and electric properties.
- (d) Manufacturing.

33. Which material is more suitable for turbine blades?

- (a) Titanium.
- (b) Steel.
- (c) Aluminium.
- (d) Copper.

34. A structural composite material ...

- (a) combines the better compression properties of the matrix with the better tension properties of the fibers.
- (b) combines the better tension properties of the matrix with the better compression properties of the fibers.
- (c) is highly anisotropic if layers of fibers are aligned in alternated directions.
- (d) None of the others.

35. Which of the following identifies the ratio of the stress along an axis to the strain (ratio of deformation over initial length)?

- (a) Young modulus.
- (b) Tensile strength.
- (c) Resilience.
- (d) Yield strength.

36. To which airplane corresponds the ICAO designation A338?

- (a) Airbus 330-800.
- (b) Antonov 33-8.
- (c) ATC 3-3-8.
- (d) Airbus 380-300.

37. According to ICAO, below which mass is defined an aircraft as "light" L in wake turbulence?

- (a) 7000 kg.
- (b) 3500 kg.
- (c) 19000 kg.
- (d) 1500 kg.

38. Identify the following aircraft



- (a) Airbus A330-200.
- (b) Boeing 767-600.
- (c) Tupolev Tu-154M.
- (d) Airbus A350-900.

39. Which aircraft manufacturer built the following airplane?



- (a) Lockheed.
- (b) Boeing.
- (c) Antonov.
- (d) Airbus.

40. Which of the following designation identifies a wide body aircraft?

- (a) B767.
- (b) A220.
- (c) B757.
- (d) A321.

BLO-5: Aerospace Propulsion

41. Sort the following engine types from minimum to maximum achievable speed.

- (a) Turboshift for helicopter application, turboprop, ramjet, scramjet.
- (b) Turbofan, turboshift for helicopter application, piston-engine-driven propeller, rocket.
- (c) Piston-engine-driven propeller, rocket, turboprop, turbofan.
- (d) Turbofan, turbojet, rocket, ramjet.

42. Which of the following statements is TRUE regarding high bypass ratio turbofans.

- (a) All are correct.
- (b) Optimal speed $M = 0.8 \sim 0.9$.
- (c) Mainly used for medium and long range airliners.
- (d) Generate a compression ratio around 30 – 40.

43. A turbofan engine has a typical fuel mass flow $m_f = 42$ t/h and a thrust $T = 840$ kN. Calculate its TSFC.

- (a) TSFC = 0.0139 (kg/s)/kN.
- (b) TSFC = 7.81 (g/s)/kN.
- (c) TSFC = 0.0313 (kg/s)/kN.
- (d) TSFC = 20 (g/s)/kN.

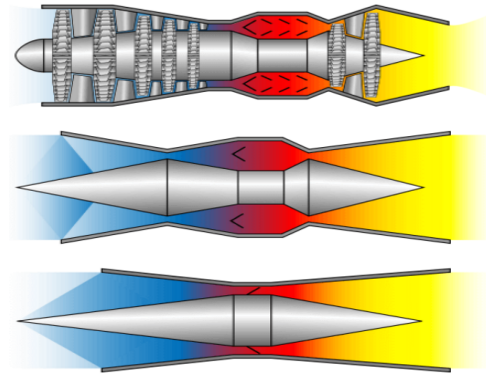
44. What can be said about propulsive power?

- (a) It's zero at the beginning of the take-off run.
- (b) It has a peak at the beginning of the take-off run.
- (c) None of the others.
- (d) It depends on thermal efficiency.

45. What is TRUE for a diffuser in a turbofan engine?

- (a) It reduces speed and recovers dynamic pressure.
- (b) It is located at the outlet of the engine.
- (c) Has a high rotation speed.
- (d) Its mission is to maintain an uniform airflow at the combustion chamber.

46. Order the aircraft engine scheme in the figure below from top to bottom.



- (a) Turbofan; Ramjet; Scramjet.
- (b) Ramjet; Scramjet; Turbofan.
- (c) Scramjet; Ramjet; Turbofan.
- (d) Turbofan; Scramjet; Ramjet.

47. Which of the following statements regarding propeller is TRUE?

- (a) Blades can be spun by acting the governor.
- (b) A turbofan uses an external propeller.
- (c) The more blades the propeller has, the faster the blades can rotate.
- (d) The efficiency of the propeller is proportional to the number of blades it has.

48. Which of the following is a component of a turbojet?

- (a) All are components of a turbojet.
- (b) Compressor.
- (c) Combustor.
- (d) Turbine.

49. Which thermodynamic cycle does alternative engines follow ideally?

- (a) Otto or Diesel.
- (b) Brayton.
- (c) Sabathé.
- (d) Combustion.

50. Which of the following turbine engines has best performance at low speed?

- (a) Turboprop.
- (b) Turbojet.
- (c) Scramjet.
- (d) Turbofan.

BLO-6: Aircraft Systems & Helicopters

51. Which of the following powers the brake system of large aircraft?
- (a) The hydraulic system.
 - (b) The electric system.
 - (c) The pneumatic system.
 - (d) None of the others.
52. Which of the following is NOT a function of the APU?
- (a) Pump fuel to engines.
 - (b) Bleed air supplier.
 - (c) Back up air conditioning system.
 - (d) Back up electrical system.
53. The hydraulic system of large aircraft can be powered by which of the following options?
- (a) All other options are possible.
 - (b) Engine driven pumps.
 - (c) The Ram Air Turbine (RAT).
 - (d) Electric pumps.
54. How is redundancy achieved for the hydraulic system of a large passenger aircraft?
- (a) By having different hydraulic circuits that can exchange pressure but not hydraulic fluid.
 - (b) By having different hydraulic circuits that can exchange hydraulic fluid.
 - (c) By having identical hydraulic circuits isolated from each other.
 - (d) By having identical hydraulic circuits that can exchange both hydraulic fluid and pressure.
55. All engine tasks are managed by which of the following systems?
- (a) FADEC.
 - (b) PTU.
 - (c) ADIRU.
 - (d) FMGC.
56. Which of the following is NOT part of the electric system?
- (a) LP compressor.
 - (b) ESS Bus.
 - (c) Battery.
 - (d) Transformer.
57. Which of the following is NOT a tail compensator type of a helicopter?
- (a) Coaxial rotor.
 - (b) NOTAR.
 - (c) Fenestron.
 - (d) Tail rotor.
58. What are the main restrictions regarding minimum allowable blade tip speed for a typical helicopter?
- (a) Kinetic energy and stall effects.
 - (b) Compressibility and noise effects.
 - (c) Compressibility and stall effects.
 - (d) None of the others.
59. Which of the following statements is WRONG for the cyclic control of a helicopter?
- (a) All other options are wrong.
 - (b) Doesn't allow control over the X-Y plane.
 - (c) It doesn't allow to modify the angle of attack of some blades.
 - (d) Is actuated by a wheel.
60. What is WRONG for a helicopter performing a lateral flight?
- (a) The helicopter has identical lift distribution along different blades.
 - (b) The helicopter does NOT have a zero specific fuel consumption.
 - (c) The helicopter requires tail compensation.
 - (d) The helicopter can move independently lateral or longitudinal.

BLO-7: Space Systems

61. In which of the following orbits is the satellite always in vertical position over the same fixed point on Earth's surface?
- (a) GEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
62. How does the Global Positioning System (GPS) provides the users position on the Earth?
- (a) Distance measuring between GNSS satellites and User receiver.
 - (b) Directly sending the user position.
 - (c) Measuring the acceleration of user receiver relative to the GNSS satellites.
 - (d) Measuring the angle between the GNSS satellites and the user receiver.
63. When comparing rocket propulsion systems, which of the following statements is TRUE?
- (a) All answers are true.
 - (b) Cold-gas thrusters provide the smallest specific impulse.
 - (c) Chemical propulsion provides higher thrust-to-weight ratios.
 - (d) Electrostatic propulsion provides higher specific impulse than chemical propulsion.
64. What is NOT the Keplerian orbital elements:
- (a) Orbital height.
 - (b) Inclination.
 - (c) Semimajor axis.
 - (d) True anomaly.
65. What is necessary in order to obtain the maximum propulsive efficiency?
- (a) Maximize the mass flow with the minimum possible acceleration.
 - (b) Maximize thermal efficiency.
 - (c) Maximize the acceleration of the exhaust gases and reduce mass flow.
 - (d) Minimize the mass flow with the maximum possible acceleration.
66. The linear speed of a circular orbit of a satellite is 7.5 km/s. Then, what is this satellite orbit?
- (a) LEO.
 - (b) HEO.
 - (c) MEO.
 - (d) GEO.
67. What is the specific impulse of a rocket engine?
- (a) The thrust per unit mass flow of ejected gases.
 - (b) The time it takes to burn all of the fuel.
 - (c) The energy contained in the fuel divided by the power.
 - (d) The maximum time it takes to reach full speed.
68. What does the Tsiolkovsky's equation deal with?
- (a) The maximum change of speed of the vehicle.
 - (b) The fuel consumption of a rocket.
 - (c) The speed of a rocket.
 - (d) The thrust time of a rocket.
69. Which of the following statements is TRUE for a rocket engine?
- (a) The ratio thrust vs weight of a solid propellant engine is higher than the ratio T/W of a liquid propellant.
 - (b) The required thrust/weight ratio for an interplanetary mission is higher than for a launching mission.
 - (c) The specific impulse of a solid propellant rocket is higher than the specific impulse of a liquid propellant.
 - (d) All other statements are wrong.
70. Which condition is preferred for rocket launch?
- (a) at close to the equator, to the east direction.
 - (b) at close to the poles, to the east direction.
 - (c) at close to the equator, to the west direction.
 - (d) at close to the poles, to the west direction.

BLO-8: Navigation & Altimetry

71. What equipment is required to determine the True Air Speed (TAS):
- (a) A pitot tube plus a static source.
 - (b) A pitot tube.
 - (c) A GNSS receiver.
 - (d) A VHF antenna.
72. Which of the following defines the ability of a system to provide timely warnings of system degradation?
- (a) Integrity.
 - (b) Accuracy.
 - (c) Continuity.
 - (d) Availability.
73. Which equipment system is NOT used for ILS (Instrument Landing System)?
- (a) DME.
 - (b) Localizer.
 - (c) Glide Slope.
 - (d) Marker Beacons.
74. What is TRUE of using the QNH calibration setting in the altimeter?
- (a) Indicated altitude is referred to an estimated sea level's pressure.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter indicates flight levels.
 - (d) All options are correct.
75. When the flight is taking off, the altimeter calibration should be changed from (1) to (2).
What is the pair of (1) and (2):
- (a) QNH-QNE.
 - (b) QNE-QFE.
 - (c) QNE-QNH.
 - (d) QFE-QNH.
76. Which of the following navigation systems is used to define airways (AWY)?
- (a) VOR.
 - (b) DME.
 - (c) ILS.
 - (d) None of the others.
77. What will the altimeter read, when an aircraft is on a particular runway and the pilot has set the altimeter to the local QFE?
- (a) 0 ft.
 - (b) The altitude of the airport.
 - (c) The altitude of the runway.
 - (d) Further data is required to anticipate the altimeter reading.
78. How does the GNSS find the receiver's position?
- (a) By measuring the pseudoranges between the GNSS satellite and the receiver.
 - (b) By communicating with GNSS satellite, receiving the position information.
 - (c) By measuring the geomagnetic north, finding the position.
 - (d) By using the record of the flight acceleration and velocity.
79. Which of the following statements is TRUE regarding the Mach number?
- (a) All the others are true.
 - (b) When descending at constant Mach, the TAS increases.
 - (c) When climbing at constant TAS, the Mach number increases.
 - (d) At constant altitude, when TAS increases the Mach number increases.
80. What is TRUE for an aircraft navigating VOR with HDG 225°.
- (a) Describes a SW track.
 - (b) Flies SE when the VOR indicator is FROM.
 - (c) Only can fly at R45 or R225.
 - (d) All are correct.

BLO-1: Flight Principles

1. Within the ISA troposphere, as altitude decreases...
 - (a) Temperature increases linearly.
 - (b) Temperature increases non-linearly.
 - (c) The speed of sound decreases non-linearly.
 - (d) None of the others.
2. What is true about an aerofoil with zero camber?
 - (a) Its zero-lift angle of attack is zero degrees.
 - (b) It produces lift at all angles of attack.
 - (c) It is not symmetric.
 - (d) It must always fly in straight flight.
3. The use of flaps...
 - (a) increases C_L and C_D
 - (b) is indicated for turns.
 - (c) generates higher downforces.
 - (d) increases C_L by reducing C_D .
4. Dihedral and sweep are...
 - (a) wing angles that affect stability.
 - (b) wing angles that affect lift.
 - (c) negative for better stability.
 - (d) positive to improve maneuverability.
5. For a fluid, Bernoulli's principle states that along a streamline...
 - (a) speed increase is associated with pressure decrease.
 - (b) speed and pressure are proportional.
 - (c) speed decrease is associated with pressure decrease.
 - (d) None of the others.
6. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 40^\circ$) and speed $V = 75 \text{ m/s}$. How long will it take to complete a 90° turn (t)? (take $g \simeq 9.8 \text{ m/s}^2$)
 - (a) $t = 14.3 \text{ s}$.
 - (b) $t = 12.2 \text{ s}$.
 - (c) $t = 11.3 \text{ s}$.
 - (d) $t = 19 \text{ s}$.
7. An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.3 + 0.4\alpha$ with $\alpha_S = 12^\circ$ and $C_D = 0.01 + 0.03C_L^2$. With full flaps, the coefficients change to: $C_L^f = 0.4 + 0.4\alpha$ with $\alpha_S = 12^\circ$ and $C_D^f = 0.02 + 0.02C_L^2$. What is the value of the angle of attack for optimal flap removal?
 - (a) $\alpha = 1.50^\circ$.
 - (b) $\alpha = 1.75^\circ$.
 - (c) $\alpha = 1.08^\circ$.
 - (d) $\alpha = 1.99^\circ$.
8. An aerobatics airplane with lift coefficient $C_L = 0.02 + 0.15\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 50 \text{ m/s}$ with $\alpha = 1^\circ$. What should the speed (V_{if}) be to perform a balanced inverted flight at the same altitude and angle of attack $\alpha = -1^\circ$?
 - (a) $V_{if} = 57.2 \text{ m/s}$.
 - (b) $V_{if} = 61.3 \text{ m/s}$.
 - (c) $V_{if} = 68.3 \text{ m/s}$.
 - (d) $V_{if} = 41.4 \text{ m/s}$.
9. Calculate the stall speed V_S of an airplane of mass $m = 1200 \text{ kg}$ and wing surface $S = 8 \text{ m}^2$ with $C_L = 0.2 + 0.2\alpha$ with $\alpha_S = 12^\circ$ at an altitude such that $\bar{\rho} = 0.65$ (take $g = 9.8 \text{ m/s}^2$ and $\rho_0 = 1.225 \text{ kg/m}^3$).
 - (a) $V_S = 37.7 \text{ m/s}^2$.
 - (b) $V_S = 25.2 \text{ m/s}^2$.
 - (c) $V_S = 42.9 \text{ m/s}^2$.
 - (d) $V_S = 31.3 \text{ m/s}^2$.
10. An airplane of mass $m = 1800 \text{ kg}$ is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 16 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8 \text{ m/s}^2$)
 - (a) $D = 1100 \text{ N}$ and $D_p \neq D_i$.
 - (b) $D = 1100 \text{ N}$ and $D_p = D_i = 551 \text{ N}$.
 - (c) $D = 2210 \text{ N}$ and $D_p = D_i = 1100 \text{ N}$.
 - (d) $D = 2210 \text{ N}$ and $D_p \neq D_i$.

BLO-2: Aircraft Stability & Control

11. Ailerons, elevator and rudder are meant to generate moments around...
- (a) roll, pitch and yaw axes, respectively.
 - (b) pitch, yaw and roll axes, respectively.
 - (c) roll, yaw and pitch axes, respectively.
 - (d) pitch, roll and yaw axes, respectively.
12. Which of the following is not considered a secondary control surface?
- (a) Elevator.
 - (b) Slats.
 - (c) Flaps.
 - (d) Trim.
13. Which of the following guarantees positive longitudinal static stability?
- (a) Center of Gravity ahead of Neutral Point.
 - (b) Center of Gravity on Neutral Point.
 - (c) Center of Gravity behind Neutral Point.
 - (d) None of the others.
14. Improving aircraft stability results in...
- (a) less manoeuvrability.
 - (b) more manoeuvrability.
 - (c) no effect on manoeuvrability.
 - (d) increased endurance.
15. What is true about the elevator trim tab?
- (a) It helps balancing the airplane in longitudinal flight without having to hold the stick or yoke permanently.
 - (b) It is the primary control surface to pilot the airplane in longitudinal flight.
 - (c) It guarantees longitudinal stability of the airplane.
 - (d) It is in charge of producing the lift to fly the airplane.
16. Which of the following is not a lateral flight mode?
- (a) Pitch oscillation.
 - (b) Dutch Roll.
 - (c) Roll subsidence.
 - (d) Spiral mode.
17. Which of the following wing parameters affect lateral stability?
- (a) Dihedral and sweep.
 - (b) Aspect ratio and camber.
 - (c) Taper and twist.
 - (d) Span and chord.
18. How should the pilot of a twin-engined airplane correct for excessive sideslip in case of left engine failure?
- (a) Press the right pedal.
 - (b) Press the left pedal.
 - (c) Switch off the right engine.
 - (d) Press both pedals simultaneously.
19. Which of the two possible straight and level flight regimes is easier to fly stably?
- (a) R1.
 - (b) R2.
 - (c) Both R1 and R2.
 - (d) Neither R1 nor R2.
20. A tailplane...
- (a) usually subtracts some lift from that generated by the wings.
 - (b) usually adds lift to that generated by the wings.
 - (c) is usually meant to generate a nose-down moment.
 - (d) is required for positive lateral stability.

BLO-3: Aircraft Performances

21. In a constant-altitude coordinated banked turn...
- lift outweighs weight.
 - lift equals weight.
 - weight outweighs lift.
 - the sense of rotation (left or right) determines whether lift or weight is higher.
22. What is true about take-off distance?
- It increases with aircraft weight.
 - It decreases with tailwind.
 - It is not affected by airport's altitude.
 - It increases with aircraft engine power.
23. Which statement is FALSE about climbs?
- The best rate of climb corresponds to maximum excess thrust.
 - The best angle of climb corresponds to maximum excess thrust.
 - Both best rate and best angle optimum speeds are dependent on engine type.
 - Both best angle and best rate of climb are zero at the absolute ceiling.
24. In minimum drag straight and level flight conditions...
- parasite drag equals induced drag.
 - the lift coefficient is maximum.
 - the drag coefficient is minimum.
 - induced drag is twice the parasite drag.
25. How is maximum gliding time achieved?
- With minimum required power.
 - With minimum required power for propeller-powered but not jet-powered aircraft.
 - With maximum drag conditions.
 - With minimum drag conditions.
26. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 35$. What is the minimum needed height (above terrain) to reach a horizontal distance of $d = 44$ km?
- $h = 1257$ m.
 - $h = 1540$ m
 - $h = 1257$ km
 - $h = 1540$ km
27. The required thrust of a jet airplane is given by $T_R = 0.32\bar{\rho}V^2 + 0.018W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1750$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D = 2600$ N.
 - $D = 3010$ N.
 - $D = 2740$ N.
 - $D = 2870$ N.
28. A certain aircraft of mass $m = 2000$ kg has a best climb angle at sea level $\gamma = 3.58^\circ$ at a speed $V = 90$ m/s. Calculate the excess power ΔP required in these conditions. (take $g \simeq 9.8$ m/s²)
- $\Delta P = 110$ kW.
 - $\Delta P = 135$ kW.
 - $\Delta P = 91$ kW.
 - $\Delta P = 149$ kW.
29. The required thrust of a jet-engined airplane of mass $m = 1600$ kg, wing area $S = 8$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.3\bar{\rho}V^2 + 0.035W^2/\bar{\rho}V^2$ (SI units), and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 8000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at an altitude such that $\bar{\rho} = 0.85$? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 51.9$ m/s
 - $V \simeq 39.8$ m/s
 - $V \simeq 47$ m/s
 - $V \simeq 33.6$ m/s
30. The required thrust of a jet airplane is given by $T_R = 0.4\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$ (SI). The airplane has mass $m = 1300$ kg. Calculate the minimum drag D_{min} flying at sea level (take $g = 9.8$ m/s², $\rho_0 = 1.225$ kg/m³)
- $D_{min} = 2548$ N.
 - $D_{min} = 2805$ N.
 - $D_{min} = 3197$ N.
 - $D_{min} = 2243$ N.

BLO-4: Aircraft Recognition / Materials & Structures

31. According to ICAO, how is an A320-200 (78 tons MTOW) classified in terms of wake turbulence?

- (a) M: Medium.
- (b) S: Super.
- (c) L: Light.
- (d) H: Heavy.

32. Choose the correct ICAO definition for a Cessna 172 Skyhawk.

- (a) L1P.
- (b) A1P.
- (c) L1J.
- (d) A2P.

33. Identify the following aircraft



- (a) Airbus A300.
- (b) Airbus 340.
- (c) Boeing DC-10.
- (d) Embraer E-175.

34. Identify the following aircraft



- (a) Boeing DC-10.
- (b) Airbus A340.
- (c) Airbus A300.
- (d) Embraer E-175.

35. Which of the following aircraft families does not include non-existing airplanes?

- (a) Airbus A300, A310, A318, A319, A320, A321, A330, A340, A350, A380.
- (b) Boeing B707, B717, B727, B737, B757, B767, B777, B787, B797.
- (c) Boeing B707, B717, B727, B737, B747, B757, B767, B777, B787, B797.
- (d) Airbus A300, A310, A318, A319, A320, A321, A322, A330, A340, A350, A360, A380.

36. Young's Modulus is related to...

- (a) elastic deformation.
- (b) plastic deformation.
- (c) the number of tension cycles a longeron can withstand before catastrophic failure.
- (d) the amount of energy absorbed in an impact.

37. The upper and lower surfaces of a wing of an airplane that is stopped on ground are subject to...

- (a) tension and compression, respectively.
- (b) compression and tension, respectively.
- (c) tension.
- (d) compression.

38. Which of the following materials has typically the highest thermal resistance?

- (a) Ceramic material.
- (b) Polymers.
- (c) Aluminum alloys.
- (d) None of the other.

39. What is the approximate melting point of aluminium?

- (a) 660°C.
- (b) 1100°C.
- (c) 1500°C.
- (d) 911°C.

40. A structural composite material...

- (a) combines the better compression properties of the matrix with the better tension properties of the fibers.
- (b) combines the better tension properties of the matrix with the better compression properties of the fibers.
- (c) is highly anisotropic if layers of fibers are aligned in alternated directions.
- (d) None of the others.

BLO-5: Aerospace Propulsion

41. Which of the following is NOT a continuous flow engine?
- (a) Alternative engine.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Turboprop.
42. In a turbofan engine, what is the by-pass ratio?
- (a) The ratio of air massflow through the *cold* duct to that entering the core engine (*hot*).
 - (b) The ratio of fan blade count to compressor blade count.
 - (c) The volumetric compression ratio of the flow.
 - (d) The ratio of maximum to minimum nozzle cross-section areas.
43. A turbofan engine has a typical fuel mass flow $m_f = 5 \text{ kg/s}$ and a thrust $T = 250 \text{ kN}$. Calculate its TSFC.
- (a) $TSFC = 20 \text{ (g/s)/kN}$
 - (b) $TSFC = 0.020 \text{ (g/s)/kN}$
 - (c) $TSFC = 50 \text{ (g/s)/kN}$
 - (d) $TSFC = 45 \text{ (g/s)/kN}$
44. Sort the following engine types by decreasing maximum achievable speed.
- (a) Ramjet, turboprop, turboshaft for helicopter application.
 - (b) Turbofan, turboshaft for helicopter application, piston-engine-driven propeller.
 - (c) Rocket, turboprop, turbofan.
 - (d) Turbofan, turbojet, ramjet.
45. Which thermodynamic cycle is usually implemented by alternative engines?
- (a) Diesel or Otto.
 - (b) Rankine.
 - (c) Ericsson.
 - (d) Brayton.
46. Turbojet engines usually have a convergent nozzle
- (a) All are correct.
 - (b) To accelerate the speed of exhaust gasses.
 - (c) To increase thrust.
 - (d) After the turbine.
47. What can be said about propulsive power?
- (a) It's zero at the beginning of the take-off run.
 - (b) It peaks at the beginning of the take-off run.
 - (c) It depends on thermal efficiency.
 - (d) None of the others.
48. What happens to turbojet thrust when external air temperature rises?
- (a) It decreases.
 - (b) It increases.
 - (c) External air temperature has no effect on thrust.
 - (d) What happens depends on the bypass ratio.
49. Which of the following statements is true regarding low bypass ratio turbofans?
- (a) Their optimal speed is $M = 1 \sim 2$.
 - (b) All are correct.
 - (c) They are mainly used for low range airliners.
 - (d) They generate a compression ratio of around 10-15.
50. Radial compressors are typically used for...
- (a) small engines.
 - (b) large engines.
 - (c) turbofan engines.
 - (d) rocket engines.

BLO-6: Aircraft Systems

51. An electro-hydraulic FCS...
- (a) is also known as Fly-by-wire.
 - (b) is heavier than a hydro-mechanical FCS.
 - (c) still requires a mechanical link to control surfaces.
 - (d) All are correct.
52. What are the primary cockpit controls on an Airbus A320?
- (a) 2 sidesticks and 2 pairs of pedals.
 - (b) 2 yokes and 2 pairs of pedals.
 - (c) 2 rudder controls and 2 elevator controls.
 - (d) 2 trims and 2 speedbrakes.
53. Which of the following is not a function of the APU?
- (a) Pump fuel to engines.
 - (b) Supply bleed air.
 - (c) Back up electrical system.
 - (d) Back up air conditioning system.
54. How many hydraulic systems/circuits does an A320 have?
- (a) 3 not fully redundant.
 - (b) 3 fully redundant.
 - (c) 2 independent + 1 redundant.
 - (d) 2 redundant + 1 independent.
55. Anti-ice system circulates bleed air through ...
- (a) wings and engine intakes.
 - (b) cabin and wings.
 - (c) engine intakes and APU.
 - (d) APU and cabin.
56. An AC electrical system is based on
- (a) AC 115/200V 400 Hz - DC 28V.
 - (b) DC 115/200V 400 Hz - AC 28V.
 - (c) AC 1 230V 50Hz - DC 12V.
 - (d) AC 3 380/50 Hz - DC 24V.
57. Which of the following systems manages all engine tasks?
- (a) FADEC.
 - (b) ADIRU.
 - (c) PTU.
 - (d) FMGC.
58. Which of the following is NOT a function of the Pneumatic System?
- (a) Fuel cooling.
 - (b) Wing anti-icing.
 - (c) Engine starting.
 - (d) Hydraulic reservoir pressurisation.
59. What is the power transfer unit (PTU) for?
- (a) To transfer pressure among hydraulic circuits.
 - (b) To transfer pressurised fluid among hydraulic circuits.
 - (c) To transfer electrical power among electrical circuits.
 - (d) To transfer pneumatic power between the two packs of the pneumatic system.
60. Which of the following is not part of the electric system?
- (a) LP compressor.
 - (b) ESS Bus.
 - (c) Battery.
 - (d) Transformer.

BLO-7: Helicopters & Space Systems

61. A rotorcraft with a unique powered rotor to provide forces in the longitudinal and vertical axes is called...
- (a) helicopter.
 - (b) aerodyne.
 - (c) gyrodyne.
 - (d) dynocopter.
62. A helicopter performing a lateral flight...
- (a) has different lift distribution along different blades.
 - (b) has a zero specific fuel consumption.
 - (c) requires no tail compensation.
 - (d) cannot stop moving laterally.
63. Tail rotor compensation...
- (a) uses around 10% of the engine power.
 - (b) requires the installation of a second dedicated engine.
 - (c) is not possible with gas turbine engine helicopters.
 - (d) is not possible on piston-based engine helicopters.
64. What is the main restriction regarding minimum allowable blade tip speed for a typical helicopter?
- (a) Kinetic energy and stall effects.
 - (b) Compressibility and noise effects.
 - (c) Compressibility and stall effects.
 - (d) None of the others.
65. What can be done in case of engine failure while flying a helicopter?
- (a) To perform an autorotation maneuver.
 - (b) There is no way out (Prayer might work for believers).
 - (c) Keep flying at maximum efficiency conditions for best glide.
 - (d) Increase lift by actuating the collective control.
66. What kind of orbit is the ISS (International Space Station) on?
- (a) LEO.
 - (b) Polar.
 - (c) MEO.
 - (d) GEO.
67. What is the specific impulse of a rocket engine?
- (a) The thrust per unit massflow of ejected gases.
 - (b) The time it takes to burn all of the fuel.
 - (c) The energy contained in the fuel divided by the power.
 - (d) The maximum time it takes to reach full speed.
68. Which propellant does ESA launcher Ariane 5 use?
- (a) Powder and $\text{LH}_2 + \text{LO}_2$.
 - (b) $\text{LH}_2 + \text{LO}_2$.
 - (c) Powder.
 - (d) Hydrazine + O_2 .
69. What is the advantage of launching space missions Equatorial locations?
- (a) ΔV is already large when launching eastwards.
 - (b) ΔV is already small when launching westwards.
 - (c) ΔV is already large when launching northwards to intercept Polar Orbits.
 - (d) All options are correct.
70. A rocket with initial mass of 20 t and final mass of 3.5 t has a specific impulse of 400 s. What is its ΔV ?
- (a) $\Delta V = 6.8 \text{ km/s}$.
 - (b) $\Delta V = 7.6 \text{ km/s}$.
 - (c) $\Delta V = 4.2 \text{ km/s}$.
 - (d) None of the others.

BLO-8: Navigation & Altimetry

71. IAS (Indicated Air Speed)...
- (a) is the raw estimation from the pitot-static tube reading.
 - (b) includes correction for compressibility but not for density.
 - (c) includes more corrections than EAS.
 - (d) is the speed of the aircraft relative to the ground.
72. "The ability of the total system to perform its function without unscheduled interruptions during the intended operation" is the definition of...
- (a) continuity.
 - (b) accuracy.
 - (c) integrity.
 - (d) availability.
73. Which of the following navigation aids does not provide azimuthal guidance?
- (a) DME.
 - (b) NDB.
 - (c) Localizer.
 - (d) GNSS.
74. What is FALSE when using the QNH calibration setting in the altimeter?
- (a) Flight levels are used instead of altitudes.
 - (b) Calibration depends on an actual measured pressure.
 - (c) The altimeter will read zero at sea level.
 - (d) It is used while below the transition altitude.
75. GPS typical accuracy is up to...
- (a) around 20 m.
 - (b) around 3°.
 - (c) around 0.1 NM.
 - (d) around 1/10 rad.
76. A SBAS augmentation system requires...
- (a) Satellite-based and ground-based extra facilities.
 - (b) Satellite-based extra facilities.
 - (c) Satellite-based and air-based extra facilities.
 - (d) Ground-based and air-based extra facilities.
77. What is true about an INS (Inertial Navigation System)?
- (a) It must be initially aligned/calibrated.
 - (b) It determines its position by using external radio signals.
 - (c) Its accuracy does not degrade with elapsed time.
 - (d) None of the others.
78. If we say that DME accuracy is 185 meters, we are saying that...
- (a) 95% of the time the navigation error is equal or smaller than 185 m.
 - (b) The standard deviation of the error is almost 200 m.
 - (c) Its accuracy is lost when the AC is at 185 m or less.
 - (d) Its integrity is lost when the AC is at 185 m or less.
79. Which statement is true about radars?
- (a) A Secondary Surveillance Radar needs data communication with the aircraft.
 - (b) A Primary Surveillance Radar assigns a 4 digit transponder number, called "squawk".
 - (c) Radars work with ILS.
 - (d) None is correct.
80. What is Dead Reckoning?
- (a) The process of calculating the current position from a previously known position and an estimation of subsequent speeds, directions and elapsed time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) An airship-based radio navigation aid.
 - (d) A generic name for ground stations.

Correct Answers

BLO-1: Flight Principles

	Permutation Number				
Question	0	1	2	3	4
1	a	b	d	c	d
2	a	a	d	b	c
3	a	b	a	b	c
4	a	b	c	b	d
5	a	a	a	b	d
6	a	c	d	b	c
7	a	b	c	b	c
8	a	a	d	c	b
9	a	d	a	a	d
10	a	a	c	d	a

BLO-2: Aircraft Stability & Control

	Permutation Number				
Question	0	1	2	3	4
11	a	a	a	a	d
12	a	d	c	d	a
13	a	a	c	a	b
14	a	a	c	b	a
15	a	c	d	b	c
16	a	a	d	b	b
17	a	b	d	d	c
18	a	b	a	a	d
19	a	b	c	b	b
20	a	d	b	b	a

BLO-3: Aircraft Performances

	Permutation Number				
Question	0	1	2	3	4
21	a	b	c	b	b
22	a	c	c	a	b
23	a	d	c	b	a
24	a	d	b	c	a
25	a	c	c	b	b
26	a	a	b	c	d
27	a	d	c	c	b
28	a	c	d	a	d
29	a	a	b	b	b
30	a	a	a	d	a

BLO-4: Aircraft Recognition / Materials & Structures

	Permutation Number				
Question	0	1	2	3	4
31	a	a	a	d	b
32	a	c	b	c	c
33	a	b	c	b	b
34	a	a	c	b	a
35	a	c	d	c	c
36	a	c	d	d	c
37	a	a	d	d	a
38	a	d	a	b	a
39	a	a	b	b	c
40	a	c	b	b	d

BLO-5: Aerospace Propulsion

	Permutation Number				
Question	0	1	2	3	4
41	a	c	b	b	b
42	a	a	b	d	d
43	a	d	c	d	c
44	a	d	d	b	d
45	a	c	d	a	b
46	a	d	b	c	a
47	a	c	a	a	b
48	a	a	c	a	c
49	a	a	d	c	b
50	a	a	a	d	c

BLO-6: Aircraft Systems

	Permutation Number				
Question	0	1	2	3	4
51	a	d	d	c	a
52	a	d	b	a	a
53	a	b	b	a	d
54	a	c	c	a	d
55	a	b	d	b	b
56	a	d	c	d	a
57	a	b	b	b	b
58	a	b	d	c	b
59	a	c	a	a	b
60	a	a	b	c	c

BLO-7: Helicopters & Space Systems

	Permutation Number				
Question	0	1	2	3	4
61	a	c	c	a	c
62	a	d	a	b	c
63	a	d	b	c	a
64	a	d	b	b	d
65	a	b	a	b	a
66	a	c	a	a	c
67	a	d	c	a	c
68	a	d	b	b	c
69	a	b	a	c	a
70	a	a	b	b	d

BLO-8: Navigation & Altimetry

	Permutation Number				
Question	0	1	2	3	4
71	a	b	c	b	c
72	a	b	a	c	a
73	a	b	b	c	d
74	a	c	b	d	b
75	a	c	a	c	b
76	a	b	b	d	c
77	a	d	d	d	d
78	a	a	d	a	d
79	a	d	d	d	b
80	a	c	b	c	b

BLO-1: Flight Principles

1. Within the ISA troposphere, as altitude increases...
 - (a) pressure decreases non-linearly.
 - (b) temperature decreases non-linearly.
 - (c) the speed of sound increases non-linearly.
 - (d) None of the others.
2. What is true about an aerofoil with non-zero camber?
 - (a) It is not symmetric.
 - (b) It produces lift at all angles of attack.
 - (c) Its zero-lift angle of attack is zero degrees.
 - (d) It must always fly nose-down.
3. The deployment of flaps...
 - (a) increases C_L at all pre-stall angles of attack.
 - (b) increases C_D at all pre-stall angles of attack.
 - (c) is indicated for turns.
 - (d) is indicated for high speed flight.
4. The angle of attack of an airfoil is defined as the angle between...
 - (a) airspeed direction and chord line.
 - (b) the horizontal plane and airspeed direction.
 - (c) chord line and the horizontal plane.
 - (d) the upper and lower surfaces.
5. For a fluid, Bernoulli's principle states that along a streamline...
 - (a) speed increase is associated with pressure decrease.
 - (b) speed and pressure are proportional.
 - (c) speed decrease is associated with pressure decrease.
 - (d) None of the others.
6. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 40^\circ$) and speed $V = 75 \text{ m/s}$. How long will it take to complete a 90° turn (t)? (take $g \simeq 9.8 \text{ m/s}^2$)
 - (a) $t = 14.3 \text{ s}$
 - (b) $t = 16.4 \text{ s}$
 - (c) $t = 11.1 \text{ s}$
 - (d) $t = 18.8 \text{ s}$
7. An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.2 + 0.3\alpha$ with $\alpha_S = 10^\circ$ and $C_D = 0.015 + 0.04C_L^2$. With full flaps, the coefficients change to: $C_L^f = 0.3 + 0.3\alpha$ with $\alpha_S = 10^\circ$ and $C_D^f = 0.025 + 0.03C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - (a) $\alpha = 2.67^\circ$
 - (b) $\alpha = 2.21^\circ$
 - (c) $\alpha = 3.08^\circ$
 - (d) $\alpha = 1.99^\circ$
8. An aircraft with a drag polar $C_D = 0.022 + 0.05C_L^2$. Which of the following conditions corresponds to maximum aerodynamic efficiency?
 - (a) All options are correct.
 - (b) $C_L = 0.663$.
 - (c) $C_D = 0.044$.
 - (d) $f = 15.1$.
9. Calculate the stall speed V_S of an airplane of mass $m = 1800 \text{ kg}$ and wing surface $S = 12 \text{ m}^2$ with $C_L = 0.3 + 0.1\alpha$ with $\alpha_S = 10^\circ$ at all altitude such that $\bar{\rho} = 0.75$ (take $g = 9.8 \text{ m/s}^2$ and $\rho_0 = 1.225 \text{ kg/m}^3$).
 - (a) $V_S = 49.61 \text{ m/s}^2$
 - (b) $V_S = 58.05 \text{ m/s}^2$
 - (c) $V_S = 42.97 \text{ m/s}^2$
 - (d) $V_S = 53.31 \text{ m/s}^2$
10. Estimate the mass of an aircraft flying in straight and level flight with aerodynamic efficiency $f = 20$ and drag $D = 1500 \text{ N}$ (take $g = 9.8 \text{ m/s}^2$).
 - (a) $m = 3060 \text{ kg}$
 - (b) $m = 3210 \text{ kg}$
 - (c) $m = 2920 \text{ kg}$
 - (d) $m = 1580 \text{ kg}$

BLO-2: Aircraft Stability & Control

11. Elevator, ailerons and rudder are meant to generate moments around the...

- (a) pitch, roll and yaw axes, respectively
- (b) pitch, yaw and roll axes, respectively
- (c) roll, yaw and pitch axes, respectively
- (d) roll, pitch and yaw axes, respectively

12. Which of the following is not considered a primary control surface?

- (a) Slot.
- (b) Ruddervator.
- (c) Elevator.
- (d) Aileron.

13. Which of the following guarantees positive longitudinal static stability?

- (a) Center of Gravity ahead of Neutral Point.
- (b) Center of Gravity on Neutral Point.
- (c) Center of Gravity behind Neutral Point.
- (d) None of the others.

14. A stable spiral mode is observed when...

- (a) The aircraft recovers from a banked turn.
- (b) The aircraft recovers from a perfect dive.
- (c) The aircraft recovers from a Roll subsidence.
- (d) The aircraft recovers from a Dutch Roll.

15. What is true about the elevator trim tab?

- (a) It helps balancing the airplane in longitudinal flight without having to hold the stick or yoke permanently.
- (b) It is the primary control surface to pilot the airplane in longitudinal flight.
- (c) It guarantees longitudinal stability of the airplane.
- (d) It is in charge of producing the lift to fly the airplane.

16. Which of the following is a lateral flight mode?

- (a) Dutch Roll.
- (b) Pitch oscillation.
- (c) Return to Straight & Level Flight.
- (d) Phugoid.

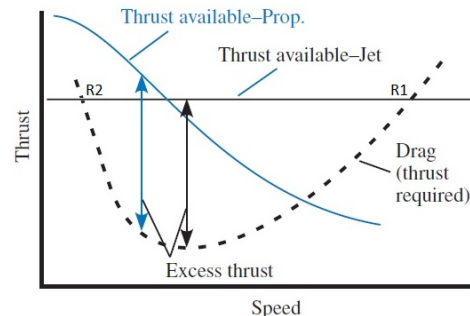
17. Which of the following is a longitudinal mode?

- (a) Phugoid.
- (b) Spiral mode.
- (c) Dutch Roll.
- (d) Roll Subsidence.

18. Which of the following Secondary Control Surfaces can be designed to be used as a backup for ailerons?

- (a) Spoilers.
- (b) Slots.
- (c) Winglets.
- (d) Elevators.

19. In which of the two possible straight and level flight regimes, stall may occur?



- (a) R2.
- (b) R1.
- (c) Both R1 and R2.
- (d) Neither R1 nor R2.

20. When a canard-configuration is chosen to stabilise longitudinal flight, what is the main contribution of canards?

- (a) Compensating a natural nose-down pitching moment.
- (b) Subtracting lift from the main wings.
- (c) Compensating a natural nose-up pitching moment.
- (d) None of the others.

BLO-3: Aircraft Performances

21. Which parameters have an influence on landing distance?
- All of them.
 - Full flaps stall speed.
 - Airport altitude.
 - Braking power.
22. What is true about optimum climb?
- The best rate of climb corresponds to maximum excess power.
 - The best climb angle corresponds to minimum drag.
 - The best climb angle corresponds to maximum excess power.
 - The best rate of climb corresponds to maximum excess thrust.
23. Which of the following performance optima are not affected by wind?
- Endurance and maximum rate of climb.
 - Range and maximum rate of climb.
 - Endurance and maximum climb angle.
 - Range and maximum climb angle.
24. Minimum drag in straight and level flight is obtained when...
- parasite drag equals induced drag.
 - lift is maximum.
 - aircraft flies at minimum height.
 - induced drag is twice the parasite drag.
25. What happens to D_{min} as a constant mass airplane with parabolic drag polar gains altitude?
- D_{min} remains constant at any altitude.
 - D_{min} increases with altitude.
 - D_{min} decreases with altitude.
 - D_{min} occurs at lower speeds as altitude increases.
26. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 42$. If its current height (above terrain) is $h = 1900$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 79.8$ km
 - $d = 43.9$ km
 - $d = 42.0$ km
 - $d = 19.0$ km
27. A jet airplane of mass $m = 4500$ kg has a required thrust given by $T_R = 0.55\bar{\rho}V^2 + 0.0283W^2/\bar{\rho}V^2$ (SI). Its available thrust is $T_A = 2.5 \times 10^4$ N. Calculate the best climb angle that the aircraft can reach at sea level. (take $g \simeq 9.8$ m/s²)
- $\gamma = 18.51^\circ$
 - $\gamma = 14.48^\circ$
 - $\gamma = 16.25^\circ$
 - $\gamma = 11.52^\circ$
28. A certain aircraft of mass $m = 2500$ kg has a best climb angle at sea level of $\gamma = 10^\circ$ at a speed of $V = 100$ m/s. Calculate the rate of climb in these conditions.
- $(R/C) = 17.36$ m/s
 - $(R/C) = 14.84$ m/s
 - $(R/C) = 23.07$ m/s
 - $(R/C) = 19.25$ m/s
29. The required thrust of a jet-engined airplane of mass $m = 1950$ kg, wing area $S = 10$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.1$, is given by $T_R = 0.27\bar{\rho}V^2 + 0.0315W^2/\bar{\rho}V^2$ (SI units), and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 10000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.75$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 61.5$ m/s
 - $V \simeq 46.6$ m/s
 - $V \simeq 70.2$ m/s
 - $V \simeq 51.7$ m/s
30. What is the flight speed that maximises range of a propeller-powered aircraft with $P_R = 0.2\bar{\rho}V^3 + 0.04W^2/\bar{\rho}V$ (SI units) and mass $m = 1300$ kg at an altitude such that $\bar{\rho} = 0.6$. (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V = 97.4$ m/s
 - $V = 72.1$ m/s
 - $V = 106.8$ m/s
 - $V = 88.6$ m/s

BLO-4: Aircraft Recognition / Materials & Structures

31. According to FAA and CAA, which aircraft is considered as "Super" in terms of wake turbulence?

- (a) A380 only.
- (b) All aircraft over 136000 kg.
- (c) All aircraft over 140000 kg.
- (d) There is no Super category in FAA or CAA.

32. As a description of an aircraft, L4T means?

- (a) Land plane, 4 engines, turboprop powered.
- (b) Light plane, 4 engines, turboprop powered
- (c) Land plane, 4 engines, turbojet powered.
- (d) Light plane, 4 engines, turbojet powered.

33. Identify the following aircraft



- (a) Cessna Skyhawk.
- (b) Beechcraft Bonanza.
- (c) Piper Cherokee.
- (d) Extra 300S.

34. Identify the following aircraft



- (a) Airbus A330.
- (b) Airbus A340.
- (c) Boeing B777.
- (d) Boeing B787.

35. Which of the following aircraft are double-deck?

- (a) Boeing 747 and Airbus 380.
- (b) Boeing 747 and Airbus 300.
- (c) Boeing 787 and Airbus 330.
- (d) Boeing 787 and Airbus 340.

36. The simplest schematisation of a wing from a structural point of view is...

- (a) a cantilevered beam with one end fixed to the fuselage and the other end free.
- (b) a beam with a fixed end at one side and an articulated end at the other.
- (c) a beam articulated at both ends.
- (d) None of the others.

37. What is true about titanium alloys?

- (a) They provide excellent corrosion resistance and high temperature behaviour.
- (b) They cannot be used in any aeronautical application due to poor mechanical performances.
- (c) They are cheap alloys.
- (d) Their specific mechanical properties are worse than those of steel.

38. What can be said about aluminium?

- (a) All options are correct.
- (b) It can be alloyed.
- (c) It is almost three times lighter than steel.
- (d) It has good corrosion resistance.

39. What is the approximate melting point of steel?

- (a) 1500°C.
- (b) 1100°C.
- (c) 660°C.
- (d) 911°C.

40. Honeycomb is a structure that might be adopted by...

- (a) composite materials.
- (b) ceramic materials.
- (c) metallic materials.
- (d) polymeric materials.

Correct Answers

BLO-1: Flight Principles

	Permutation Number				
Question	0	1	2	3	4
1	a	c	c	a	a
2	a	a	b	c	d
3	a	b	b	b	c
4	a	c	a	d	d
5	a	d	d	c	d
6	a	c	a	d	a
7	a	a	d	d	a
8	a	b	c	c	d
9	a	a	d	c	b
10	a	c	d	d	c

BLO-2: Aircraft Stability & Control

	Permutation Number				
Question	0	1	2	3	4
11	a	c	d	a	a
12	a	d	b	b	a
13	a	a	d	c	d
14	a	b	d	b	a
15	a	b	c	c	c
16	a	b	b	c	b
17	a	b	b	b	c
18	a	c	b	d	c
19	a	c	c	b	a
20	a	d	c	d	d

BLO-3: Aircraft Performances

	Permutation Number				
Question	0	1	2	3	4
21	a	a	a	b	a
22	a	d	b	b	a
23	a	c	d	c	a
24	a	d	d	b	b
25	a	b	b	d	a
26	a	d	d	b	c
27	a	a	b	a	d
28	a	d	c	d	a
29	a	c	c	c	c
30	a	a	d	c	d

BLO-4: Aircraft Recognition / Materials & Structures

	Permutation Number				
Question	0	1	2	3	4
31	a	a	b	d	a
32	a	b	b	a	b
33	a	d	a	c	d
34	a	c	b	c	a
35	a	b	c	b	c
36	a	c	d	d	a
37	a	d	a	c	a
38	a	b	b	d	a
39	a	c	b	b	c
40	a	a	c	c	d

BLO-1: Flight Principles

1. Dihedral and sweep are...
 - (a) angles defining the wing configuration.
 - (b) angles defining the airfoil configuration.
 - (c) lengths defining the wing configuration.
 - (d) lengths defining the airfoil configuration.
2. What is true about the ISA atmosphere?
 - (a) Temperature is constant in the low stratosphere.
 - (b) Temperature grows linearly in the troposphere at a rate of $5^\circ\text{C}/\text{km}$.
 - (c) Pressure grows linearly at a rate of $6.5\text{ Pa}/\text{km}$.
 - (d) None of the others.
3. A symmetric airfoil...
 - (a) has a $C_{L_0} = 0$
 - (b) produces more lift than a cambered airfoil at the same angle of attack.
 - (c) withstands higher angles of attack than a cambered airfoil.
 - (d) is the usual choice for aircraft wings.
4. An airfoil with parabolic drag polar has its minimum drag coefficient in conditions of...
 - (a) zero lift coefficient.
 - (b) minimum aerodynamic efficiency.
 - (c) zero angle of attack.
 - (d) None of the others.
5. Which of the following statements is correct about aerodynamic efficiency?
 - (a) It is the lift-to-drag ratio.
 - (b) Greatly increases with altitude.
 - (c) Gliders are designed for low aerodynamic efficiency.
 - (d) It cannot exceed unity.
6. What is the lift coefficient ($C_{L_{opt}}$) for the optimal condition providing maximum aerodynamic efficiency (f_{max}) for an airplane with drag polar $C_D = 0.028 + 0.145C_L^2$?
 - (a) $C_{L_{opt}} = 0.439$
 - (b) $C_{L_{opt}} = 0.418$
 - (c) $C_{L_{opt}} = 0.482$
 - (d) $C_{L_{opt}} = 0.375$
7. An aerobatics airplane of mass $m = 1100\text{ kg}$ and wing surface $S = 14\text{ m}^2$ with drag polar $C_D = 0.075 + 0.15C_L^2$ engages in a perfect dive (vertical descent) with its engine at idle. What will the terminal dive speed (V) be? (take $g = 9.8\text{ m/s}^2$ and $\rho = 1.225\text{ kg/s}$)
 - (a) $V = 129\text{ m/s}$
 - (b) $V = 118\text{ m/s}$
 - (c) $V = 108\text{ m/s}$
 - (d) $V = 167\text{ m/s}$
8. A small airplane of unknown mass is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}). We know its speed is $V = 2.5 V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max})?
 - (a) $\phi_{max} = 80.8^\circ$
 - (b) $\phi_{max} = 82^\circ$
 - (c) $\phi_{max} = 66.4^\circ$
 - (d) It depends on the mass of the airplane.
9. The lift coefficient and drag polar of an airplane changes from $C_L = 0.18 + 0.11\alpha$ and $C_D = 0.05 + 0.075C_L^2$ with full flaps to $C_L = 0.1 + 0.11\alpha$ and $C_D = 0.04 + 0.15C_L^2$ in clean configuration. What is the value of the angle of attack (α) just before optimal flap removal?
 - (a) $\alpha = 1.68^\circ$
 - (b) $\alpha = 2.41^\circ$
 - (c) $\alpha = 2.17^\circ$
 - (d) $\alpha = 1.92^\circ$
10. An airplane of mass $m = 1900\text{ kg}$ is cruising in straight and level flight with optimal aerodynamic efficiency $f = 23 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8\text{ m/s}^2$)
 - (a) $D = 810\text{ N}$ and $D_p = D_i = 405\text{ N}$.
 - (b) $D = 810\text{ N}$ and $D_p > D_i$.
 - (c) $D = 405\text{ N}$ and $D_p = D_i = 202\text{ N}$.
 - (d) $D = 405\text{ N}$ and $D_p > D_i$.

BLO-2: Aircraft Stability & Control

11. Elevator, ailerons and rudder are meant to generate moments around which axes?
- (a) Pitch, roll and yaw, respectively.
 - (b) Pitch, yaw and roll, respectively.
 - (c) Roll, yaw and pitch, respectively.
 - (d) Roll, pitch and yaw, respectively.
12. What is the usual contribution of the tailplane to total lift and pitching moment of a stable airplane?
- (a) It subtracts lift and generates a nose-up moment.
 - (b) It adds lift and generates a nose-down moment.
 - (c) It subtracts lift and generates a nose-down moment.
 - (d) It adds lift and generates a nose-up moment
13. What is true of longitudinal static instability?
- (a) All options are correct.
 - (b) In combination with automatic stabilisation, it provides good manoeuvrability.
 - (c) It implies dynamic instability.
 - (d) It occurs whenever the neutral point is ahead of the center of gravity.
14. The dutch roll mode couples oscillations in...
- (a) roll and yaw.
 - (b) yaw and pitch.
 - (c) pitch and roll.
 - (d) roll, pitch and yaw.
15. What is the task of the elevator trim tab?
- (a) Balancing the airplane at the desired angle of attack without having to hold the elevator.
 - (b) Piloting the airplane in lateral flight.
 - (c) Stabilising the airplane.
 - (d) Producing the lift required to sustain flight.
16. Which of the following actions modifies the equilibrium velocity in longitudinal flight, once the pitch oscillation and the phugoid have been damped?
- (a) Acting on the stick.
 - (b) Acting on the thrust lever.
 - (c) Both acting on the stick or on the thrust lever.
 - (d) Neither acting on the stick nor on the thrust lever.
17. Which one of the following is an existing lateral mode?
- (a) Spiral mode.
 - (b) Phugoid.
 - (c) Roll oscillation.
 - (d) Yaw subsidence.
18. Which one of the following modes may be left unstable with no risk for aircraft integrity?
- (a) Roll subsidence.
 - (b) Pitch oscillation.
 - (c) Return to straight and level flight.
 - (d) Phugoid.
19. Which of the following is a suitable name for the coefficient $C_{n\delta_i}$?
- (a) Adverse yaw coefficient.
 - (b) Roll damping coefficient.
 - (c) Directional stability coefficient.
 - (d) Dihedral effect coefficient.
20. In which of the following situations is crosswind straight flight implemented?
- (a) Crosswind landing.
 - (b) Headwind cruise.
 - (c) Tailwind cruise.
 - (d) Whenever a pilot suffering from a stiff neck needs to look through the side window.

BLO-3: Aircraft Performances

21. Which statement is false about climbs?
- The best rate of climb corresponds to maximum excess thrust.
 - The best angle of climb corresponds to maximum excess thrust.
 - Cruise climb is related to keeping cruise optimal as the mass decreases due to fuel consumption.
 - Both best angle and best rate of climb are zero at the absolute ceiling.
22. Which of the following conditions maximises endurance (exactly or approximately) in cruising flight?
- All are correct.
 - Minimum hourly fuel consumption.
 - Minimum drag for turbojet engines.
 - Minimum drag power for propeller engines.
23. Which of the following parameters is not a limiting factor in an extreme turn?
- All are limiting factors.
 - Maximum engine power/thrust.
 - Maximum load factor.
 - Stall speed.
24. The minimum of the drag curve...
- is roughly independent of altitude.
 - is roughly independent of airplane weight.
 - mainly consists of parasitic drag.
 - mainly consists of lift-induced drag.
25. Which of the following factors reduces takeoff distance?
- Strong headwind.
 - High airport altitude.
 - Low engine power.
 - High aircraft takeoff weight.
26. The required thrust of a jet airplane is given by $T_R = 0.24\bar{\rho}V^2 + 0.021W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1600$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.3$, what is the optimal velocity for maximum range (V_R)? (take $g \simeq 9.8$ m/s²)
- $V_R = 164$ m/s
 - $V_R = 124$ m/s
 - $V_R = 94.5$ m/s
 - It depends on the available thrust T_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1400$ kg has a required power curve $P_R = 0.27\bar{\rho}V^3 + 0.031W^2/\bar{\rho}V$. What is the maximum rate of climb $((R/C)_{max})$ at sea level ($\bar{\rho} = 1$) if the available power is $P_A = 320\bar{\rho}\delta_x$ kW? (take $g \simeq 9.8$ m/s²)
- $(R/C)_{max} = 12.4$ m/s
 - $(R/C)_{max} = 10.8$ m/s
 - $(R/C)_{max} = 8.92$ m/s
 - $(R/C)_{max} = 7.78$ m/s
28. An airplane with drag polar $C_D = 0.03 + 0.14C_L^2$, wing surface $S = 35$ m² and mass $m = 1800$ kg is flying at a speed $V = 125$ m/s. What is the engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.5$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $T_R = 4420$ N
 - $T_R = 4390$ N
 - $T_R = 4360$ N
 - $T_R = 4320$ N
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 48$. If its current height (above terrain) is $h = 900$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 43.2$ km
 - $d = 18.8$ km
 - $d = 0.9$ km
 - $d = 86.4$ km
30. The required thrust of a jet-engined airplane of mass $m = 1700$ kg, wing area $S = 18$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.5$, is given by $T_R = 0.35\bar{\rho}V^2 + 0.026W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 15000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.4$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 57$ m/s
 - $V \simeq 50.2$ m/s
 - $V \simeq 53.2$ m/s
 - $V \simeq 61.3$ m/s

31. Young's Modulus has to do with...

- (a) elastic deformation.
- (b) plastic deformation.
- (c) the number of tension cycles a longeron can withstand before catastrophic failure.
- (d) the amount of energy absorbed in an impact.

32. The S-N curve is related to...

- (a) fatigue.
- (b) plasticity.
- (c) elasticity.
- (d) hardness.

33. What can be said about steel?

- (a) It is almost three times denser than Al.
- (b) It is not to be used in parts of airframe structures that are subject to high mechanical loading.
- (c) Its crystallographic structure is HCP.
- (d) It is used in over 80% of modern airframe structures.

34. Which of the following materials is commonly used as reinforcement fibers in composites?

- (a) Carbon fiber.
- (b) Polyester resin.
- (c) Vynilester resin.
- (d) Wood.

35. What can be said about fiber reinforced composites?

- (a) They have a very good resistance to fatigue.
- (b) They have low production costs.
- (c) They never experience delamination damage.
- (d) They exhibit bad resistance to corrosion.

36. To which airplane corresponds ICAO designation AT43?

- (a) ATR 42-300.
- (b) Cessna 40-300.
- (c) Boeing 747-300.
- (d) Airbus 400-300.

37. Regarding approach speed in landing procedures, which category designates helicopters?

- (a) Category H.
- (b) Category A.
- (c) Category B.
- (d) Category C.

38. Identify the following aircraft



- (a) Boeing 777.
- (b) Boeing 737.
- (c) Airbus A320.
- (d) Airbus A340.

39. Identify the following jet fighter.



- (a) BAE Sea Harrier FA2.
- (b) MiG 29.
- (c) Sukhoi SU-34.
- (d) LM F-35 Lightning.

40. Which of the following aircraft is larger?

- (a) Antonov AN-225.
- (b) Boeing 717.
- (c) MD 80.
- (d) Embraer ERJ-175.

BLO-5: Aerospace Propulsion

41. In which type of engine part of the power generated by the turbine is used to drive a propeller?
- (a) Turboprop.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Alternative engine.
42. What happens to the the thrust of a turbojet when ambient pressure rises?
- (a) It increases.
 - (b) It decreases.
 - (c) It remains approximately constant.
 - (d) It increases for centrifugal-compressor turbojets and decreases for axial-compressor turbojets.
43. Which one of the following gas turbine engines can operate at higher speed?
- (a) Turbojet.
 - (b) High bypass ratio turbofan.
 - (c) Turboshift.
 - (d) Alternative engine.
44. What can be said about turbojet engine nozzles?
- (a) All options are correct.
 - (b) They increase airflow velocity and reduce pressure.
 - (c) They can accomodate afterburning.
 - (d) Supersonic nozzles end in a divergent section.
45. Propulsive efficiency of turbojets...
- (a) depends on aircraft speed and exhaust jet velocity.
 - (b) depends on compressor type.
 - (c) is higher with tubular combustion chambers.
 - (d) reaches its maximum at very low speed.
46. Which of the following statements related to turbojet engines is true?
- (a) Airflow speed decreases and pressure increases through the intake.
 - (b) Compressor extracts energy from the air flow to drive the turbine.
 - (c) Annular is the heaviest type of combustion chamber.
 - (d) Airflow speed decreases and pressure increases through the nozzle.
47. Which of the following is not a type of combustion chamber?
- (a) Centrifugal.
 - (b) Tubular.
 - (c) Annular.
 - (d) Turbo-annular.
48. What happens to turbojet thrust when external air temperature rises?
- (a) It decreases.
 - (b) It increases.
 - (c) External air temperature has no effect on thrust.
 - (d) What happens depends on the bypass ratio.
49. A supersonic inlet is...
- (a) convergent.
 - (b) divergent.
 - (c) of constant area.
 - (d) convergent for high bypass turbofan and divergent for low bypass turbofan.
50. Which of the following turbine engines performs better at low speed?
- (a) Turboprop.
 - (b) Turbojet.
 - (c) Scramjet.
 - (d) Turbofan.

BLO-6: Aircraft Systems

51. Which of the following is not a type of flight control system configuration?
- (a) AC/DC.
 - (b) Mechanical.
 - (c) Hydro-mechanical.
 - (d) Electro-hydraulic.
52. In a fly-by-wire flight control system, the link between cockpit controls and control surfaces is...
- (a) electric.
 - (b) mechanical.
 - (c) hydro-mechanical.
 - (d) pneumatic.
53. Which of the following can be a pressurisation source for the hydraulic system?
- (a) RAT.
 - (b) Bleed valve.
 - (c) Static inverter.
 - (d) CVR.
54. Which of the following components is part of the hydraulic system?
- (a) Reservoir.
 - (b) Transformer Rectifier.
 - (c) FADEC.
 - (d) Spoiler.
55. Which of the following components is part of the electric system?
- (a) Battery.
 - (b) Oil tank.
 - (c) Hydraulic accumulator.
 - (d) ADIRU.
56. Which of the following is not part of the electric system?
- (a) LP compressor.
 - (b) ESS Bus.
 - (c) Battery.
 - (d) Transformer.
57. Which of the following systems is not supplied by the pneumatic system?
- (a) Landing lights.
 - (b) Air conditioning.
 - (c) Engine starting.
 - (d) Wing anti-icing or de-icing.
58. Which of the following is not a function of the fuel system?
- (a) Reduce fuel weight during take off.
 - (b) Defuel into the atmosphere if required.
 - (c) Pump fuel to the engines.
 - (d) Store fuel.
59. Which system will start working after a cabin depressurisation at cruise altitude following a failure in the air conditioning system?
- (a) Oxygen system.
 - (b) Fire protection system.
 - (c) Autoflight system.
 - (d) Light system.
60. Which of the following is not a subsystem of the power plant?
- (a) Oxygen system.
 - (b) Thrust reverser system.
 - (c) Oil system.
 - (d) Ignition and starting system.

BLO-7: Helicopters & Space Systems

61. An aircraft with independent propulsion in the longitudinal and vertical axes is called...
- (a) a gyrodyne.
 - (b) an aerodyne.
 - (c) an helicopter.
 - (d) a dynocopter.
62. A helicopter performing a hovering manoeuvre...
- (a) has the same lift distribution along all blades.
 - (b) has a zero specific fuel consumption.
 - (c) requires no tail compensation.
 - (d) cannot start moving laterally.
63. What is the maximum allowable blade tip speed in a typical helicopter
- (a) Close to $M = 1$.
 - (b) $M = 0.5$.
 - (c) 340 m/s.
 - (d) $0.3 \Omega R$.
64. A NOTAR helicopter...
- (a) All options are correct.
 - (b) still needs to compensate the main rotor torque.
 - (c) is only feasible with turboshaft power plants.
 - (d) circulates exhaust gases through the helicopter's tail
65. The helicopter is controlled...
- (a) by cyclic, collective and anti-torque actuators
 - (b) longitudinally by an anti-torque system.
 - (c) vertically by a cyclic action.
 - (d) by autorotation.
66. In which of the following orbits the satellite is always in the vertical of a same fixed point on the Earth surface?
- (a) GEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
67. What is the specific impulse of a rocket engine in seconds?
- (a) The thrust per unit mass flow of ejected gases.
 - (b) The time it takes to burn all of the fuel.
 - (c) The energy contained in the fuel divided by the power.
 - (d) The maximum time it takes to reach full speed.
68. Which one of the following orbits allows observation of all regions on the Earth?
- (a) Polar.
 - (b) LEO.
 - (c) GEO.
 - (d) HEO.
69. What is the advantage of launching space missions from sites on the Equator?
- (a) ΔV is already large when launching eastwards.
 - (b) ΔV is already small when launching westwards.
 - (c) ΔV is already large when launching northwards to intercept Polar Orbits.
 - (d) All options are correct.
70. A rocket with initial mass of 20 t and final mass of 3.5 t has a specific impulse of 400 s. What is its ΔV ?
- (a) $\Delta V = 6.8$ km/s.
 - (b) $\Delta V = 7.6$ km/s.
 - (c) $\Delta V = 4.2$ km/s.
 - (d) None of the others.

BLO-8: Navigation & Altimetry

71. What is the name of the radionavigation aid system that provides zenithal/elevation guidance in the approach to a runway?
- (a) Glide Slope.
 - (b) Distance Measurement Equipment.
 - (c) VHF Omnidirectional Range.
 - (d) Automatic Direction Finder.
72. What does accuracy refer to in air navigation?
- (a) To the conformance between calculated and actual position of the aircraft.
 - (b) To the issuing of an alarm whenever the aircraft cannot be guaranteed as being inside the outer tunnel.
 - (c) To the guarantee that the outer tunnel will not disappear during operation.
 - (d) To the guarantee that the tunnel will be in place when the operation starts.
73. Which of the following radionavigation aids provide azimuthal guidance with directional information?
- (a) VOR and ILS/LLZ.
 - (b) ADF/NDB.
 - (c) ILS/GS.
 - (d) DME.
74. What is true of using the STD calibration setting in the altimeter?
- (a) Indicated altitude is not real altitude.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter can stop working at random.
 - (d) All options are correct.
75. VOR accuracy is...
- (a) $\pm 1.4^\circ$.
 - (b) Less than 1° .
 - (c) 1 nautical mile.
 - (d) ± 90 m.
76. TAS speed corrects indicated speed for...
- (a) instrument installation, compressibility and density.
 - (b) instrument installation and compressibility.
 - (c) compressibility and density.
 - (d) instrument installation and density.
77. How are cruise altitudes usually given when using the STD altimeter setting?
- (a) In flight levels (FLxxx).
 - (b) In feet.
 - (c) In meters.
 - (d) In hPa.
78. Which of the following statements is false about the barometer-based altimeter calibration?
- (a) The QNH setting assumes 1013 hPa and 15°C at sea level.
 - (b) With the QNH setting, the altimeter provides the elevation of the airfield above mean sea level while the aircraft is at runway threshold.
 - (c) With the QFE setting, the altimeter provides the height above the airfield while flying in the vicinity of that airfield.
 - (d) A flight level unit is equivalent to 100 ft altitude separation as measured by the altimeter.
79. The imaginary line used to change the altimeter setting from STD to QNH is called...
- (a) Transition Level.
 - (b) Transition Altitude.
 - (c) QNH line.
 - (d) STD line.
80. What is Dead Reckoning?
- (a) The process of calculating the current position from a previously known position and an estimation of subsequent speeds, directions and elapsed time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) An airship-based radio navigation aid.
 - (d) A generic name for ground stations.

Correct Answers

BLO-1: Flight Principles

	Permutation Number				
Question	0	1	2	3	4
1	a	d	c	c	d
2	a	d	c	d	d
3	a	b	c	b	c
4	a	c	b	b	a
5	a	b	b	a	c
6	a	a	b	a	a
7	a	d	d	d	c
8	a	b	d	b	c
9	a	c	d	a	a
10	a	b	b	c	c

BLO-2: Aircraft Stability & Control

	Permutation Number				
Question	0	1	2	3	4
11	a	c	d	b	c
12	a	a	a	b	d
13	a	a	d	c	a
14	a	a	a	a	c
15	a	a	d	c	c
16	a	a	c	a	a
17	a	a	d	d	d
18	a	a	d	b	c
19	a	c	a	a	b
20	a	b	d	b	a

BLO-3: Aircraft Performances

	Permutation Number				
Question	0	1	2	3	4
21	a	b	d	b	d
22	a	c	a	d	a
23	a	c	c	c	d
24	a	b	c	c	c
25	a	b	d	b	a
26	a	d	a	b	d
27	a	c	c	b	d
28	a	a	c	d	b
29	a	b	a	d	d
30	a	c	a	a	c

BLO-4: Materials & Structures / Aircraft Recognition

	Permutation Number				
Question	0	1	2	3	4
31	a	c	d	c	a
32	a	b	a	a	d
33	a	c	c	b	d
34	a	a	b	c	b
35	a	d	d	a	b
36	a	a	c	a	d
37	a	c	a	d	d
38	a	c	b	c	d
39	a	c	c	c	c
40	a	d	d	b	d

BLO-5: Aerospace Propulsion

	Permutation Number				
Question	0	1	2	3	4
41	a	a	d	a	c
42	a	c	b	b	d
43	a	a	c	b	b
44	a	d	b	b	a
45	a	b	b	d	b
46	a	a	d	a	c
47	a	a	b	a	a
48	a	c	d	d	d
49	a	a	b	d	d
50	a	d	a	c	d

BLO-6: Aircraft Systems

	Permutation Number				
Question	0	1	2	3	4
51	a	c	c	c	b
52	a	d	c	d	b
53	a	b	a	b	c
54	a	c	a	c	b
55	a	a	c	d	c
56	a	d	d	d	c
57	a	b	b	a	a
58	a	c	d	b	b
59	a	d	b	c	c
60	a	d	d	d	b

BLO-7: Helicopters & Space Systems

	Permutation Number				
Question	0	1	2	3	4
61	a	d	d	b	a
62	a	d	d	c	b
63	a	a	d	b	c
64	a	c	c	c	b
65	a	c	c	a	d
66	a	d	a	b	a
67	a	a	c	c	c
68	a	a	d	d	d
69	a	a	d	d	c
70	a	d	c	c	c

BLO-8: Navigation & Altimetry

	Permutation Number				
Question	0	1	2	3	4
71	a	d	a	d	d
72	a	c	c	a	d
73	a	d	c	a	c
74	a	c	d	b	c
75	a	b	c	d	b
76	a	a	c	c	d
77	a	c	c	a	b
78	a	d	d	c	a
79	a	b	c	c	b
80	a	a	c	d	d

BLO-1: Flight Principles

1. Which of the following surfaces is the most commonly used high-lift device?
 - (a) Flaps.
 - (b) Slots.
 - (c) Slats.
 - (d) Flops.
2. How is the wing tip placed/oriented with respect to wing root for a plane with positive dihedral?
 - (a) Higher up.
 - (b) Further back.
 - (c) At higher incidence.
 - (d) None of the others.
3. High angles of attack...
 - (a) can induce stall.
 - (b) are desirable to fly at high altitude cruise conditions.
 - (c) correspond to low induced drag.
 - (d) are necessary to fly at high speed.
4. Which of the following statements about ISA is false?
 - (a) The speed of sound increases with altitude.
 - (b) Sea level pressure is considered the same in the Equator and the North Pole.
 - (c) Air density decreases with altitude.
 - (d) Temperature is considered constant for some altitude above the Tropopause.
5. Minimum drag coefficient is obtained for conditions of...
 - (a) None of the others.
 - (b) maximum aerodynamic efficiency.
 - (c) maximum lift coefficient.
 - (d) maximum angle of attack.
6. What is the drag coefficient ($C_{D_{opt}}$) for the optimal condition that maximises aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.02 + 0.12C_L^2$?
 - (a) $C_{D_{opt}} = 0.04$
 - (b) $C_{D_{opt}} = 0.02$
 - (c) $C_{D_{opt}} = 0.408$
 - (d) $C_{D_{opt}} = 0.05$
7. An aerobatics airplane with lift coefficient $C_L = 0.02 + 0.15\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 50$ m/s with $\alpha = 1^\circ$. What should the speed (V_{if}) be to perform a balanced inverted flight at the same altitude and angle of attack $\alpha = -1^\circ$?
 - (a) $V_{if} = 57.2$ m/s
 - (b) $V_{if} = 61.8$ m/s
 - (c) $V_{if} = 66.3$ m/s
 - (d) $V_{if} = 42.8$ m/s
8. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 40^\circ$) and speed $V = 75$ m/s. How long will it take to complete a 90° turn (t)? (take $g \simeq 9.8$ m/s²)
 - (a) $t = 14.3$ s
 - (b) $t = 12.5$ s
 - (c) $t = 11.4$ s
 - (d) $t = 19.1$ s
9. The drag polar of an airplane changes from $C_D = 0.024 + 0.122C_L^2$ in clean configuration to $C_D = 0.036 + 0.086C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.577$
 - (b) $C_L = 0.727$
 - (c) $C_L = 0.832$
 - (d) $C_L = 0.217$
10. An airplane of mass $m = 1800$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 16 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
 - (a) $D = 1100$ N and $D_p \neq D_i$.
 - (b) $D = 1100$ N and $D_p = D_i = 551$ N.
 - (c) $D = 2210$ N and $D_p = D_i = 1100$ N.
 - (d) $D = 2210$ N and $D_p \neq D_i$.

BLO-2: Aircraft Stability & Control

11. An airplane that is statically stable around a certain axis...
- (a) May be dynamically stable around that axis.
 - (b) Is definitely dynamically stable around that axis.
 - (c) Is definitely dynamically unstable around that axis.
 - (d) Is definitely statically stable around all other axes.
12. What name receives the optimisation of the dynamical response of an aircraft?
- (a) Handling & flying qualities.
 - (b) Performances.
 - (c) Manoeuvrability.
 - (d) Stability.
13. Rudder, elevator and ailerons are meant to generate moments around which axes?
- (a) yaw, pitch and roll, respectively.
 - (b) pitch, yaw and roll, respectively.
 - (c) roll, yaw and pitch, respectively.
 - (d) pitch, roll and yaw, respectively.
14. What can be said about a V-tail configuration?
- (a) It usually requires the use of ruddervators.
 - (b) It is often used in aerobatic airplanes.
 - (c) It substitutes ailerons.
 - (d) It is the only choice for supersonic flight.
15. What is true about canards?
- (a) All options are correct.
 - (b) They may help in lift generation.
 - (c) They may be used to stabilise a longitudinally unstable airplane.
 - (d) Whenever used, they are mounted ahead of the wings.
16. Which of the following hypotheses is not a requirement for longitudinal flight?
- (a) The angle of attack must be small.
 - (b) Propulsion moments must be orthogonal to the symmetry plane of the aircraft.
 - (c) Aerodynamics forces must be parallel to the symmetry plane of the aircraft.
 - (d) Aircraft reflexion-symmetry plane must be vertical.
17. Which of the following is a longitudinal mode?
- (a) Phugoid.
 - (b) Spiral mode.
 - (c) Dutch Roll.
 - (d) Roll Subsidence.
18. Which one of the following modes is usually the first to be completely damped in stable airplanes?
- (a) Pitch oscillation.
 - (b) Phugoid.
 - (c) Return to straight and level flight.
 - (d) All of them are damped equally fast.
19. Which of the following is a suitable name for the coefficient C_{l_p} ?
- (a) Roll damping coefficient.
 - (b) Directional stability coefficient.
 - (c) Dihedral effect coefficient.
 - (d) Adverse yaw coefficient.
20. Which of the following manoeuvres can be considered a lateral equilibrium?
- (a) All are lateral equilibria.
 - (b) Crosswind straight flight.
 - (c) Coordinated banked turn.
 - (d) Operational banked turn.

BLO-3: Aircraft Performances

21. What happens to the minimum of D as a constant mass airplane loses altitude?
- It occurs at lower speeds (moves left) but remains fairly constant (neither goes up nor down).
 - It occurs at higher speeds (moves right) and decreases (moves down).
 - It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - It occurs at lower speeds (moves left) and increases (moves up).
22. How do absolute and operational ceilings compare?
- Absolute is higher than operational.
 - Operational is higher than absolute.
 - Which one is higher depends on the type of engine.
 - They are synonyms.
23. Which statement is false about climbs?
- The best angle of climb corresponds to minimum drag regardless of engine type.
 - The best rate of climb corresponds to maximum excess power.
 - The best angle of climb corresponds to maximum excess thrust.
 - Both best angle and best rate of climb are zero at the absolute ceiling.
24. What is essentially true about maximum endurance conditions at cruise?
- For a propeller-powered airplane, it roughly corresponds to minimum required power.
 - Tail wind increases endurance.
 - High altitude is better.
 - None of the others.
25. How is maximum distance achieved in gliding flight?
- For minimum drag conditions.
 - For minimum descent rate conditions.
 - For minimum required power conditions.
 - For minimum aerodynamic efficiency.
26. The required thrust of a jet airplane is given by $T_R = 0.23\bar{\rho}V^2 + 0.022W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1600$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.3$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D = 2230$ N
 - $D = 2580$ N
 - $D = 2120$ N
 - $D = 2460$ N
27. A propeller-powered airplane of mass $m = 1750$ kg has a required power curve $P_R = 0.31\bar{\rho}V^3 + 0.028W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s²)
- $V_{(R/C)_{max}} = 54.6$ m/s
 - $V_{(R/C)_{max}} = 71.8$ m/s
 - $V_{(R/C)_{max}} = 94.5$ m/s
 - $V_{(R/C)_{max}} = 41.4$ m/s
28. A propeller-powered airplane with drag polar $C_D = 0.025 + 0.125C_L^2$, wing surface $S = 24$ m² and mass $m = 1400$ kg is flying at a speed $V = 80$ m/s. The available engine power at sea level is $(P_A)_{SL} = 500$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.5$ kg/m³? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $\delta_x = 0.617$
 - $\delta_x = 0.671$
 - $\delta_x = 0.476$
 - $\delta_x = 0.252$
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 48$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 50$ km of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- $h = 1040$ m
 - $h = 2080$ m
 - $h = 2400$ m
 - $h = 960$ m
30. The required thrust of a jet-engined airplane of mass $m = 1800$ kg, wing area $S = 16$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.35\bar{\rho}V^2 + 0.028W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 16000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.4$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 60.9$ m/s
 - $V \simeq 56.7$ m/s
 - $V \simeq 59.2$ m/s
 - $V \simeq 63.2$ m/s

31. Young's Modulus...

- (a) is the stress-strain ratio in the elastic regime.
- (b) is the stress-strain ratio in the plastic regime.
- (c) is a measure of crack propagation.
- (d) is the number of tension cycles before catastrophic failure of the material.

32. What can be said about composites?

- (a) They are made of two or more different materials.
- (b) They are a type of titanium alloy.
- (c) Carbon fiber cannot be one of their constituents.
- (d) Their importance has steadily declined as airframe structural materials in the last 30 years.

33. Which of the following is not a manufacturing selection criteria for materials?

- (a) Density.
- (b) Malleability.
- (c) Ductility.
- (d) Castability.

34. What is fatigue in materials science?

- (a) The damage of the material caused by cyclic stress.
- (b) The amount of energy absorbed by a material during an impact.
- (c) A property of no importance to aircraft structure design.
- (d) All options are correct.

35. What is true about titanium?

- (a) It has an allotropic transformation from HCP (alpha) to BCC (beta).
- (b) It is widely used in aeronautics due to its low price.
- (c) It has poor corrosion behavior.
- (d) It can be obtained in the form of Cementite.

36. To which airplane corresponds ICAO designation C172?

- (a) Cessna 172.
- (b) Cessna 170-200.
- (c) Boeing 717-200.
- (d) Airbus 370-200.

37. Under which circumstances must the minimum radar separation be larger for successive landing procedures?

- (a) Super followed by small airplane.
- (b) Small followed by super airplane.
- (c) Super followed by super airplane.
- (d) Small followed by small airplane.

38. Identify the following aircraft



- (a) Boeing 737.
- (b) Boeing 777.
- (c) Airbus A320.
- (d) Airbus A340.

39. Which aircraft manufacturer builds the following airplane?



- (a) Embraer.
- (b) Antonov.
- (c) Airbus.
- (d) Boeing.

40. Which of the following aircraft has three engines?

- (a) DC-10.
- (b) Boeing 717.
- (c) MD 80.
- (d) All of them.

Correct Answers

BLO-1: Flight Principles

	Permutation Number				
Question	0	1	2	3	4
1	a	d	d	d	c
2	a	a	c	d	b
3	a	c	c	b	d
4	a	d	c	d	d
5	a	c	a	d	a
6	a	b	b	c	a
7	a	b	b	d	a
8	a	b	b	b	a
9	a	d	d	a	c
10	a	d	b	d	c

BLO-2: Aircraft Stability & Control

	Permutation Number				
Question	0	1	2	3	4
11	a	b	a	d	a
12	a	a	a	b	a
13	a	d	a	b	d
14	a	c	a	d	b
15	a	a	c	a	d
16	a	b	b	d	b
17	a	b	c	b	c
18	a	a	d	c	c
19	a	d	d	a	d
20	a	a	c	c	b

BLO-3: Aircraft Performances

	Permutation Number				
Question	0	1	2	3	4
21	a	c	a	c	c
22	a	a	b	b	d
23	a	d	d	b	d
24	a	c	a	d	b
25	a	c	c	d	c
26	a	d	b	a	a
27	a	b	b	c	c
28	a	a	c	d	c
29	a	c	c	d	b
30	a	c	d	d	a

BLO-4: Materials & Structures / Aircraft Recognition

	Permutation Number				
Question	0	1	2	3	4
31	a	b	c	a	a
32	a	d	c	d	c
33	a	d	a	b	b
34	a	a	a	d	b
35	a	c	d	c	a
36	a	c	c	a	c
37	a	b	b	a	a
38	a	b	a	b	d
39	a	a	c	a	b
40	a	d	b	c	c

BLO-1: Concepts on flight principles and aircraft performance

1. Which of the following devices are specifically conceived to reduce the wing lift?
 - (a) Spoilers
 - (b) Flaps
 - (c) Winglets
 - (d) Slats
2. The angle of attack of an airfoil is defined as the angle between what and what?
 - (a) Airspeed and chord
 - (b) Airspeed and chamber line
 - (c) Trajectory and chord
 - (d) Trajectory and chamber line
3.
 - (a) has separation flow at $C_{L_{max}}$.
 - (b) has maximum efficiency at $C_{D_{min}}$.
 - (c) has maximum C_D at $C_{L_{max}}$.
 - (d) has minimum C_L at $C_{D_{stall}}$.
4.
 - (a) $\Phi_h < \Phi_{SL}$
 - (b) $\Phi_h > \Phi_{SL}$
 - (c) $\Phi_h = \Phi_{SL}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
5. Which of the following parameters of an airplane is, approximately, independent of the altitude and speed at which it is flying?
 - (a) The angle of attack at which stall occurs.
 - (b) The speed at which stall occurs.
 - (c) The pitch angle at which stall occurs.
 - (d) The bank angle at which stall occurs.
6. Which of the following concepts is not directly related to takeoff ?
 - (a) Bank angle.
 - (b) Decision Speed.
 - (c) Rounding.
 - (d) Lift off.
7. Which component of Drag decreases when speed increases?
 - (a) Induced drag.
 - (b) Parasite Drag.
 - (c) Parasite Drag and Induced Drag.
 - (d) None.
8. What is Maximum Climb Rate?
 - (a) Maximum altitude gain per unit time.
 - (b) Maximum climb slope.
 - (c) Cruise Climb.
 - (d) Thermal Climb.
9. Which of the following performance optima are not affected by wind?
 - (a) Endurance and maximum rate of climb.
 - (b) Range and maximum rate of climb.
 - (c) Endurance and maximum climb angle.
 - (d) Range and maximum climb angle.
10. Minimum drag is obtained when...
 - (a) ...Parasite Drag equals Induced Drag.
 - (b) ...Load factor is maximum.
 - (c) ...aircraft flies at minimum height.
 - (d) ...Induced drag is two times Parasite Drag .

BLO-2: Flight principles related problems

11. An airplane is gliding. The parameters of this airplane are: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$, $\alpha_s = 11^\circ$, $m = 9000 \text{ kg}$, $S = 32 \text{ m}^2$, $\text{span} = 20 \text{ m}$. The airplane is at sea level. The optimal descent angle is...
- (a) 2.29°
 - (b) 1.05°
 - (c) 2.01°
 - (d) None of the others is correct
12. An airplane is gliding. The parameters of this airplane are: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$, $\alpha_s = 11^\circ$, $m = 9000 \text{ kg}$, $S = 32 \text{ m}^2$, $\text{span} = 20 \text{ m}$. The airplane is at sea level. The optimal glide speed is...
- (a) 95 m/s
 - (b) 155 m/s
 - (c) 131 m/s
 - (d) None of the others is correct
13. An airfoil has the following characteristics: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$. Its maximum efficiency is...
- (a) 25
 - (b) 17.3
 - (c) 22.4
 - (d) None of the others is correct
14. The drag polar of an airplane changes from $C_D = 0.025 + 0.12C_L^2$ in clean configuration to $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.07$
 - (b) $C_D = 0.0843$
 - (c) $C_D = 0.101$
 - (d) $C_D = 0.0133$
15. A $m = 1000 \text{ kg}$ airplane is engaged in a coordinated banked turn with bank angle ($\Phi = 45^\circ$) at a speed $V = 90 \text{ m/s}$. What is the turn radius (R_t)? (Take $g = 9.8 \text{ m/s}^2$)
- (a) 827 m
 - (b) 783 m
 - (c) 695 m
 - (d) None of the others is correct.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is not required by dynamic stability around an axis?
- (a) High Manoeuvrability.
 - (b) Static Stability.
 - (c) Restoring Force.
 - (d) Damping Force.
17. Which of the following Primary Control Surfaces is commanded moving the yoke left and right?
- (a) Ailerons.
 - (b) Slat.
 - (c) Elevator.
 - (d) Rudder.
18. Flight equilibria requires...
- (a) All options are correct.
 - (b) Summation of all forces over the aircraft equals zero.
 - (c) All moments over the aircraft are balanced.
 - (d) linear and angular acceleration of the aircraft are zero.
19. Which of the following is a high-lift secondary control surface...
- (a) Flaps.
 - (b) Air Brakes.
 - (c) Winglets.
 - (d) Throttle.
20. Which of the following relative positions of the Center of Gravity and Neutral Point of an airplane produces Positive Stability?
- (a) Center of Gravity ahead of Neutral Point.
 - (b) Center of Gravity on Neutral Point.
 - (c) Center of Gravity after Neutral Point.
 - (d) All options are correct.
21. How does an aircraft recover from a Roll perturbation?
- (a) Lift asymmetry between wings due to increased angle of attack in down-going wing and reduced angle of attack in up-going wing.
 - (b) Inertial forces due to fuel movement in wing integral fuel tank.
 - (c) Spoilers commanded by Automatic Control System of the aircraft.
 - (d) Pilot acting on ailerons trim tabs.
22. What can be said about the Lateral Stability of an airplane?
- (a) Roll and Yaw are strongly coupled.
 - (b) A restoring moment appears, which depends on the size of the fin.
 - (c) A dampening moment appears, which depends on the deployed angle of the flaps.
 - (d) A restoring moment appears, which depends on the size of the horizontal stabiliser.
23. Which of the following is not a lateral mode?
- (a) Phugoid.
 - (b) Roll subsidence.
 - (c) Dutch roll.
 - (d) Spiral mode.
24. What can be said about Dutch roll mode?
- (a) all options are correct.
 - (b) It has a short period.
 - (c) Yaw dumpers are installed in some aircrafts to dampen it.
 - (d) Size of vertical stabilizer affects damping.
25. Which of the following is not a stabilizing contributor to Directional Stability in an airplane?
- (a) Forward wing sweep.
 - (b) Backward wing sweep.
 - (c) Vertical Stabilizer size.
 - (d) Centre of Gravity to empenage distance.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.28\bar{\rho}V^3 + 0.02W^2/\bar{\rho}V$. If the airplane has mass $m = 2500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.8$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $V_E = 68.7$ m/s
 - $V_E = 90.5$ m/s
 - $V_E = 119$ m/s
 - $V_E = 61.6$ m/s
27. A jet-powered airplane of mass $m = 2000$ kg has a required thrust curve $T_R = 0.28\bar{\rho}V^2 + 0.03W^2/\bar{\rho}V^2$. What is the velocity for best climb angle ($V_{\gamma_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s²)
- $V_{\gamma_{max}} = 80.1$ m/s
 - $V_{\gamma_{max}} = 60.9$ m/s
 - $V_{\gamma_{max}} = 105$ m/s
 - $V_{\gamma_{max}} = 46.2$ m/s
28. An airplane with drag polar $C_D = 0.03 + 0.18C_L^2$, wing surface $S = 40$ m² and mass $m = 2500$ kg is flying at a speed $V = 150$ m/s. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.32$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $T_R = 5070$ N
 - $T_R = 5150$ N
 - $T_R = 4940$ N
 - $T_R = 4840$ N
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 43$. What is the descent angle (γ)?
- $\gamma = -1.33^\circ$
 - $\gamma = -1.17^\circ$
 - $\gamma = -1.61^\circ$
 - $\gamma = -0.931^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 2300$ kg, wing area $S = 34$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.35$, is given by $T_R = 0.65\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 24000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.35$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 62.1$ m/s
 - $V \simeq 47.9$ m/s
 - $V \simeq 40.4$ m/s
 - $V \simeq 71.7$ m/s

Concepts on aerospace propulsion, helicopters and space systems

31. About turbojet compressor,
- the compressor gets power from the turbine and transfers it to the gases.
 - the compressor gets power from the nozzle and transfers it to the gases.
 - an axial compressor is less efficient than a radial compressor.
 - the compressor expands the gases providing the thrust of the engine.
32. Which of the following engines does NOT have bypass ratio?
- rocket engine
 - turbo propeller engine
 - turbofan engine
 - all the engines have a by-pass ratio
33. The Ministry of Industry plans to reduce the emissions of road transport systems. Considering parity of horsepower, they should know that...
- Diesel engines emit less CO_2 than Otto engine.
 - Electric engines emit less CO_2 if the electricity comes from a coal power plant.
 - Diesel engines emit more NO_x than Otto engine.
- I and III
 - III
 - II and III
 - II
34. ¿From the entrance to the exit of the gases, a 2 spool turbojet is composed by
- diffuser, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - nozzle, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, diffuser
 - diffuser, high pressure compressor, low pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - nozzle, high pressure compressor, low pressure compressor, combustion chamber, low pressure turbine, high pressure turbine, diffuser
35. What kind air to fuel mixture does an Otto engine has?
- Stoichiometric mixture
 - Oxygen rich in oxygen mixture
 - Fuel rich mixture
 - Bacon and peperoni mixture
36. Which of the following statements is correct regarding the controls of a helicopter,
- the tail rotor compensates the torque of the main rotor.
 - the collective action changes the pitch of all the rotor blades.
 - the cyclic rotor changes the pitch of the blades according to their position.
- All three statements are true.
 - Only I
 - Only I and II
 - Only I and III
37. Which of the following statements are true?
- An autogiro can perform stationary flight.
 - The rotors of tandem helicopter are counter-rotating.
 - The rotors of a coaxial helicopter are contra-rotating.
- Only II and III
 - Only I
 - Only II
 - Only I and III
38. An orbit mostly used for navigation satellites purposes is...
- a medium Earth orbit
 - a sun synchronous polar orbit
 - a geostationary orbit
 - a highly elliptic orbit
39. The mass of a rocket engine is $80Tn$. The mass flow of a rocket engine is $2000kg/s$. The nozzle of such rocket is adapted, and the exhaust gases speed at the exit is $3400m/s$. The rocket is a...
- Solid propellant chemical engine
 - Solar sail
 - Magneto-plasma engine
 - Mono propellant chemical engine
40. To launch a satellite into geostationary orbit, the launcher flies to the
- East.
 - West.
 - North-East.
 - North-West

41. What Dead Reckoning is?
- (a) The process of calculating the current position from an initial known position and subsequent speed, direction and time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) A air ship based radio navigation aid.
 - (d) A generic name for ground stations.
42. Which navigation systems uses ground stations that broadcast a non directional signal at afrequency between 190 and 1750 kHz)?
- (a) ADF.
 - (b) VOR.
 - (c) DME.
 - (d) GLONASS.
43. Which navigation system gives the angle referred to the true north of the line between the aircraft and a ground station?
- (a) VOR.
 - (b) WAAS.
 - (c) NDB.
 - (d) GPS.
44. One of the following navigation systems gives the distance between the aircraft and a ground station which is interrogated by the transponder:
- (a) DME.
 - (b) Galileo.
 - (c) VOR.
 - (d) IRS.
45. Which navigation systems allows a safe landing giving the relative position of an aircraft to the glideslope and runway axis?:
- (a) ILS.
 - (b) NDB.
 - (c) EGNOS.
 - (d) RNAV.
46. Which autonomous navigation systems uses accelerometers and gyroscopes to determine the position of an aircraft?
- (a) INS.
 - (b) VOR.
 - (c) GPS.
 - (d) NDB.
47. Which of the following is not a GNSS?
- (a) NDB.
 - (b) Galileo.
 - (c) GLONASS.
 - (d) GPS.
48. What parameter is measured with a simple Pitot tube?
- (a) Total pressure.
 - (b) Dynamic pressure.
 - (c) Static pressure.
 - (d) Air temperature
49. Which pressure setting is defined as the barometric pressure adjusted to sea level?
- (a) QNH.
 - (b) QFE.
 - (c) Standard.
 - (d) 1013 mb.
50. What can be read in the altimeter if it has been set to QFE and the aircraft is on ground at the threshold of the runway of an airfield?
- (a) Zero altitude.
 - (b) The altitude of the airport.
 - (c) FL310.
 - (d) ISA+10.

BLO-7: Aircraft recognition and materials & structures

51. To which airplane corresponds ICAO designation B735?

- (a) Boeing B737-500.
- (b) Airbus A306.
- (c) Airbus A320-600.
- (d) None of the others.

52. Which of the following is not a narrow body?

- (a) Airbus A380.
- (b) Boeing 737.
- (c) Airbus A320.
- (d) Boeing 757.

53. Identify the following aircraft



- (a) Airbus A380.
- (b) Airbus A340.
- (c) Boeing 747.
- (d) Boeing 767.

54. Identify the following aircraft



- (a) Airbus A318-100.
- (b) Antonov An-2.
- (c) Gulfstream G550.
- (d) Boeing 747.

55. Which one of the following aircraft is the shortest?

- (a) Boeing 747-SP.
- (b) Boeing 747-200.
- (c) Boeing 747-300.
- (d) Boeing 747-400.

56. Elastic limit or yield point...

- (a) is the maximum stress for elastic deformation of a material.
- (b) is the stress-deformation ratio.
- (c) is the minimum number of cycles of the S-N curve.
- (d) is the stress at which a material breaks.

57. S-N curve is related to...

- (a) Fatigue.
- (b) Plasticity.
- (c) Elasticity.
- (d) Hardness.

58. What does 2024-T3 refers to?

- (a) To an Aluminium alloy.
- (b) To a Titanium alloy.
- (c) To a fighter aircraft.
- (d) To a glass fiber-epoxy composite.

59. What can be said about Titanium?

- (a) It has good resistance to corrosion.
- (b) It has no allotropic transformation.
- (c) It is cheaper than Aluminium.
- (d) It has worst specific mechanical properties than Aluminium.

60. What can be said about fiber reinforced composites?

- (a) They have a very good resistance to fatigue.
- (b) They have lower production cost.
- (c) They cannot show delamination damage.
- (d) They have bad resistance to corrosion.

BLO-8: Aircraft systems

61. Which of the following is not a part of the flight control system?
- (a) APU.
 - (b) Rudder pedals.
 - (c) Ailerons.
 - (d) Spoilers.
62. What is a hydro-mechanical flight control system?
- (a) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made with a system of rods, levers, cables, pulleys, springs and/or dampers.
 - (b) A flight control system in which the linkage between cockpit controls and control surfaces is made with a system of rods, levers, cables, pulleys, springs, dampers and no additional power source.
 - (c) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made through an electronic system only.
 - (d) A flight control system for seaplanes.
63. Which of the following is not a source of pressure of a hydraulic system?
- (a) DC Essential bus.
 - (b) Electric pump.
 - (c) Power Transfer Unit.
 - (d) Engine-driven pump.
64. Which of the following is not a part of a hydraulic system?
- (a) Inverter.
 - (b) Accumulator.
 - (c) Reservoir.
 - (d) Filter.
65. Which of the following is part of an electric system?
- (a) Battery.
 - (b) Yoke.
 - (c) FDR.
 - (d) Oxygen mask.
66. Which of the following is not part of the electric system?
- (a) Cross feed valve.
 - (b) AC bus.
 - (c) DC bus.
 - (d) Emergency generator.
67. Which of the following is a function of the APU?
- (a) Back up air conditioning system in flight.
 - (b) Remove power from secondary systems in the event of an emergency.
 - (c) Remove ice from the wing.
 - (d) Pump fuel to engines.
68. Which of the following is not part of a fuel system?
- (a) FADEC.
 - (b) Tanks.
 - (c) Transfer valve.
 - (d) Fuel pump.
69. Which system will start working after a depressurization at cruise altitude due to a failure in air conditioning system?
- (a) Oxygen system.
 - (b) Fire protection system.
 - (c) Autoflight system.
 - (d) Light system.
70. Which of the following systems is not fed by the pneumatic system?
- (a) Light system.
 - (b) Air Conditioning System.
 - (c) Wing anti-ice system.
 - (d) Potable water tank pressurization system.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	c	c	a
2	a	d	a	b	a
3	a	a	d	a	a
4	a	c	d	a	c
5	a	a	b	a	c
6	a	b	a	d	b
7	a	c	d	d	d
8	a	c	d	d	b
9	a	a	c	a	b
10	a	b	c	c	c

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	a	a	b
12	a	a	b	a	b
13	a	a	b	c	b
14	a	a	c	b	d
15	a	c	a	d	d

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	d	c	a	a
17	a	c	a	b	d
18	a	a	b	d	b
19	a	d	d	c	c
20	a	b	c	c	a
21	a	c	c	b	a
22	a	d	d	b	c
23	a	b	a	c	c
24	a	d	c	d	a
25	a	b	b	c	c

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	a	d	a
27	a	d	a	c	d
28	a	d	a	a	a
29	a	d	b	c	b
30	a	b	c	a	b

Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	a	d	c
32	a	a	b	c	b
33	a	b	a	b	b
34	a	d	d	d	b
35	a	d	d	b	d
36	a	d	c	b	a
37	a	b	a	c	c
38	a	a	a	b	c
39	a	d	a	d	a
40	a	a	b	a	a

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
41	a	c	d	d	d
42	a	b	d	b	d
43	a	a	b	b	c
44	a	c	a	b	a
45	a	c	c	c	c
46	a	c	b	d	b
47	a	b	a	c	b
48	a	b	c	b	b
49	a	d	a	a	a
50	a	d	c	c	d

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
51	a	d	d	b	c
52	a	d	b	d	a
53	a	c	b	b	d
54	a	b	a	c	c
55	a	b	c	a	a
56	a	b	a	d	c
57	a	c	c	c	b
58	a	b	a	a	c
59	a	c	c	c	a
60	a	d	d	c	a

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	d	c	a	c
62	a	c	a	c	d
63	a	d	c	c	a
64	a	c	b	c	b
65	a	a	c	b	b
66	a	d	a	d	b
67	a	b	d	d	b
68	a	d	b	d	b
69	a	c	b	a	a
70	a	b	b	c	a

Concepts on aerospace propulsion, helicopters and space systems

1. About turbojet turbines,
 - (a) the turbine gets power from the gases and transfers it to the compressor.
 - (b) the turbine gets power from the gases and transfers it to the diffuser.
 - (c) the turbine does not need any refrigeration.
 - (d) the turbine expands the gases providing the thrust of the engine.
2. Which of the following engines has a highest bypass ratio?
 - (a) Turboprop
 - (b) Turbojet
 - (c) Low bypass ratio turbofan
 - (d) High bypass ratio turbofan
3. The Ministry of Industry plans to reduce the emissions of road transport systems. Considering parity of horsepower, they should know that...
 - I Diesel engines emit less CO_2 than Otto engine.
 - II Electric engines emit less CO_2 if the electricity comes from a coal power plant.
 - III Diesel engines emit more NO_x than Otto engine.
 - (a) I and III
 - (b) III
 - (c) II and III
 - (d) II
4. From the entrance to the exit of the gases, a 2 spool turbojet is composed by
 - (a) diffuser, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (b) nozzle, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, diffuser
 - (c) diffuser, high pressure compressor, low pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (d) nozzle, high pressure compressor, low pressure compressor, combustion chamber, low pressure turbine, high pressure turbine, diffuser
5. What kind of flame typology does an Otto engine has?
 - (a) Premixed flame
 - (b) Diffusive flame
 - (c) Pyrotechnic flame
 - (d) Gasoline flame
6. A helicopter in horizontal flight,
 - (a) the distribution of the main rotor speeds induced a roll momentum in the helicopter.
 - (b) the distribution of the main rotor speeds is proportional to the radius.
 - (c) the retreating blade moves up to compensate the reverse flow zone.
 - (d) there is no reverse flow in the main rotor.
7. Which of the following statements are true?
 - I An autogiro can perform stationary flight.
 - II The rotors of tandem helicopter are counter-rotating.
 - III The rotors of a coaxial helicopter are contra-rotating.
 - (a) II and III
 - (b) I
 - (c) II
 - (d) I and III
8. An orbit mostly used for spying purposes is...
 - (a) a highly elliptic orbit
 - (b) a sun synchronous polar orbit
 - (c) a geostationary orbit
 - (d) a medium Earth orbit
9. The mass of a rocket engine is $80Tn$. The mass flow of a rocket engine is $2000kg/s$. The nozzle of such rocket is adapted, and the exhaust gases speed at the exit is $3400m/s$. The rocket is a...
 - (a) Solid propellant chemical engine
 - (b) Solar sail
 - (c) Magneto-plasma engine
 - (d) Mono propellant chemical engine
10. To launch a satellite into geostationary orbit, the launcher flies to the
 - (a) East.
 - (b) West.
 - (c) North-East.
 - (d) North-West

BLO-6: Navigation & altimetry

11. What Dead Reckoning is?
- (a) The process of calculating the current position from an initial known position and subsequent speed, direction and time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) A air ship based radio navigation aid.
 - (d) A generic name for ground stations.
12. Which of the following is true about ADF?
- (a) Gives the direction to a ground station to the pilot.
 - (b) Its signal is highly resistant to atmospheric conditions.
 - (c) Works in 250-300 MHz band.
 - (d) Gives the distance to a ground station of the aircraft.
13. Which of the following is not true about VOR
- (a) Aircraft transponder interrogates the VOR ground station.
 - (b) Transmits radials referenced to magnetic north.
 - (c) Multipurpose ILS and VOR equipment are common.
 - (d) Ground stations broadcast a non directional signal (108-117.95 MHz) and a rotating directional beam (9960+/-460 Hz).
14. What info is given by DME?
- (a) Distance to a ground station.
 - (b) Direction to a ground station.
 - (c) Weather forecast.
 - (d) Landing glide slope.
15. Which of the following is not part of ILS?
- (a) Range calculator.
 - (b) Glide slope.
 - (c) Localizer.
 - (d) Marker Beacon.
16. Which of the following cannot be said about INS?
- (a) It depends on communication with other aircrafts.
 - (b) It does not depend on aircraft external equipment or ground stations.
 - (c) Uses accelerometers and gyroscopes.
 - (d) Needs to be initially aligned.
17. Which of the following is the EU Global Navigation Satellite System?
- (a) Galileo.
 - (b) GPS.
 - (c) GLONASS.
 - (d) WASS
18. What parameter is measured with the static source.
- (a) Static pressure.
 - (b) Dynamic pressure.
 - (c) Total pressure.
 - (d) Air temperature.
19. Which of the following airspeeds definitions is the actual speed of the aircraft relative to the air?
- (a) TAS.
 - (b) IAS.
 - (c) RAS.
 - (d) CAS.
20. What can be read in the altimeter if it has been set to QNH and the aircraft is on ground at the threshold of the runway of an airfield?
- (a) The altitude of the airport.
 - (b) Zero altitude.
 - (c) FL270.
 - (d) ISA+10.

21. To which airplane corresponds ICAO designation B744?

- (a) Boeing 747-400.
- (b) Boeing 737-440.
- (c) Boeing 740-400.
- (d) None of the others.

22. Which of the following is not a wide body?

- (a) Airbus A319.
- (b) Boeing 747.
- (c) Airbus A330.
- (d) Boeing 777.

23. Identify the following aircraft



- (a) Boeing 777.
- (b) Boeing 767.
- (c) Boeing 747.
- (d) Boeing 787.

24. Which aircraft manufacturer builds the following airplane (the one below the buran space shuttle)?



- (a) Antonov.
- (b) Embraer.
- (c) Airbus.
- (d) Boeing.

25. Which one of the following aircraft is the shortest?

- (a) Airbus A318.
- (b) Boeing 737-900.
- (c) Boeing 737-400.
- (d) Airbus A320.

26. S-N curve is related to...

- (a) Fatigue.
- (b) Plasticity.
- (c) Elasticity.
- (d) Hardness.

27. Which of the following is a selection criteria for materials?

- (a) All options are correct.
- (b) Cost.
- (c) Thermal and electric properties.
- (d) Corrosion resistance.

28. Which of the following crystal structures can Aluminum show?

- (a) FCC.
- (b) BCC.
- (c) HC.
- (d) All the options are allotropes of Al.

29. Which of the following is not a component of steel?

- (a) Alpha alloy.
- (b) Pearlite.
- (c) Ferrite.
- (d) Austenite.

30. Which of the following material is commonly used as reinforcement fibers in composites?

- (a) Carbon fiber.
- (b) Epoxy resin.
- (c) Beryllium alloys.
- (d) Nickel super-alloys.

BLO-8: Aircraft systems

31. In a fly-by-wire flight control system...
- (a) ...there is no mechanical link between the cockpit controls and the control surfaces.
 - (b) ...there is mechanical link between the cockpit controls and the control surfaces.
 - (c) ...there is hydro-mechanical link between the cockpit controls and the control surfaces.
 - (d) ...there is pneumatic link between the cockpit controls and the control surfaces.
32. Which of the following devices can be part of a hydro-mechanical flight control system?
- (a) Hydraulic actuator.
 - (b) Bleed valve.
 - (c) De-ice boot.
 - (d) APU Generator.
33. Which of the following is a source of pressure of a hydraulic system?
- (a) Engine-driven pump.
 - (b) Thrust reverser system.
 - (c) Inverter.
 - (d) DC Essential bus.
34. Which of the following components is a part of a hydraulic system?
- (a) Reservoir.
 - (b) Transformer Rectifier.
 - (c) FADEC.
 - (d) Spoiler.
35. Which of the following components is a part of an electric system?
- (a) Battery.
 - (b) Oil tank.
 - (c) Hydraulic accumulator.
 - (d) ADIRU.
36. Which of the following components is not a source of electric power?
- (a) Fuel transfer valve.
 - (b) Battery.
 - (c) Main Generator.
 - (d) Emergency generator.
37. Which of the following is not a high pressure air source for the Pneumatic System?
- (a) Static inverter.
 - (b) APU load compressor.
 - (c) Engine bleed system.
 - (d) HP ground connection.
38. Which of the following is not a Power Plant subsystem?
- (a) Oxygen system.
 - (b) Thrust reverser system.
 - (c) Oil system.
 - (d) Ignition and starting system.
39. Which component of the power plant system provides with mechanical power to other aircraft systems?
- (a) Accessory gearbox.
 - (b) FADEC.
 - (c) Essential Bus.
 - (d) Fuel filter.
40. Which of the following systems is not fed by the pneumatic system?
- (a) Communications system.
 - (b) Air Conditioning System.
 - (c) Engine Starting System.
 - (d) Anti-ice system.

Correct Answers

Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	d	c	c	d
2	a	d	c	c	a
3	a	d	a	a	a
4	a	c	d	c	a
5	a	a	a	b	a
6	a	d	b	a	a
7	a	b	d	a	c
8	a	b	a	a	b
9	a	b	d	c	c
10	a	d	c	d	d

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	b	b	b	c
12	a	a	d	c	b
13	a	a	a	d	c
14	a	d	d	b	a
15	a	a	c	b	d
16	a	b	d	d	c
17	a	d	a	d	d
18	a	b	b	a	b
19	a	c	a	a	d
20	a	a	b	c	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	d	d	d	a
22	a	d	a	c	b
23	a	b	c	a	b
24	a	b	b	b	a
25	a	b	a	c	d
26	a	b	a	c	d
27	a	b	c	d	b
28	a	a	d	d	d
29	a	b	c	c	a
30	a	b	c	a	b

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	b	a	d	d
32	a	c	b	a	a
33	a	c	a	a	c
34	a	b	d	d	a
35	a	c	b	c	c
36	a	d	d	b	b
37	a	b	a	c	b
38	a	b	b	c	c
39	a	d	d	c	d
40	a	c	a	b	d

BLO-1: Concepts on flight principles and aircraft performance

1. Which of the following devices are specifically conceived to reduce the wing lift?
 - (a) Spoilers
 - (b) Flaps
 - (c) Winglets
 - (d) Slats
2. The angle of attack of an airfoil is defined as the angle between what and what?
 - (a) Airspeed and chord
 - (b) Airspeed and chamber line
 - (c) Trajectory and chord
 - (d) Trajectory and chamber line
3. Which of the following statements is true? An airfoil...
 - (a) has separation flow at $C_{L_{max}}$.
 - (b) has maximum efficiency at $C_{D_{min}}$.
 - (c) has maximum C_D at $C_{L_{max}}$.
 - (d) has minimum C_L at $C_{D_{stall}}$.
4. How does maximum bank angle compare between an airplane flying high at an altitude h_h (Φ_h) and the same airplane, at the same speed, flying at sea level at h_{SL} (Φ_{SL})?
 - (a) $\Phi_h < \Phi_{SL}$
 - (b) $\Phi_h > \Phi_{SL}$
 - (c) $\Phi_h = \Phi_{SL}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
5. Which of the following parameters of an airplane is, approximately, independent of the altitude and speed at which it is flying?
 - (a) The angle of attack at which stall occurs.
 - (b) The speed at which stall occurs.
 - (c) The pitch angle at which stall occurs.
 - (d) The bank angle at which stall occurs.
6. Which component of Drag grows when speed increases?
 - (a) Parasite drag.
 - (b) Induced Drag.
 - (c) Parasite Drag and Induced Drag.
 - (d) None.
7. Which of the following performance optima is affected by wind?
 - (a) Best angle of climb.
 - (b) Endurance.
 - (c) Maximum rate of climb.
 - (d) Minimum rate of descent.
8. Which of the following conditions is generally adequate for maximising the endurance of a jet airplane?
 - (a) Fly at minimum required thrust.
 - (b) Fly at minimum altitude.
 - (c) Fly at minimum required power.
 - (d) All options are correct.
9. Minimum drag is obtained when...
 - (a) ...Parasite Drag equals Induced Drag.
 - (b) ...Load factor is maximum.
 - (c) ...aircraft flies at minimum height.
 - (d) ...Induced drag is two times Parasite Drag .
10. Which of the following is true regarding decision speed or V1 during takeoff phase?
 - (a) If one engine fails after reaching V1 takeoff will continue.
 - (b) If one engine fails before reaching V1 takeoff will continue.
 - (c) If wind speed is below V1 takeoff will be headwind.
 - (d) If wind speed is over V1 no flaps are required to take off.

BLO-2: Flight principles related problems

11. An airplane is gliding. The parameters of this airplane are: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$, $\alpha_s = 11^\circ$, $m = 9000 \text{ kg}$, $S = 32 \text{ m}^2$, $\text{span} = 20 \text{ m}$. The airplane is at sea level. The optimal descent angle is...
- (a) 2.29°
 - (b) 1.05°
 - (c) 2.01°
 - (d) None of the others is correct
12. An airplane is gliding. The parameters of this airplane are: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$, $\alpha_s = 11^\circ$, $m = 9000 \text{ kg}$, $S = 32 \text{ m}^2$, $\text{span} = 20 \text{ m}$. The airplane is at sea level. The optimal glide speed is...
- (a) 95 m/s
 - (b) 155 m/s
 - (c) 131 m/s
 - (d) None of the others is correct
13. An airfoil has the following characteristics: $C_L = 0.4 + 0.2\alpha$, $C_D = 0.01 + 0.04C_L^2$. Its maximum efficiency is...
- (a) 25
 - (b) 17.3
 - (c) 22.4
 - (d) None of the others is correct
14. The drag polar of an airplane changes from $C_D = 0.025 + 0.12C_L^2$ in clean configuration to $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.07$
 - (b) $C_D = 0.0843$
 - (c) $C_D = 0.101$
 - (d) $C_D = 0.0133$
15. A $m = 1000 \text{ kg}$ airplane is engaged in a coordinated banked turn with bank angle ($\Phi = 45^\circ$) at a speed $V = 90 \text{ m/s}$. What is the turn radius (R_t)? (Take $g = 9.8 \text{ m/s}^2$)
- (a) 827 m
 - (b) 783 m
 - (c) 695 m
 - (d) None of the others is correct.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is not an stability condition?
- (a) Aural.
 - (b) Unstable.
 - (c) Stable.
 - (d) Neutral.
17. Which of the following Primary Control Surfaces is commanded moving the yoke forward and backward?
- (a) Elevator.
 - (b) Slat.
 - (c) Ailerons.
 - (d) Rudder.
18. Which of the following Secondary Control Surfaces can be designed as a backup for ailerons?
- (a) Spoilers.
 - (b) Slots.
 - (c) Winglets.
 - (d) Vortex generators.
19. Which of the following modes is an exchange of height and speed at constant angle of attack?
- (a) Phugoid.
 - (b) Short Period.
 - (c) Pitch Oscillation.
 - (d) Return to Straight and Level Flight.
20. Which of the following control surfaces is used to control pitch?
- (a) Elevator.
 - (b) Ailerons.
 - (c) Rudder.
 - (d) All answers are correct.
21. What is adverse yaw?
- (a) Sideslip produced when ailerons are operated.
 - (b) Rolling movement produced by rudder-elevator coupling.
 - (c) Sideslip produced by different fuel flow rate between wing tanks.
 - (d) Pilot acting on rudder trim tab.
22. What is Lateral Stability of an airplane?
- (a) Tendency of the airplane to recover from a Roll perturbation.
 - (b) Tendency of the airplane to recover from a Yaw perturbation.
 - (c) Tendency of the airplane to recover from a Pitch perturbation.
 - (d) Tendency of the airplane to recover from a Speed perturbation.
23. Which of the following is a stabilising contributor to Directional Stability in an airplane?
- (a) Vertical Stabiliser.
 - (b) Ailerons size.
 - (c) Forward wing sweep.
 - (d) Canard.
24. How is Roll Subsidence Mode produced?
- (a) Through lift asymmetry between wings lift due to increased angle of attack in down-going wing and reduced angle of attack in up-going wing.
 - (b) angle of attack variation at constant height.
 - (c) Wing dihedral.
 - (d) Manual throttle control.
25. Which of the following is not a lateral mode?
- (a) Short Period.
 - (b) Spiral mode.
 - (c) Dutch Roll.
 - (d) Roll Subsidence.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.33\bar{\rho}V^3 + 0.018W^2/\bar{\rho}V$. If the airplane has mass $m = 2000\text{ kg}$ and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.6$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8\text{ m/s}^2$)
- $V_E = 66.4\text{ m/s}$
 - $V_E = 87.3\text{ m/s}$
 - $V_E = 115\text{ m/s}$
 - $V_E = 74.1\text{ m/s}$
27. A jet-powered airplane of mass $m = 1800\text{ kg}$ has a required thrust curve $T_R = 0.3\bar{\rho}V^2 + 0.026W^2/\bar{\rho}V^2$. What is the velocity for best climb angle ($V_{\gamma_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8\text{ m/s}^2$)
- $V_{\gamma_{max}} = 72.1\text{ m/s}$
 - $V_{\gamma_{max}} = 54.8\text{ m/s}$
 - $V_{\gamma_{max}} = 94.8\text{ m/s}$
 - $V_{\gamma_{max}} = 125\text{ m/s}$
28. An airplane with drag polar $C_D = 0.025 + 0.17C_L^2$, wing surface $S = 30\text{ m}^2$ and mass $m = 1500\text{ kg}$ is flying at a speed $V = 130\text{ m/s}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.35\text{ kg/m}^3$? (take $g \simeq 9.8\text{ m/s}^2$)
- $T_R = 2630\text{ N}$
 - $T_R = 2680\text{ N}$
 - $T_R = 2540\text{ N}$
 - $T_R = 2750\text{ N}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 45$. What is the descent angle (γ)?
- $\gamma = -1.27^\circ$
 - $\gamma = -1.42^\circ$
 - $\gamma = -1.01^\circ$
 - $\gamma = -0.845^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 1800\text{ kg}$, wing area $S = 30\text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.3$, is given by $T_R = 0.7\bar{\rho}V^2 + 0.018W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 22000\text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 9.8\text{ m/s}^2$ and $\rho_0 = 1.225\text{ kg/m}^3$)
- $V \simeq 50.4\text{ m/s}$
 - $V \simeq 47.3\text{ m/s}$
 - $V \simeq 48.7\text{ m/s}$
 - $V \simeq 48.3\text{ m/s}$

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	a	a	b	d
2	a	a	c	a	b
3	a	c	c	d	d
4	a	b	b	d	c
5	a	b	a	d	b
6	a	c	d	a	c
7	a	b	d	d	c
8	a	a	a	a	b
9	a	c	b	d	b
10	a	b	d	b	b

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	b	d	a	c
12	a	b	c	d	d
13	a	c	b	c	b
14	a	b	d	a	b
15	a	c	d	d	c

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	d	c	b
17	a	a	d	a	b
18	a	b	c	a	c
19	a	d	d	a	d
20	a	b	d	c	d
21	a	b	b	d	b
22	a	b	d	c	c
23	a	c	c	d	d
24	a	a	c	a	a
25	a	d	d	b	a

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	a	b	c	b
27	a	b	d	c	b
28	a	c	a	b	c
29	a	a	a	a	a
30	a	b	d	c	d

BLO-1: Concepts on flight principles and aircraft performances

1. What makes an helicopter essentially different from an autogyro?
 - (a) The powered main rotor.
 - (b) Its having an engine.
 - (c) Its being a rotorcraft.
 - (d) Its being an aerodyne.
2. The (usually rearward) angle of a wing's leading edge with respect to a plane orthogonal to the fuselage is called...
 - (a) sweep.
 - (b) dihedral.
 - (c) angle of attack.
 - (d) pitch.
3. What is the main effect on the lift characteristics ($C_L(\alpha) = C_{L_0} + C_{L_\alpha} \alpha$ for $\alpha < \alpha_s$) of a high lift device that merely increases the chord of the airfoil?
 - (a) It increases the slope ($C_{L_\alpha} \uparrow$).
 - (b) It shifts the curve up ($C_{L_0} \uparrow$).
 - (c) It retards stall ($\alpha_s \uparrow$).
 - (d) None of the others.
4. Which of the following assertions is not true about lift production of an airfoil?
 - (a) It is a consequence of air compressibility.
 - (b) It is a consequence of viscosity.
 - (c) It is a consequence of deflecting air downwards.
 - (d) It is a consequence of creating a pressure difference between upper and lower surfaces.
5. How is lift-induced drag minimised at the design stage of a wing?
 - (a) Aiming at an elliptical lift distribution.
 - (b) With a low aspect ratio design.
 - (c) By introducing dihedral.
 - (d) By introducing a positive twist to transfer lift production from root to tip.
6. Which type of drag, parasitic (D_p) or lift-induced (D_i), dominates at the velocity for minimum drag $V_{D_{min}}$?
 - (a) $D_p = D_i$.
 - (b) $D_p > D_i$.
 - (c) $D_p < D_i$.
 - (d) It depends on altitude.
7. Which of the following conditions is required to achieve the maximum rate of climb for any given aircraft?
 - (a) All are necessary requirements.
 - (b) Full throttle.
 - (c) Maximum excess power.
 - (d) Minimum weight.
8. What must a pilot do to engage in a coordinated banked turn at constant altitude and angle of attack?
 - (a) Increase speed through increasing power.
 - (b) Reduce speed through increasing power.
 - (c) Increase speed through reducing power.
 - (d) Reduce speed through reducing power.
9. Which of the following conditions is a requirement, exact or approximate, for maximising endurance of a propeller-powered airplane?
 - (a) Minimum flight altitude.
 - (b) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (c) Minimum drag.
 - (d) Minimum drag over speed.
10. The speed beyond which an airplane has no longer enough runway to stop in case of one engine failure is called...
 - (a) decision speed.
 - (b) stall speed.
 - (c) takeoff speed.
 - (d) no-go speed.

BLO-2: Flight principles related problems

11. What is the lift coefficient ($C_{L_{opt}}$) for the optimal condition providing maximum aerodynamic efficiency (f_{max}) for an airplane with drag polar $C_D = 0.04 + 0.14C_L^2$?
 - (a) $C_{L_{opt}} = 0.535$
 - (b) $C_{L_{opt}} = 0.557$
 - (c) $C_{L_{opt}} = 0.584$
 - (d) $C_{L_{opt}} = 0.616$

12. An aerobatics airplane with lift coefficient $C_L = 0.15 + 0.12\alpha$ with α in $^\circ$ engages in a perfect dive (vertical descent). What should the angle of attack (α) be?
 - (a) $\alpha = -1.25^\circ$
 - (b) $\alpha = 1.25^\circ$
 - (c) $\alpha = 0.8^\circ$
 - (d) $\alpha = -0.8^\circ$

13. A small airplane of unknown mass is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}). We know its speed is $V = 2.5 V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max})?
 - (a) $\phi_{max} = 80.8^\circ$
 - (b) $\phi_{max} = 82^\circ$
 - (c) $\phi_{max} = 66.4^\circ$
 - (d) It depends on the mass of the airplane.

14. The lift coefficient and drag polar of an airplane changes from $C_L = 0.12 + 0.14\alpha$ and $C_D = 0.04 + 0.14C_L^2$ in clean configuration to $C_L = 0.18 + 0.14\alpha$ and $C_D = 0.06 + 0.08C_L^2$ with full flaps. What is the value of the angle of attack (α) just before optimal flap deployment?
 - (a) $\alpha = 3.27^\circ$
 - (b) $\alpha = 2.84^\circ$
 - (c) $\alpha = 3.13^\circ$
 - (d) $\alpha = 2.7^\circ$

15. An airplane of mass $m = 1600$ kg is cruising in straight and level flight with suboptimal aerodynamic efficiency $f = 22 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
 - (a) $D = 713$ N and $D_p \neq D_i$.
 - (b) $D = 713$ N and $D_p = D_i = 356$ N.
 - (c) $D = 356$ N and $D_p = D_i = 178$ N.
 - (d) $D = 356$ N and $D_p \neq D_i$.

BLO-3: Concepts on aircraft stability and control

16. Dynamic stability requires...
- (a) both a restoring and a damping force/moment.
 - (b) a restoring force/moment only.
 - (c) a damping force/moment only.
 - (d) Neither a restoring nor a damping force/moment.
17. The primary control surfaces are...
- (a) elevator, ailerons and rudder.
 - (b) rudder, canards and spoilers.
 - (c) ailerons, spoilers and flaps.
 - (d) flaps, canards and elevator.
18. Which of the following conditions is not a requirement for longitudinal flight?
- (a) Zero pitch: $\theta = 0$.
 - (b) Aircraft reflection symmetry plane is vertical: $\phi = 0$.
 - (c) No roll or yaw velocities: $p = r = 0$.
 - (d) All are necessary requirements.
19. Which of the following control surfaces is meant for introducing roll moment?
- (a) Ailerons.
 - (b) Elevator.
 - (c) Rudder.
 - (d) Flap.
20. In longitudinal flight, what will the effect have been on the new (pseudo)equilibrium, once the pitch oscillation and phugoid modes have been damped, of pulling the stick (δ_m)?
- (a) Changing both speed (V) and trajectory angle (γ).
 - (b) Changing speed (V) only.
 - (c) Changing the trajectory angle (γ) only.
 - (d) Changing neither speed (V) nor trajectory angle (γ).
21. Which airplane element is responsible for static lateral stability?
- (a) None. Lateral stability is purely dynamical.
 - (b) The vertical stabiliser or fin.
 - (c) The horizontal stabiliser or tailplane.
 - (d) The empennage.
22. Which of the following statements is true about the pitch oscillation mode?
- (a) The horizontal stabiliser size and position is the main responsible for its damping.
 - (b) The angle of attack is constant throughout.
 - (c) It has a long period in the order of minutes.
 - (d) It requires the aerodynamic centre to be ahead of the centre of gravity for stability.
23. Which of the following is a suitable name for the coefficient C_{l_p} ?
- (a) Roll damping coefficient.
 - (b) Dihedral effect coefficient.
 - (c) Directional stability coefficient.
 - (d) Adverse yaw coefficient.
24. How is excessive sideslip corrected for in the event of left engine failure in a twin-engine aircraft?
- (a) Pressing the right pedal.
 - (b) Pressing the left pedal.
 - (c) Switching off the right engine.
 - (d) Sideslip is not corrected upon engine failure.
25. Why can the spiral mode be left unstable for some aircraft?
- (a) Because it is very slow.
 - (b) It can never be left unstable.
 - (c) Because it is very fast.
 - (d) Because it is oscillatory.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.23\bar{\rho}V^2 + 0.022W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1400$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.6$, what is the drag (D) at the optimal velocity for maximum range (V_R)? (take $g \simeq 9.8$ m/s²)
- $D = 2250$ N
 - $D = 1950$ N
 - $D = 1850$ N
 - $D = 2350$ N
27. A propeller-powered airplane of mass $m = 1600$ kg has a required power curve $P_R = 0.28\bar{\rho}V^3 + 0.032W^2/\bar{\rho}V$. What is the minimum engine power needed at sea level ($\bar{\rho} = 1$) to achieve a rate of climb $\dot{h} = 6$ m/s? (take $g \simeq 9.8$ m/s²)
- $P_A = 284$ kW
 - $P_A = 316$ kW
 - $P_A = 333$ kW
 - $P_A = 200$ kW
28. An airplane with drag polar $C_D = 0.04 + 0.16C_L^2$, wing surface $S = 30$ m² and mass $m = 1700$ kg is flying at a speed $V = 110$ m/s. What is the engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.8$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $T_R = 6110$ N
 - $T_R = 6140$ N
 - $T_R = 6050$ N
 - $T_R = 6210$ N
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 43$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 60$ km of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- $h = 1400$ m
 - $h = 2790$ m
 - $h = 2580$ m
 - $h = 717$ m
30. The required thrust of a jet-engined airplane of mass $m = 1000$ kg, wing area $S = 16$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.1$, is given by $T_R = 0.36\bar{\rho}V^2 + 0.022W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 4000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.5$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 53.3$ m/s
 - $V \simeq 42.6$ m/s
 - $V \simeq 37.4$ m/s
 - $V \simeq 46.5$ m/s

31. In which type of jet engine part of the airflow is driven through a bypass?
- (a) Turbofan.
 - (b) Turboprop.
 - (c) Ramjet.
 - (d) Alternative Engine.
32. Which one of the following engines has the best performances during take off, at low speed and low altitude?
- (a) Turboprop.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Scramjet.
33. Which one of the following gas turbine engines can operate at higher speed?
- (a) Turbojet.
 - (b) High bypass ratio turbofan.
 - (c) Turbofan.
 - (d) Alternative engine.
34. Which of the following is not a type of combustion chamber?
- (a) Centrifugal.
 - (b) Can (or tubular).
 - (c) Annular.
 - (d) Can-annular (or tubo-annular).
35. Propulsive efficiency of turbojets...
- (a) depends on the aircraft speed and the exhaust air speed.
 - (b) depends on the type of compressor.
 - (c) is higher with tubular combustion chambers.
 - (d) is highest at very low speed.
36. Which of the following statements is true about helicopters?
- (a) The rotor is designed for vertical, longitudinal and lateral control.
 - (b) Helicopters are inherently stable aircrafts.
 - (c) Vertical control is achieved through control surfaces in the tail.
 - (d) Helicopters are used because of their low operational cost.
37. In helicopters, what is the reverse flow region of a rotor?
- (a) The area of the rotor disc in which the air around the blade airfoil flows from trailing edge to leading edge.
 - (b) The area of the rotor disc in which the air around the blade airfoil flows from leading edge to trailing edge.
 - (c) The area over the top of a mountain in which the flight of an helicopter is impossible due to strong vertical wind.
 - (d) The area of the rotor disc in which the air speed is maximum.
38. Which one of the following does not constitute a control system in helicopters?
- (a) Thrust reverser control system.
 - (b) Cyclic pitch control system.
 - (c) Collective pitch control system.
 - (d) Tail rotor pitch control system.
39. In which of the following orbits the satellite is always in the vertical of a same fixed point on the Earth surface?
- (a) GEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
40. Which of the following is not a function of propulsive systems in space vehicles?
- (a) Communications.
 - (b) Lift the vehicle and the payload from the launch pad and place the payload into a low-Earth orbit.
 - (c) Transfer payloads from low-Earth orbits into higher orbits or inter-planetary trajectories.
 - (d) Attitude control and orbit correction.

BLO-6: Navigation & altimetry

41. Which of the following statements on Required Navigation Performance (RNP) is FALSE:
- (a) The RNP “configures the airspace to fit the least capable aircraft”, rather than “designs aircraft to fit the airspace”.
 - (b) The RNP concept was developed by the International Civil Aviation Organization (ICAO) in the 90s.
 - (c) The RNP central idea is the aircraft containment surface, or tunnel.
 - (d) The RNP allows ICAO to only define airspace requirements, and not specifying required equipment.
42. Which of the following statements on the Tunnel concept is FALSE:
- (a) The inner tunnel dimensions contain 50% of aircraft deviations from assigned path.
 - (b) The outer tunnel determines the standard separation from terrain, obstacles, other aircraft, etc.
 - (c) Around the flight path an aircraft containment surface (the tunnel) is built.
 - (d) A tunnel incident happens when the aircraft unintentionally leaves the tunnel.
43. Which of the following statements on the Target Level of Safety is correct:
- (a) Historically, the accident-to-incident ratio is 1/10.
 - (b) A typical tunnel incident risk is 10^{-3} .
 - (c) The number of Hull Loss accidents are higher in the en-route phase of flight.
 - (d) The number of Hull Loss accidents are mostly associated to maintenance cause factor.
44. Which of the following statements on Accuracy is FALSE:
- (a) Accuracy is the ability to stay within the outer tunnel.
 - (b) Accuracy is the ability of the total system to maintain the aircraft position within a given Total System Error (TSE) with a 95% probability.
 - (c) Accuracy relates to nominal conditions.
 - (d) The Total System Error (TSE) is related to the inner tunnel.
45. Which of the following statements on Integrity is FALSE:
- (a) Integrity relates to ordinary conditions.
 - (b) Integrity is a measure of the trust which can be placed in the correctness of the information supplied by the total system.
 - (c) If not inside the outer tunnel, the user must quickly receive an alarm.
 - (d) Integrity provides some guarantee of staying within the outer tunnel.
46. Which of the following statements on Continuity is TRUE:
- (a) Continuity is the ability of the total system to perform its function without unscheduled interruptions during the intended operation.
 - (b) Continuity is a kind of long-term reliability.
 - (c) Continuity is compatible with the non-availability of the tunnel while we are carrying out the operation.
 - (d) Continuity is automatically accomplished once integrity is fulfilled.
47. Determine which of the following assumptions or auxiliary equations are NOT required to derive the pressure, density and altitude relationship:
- (a) The Maxwell equations.
 - (b) The hydrostatic equilibrium equation.
 - (c) Assuming a constant gravity field.
 - (d) The ideal gas law.
48. Which of the following statements related with the barometer-based altimeter calibration is true:
- (a) The STD adjustment assumes 1013 hPa and 15 C at sea level.
 - (b) The QNH adjustment guarantees the vertical separation between flight levels.
 - (c) The QFE adjustment provides the true height above the geoid for the airfield.
 - (d) One flight level is equivalent to 100 meters of altitude given by the barometer-based altimeter.
49. Which of the following kind of components does NOT belong to the definition of avionics?
- (a) Cattering.
 - (b) Navigation.
 - (c) Flight control.
 - (d) Engine control.
50. What equipment is required to determine the True Air Speed (TAS):
- (a) A Pitot tube plus a Static source.
 - (b) A Pitot tube.
 - (c) A Static source.
 - (d) A GNSS receiver.

BLO-7: Aircraft recognition and materials & structures

51. To which airplane corresponds ICAO designation A358?

- (a) Airbus A350-800.
- (b) Airbus A300-580.
- (c) Airbus A358-100.
- (d) None of the others.

52. In which of the following situations will the minimum radar separation need to be larger?

- (a) Super aircraft preceding small aircraft.
- (b) Small aircraft preceding super aircraft.
- (c) Small aircraft preceding small aircraft.
- (d) Super aircraft preceding super aircraft.

53. Identify the following aircraft



- (a) Boeing 737.
- (b) Boeing 717.
- (c) Airbus A320.
- (d) Airbus A340.

54. Identify the following fighter aircraft



- (a) Sukhoi SU-34.
- (b) MiG 29.
- (c) Joint Strike Fighter (LM F-35 Lightning).
- (d) BAe Sea Harrier FA2.

55. Which one of the following aircraft is the largest in terms of cargo/passenger capacity?

- (a) Airbus A380.
- (b) Airbus A340.
- (c) Boeing 777.
- (d) Boeing 747.

56. The ratio of the stress along an axis to the strain (ratio of deformation over initial length) is called...

- (a) Young modulus.
- (b) elastic limit.
- (c) fatigue.
- (d) resilience.

57. Which of the following is a valid material selection criterion for aeronautical applications?

- (a) All options are correct.
- (b) Cost.
- (c) Thermal and electric properties.
- (d) Corrosion resistance.

58. Aluminum alloys can highly improve the mechanical properties of pure aluminium but...

- (a) will deteriorate its corrosion behaviour.
- (b) will turn it into an electrical insulator.
- (c) will create a lot of cementite.
- (d) All options are correct.

59. Which of the following can be a component of steel?

- (a) All the options are correct.
- (b) Martensite.
- (c) Pearlite.
- (d) Ferrite.

60. Which of the following is not an advantage of carbon fiber-epoxy parts?

- (a) Protect adjacent Aluminum parts from corrosion.
- (b) Very good fatigue behaviour.
- (c) Corrosion resistant.
- (d) Tailored mechanical properties.

BLO-8: Aircraft systems

61. Which of the following is not a part of the flight control system?
- (a) Fuel Pump.
 - (b) Side stick.
 - (c) Ailerons.
 - (d) Rudder.
62. A fly-by-wire flight control system is a flight control system in which...
- (a) the linkage between cockpit controls and the hydraulic actuator system that operates the control surfaces is made through an electronic system only.
 - (b) the linkage between cockpit controls and the hydraulic actuator system that operates the control surfaces is made with a system of rods, levers, cables, pulleys, springs and/or dampers.
 - (c) the linkage between cockpit controls and control surfaces consists of rods, levers, cables, pulleys, springs, dampers and no additional power source.
 - (d) the pilot pulls strings to deflect control surfaces.
63. Which of the following can be a source of pressure for the hydraulic system?
- (a) RAT.
 - (b) Bleed valve.
 - (c) Static inverter.
 - (d) CVR.
64. Which of the following is not a part of a hydraulic system?
- (a) DC Bus.
 - (b) Power Transfer Unit.
 - (c) Reservoir.
 - (d) Filter.
65. Which of the following is a part of an electric system?
- (a) Inverter.
 - (b) Accumulator.
 - (c) Slat.
 - (d) Fuel vent.
66. Which of the following is not a part of the electric system?
- (a) LP compressor.
 - (b) ESS Bus.
 - (c) Battery.
 - (d) Transformer.
67. Which of the following is a function of the APU?
- (a) Back up air conditioning system in flight.
 - (b) Remove power from secondary systems in the event of an emergency.
 - (c) Remove ice from the wing.
 - (d) Pump fuel to the engines.
68. Which of the following is not a function of the fuel system?
- (a) Reduce fuel weight during take off.
 - (b) Eject fuel to the atmosphere if necessary.
 - (c) Pump fuel to the engines.
 - (d) Store fuel.
69. Which system will start working after a cabin depressurisation at cruise altitude due to a failure in the air conditioning system?
- (a) Oxygen system.
 - (b) Fire protection system.
 - (c) Autoflight system.
 - (d) Light system.
70. Which of the following systems is fed by the hydraulic system?
- (a) Brake system.
 - (b) Engine nacelle anti-ice system.
 - (c) Wing anti-ice system.
 - (d) Potable water tank pressurisation system.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performances

	Permutation Number				
Question	0	1	2	3	4
1	a	b	d	c	b
2	a	b	a	a	a
3	a	a	d	c	c
4	a	b	b	c	c
5	a	a	a	b	a
6	a	c	c	c	d
7	a	b	a	d	d
8	a	a	a	b	a
9	a	d	d	b	a
10	a	c	a	d	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	c	a	c	a
12	a	d	b	c	d
13	a	b	c	a	a
14	a	c	d	d	a
15	a	b	b	b	b

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	a	c	d
17	a	d	d	b	b
18	a	c	c	a	a
19	a	d	b	d	a
20	a	a	d	a	c
21	a	c	b	b	a
22	a	a	c	c	b
23	a	a	c	c	a
24	a	d	c	b	d
25	a	c	d	c	b

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	b	c	c
27	a	c	d	b	c
28	a	d	a	a	d
29	a	c	c	c	c
30	a	d	b	a	b

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	d	b	d
32	a	d	b	c	c
33	a	c	b	d	b
34	a	d	d	a	a
35	a	b	c	b	b
36	a	c	d	b	d
37	a	d	d	c	b
38	a	a	c	d	b
39	a	d	a	b	a
40	a	d	d	c	c

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
41	a	d	b	a	d
42	a	d	c	b	a
43	a	c	c	a	b
44	a	a	c	d	c
45	a	d	b	a	d
46	a	c	b	d	c
47	a	d	d	c	a
48	a	c	b	d	b
49	a	c	b	a	b
50	a	a	b	a	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
51	a	a	d	b	c
52	a	c	a	a	d
53	a	c	b	b	d
54	a	b	a	c	c
55	a	c	a	b	c
56	a	c	a	b	c
57	a	b	c	d	a
58	a	d	b	d	b
59	a	d	b	c	a
60	a	b	c	a	b

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	b	c	c	a
62	a	b	d	d	a
63	a	c	a	d	b
64	a	b	b	c	a
65	a	a	a	b	a
66	a	d	a	a	d
67	a	c	a	d	c
68	a	c	d	a	d
69	a	c	a	a	b
70	a	a	d	c	b

1. Which of the following jet engines has a bypass?
 - (a) Turbofan.
 - (b) Turboprop.
 - (c) Ramjet.
 - (d) Alternative Engine.
2. What happens to the the thrust of a turbojet when air pressure raises?
 - (a) Increases.
 - (b) Decreases.
 - (c) Remains approximately constant.
 - (d) Increases with centrifugal compressors and decreases with axial ones.
3. Supersonic intake are...
 - (a) convergent.
 - (b) divergent.
 - (c) constant in area.
 - (d) convergent in high bypass turbofans and divergent in low bypass turbofans.
4. Which of the following statements is false regarding axial compressors?
 - (a) They can increase the pressure of the airflow three-fold per stage.
 - (b) They retain highest efficiency only within a narrow rpm range.
 - (c) They require a larger starting torque than centrifugal compressors.
 - (d) They have a loss of compresion efficiency due to temperature rise at high speed.
5. Subsonic nozzles are...
 - (a) convergent.
 - (b) divergent.
 - (c) equipped with an afterburner only in turbofans.
 - (d) obsolete in modern aircraft.
6. Which of the following sentences is not true about helicopters?
 - (a) Direct operational cost of helicopters is cheaper than any other aircraft.
 - (b) Landing gear of helicopters can be a skids, wheels and/or floats.
 - (c) Helicopters are capable of all-direction-flight.
 - (d) Helicopters are capable of vertical take-off and landing.
7. In helicopters, which of the following is not a rotor type?
 - (a) NOTAR.
 - (b) Articulated.
 - (c) Rigid.
 - (d) Semirigid.
8. How is directional control achieved in conventional helicopters?
 - (a) Modifying the tail anti-torque rotor pitch.
 - (b) Deflecting the tail control surfaces.
 - (c) Deflecting ailerons.
 - (d) Differential thrust.
9. Which of the following is not an orbital element?
 - (a) Thrust.
 - (b) Argument of periapsis.
 - (c) Semimajor axis.
 - (d) Inclination.
10. The specific impulse of a rocket with an adapted nozzle is...
 - (a) The exhaust gas speed divided by the Earth surface gravity acceleration.
 - (b) The structural mass index times the payload.
 - (c) The thrust divided by payload.
 - (d) The exhaust gas speed times the payload index.

BLO-6: Navigation & altimetry

11. The internal tunnel, defined in Required Navigation Performance (RNP), is directly connected with...
- (a) accuracy.
 - (b) integrity.
 - (c) availability.
 - (d) continuity.
12. The line Y (vertical) = X (horizontal) in the Stanford plot represents...
- (a) the points for which estimated positioning error equals to the actual positioning error.
 - (b) the points for which estimated positioning error equals to the actual position.
 - (c) the points for which estimated position equals to the actual positioning error.
 - (d) the points for which estimated position equals to the actual position.
13. Regarding the accuracy and precision concepts in navigation systems for civil aviation...
- (a) the accuracy is more important than the precision.
 - (b) the precision is more important than the accuracy.
 - (c) the target is to achieve the same precision than the accuracy.
 - (d) the target is to get an accuracy value equals to 95% of the precision value.
14. The way in which the integrity is implemented in civil aviation navigation is...
- (a) None of the others.
 - (b) by disconnecting the service, after detecting the failure, in Global Navigation Satellite Systems like GPS.
 - (c) by an alarm sound in the cabin, transmitted from the conventional radio aids.
 - (d) by an alarm sound in the cabin, transmitted from the Global Navigation Satellite Systems like GPS.
15. Regarding the Global Positioning System (GPS) receiver, it is true that...
- (a) it can be initially considered a sort of chronometer estimating travel times between each satellite in view and the user.
 - (b) it can be considered a sort of radar estimating two-way travel times between each satellite in view to the user and to the same satellite again.
 - (c) it carries an atomic clock.
 - (d) it transmits the user position to the GPS satellite which broadcast it.
16. Which of the following statements on avionics is false?
- (a) It comprises the optimal organization of the aircrews.
 - (b) It deals with the processing of barometric data.
 - (c) It deals with the airplane communications.
 - (d) It comprises a central processing computer on the airplane with safety redundancies.
17. A pilot is observing how the height estimates provided by the Satellite Based Augmentation System (SBAS) GPS receiver, and without the notification of any tunnel incident, are constant, flying at a constant flight level. In this situation you can infer that...
- (a) there are likely no horizontal pressure and temperature gradients on the ground, under Standard (STD/QNE) altimeter calibration.
 - (b) the temperature and pressure on sea level are likely 15°C and 1013.25 hPa respectively under Standard (STD/QNE) barometer calibration.
 - (c) a QFE altimeter calibration is being used .
 - (d) a Standard (STD/QNE) altimeter calibration is being used.
18. During a certain phase of the flight, the airplane altimeter and the SBAS GPS receiver height are in good agreement. Then you can infer that...
- (a) None of the others.
 - (b) you are either landing or taking off.
 - (c) you are en route with good weather.
 - (d) you are using the QFE altimeter calibration.
19. Regarding the measurement of the true air speed TAS of a low speed aircraft, it is false that...
- (a) the Saint-Venant equation is required.
 - (b) the Pitot tube and static port are both needed.
 - (c) the Bernoulli equation is used as main model.
 - (d) the knowledge of the wind velocity is not required.
20. Regarding the usage of inertial positioning systems in civil aviation, it is true that...
- (a) the initial velocity and position vectors are required.
 - (b) it is less accurate in general than the dead-reckoning technique.
 - (c) the acceleration is time-integrated once to estimate the position.
 - (d) the acceleration error bias propagates linearly in time in terms of positioning error.

21. To which airplane corresponds ICAO designation A306?

- (a) Airbus A300-600.
- (b) Airbus A306.
- (c) Airbus A320-600.
- (d) None of the others.

22. Which of the following is not a wide body?

- (a) Airbus A319.
- (b) Boeing 747.
- (c) Airbus A330.
- (d) Boeing 777.

23. Identify the following aircraft



- (a) Airbus A320.
- (b) Airbus A340.
- (c) Boeing 737.
- (d) Boeing 767.

24. Which aircraft manufacturer builds the following airplane?



- (a) Bombardier.
- (b) Airbus.
- (c) Antonov.
- (d) Douglas Corporation.

25. Which one of the following aircraft is the shortest?

- (a) Boeing 737-600.
- (b) Boeing 737-900.
- (c) Airbus A319.
- (d) Airbus A320.

26. The maximum stress for a material before breaking is called...

- (a) ultimate strength.
- (b) Young modulus.
- (c) resilience.
- (d) hardness.

27. The damage of a material under a cyclic stress is called...

- (a) fatigue.
- (b) resilience.
- (c) corrosion.
- (d) abrasion.

28. What is 7075-T6?

- (a) An Aluminum alloy.
- (b) A hardness scale.
- (c) A Titanium jet fighter.
- (d) A type of landing gear fitting.

29. Which of the following is false about Titanium?

- (a) It has a poor behaviour at high temperatures.
- (b) It has a high corrosion resistance.
- (c) It is expensive.
- (d) It has an allotropic transformation at 882.5°C from HCP to BCC.

30. Which of the following materials is not used in composites as the matrix?

- (a) Kevlar.
- (b) Polyester.
- (c) Vinylester.
- (d) Epoxy.

BLO-8: Aircraft systems

31. Which of the following is not a part of the flight control system?
- (a) RAT.
 - (b) Elevator.
 - (c) Flap lever.
 - (d) Rudder.
32. What is a hydro-mechanical flight control system?
- (a) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made with a system of rods, levers, cables, pulleys, springs and/or dampers.
 - (b) A flight control system in which the linkage between cockpit controls and control surfaces is made with a system of rods, levers, cables, pulleys, springs, dampers and no auxiliary power source.
 - (c) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made through electronic components only.
 - (d) A flight control system for seaplanes.
33. Which of the following can be a source of pressure of the hydraulic system?
- (a) Electric pump.
 - (b) Oil pump.
 - (c) Pitot-static system.
 - (d) AC bus.
34. Which of the following is not a part of a hydraulic system?
- (a) HP bleed valve.
 - (b) Accumulator.
 - (c) Reservoir.
 - (d) Engine-driven pump.
35. Which of the following is part of an electric system?
- (a) Battery.
 - (b) Yoke.
 - (c) FDR.
 - (d) Oxygen mask.
36. Which of the following systems is not part of the electric system?
- (a) Thrust reverser.
 - (b) Circuit breaker.
 - (c) APU Generator.
 - (d) ESS Bus.
37. Which of the following is a function of the APU?
- (a) Provide power to start the engines.
 - (b) Wipe rain drops from the windshield.
 - (c) Back up VHF communications in flight.
 - (d) Activate wheel brakes.
38. Which of the following is not a function of the fuel system?
- (a) Inject fuel in the engines combustion chamber.
 - (b) Indicate fuel quantity on board.
 - (c) Circulate fuel among tanks.
 - (d) Defuel the aircraft.
39. Which of the following aircraft parts is usually equipped with an anti-ice system?
- (a) Engine inlet lips.
 - (b) Engine nozzle.
 - (c) Main generators.
 - (d) FDR.
40. Which of the following systems provides the de-ice system with power?
- (a) Pneumatic system.
 - (b) Hydraulic system.
 - (c) Brake system.
 - (d) Oxygen system.

Correct Answers

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	b	d	a	d
2	a	d	a	c	c
3	a	c	b	b	a
4	a	a	d	a	a
5	a	c	d	b	c
6	a	d	b	d	d
7	a	a	a	a	a
8	a	c	c	b	c
9	a	c	c	c	d
10	a	b	a	a	d

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	b	b	d	c
12	a	a	c	d	d
13	a	d	a	c	d
14	a	b	c	d	c
15	a	b	c	a	b
16	a	c	c	a	b
17	a	d	b	a	b
18	a	a	a	c	d
19	a	a	d	c	a
20	a	d	d	c	a

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	c	b	c	a
22	a	d	d	a	c
23	a	b	c	b	a
24	a	a	c	d	b
25	a	d	a	a	c
26	a	d	b	d	b
27	a	c	b	d	a
28	a	b	a	a	c
29	a	a	c	c	c
30	a	d	a	b	c

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	d	d	a
32	a	b	a	b	d
33	a	d	d	c	a
34	a	c	c	a	d
35	a	a	b	c	a
36	a	d	b	b	d
37	a	d	b	b	b
38	a	b	a	c	c
39	a	d	c	d	a
40	a	c	b	d	d

1. Which of the following jet engines has a propeller?
 - (a) Turboprop.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Scramjet.
2. What happens to the the maximum thrust of a turbojet when air temperature raises?
 - (a) Decreases.
 - (b) Increases.
 - (c) Remains approximately constant.
 - (d) Increases in two spools turbojet and decreases in single spool ones.
3. Subsonic intakes are...
 - (a) divergent.
 - (b) convergent.
 - (c) convergent in high bypass turbofan and divergent in low bypass turbofan.
 - (d) smaller than supersonic intakes.
4. Which of the following statements is false regarding centrifugal compressors?
 - (a) Pressure increase is only around 10 fold
 - (b) They retain good performances in a wide rpm range.
 - (c) They expose a larger frontal area than axial compressors for the same air flow.
 - (d) Multi-stage designs suffer from efficiency losses due to inter-stage ducts.
5. Supersonic nozzle exits are...
 - (a) divergent.
 - (b) convergent.
 - (c) bigger for axial compressors.
 - (d) divergent or convergent depending on the type of turbine blades.
6. Which of the following statements is not true about helicopters?
 - (a) Tail horizontal stabilizer control in helicopters provides longitudinal control.
 - (b) Helicopters are VTOL capable.
 - (c) The main rotor provides both thrust and control in helicopters.
 - (d) Helicopters are capable of stationary flight.
7. In helicopters, which of the following is not a main rotor blade movement?
 - (a) Slat.
 - (b) Drag.
 - (c) Flap.
 - (d) Pitch.
8. How is vertical control achieved in helicopters?
 - (a) Through collective pitch control.
 - (b) Deflecting the tail control surfaces.
 - (c) With NOTAR.
 - (d) Through differential thrust.
9. Which of the following is not an orbital element?
 - (a) Specific impulse.
 - (b) Eccentricity.
 - (c) Longitude of the ascending node.
 - (d) Mean anomaly.
10. In a rocket with adapted nozzle, thrust is...
 - (a) the gas mass-flow times the exhaust gas speed.
 - (b) the difference of the inner and outer pressure times the nozzle area.
 - (c) Depends on the weight of the vehicle.
 - (d) Maximum in the Equator.

BLO-6: Navigation & altimetry

11. The external tunnel, defined in Required Navigation Performance (RNP), is directly connected with...
- (a) integrity
 - (b) accuracy
 - (c) availability
 - (d) continuity
12. The Y (vertical) and X (horizontal) axis of the Stanford plot represent, respectively,...
- (a) the estimated positioning error vs. the actual positioning error.
 - (b) the actual positioning error vs. the estimated positioning error.
 - (c) the actual position vs. the estimated position.
 - (d) the estimated position vs. the actual position.
13. Regarding the accuracy and precision concepts in navigation systems...
- (a) accuracy is related to the unbiased error property and precision to the repeatability property.
 - (b) both accuracy and precision are equivalent concepts.
 - (c) accuracy implies precision.
 - (d) precision implies accuracy.
14. The way in which the integrity is implemented in civil aviation navigation is...
- (a) by disconnecting the service, after detecting the failure, in conventional radio aids.
 - (b) by providing a reliable error estimate, almost always slightly higher than the actual one, in conventional radio aids.
 - (c) by disconnecting the service, after detecting the failure, in Global Navigation Satellite Systems like GPS.
 - (d) None of the others.
15. Regarding the Global Positioning System (GPS) functioning foundations, they can be briefly and roughly described as:
- (a) The user positioning is the result of the intersection of the spheres centered at each satellite in view and with radius each pseudorange measurement.
 - (b) The pseudorange is measured thanks to the two-way travel time between satellite and user.
 - (c) The pseudorange is measured thanks to the travel time between two satellites in view.
 - (d) The user positioning is the result of the double integration in time of the user acceleration.
16. Which of the following statements on avionics is true?
- (a) It comprises the electronics required for the normal airplane operation.
 - (b) It comprises the scientific fundamentals required for the normal airplane operation.
 - (c) It consists on the usage of airplanes for sport competition.
 - (d) It studies all the electronics and mechanical aspects affecting small airplanes.
17. A pilot is observing how the height estimates provided by the Satellite Based Augmentation System (SBAS) GPS receiver is showing a positive deviation first, and a negative deviation later on, flying at a constant flight level. In this situation it can be inferred that...
- (a) None of the others.
 - (b) the barometer system is not working well
 - (c) the SBAS GPS receiver is not working well
 - (d) the pressure at sea level is not 1013.25 hPa
18. During a certain phase of the flight, the airplane altimeter and the SBAS GPS receiver height are in good agreement. Then it can be inferred that...
- (a) None of the others.
 - (b) the airplane is landing or taking off.
 - (c) the airplane is en route with good weather.
 - (d) the airplane is using the QFE altimeter calibration.
19. Regarding the measurement of the airplane velocity for a commercial airplane, it is true that...
- (a) the Pitot tube and static port are both needed.
 - (b) the Pitot tube is only needed.
 - (c) the Static port is only needed.
 - (d) the Bernoulli equation is used as main model.
20. Regarding the usage of inertial positioning systems in civil aviation, it is false that...
- (a) it is less accurate in general than the dead-reckoning technique.
 - (b) the acceleration is double-time integrated to estimate the position.
 - (c) the acceleration error bias propagates quadratically in time with respect to positioning error.
 - (d) the initial velocity and position vectors are required.

BLO-7: Aircraft recognition and materials & structures

21. To which airplane corresponds ICAO designation B743?

- (a) Boeing 747-300.
- (b) Boeing 737-400.
- (c) Boeing 740-300.
- (d) None of the others.

22. Which of the following is not a wide body?

- (a) Boeing 737.
- (b) Boeing 767.
- (c) Airbus A300.
- (d) Airbus A340..

23. Identify the following aircraft



- (a) Boeing 777.
- (b) Boeing 767.
- (c) Airbus A340.
- (d) Airbus A330.

24. Which aircraft manufacturer builds the following airplane?



- (a) Antonov.
- (b) Embraer.
- (c) Airbus.
- (d) Boeing.

25. Which one of the following aircraft is the longest?

- (a) Airbus A321.
- (b) Boeing 737-600.
- (c) Boeing 737-500.
- (d) Airbus A319.

26. The maximum stress for elastic deformation of a material is called...

- (a) elastic limit.
- (b) ultimate strenght.
- (c) resilience.
- (d) hardness.

27. Which of the following is not a manufacturing selection criteria for materials?

- (a) Density.
- (b) Malleability.
- (c) Ductility.
- (d) Castability.

28. What is true about Aluminum?

- (a) Its crystal structure is FCC.
- (b) It is rarely used in aerospace structures.
- (c) It features a very high melting point.
- (d) In pure state has its best mechanical behaviour.

29. Which of the following statements is false about steel?

- (a) It cannot be used in parts with high mechanical responsibility because of its poor behaviour.
- (b) Its main components are Fe and C.
- (c) Its crystal structure is BCC below 911°C.
- (d) It is around three times denser than Aluminum.

30. Which of the following materials is not used in composites as a reinforcement fiber?

- (a) Vinylester.
- (b) Spectra.
- (c) Boron.
- (d) Kevlar.

BLO-8: Aircraft systems

31. Which of the following is not a part of the flight control system?
- (a) APU.
 - (b) Rudder pedals.
 - (c) Ailerons.
 - (d) Spoilers.
32. What is a mechanical flight control system?
- (a) A flight control system in which the linkage between cockpit controls and control surfaces is made with a system of rods, levers, cables, pulleys, springs, dampers and no auxiliary power source.
 - (b) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made with a system of rods, levers, cables, pulleys, springs and/or dampers.
 - (c) A flight control system in which the linkage between cockpit controls and the hydraulic actuator system which operates the control surfaces is made through electronic components only.
 - (d) A flight control system with limited self-repair capacity.
33. Which of the following can be a source of pressure of the hydraulic system?
- (a) Engine-driven pump.
 - (b) Transformer rectifier.
 - (c) Fuel pump.
 - (d) De-ice boot.
34. Which of the following is not a part of a hydraulic system?
- (a) APU Generator.
 - (b) Accumulator.
 - (c) Valve.
 - (d) Filter.
35. Which of the following is part of an electric system?
- (a) APU generator.
 - (b) Oil pump.
 - (c) Air conditioning pack.
 - (d) Landing gear.
36. Which of the following is not part of the electric system?
- (a) Cross feed valve.
 - (b) AC bus.
 - (c) DC bus.
 - (d) Emergency generator.
37. Which of the following is a function of the APU?
- (a) Back up the electrical system in flight.
 - (b) Operate engine anti-ice system.
 - (c) Transfer fuel between tanks.
 - (d) Provide the crew and passengers with Oxygen in the event of a depressurization.
38. Which of the following is not a function of the fuel system?
- (a) Calculate fuel required for the flight.
 - (b) Vent fuel tanks.
 - (c) Supply fuel to engines.
 - (d) Refuel the aircraft.
39. Which of the following aircraft parts is usually equipped with an anti-ice system?
- (a) Wing leading edge.
 - (b) Wing trailing edge.
 - (c) Fuel tanks.
 - (d) Rudder pedals.
40. Which of the following systems provides the brake system with power?
- (a) Hydraulic system.
 - (b) Electric system.
 - (c) Oxygen system.
 - (d) Potable water tank pressurization system.

Correct Answers

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	b	d	d	c
2	a	a	a	c	a
3	a	b	b	b	d
4	a	a	b	a	d
5	a	d	b	b	b
6	a	a	c	a	a
7	a	a	c	c	a
8	a	b	c	a	b
9	a	b	b	b	b
10	a	b	a	a	c

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	a	a	a	c
12	a	b	a	b	b
13	a	c	a	d	d
14	a	a	c	d	a
15	a	a	b	b	a
16	a	c	c	d	c
17	a	d	a	b	d
18	a	b	b	a	d
19	a	c	b	b	a
20	a	b	b	d	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	a	b	b	c
22	a	a	a	c	d
23	a	a	a	d	a
24	a	c	a	c	b
25	a	b	c	a	a
26	a	c	b	b	a
27	a	d	d	d	d
28	a	c	b	d	b
29	a	d	b	b	c
30	a	c	a	b	d

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	a	c	b
32	a	b	a	d	a
33	a	a	d	d	b
34	a	d	a	b	a
35	a	a	b	d	d
36	a	c	a	b	b
37	a	a	a	c	c
38	a	d	b	b	d
39	a	c	a	a	b
40	a	b	d	d	d

BLO-1: Concepts on flight principles and aircraft performance

1. Which of the following aircraft types is better suited for rescue missions in regions of difficult access where landing is not an option?
 - (a) Rotorcraft.
 - (b) Fixed wing aerodyne.
 - (c) Space vehicle.
 - (d) Rocket.
2. Which of the following is not an advantage for low aspect-ratio wings?
 - (a) higher aerodynamic efficiency.
 - (b) higher manoeuvrability.
 - (c) lower structural weight.
 - (d) higher speeds achievable.
3. A wing that has a variable chord along its span is called...
 - (a) tapered.
 - (b) swept.
 - (c) pitched.
 - (d) twisted.
4. Which of the following assertions is false about the boundary layer?
 - (a) It is always laminar.
 - (b) It is a consequence of viscosity.
 - (c) It is not necessary to "model" lift, as long as viscous effects on circulation are somehow taken into account.
 - (d) It is characterised by strong transversal velocity gradients.
5. What is true about two identical airplanes performing a straight and level flight at different altitudes?
 - (a) They experience the same lift.
 - (b) Their speed must be the same.
 - (c) Their angle of attack must be the same.
 - (d) Their thrust must be the same.
6. What happens to the minimum of $D/V(V)$ as a constant mass airplane gains altitude?
 - (a) It occurs at higher speeds (moves right) and decreases (moves down).
 - (b) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at lower speeds (moves left) but remains fairly constant (neither goes up nor down).
 - (d) It occurs at lower speeds (moves left) and increases (moves up).
7. How is a glider flown for optimality?
 - (a) Minimum descent angle while headed towards a column of ascending air and minimum rate of descent once in it.
 - (b) Minimum rate of descent while headed towards a column of ascending air and minimum descent angle once in it.
 - (c) Minimum rate of descent always.
 - (d) Minimum descent angle always.
8. An airplane flying on full throttle engages in a coordinated banked turn at a desired bank angle. What must be done and what will happen to speed if altitude is to be preserved?
 - (a) Increase angle of attack; speed will decrease.
 - (b) Increase angle of attack; speed will increase.
 - (c) Reduce angle of attack; speed will increase.
 - (d) Reduce angle of attack; speed will decrease.
9. Which of the following conditions is a requirement for maximising cruise range of any airplane?
 - (a) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (b) Minimum flight altitude.
 - (c) Minimum drag.
 - (d) Minimum drag over speed.
10. Which parameters influence landing distance?
 - (a) All of them.
 - (b) Braking power.
 - (c) Airport altitude.
 - (d) Full flaps stall speed.

BLO-2: Flight principles related problems

11. What is the optimal angle of attack (α_{opt}) that maximises aerodynamic efficiency (f_{max}) for an airplane with lift coefficient $C_L = 0.2 + 0.12\alpha$ for $\alpha < \alpha_S = 12^\circ$ and drag polar $C_D = 0.025 + 0.15 C_L^2$?
- (a) $\alpha_{opt} = 1.74^\circ$
 - (b) $\alpha_{opt} = 12^\circ$
 - (c) $\alpha_{opt} = 1.6^\circ$
 - (d) $\alpha_{opt} = 1.35^\circ$
12. An aerobatics airplane with lift coefficient $C_L = 0.05 + 0.1\alpha$ with α in $^\circ$ is performing a balanced inverted flight. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.2$?
- (a) $\alpha = -2.5^\circ$
 - (b) $\alpha = 1.5^\circ$
 - (c) $\alpha = -0.5^\circ$
 - (d) $\alpha = 2.5^\circ$
13. A small airplane engages in a constant altitude coordinated banked turn with bank angle ($\phi = 30^\circ$) and speed $V = 60$ m/s. What is the turn radius (R_t)? (take $g \simeq 9.8$ m/s²)
- (a) $R_t = 636$ m
 - (b) $R_t = 729$ m
 - (c) $R_t = 504$ m
 - (d) $R_t = 838$ m
14. The drag polar of an airplane changes from $C_D = 0.025 + 0.1C_L^2$ in clean configuration to $C_D = 0.035 + 0.08C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.075$
 - (b) $C_D = 0.0651$
 - (c) $C_D = 0.101$
 - (d) $C_D = 0.114$
15. An airplane of mass $m = 1100$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 16 = f_{max}$. Its velocity is $V = 70$ m/s. What can we say about the total (P_R), induced (P_{R_i}) and parasitic (P_{R_p}) required power? (take $g \simeq 9.8$ m/s²)
- (a) $P_R = 47.2$ kW and $P_{R_p} = P_{R_i} = 23.6$ kW.
 - (b) $P_R = 674$ kW and $P_{R_p} = P_{R_i} = 337$ kW.
 - (c) $P_R = 674$ kW and $P_{R_p} > P_{R_i}$.
 - (d) $P_R = 47.2$ kW and $P_{R_p} \neq P_{R_i}$.

BLO-3: Concepts on aircraft stability and control

16. Why would one tend to design unstable airplanes that are automatically stabilised?
- (a) For better manoeuvrability.
 - (b) For cheaper design costs.
 - (c) For marketing purposes.
 - (d) None of the others.
17. Which of the following combined control surfaces makes no sense whatsoever?
- (a) All may be implemented in certain aircraft.
 - (b) Flaperons.
 - (c) Stabilators.
 - (d) Ruddervators.
18. What is the main role of the elevator trim tab?
- (a) Allow setting any desired angle of attack without having to hold the stick.
 - (b) Produce extra lift for landing.
 - (c) Reduce wing tip vortices.
 - (d) Trim the pilot's beard while in the elevator.
19. Which of the following control surfaces is meant for introducing yaw moment?
- (a) Rudder.
 - (b) Elevator.
 - (c) Ailerons.
 - (d) Flap.
20. What do spoilers do when deployed?
- (a) Reduce wing lift.
 - (b) Increase wing lift.
 - (c) Increase wing stall angle.
 - (d) Reduce wing drag.
21. Longitudinal static instability entails...
- (a) longitudinal dynamic instability.
 - (b) directional dynamic instability.
 - (c) directional static instability.
 - (d) nothing can be anticipated.
22. What cannot be said about the return to straight and level flight mode?
- (a) It couples rotations about two axes.
 - (b) It is aperiodic.
 - (c) It is related to density decrease with altitude.
 - (d) It is the reason for prefixing with *pseudo* the word equilibrium when referring to a straight climb.
23. Which of the following is a suitable name for the coefficient C_{l_β} ?
- (a) Dihedral effect coefficient.
 - (b) Directional stability coefficient.
 - (c) Roll damping coefficient.
 - (d) Adverse yaw coefficient.
24. Twin-engine vertical stabilisers are big because...
- (a) they must compensate a huge torque in case of failure of one of the engines.
 - (b) they need to compensate for huge lateral forces due to engine corotation.
 - (c) twin-engined aircraft spontaneously tend to be less directionally stable.
 - (d) None of the others.
25. Automatic control of which surface can be used for damping the dutch roll mode in large aircraft?
- (a) Rudder trim tab.
 - (b) Elevator trim tab.
 - (c) Canards.
 - (d) Elevator.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.25\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $V_E = 91.2$ m/s
 - $V_E = 87.7$ m/s
 - $V_E = 80.5$ m/s
 - $V_E = 75.5$ m/s
27. A jet-powered airplane of mass $m = 1800$ kg has a required thrust curve $T_R = 0.25\bar{\rho}V^2 + 0.05W^2/\bar{\rho}V^2$. What is the velocity for best climb angle ($V_{\gamma_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s²)
- $V_{\gamma_{max}} = 88.8$ m/s
 - $V_{\gamma_{max}} = 67.5$ m/s
 - $V_{\gamma_{max}} = 117$ m/s
 - $V_{\gamma_{max}} = 154$ m/s
28. A propeller-powered airplane with drag polar $C_D = 0.03 + 0.12C_L^2$, wing surface $S = 20$ m² and mass $m = 1200$ kg is flying at a speed $V = 80$ m/s. The available engine power at sea level is $(P_A)_{SL} = 400$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.6$ kg/m³? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $\delta_x = 0.647$
 - $\delta_x = 0.583$
 - $\delta_x = 0.774$
 - $\delta_x = 0.317$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 46$. What is the descent angle (γ)?
- $\gamma = -1.25^\circ$
 - $\gamma = -1.43^\circ$
 - $\gamma = -1.54^\circ$
 - $\gamma = -0.819^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 900$ kg, wing area $S = 16$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.1$, is given by $T_R = 0.4\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 8000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.7$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 34.2$ m/s
 - $V \simeq 20.1$ m/s
 - $V \simeq 26$ m/s
 - $V \simeq 15.4$ m/s

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	c	b	a	c
2	a	b	a	d	b
3	a	c	c	a	c
4	a	c	d	b	b
5	a	b	b	c	d
6	a	c	d	c	c
7	a	d	c	c	b
8	a	a	c	d	b
9	a	c	d	d	b
10	a	c	b	a	a

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	b	c	c	b
12	a	c	c	a	a
13	a	d	c	c	a
14	a	b	b	d	d
15	a	d	d	c	a

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	a	d	c	b
17	a	c	d	a	c
18	a	c	a	a	a
19	a	a	b	a	d
20	a	a	d	b	c
21	a	a	b	a	d
22	a	b	d	d	a
23	a	d	a	c	b
24	a	d	d	b	d
25	a	b	b	d	c

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	d	a	b
27	a	c	a	a	d
28	a	a	d	d	a
29	a	b	c	a	a
30	a	a	c	a	c

BLO-1: Concepts on flight principles and aircraft performance

1. Reentry vehicles are...
 - (a) specific space vehicles.
 - (b) aerodynes.
 - (c) aerostats.
 - (d) rockets.
2. Which of the following devices are specifically devised to reduce wing tip vortices?
 - (a) Winglets.
 - (b) Flaps.
 - (c) Spoilers.
 - (d) Slats.
3. The chord variation along the span of a wing is called...
 - (a) taper.
 - (b) sweep.
 - (c) dihedral.
 - (d) twist.
4. Lift is the result of...
 - (a) All options are correct.
 - (b) a pressure difference between the inner and outer surfaces of the wing.
 - (c) the reaction to deflecting the airstream downwards.
 - (d) a net circulation around the airfoil.
5. How does the load factor compare between an airplane flying high in straight and level flight at an altitude h_h (n_h) and the same airplane doing the same at sea level at h_{SL} (n_{SL})?
 - (a) $n_h = n_{SL}$
 - (b) $n_h > n_{SL}$
 - (c) $n_h < n_{SL}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
6. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the airplane burns fuel during cruise if altitude is kept constant?
 - (a) It occurs at lower speeds (moves left) and decreases (moves down).
 - (b) It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. What happens to the best rate of climb in the presence of head wind?
 - (a) It does not change.
 - (b) It improves.
 - (c) It deteriorates.
 - (d) It will improve or deteriorate depending on engine type.
8. To perform a violent coordinated turn at constant altitude and speed, which of the following will never be a limiting parameter?
 - (a) All three parameters may limit the manoeuvre.
 - (b) Stall.
 - (c) Engine power.
 - (d) Load factor.
9. As a first approximation, which of the following conditions is not a requirement for maximising cruise range of a propeller airplane flying in a still atmosphere?
 - (a) Maximum flight altitude.
 - (b) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (c) Minimum drag.
 - (d) All options are essential requirements.
10. High altitude airports that intend to operate with large commercial airplanes...
 - (a) require longer runways for both takeoff and landing.
 - (b) require longer runways for takeoff but not necessarily for landing.
 - (c) require longer runways for landing but not necessarily for takeoff.
 - (d) can do with shorter runways.

BLO-2: Flight principles related problems

11. What is the optimal angle of attack (α_{opt}) that maximises aerodynamic efficiency (f_{max}) for an airplane with lift coefficient $C_L = 0.1 + 0.12\alpha$ for $\alpha < \alpha_S = 10^\circ$ and drag polar $C_D = 0.025 + 0.15 C_L^2$?
- (a) $\alpha_{opt} = 2.57^\circ$
 - (b) $\alpha_{opt} = 10^\circ$
 - (c) $\alpha_{opt} = 2.37^\circ$
 - (d) $\alpha_{opt} = 2.04^\circ$
12. An aerobatics airplane of mass $m = 900$ kg and wing surface $S = 11$ m² with drag polar $C_D = 0.08 + 0.14 C_L^2$ engages in a perfect dive (vertical descent) with its engine at idle. What will the terminal dive speed (V) be? (take $g = 9.8$ m/s² and $\rho = 1.225$ kg/s)
- (a) $V = 128$ m/s
 - (b) $V = 138$ m/s
 - (c) $V = 149$ m/s
 - (d) $V = 161$ m/s
13. A small airplane of unknown mass is flying in straight and level flight at a speed $V_{n=1} = 50$ m/s. The airplane engages in a coordinated banked turn at constant altitude with bank angle ($\phi = 30^\circ$). What is the new speed V if the pilot keeps the same angle of attack as in straight and level flight?
- (a) $V = 53.7$ m/s
 - (b) $V = 48.4$ m/s
 - (c) $V = 64.5$ m/s
 - (d) It depends on the mass of the airplane.
14. The drag polar of an airplane changes from $C_D = 0.02 + 0.12 C_L^2$ in clean configuration to $C_D = 0.04 + 0.1 C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.14$
 - (b) $C_D = 0.12$
 - (c) $C_D = 0.0869$
 - (d) $C_D = 0.217$
15. An airplane of mass $m = 1800$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 25 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
- (a) $D = 706$ N and $D_p \neq D_i$.
 - (b) $D = 706$ N and $D_p = D_i = 353$ N.
 - (c) $D = 353$ N and $D_p = D_i = 176$ N.
 - (d) $D = 353$ N and $D_p \neq D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is a stability condition?
- (a) All options are correct.
 - (b) Unstable.
 - (c) Neutral.
 - (d) Stable.
17. Dynamic stability requires...?
- (a) All options are correct.
 - (b) Static stability.
 - (c) A restoring force.
 - (d) A damping force.
18. Which of the following control surfaces is commanded through the yoke or stick?
- (a) Ailerons.
 - (b) Rudder.
 - (c) Throttle.
 - (d) Slats.
19. Which of the following devices is not meant for high-lift?
- (a) Elevator.
 - (b) flaps.
 - (c) slots.
 - (d) Slats.
20. Flight equilibria requires...
- (a) All options are correct.
 - (b) Summation of all forces over the aircraft equal zero.
 - (c) All moments over the aircraft are balanced.
 - (d) Linear and angular accelerations of the aircraft equal zero.
21. Which of the following is not a longitudinal mode?
- (a) Spiral mode.
 - (b) Pitch oscillation.
 - (c) Phugoid.
 - (d) Return to straight and level flight.
22. What cannot be said about pitch oscillation?
- (a) It is damped by the vertical stabilizer.
 - (b) It has a short period.
 - (c) It is damped by horizontal stabilizer and wing.
 - (d) Altitude remains approximately constant.
23. How is the damping moment of Roll Subsidence produced?
- (a) By the lift asymmetry between wings due to increased angle of attack in down-going wing and reduced angle of attack in up-going wing.
 - (b) By the lift symmetry between wings due to constant angle of attack.
 - (c) By a stability augmentation system acting on the rudder tab.
 - (d) By a positive sweep wing design.
24. What can be said about Dutch Roll?
- (a) It is a yaw oscillation with roll coupling.
 - (b) It is a pitch oscillation with sideslip coupling.
 - (c) It is an altitude oscillation with sideslip coupling.
 - (d) It is a roll oscillation with pitch coupling.
25. Which of the following is not a lateral-directional mode?
- (a) Short period.
 - (b) Roll subsidence.
 - (c) Dutch roll.
 - (d) Spiral mode.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.25\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1300$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- (a) $V_E = 89.8$ m/s
 - (b) $V_E = 118$ m/s
 - (c) $V_E = 68.2$ m/s
 - (d) It depends on the available thrust T_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1250$ kg has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.03W^2/\bar{\rho}V$. What is the minimum engine power needed at sea level ($\bar{\rho} = 1$) to achieve a rate of climb $\dot{h} = 6$ m/s? (take $g \simeq 9.8$ m/s²)
- (a) $P_A = 200$ kW
 - (b) $P_A = 180$ kW
 - (c) $P_A = 166$ kW
 - (d) $P_A = 146$ kW
28. An airplane with drag polar $C_D = 0.04 + 0.14C_L^2$, wing surface $S = 40$ m² and mass $m = 1500$ kg is flying at a speed $V = 90$ m/s. What is the engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.6$ kg/m³? (take $g \simeq 9.8$ m/s²)
- (a) $T_R = 4200$ N
 - (b) $T_R = 4160$ N
 - (c) $T_R = 4130$ N
 - (d) $T_R = 4100$ N
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 30$. What is the descent angle (γ)?
- (a) $\gamma = -1.91^\circ$
 - (b) $\gamma = -2.16^\circ$
 - (c) $\gamma = -2.35^\circ$
 - (d) $\gamma = -2.52^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 2000$ kg is given by $T_R = 0.5\bar{\rho}V^2 + 0.012W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 12000$ N. What is the maximum speed this airplane can achieve in straight and level flight at an altitude such that $\bar{\rho} = \rho/\rho_0 = 0.3$? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- (a) $V \simeq 136$ m/s
 - (b) $V \simeq 149$ m/s
 - (c) $V \simeq 157$ m/s
 - (d) $V \simeq 173$ m/s

BLO-5: Aerospace propulsion, helicopters and space systems

31. Which of the following aerodyne types is a fixed-wing aircraft?
- (a) Glider.
 - (b) Gyrodyne.
 - (c) Helicopters.
 - (d) Autogyro.
32. Which of the following is not a part of an helicopter?
- (a) Supersonic nozzle.
 - (b) Rotor.
 - (c) Fuselage.
 - (d) Anti-torque tail rotor.
33. How directional control can be achieved in helicopters?
- (a) Modifying the tail anti-torque rotor pitch.
 - (b) Deflecting the ailerons.
 - (c) Through collective pitch control.
 - (d) Differential thrust.
34. Which of the following is not true about Geostationary orbits?
- (a) Their shape is a high eccentrical ellipse.
 - (b) They have a 24 hour period.
 - (c) Their shape is circular.
 - (d) Their nadir on Earth is a fixed point.
35. Which of the following is a rocket engine propulsive parameter?
- (a) Specific impulse.
 - (b) Aerodynamic efficiency.
 - (c) Payload.
 - (d) Maneuverability.
36. In which type of jet engine part of the power from the turbine is used to drive a ducted fan?
- (a) Turbofan.
 - (b) Turboprop.
 - (c) Scramjet.
 - (d) Alternative Engine.
37. What happens to the pressure of the airflow through the nozzle of a turbojet
- (a) Decreases.
 - (b) Increases.
 - (c) Remains approximately constant.
 - (d) Equals the pressure at the end of the compression stage.
38. Which of the following turbine engines has better performance at higher altitude and speed operation?
- (a) Turbojet.
 - (b) High bypass ratio turbofan.
 - (c) Turbofan.
 - (d) Low bypass ratio turbofan.
39. Which of the following statements is true about jet engine inlets?
- (a) They are designed to reduce the airflow speed and increase pressure.
 - (b) Subsonic inlets are convergent.
 - (c) They are designed to increase the airflow speed and reduce pressure.
 - (d) They are between combustion chamber and nozzle in jet engines.
40. Which of the following is not true about jet engine compressors?
- (a) Turbines are driven with them.
 - (b) They can be centrifugal.
 - (c) They can be axial.
 - (d) They can have bleed ports for other systems.

BLO-6: Navigation & altimetry

41. What is Dead Reckoning?
- (a) The process of calculating the current position from an initial known position and subsequent speed, direction and time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) An airship based radio navigation aid.
 - (d) A generic name for ground stations.
42. Which of the following is true about ADF?
- (a) Gives the direction to a ground station to the pilot.
 - (b) Its signal is highly resistant to atmospheric conditions.
 - (c) Works in the 250-300 MHz band.
 - (d) Gives the distance to a ground station of the aircraft.
43. Which of the following is not true about VOR?
- (a) Aircraft transponder interrogates the VOR ground station.
 - (b) Provides radials referenced to the magnetic north.
 - (c) Multipurpose ILS and VOR equipment are common.
 - (d) Ground stations broadcast a non directional signal (108-117.95 MHz) and a rotating directional beam (9960+/-460 Hz).
44. What info is given by DME?
- (a) Distance to a ground station.
 - (b) Direction to a ground station.
 - (c) Weather forecast.
 - (d) Landing glide slope.
45. Which of the following is not part of ILS?
- (a) Range calculator.
 - (b) Glide slope.
 - (c) Localizer.
 - (d) Marker Beacon.
46. Which of the following cannot be said about INS?
- (a) It depends on communications with other aircrafts.
 - (b) It does not depend on aircraft external equipment or ground stations.
 - (c) Uses accelerometers and gyroscopes.
 - (d) Needs to be initially aligned.
47. Which of the following is the EU Global Navigation Satellite System?
- (a) Galileo.
 - (b) GPS.
 - (c) GLONASS.
 - (d) WASS.
48. What parameter is measured with a static probe.
- (a) Static pressure.
 - (b) Dynamic pressure.
 - (c) Total pressure.
 - (d) Air temperature.
49. Which of the following airspeed definitions is the actual speed of the aircraft relative to the air?
- (a) TAS.
 - (b) IAS.
 - (c) RAS.
 - (d) CAS.
50. What can be read in the altimeter if it has been set to QNH and the aircraft is on ground at the threshold of the runway of an airfield?
- (a) The altitude of the airport.
 - (b) Zero altitude.
 - (c) FL270.
 - (d) ISA+10.

51. The S-N curve is related to...

- (a) Fatigue.
- (b) Plasticity.
- (c) Elasticity.
- (d) Hardness.

52. Which of the following is a selection criterion for materials?

- (a) All options are correct.
- (b) Cost.
- (c) Thermal and electric properties.
- (d) Corrosion resistance.

53. Which of the following crystal structures can Aluminum show?

- (a) FCC.
- (b) BCC.
- (c) HC.
- (d) All the options are allotropes of Al.

54. Which of the following is not a constituent of steel?

- (a) Alpha alloy.
- (b) Pearlite.
- (c) Ferrite.
- (d) Austenite.

55. Which of the following materials is commonly used as reinforcement fibers in composites?

- (a) Carbon fiber.
- (b) Epoxy resin.
- (c) Beryllium alloys.
- (d) Nickel super-alloys.

56. Identify the following aircraft:



- (a) Airbus A300-600ST Beluga
- (b) Airbus A340-600
- (c) Airbus A380-800F
- (d) Airbus A320-200

57. Which of the following aircraft is not in the wide body category?

- (a) A320
- (b) B777
- (c) A380
- (d) B747

58. Identify the following aircraft: A346 L4J

- (a) Airbus 346: Land Plane, 4 engines, jet type
- (b) Boeing 346: Light Plane, 4 engines, jet type
- (c) Lockheed Martin 4J: Amphibia Plane, 346 passengers
- (d) Airbus L4J: Sea Plane, 3 engines, 46 passengers

59. Identify the aircraft in the foreground:



- (a) Boeing B777
- (b) Airbus A380
- (c) Airbus A340
- (d) Boeing B747

60. Which of the following is not an aircraft manufacturer?

- (a) Pratt and whitney
- (b) Lockheed Martin
- (c) Embraer
- (d) Bombardier

BLO-8: Aircraft systems

61. In a fly-by-wire flight control system...
- (a) ...there is no mechanical link between cockpit commands and control surfaces.
 - (b) ...there is mechanical link between cockpit commands and control surfaces.
 - (c) ...there is hydro-mechanical link between cockpit commands and control surfaces.
 - (d) ...there is pneumatic link between cockpit commands and control surfaces.
62. Which of the following devices can be part of a hydro-mechanical flight control system?
- (a) Hydraulic actuator.
 - (b) Bleed valve.
 - (c) De-ice boot.
 - (d) APU Generator.
63. Which of the following is a potential source of power for a hydraulic system?
- (a) Engine-driven pump.
 - (b) Thrust reverser system.
 - (c) Inverter.
 - (d) DC Essential bus.
64. Which of the following components is part of a hydraulic system?
- (a) Reservoir.
 - (b) Transformer Rectifier.
 - (c) FADEC.
 - (d) Spoiler.
65. Which of the following components is part of an electric system?
- (a) Battery.
 - (b) Oil tank.
 - (c) Hydraulic accumulator.
 - (d) ADIRU.
66. Which of the following components is not a source of electric power?
- (a) Fuel transfer valve.
 - (b) Battery.
 - (c) Main Generator.
 - (d) Emergency generator.
67. Which of the following is not a high pressure air source for the Pneumatic System?
- (a) Static inverter.
 - (b) APU load compressor.
 - (c) Engine bleed system.
 - (d) HP ground connection.
68. Which of the following is not a power plant subsystem?
- (a) Oxygen system.
 - (b) Thrust reverser system.
 - (c) Oil system.
 - (d) Ignition and starting system.
69. Which component of the power plant system provides with mechanical power other aircraft systems?
- (a) Accessory gearbox.
 - (b) FADEC.
 - (c) Essential Bus.
 - (d) Fuel filter.
70. Which of the following systems is not fed by the pneumatic system?
- (a) Communications system.
 - (b) Air Conditioning System.
 - (c) Engine Starting System.
 - (d) Anti-ice system.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	a	d	c
2	a	d	a	d	b
3	a	a	a	a	d
4	a	b	d	d	d
5	a	c	c	b	a
6	a	c	c	c	b
7	a	a	b	c	a
8	a	b	c	a	a
9	a	a	d	d	b
10	a	a	a	d	a

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	c	a	a
12	a	c	c	d	c
13	a	c	a	b	d
14	a	c	b	a	a
15	a	c	b	a	a

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	c	d	a
17	a	b	b	c	d
18	a	d	c	a	c
19	a	c	c	a	c
20	a	b	b	d	c
21	a	a	d	a	c
22	a	c	b	d	d
23	a	c	a	c	d
24	a	a	d	c	d
25	a	a	c	c	a

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	b	c	d	b
27	a	d	c	d	b
28	a	a	d	d	c
29	a	b	c	a	d
30	a	c	d	c	d

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	c	c	c
32	a	c	d	c	b
33	a	d	b	b	b
34	a	a	b	b	a
35	a	c	b	d	b
36	a	b	c	c	a
37	a	a	a	c	b
38	a	d	b	c	b
39	a	c	c	a	b
40	a	a	c	d	b

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
41	a	d	b	a	a
42	a	c	a	b	a
43	a	a	b	b	a
44	a	b	b	d	a
45	a	d	a	c	a
46	a	a	a	a	d
47	a	b	a	b	d
48	a	a	c	d	c
49	a	a	d	c	c
50	a	b	c	d	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
51	a	a	d	a	c
52	a	c	a	d	d
53	a	d	b	a	b
54	a	b	a	a	c
55	a	d	b	d	c
56	a	a	c	a	d
57	a	c	d	c	d
58	a	d	d	c	a
59	a	d	d	d	b
60	a	b	b	b	a

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	d	c	c	c
62	a	c	c	d	a
63	a	b	a	d	c
64	a	d	d	b	a
65	a	a	b	d	a
66	a	d	a	b	d
67	a	b	a	b	a
68	a	a	a	a	b
69	a	b	b	a	b
70	a	a	c	c	d

BLO-5: Aerospace propulsion, helicopters and space systems

1. Which of the following sentences is not true about helicopters?
 - (a) Helicopters are designed for heavy payloads and long range operation.
 - (b) Lift is generated by rotor.
 - (c) Helicopters are capable of stationary flight (hovering).
 - (d) Helicopters are capable of vertical take-off and landing.
2. In helicopters, which of the following is not an anti-torque feature ?
 - (a) Collective pitch control.
 - (b) NOTAR.
 - (c) Tandem or coaxial counter-rotating rotors.
 - (d) Anti-torque tail rotor.
3. How lateral and longitudinal control is achieved in helicopters?
 - (a) Through cyclic pitch control.
 - (b) Deflecting the tail control surfaces.
 - (c) Modifying the tail anti-torque rotor pitch.
 - (d) Differential thrust.
4. Which of the following orbits is normally used for navigation satellites?
 - (a) MEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
5. Which of the following is true regarding rocket engines?
 - (a) They carry fuel and oxidizer.
 - (b) Atmospheric air is accelerated through them to get thrust.
 - (c) Payload is heavier than fuel weight in the Ariane family.
 - (d) They can only be used for communications in locations situated in temperate longitudes.
6. Which jet engine type uses part of the turbine power to drive a propeller?
 - (a) Turboprop.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Alternative Engine.
7. In a turbojet nozzle the speed of the airflow...
 - (a) increases.
 - (b) decreases.
 - (c) remains approximately constant.
 - (d) increases in twin-spool turbojets and decreases in single-spool turbojets.
8. Which of the following turbine engines offers better take-off performances?
 - (a) Turboprop.
 - (b) High bypass ratio turbofan.
 - (c) Turbojet.
 - (d) Low bypass ratio turbofan.
9. Which of the following statements related to turbojet engines is true?
 - (a) Inlets are designed to reduce air speed and increase pressure.
 - (b) Ramjets can only operate at low speed.
 - (c) Compressor extracts power from the airflow to drive the turbine.
 - (d) Supersonic nozzles are convergent.
10. Propulsive efficiency of turbojets...
 - (a) depends on the aircraft speed and the exhaust air speed.
 - (b) depends on the type of compressor.
 - (c) is higher with tubular combustion chambers.
 - (d) is maximum at take-off.

BLO-6: Navigation & altimetry

11. Which navigation systems uses ground stations that broadcast a non-directional signal at a frequency between 190 and 1750 kHz)?
 - (a) ADF.
 - (b) VOR.
 - (c) DME.
 - (d) GLONASS.
12. Which navigation system provides the angle referred to true north of a line connecting the aircraft and a tuned ground station?
 - (a) VOR.
 - (b) WAAS.
 - (c) NDB.
 - (d) GPS.
13. Which one of the following navigation systems provides the distance between the aircraft and a ground station that is interrogated by the transponder?
 - (a) DME.
 - (b) Galileo.
 - (c) VOR.
 - (d) IRS.
14. Which navigation systems allows a safe landing giving the relative position of an aircraft to the glideslope and runway axis?
 - (a) ILS.
 - (b) NDB.
 - (c) EGNOS.
 - (d) RNAV.
15. Which autonomous navigation system uses accelerometers and gyroscopes to determine the position of an aircraft?
 - (a) INS.
 - (b) VOR.
 - (c) GPS.
 - (d) NDB.
16. Which of the following is not a GNSS?
 - (a) NDB.
 - (b) Galileo.
 - (c) GLONASS.
 - (d) GPS.
17. Which parameter is measured with a simple Pitot tube?
 - (a) Total pressure.
 - (b) Dynamic pressure.
 - (c) Static pressure.
 - (d) Air temperature
18. Which of the following airspeed definitions takes into account corrections for instrument position and installation errors?
 - (a) CAS.
 - (b) IAS.
 - (c) EAS.
 - (d) AoA.
19. Which pressure setting is defined as the barometric pressure adjusted to sea level?
 - (a) QNH.
 - (b) QFE.
 - (c) Standard.
 - (d) 1013 mb.
20. What can be read in the altimeter if it has been set to QFE and the aircraft is on ground at the threshold of the runway of an airfield?
 - (a) Zero altitude.
 - (b) The altitude of the airport.
 - (c) FL310.
 - (d) ISA+10.

BLO-7: Aircraft recognition and materials & structures

21. The elastic limit or yield point...

- (a) is the maximum stress for elastic deformation of a material.
- (b) is the stress-deformation ratio.
- (c) is the minimum number of cycles of the S-N curve.
- (d) is the stress at which a material breaks.

22. What is 2024-T3?

- (a) An aluminum alloy.
- (b) A titanium alloy.
- (c) A copper alloy
- (d) A magnesium alloy

23. What is true about titanium?

- (a) It has an allotropic transformation from HCP (alpha) to BCC (beta).
- (b) It is widely used in aeronautics because of the low price.
- (c) It has poor corrosion properties.
- (d) It can be in the form of Cementite.

24. Which of the following is not a reinforcement fibre used in composites?

- (a) Epoxy resin.
- (b) Carbon.
- (c) Kevlar.
- (d) Spectra.

25. Which of the following alloys have good anti-seize properties and therefore used in bushings and bearings?

- (a) Copper alloys.
- (b) Magnesium alloys.
- (c) Beryllium alloys.
- (d) Nickel super-alloys.

26. Identify the following aircraft:



- (a) Boeing
- (b) Airbus
- (c) Bombardier
- (d) Gulfstream

27. Which of the following aircraft does not mount four engines?

- (a) B777
- (b) A340
- (c) A380
- (d) B747

28. Identify the following aircraft: B735 L2J

- (a) Boeing 735: Land Plane, 2 engines, jet type
- (b) Boeing L2J: Conventional Plane, 735 passengers, jet type
- (c) Lockheed Martin 2J: Land Plane, 735 passengers, bomber type
- (d) Airbus 735: Conventional Plane, 2 engines, jet type

29. Identify the following aircraft:



- (a) Airbus A380
- (b) Airbus A340
- (c) Boeing B747
- (d) Boeing B787

30. Identify the following aircraft type:



- (a) Autogyro
- (b) Helicopter
- (c) Vertalot
- (d) Tilt-rotor

31. Which of the following is not a flight control system type?
- (a) AC/DC.
 - (b) Mechanical.
 - (c) Hydro-mechanical.
 - (d) Electro-hydraulic.
32. Which of the following devices cannot be part of a mechanical flight control system?
- (a) Hydraulic actuator.
 - (b) Lever.
 - (c) Pulley.
 - (d) Cable.
33. Which of the following is not a pressure source of an hydraulic system?
- (a) DC Essential bus.
 - (b) Engine-driven pump.
 - (c) Electric pump.
 - (d) Power Transfer Unit.
34. Which of the following is not a part of the hydraulic system?
- (a) Inverter.
 - (b) Accumulator.
 - (c) Reservoir.
 - (d) Filter.
35. Which of the following is not part of an electric system?
- (a) Bleed valve.
 - (b) Main generator.
 - (c) Transformer rectifier.
 - (d) Batteries.
36. Which of the following systems is not supplied by the pneumatic system?
- (a) Landing lights.
 - (b) Air conditioning.
 - (c) Engine starting.
 - (d) Wing anti-icing or de-icing.
37. Which of the following is not a function of the APU?
- (a) Vent fuel tanks.
 - (b) Start engines.
 - (c) Back up electrical system in flight.
 - (d) Back up air conditioning system in flight.
38. Which of the following is not part of the fuel system?
- (a) FADEC.
 - (b) Tanks.
 - (c) Transfer valve.
 - (d) Fuel pump.
39. Which system will start working after a depressurization at cruise altitude due to a failure in the air conditioning system?
- (a) Oxygen system.
 - (b) Fire protection system.
 - (c) Autoflight system.
 - (d) Light system.
40. Which of the following systems is fed by the hydraulic system?
- (a) Brake system.
 - (b) Engine nacelle anti-ice system.
 - (c) Wing anti-ice system.
 - (d) Potable water tank pressurization system.

Correct Answers

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	a	d	b	c
2	a	d	c	a	d
3	a	d	a	a	a
4	a	d	a	d	b
5	a	d	c	b	c
6	a	c	b	b	d
7	a	a	a	d	c
8	a	d	b	c	a
9	a	c	c	d	d
10	a	c	b	d	d

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	d	b	d	b
12	a	d	b	d	b
13	a	d	b	d	a
14	a	a	a	b	d
15	a	a	c	d	a
16	a	c	c	b	b
17	a	d	a	c	d
18	a	a	c	a	d
19	a	d	a	b	d
20	a	a	b	a	a

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	c	b	a	a
22	a	d	c	b	d
23	a	a	b	a	c
24	a	c	a	d	b
25	a	d	b	d	c
26	a	d	a	d	b
27	a	b	d	a	d
28	a	d	a	c	b
29	a	c	b	d	d
30	a	d	d	d	c

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	c	d	b
32	a	c	c	d	c
33	a	b	a	d	b
34	a	c	b	d	a
35	a	d	a	b	d
36	a	a	d	d	a
37	a	d	a	a	c
38	a	b	c	d	d
39	a	d	c	c	b
40	a	b	a	a	a

BLO-1: Concepts on flight principles and aircraft performance

1. Which of the following is false about an autogyro?
 - (a) It can perform stationary fly.
 - (b) It is a rotorcraft.
 - (c) It is an aerodyne.
 - (d) The lifting rotor is not powered.
2. Wing tip vortices result in...
 - (a) increased drag.
 - (b) increased lift.
 - (c) increased aerodynamic efficiency.
 - (d) All options are correct.
3. The backward (or, rarely, forward) angle of the wing's leading edge with respect to a plane perpendicular to the fuselage axis is called...
 - (a) sweep.
 - (b) taper.
 - (c) dihedral.
 - (d) twist.
4. Air viscosity is ultimately responsible for...
 - (a) All options are correct.
 - (b) friction drag.
 - (c) lift.
 - (d) boundary layer separation.
5. How does stall speed compare between an airplane flying high at an altitude h_h (V_{S_h}) and the same airplane flying at sea level at h_{SL} ($V_{S_{SL}}$)?
 - (a) $V_{S_h} > V_{S_{SL}}$
 - (b) $V_{S_h} < V_{S_{SL}}$
 - (c) $V_{S_h} = V_{S_{SL}}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
6. What happens to the minimum ($P_{R_{min}}$) of the power curve ($P_R(V)$) as the airplane burns fuel during cruise if altitude is kept constant?
 - (a) It occurs at lower speeds (moves left) and decreases (moves down).
 - (b) It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. What happens to the best climb angle in the presence of head wind?
 - (a) It improves.
 - (b) It deteriorates.
 - (c) It does not change.
 - (d) It will improve or deteriorate depending on engine type.
8. To perform a turn in a tight valley, which of the following flight parameters must be optimised?
 - (a) Minimum turn radius.
 - (b) Maximum turning rate.
 - (c) Maximum velocity.
 - (d) Minimum angle of attack.
9. Which of the following conditions is a requirement for maximising cruise endurance of a propeller airplane?
 - (a) Minimum flight altitude.
 - (b) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (c) Minimum drag.
 - (d) All options are correct.
10. To which manoeuvre does the flare stage correspond?
 - (a) Landing.
 - (b) Takeoff.
 - (c) Initial Climb.
 - (d) Glide.

BLO-2: Flight principles related problems

11. What is the maximum aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.01 + 0.13C_L^2$?
- $f_{max} = 13.9$
 - $f_{max} = 14.5$
 - $f_{max} = 15.3$
 - $f_{max} = 11.9$
12. An aerobatics airplane of mass $m = 900$ kg and wing surface $S = 11 \text{ m}^2$ with drag polar $C_D = 0.08 + 0.14C_L^2$ engages in a perfect dive (vertical descent) with its engine at idle. What will the terminal dive speed (V) be? (take $g = 9.8 \text{ m/s}^2$ and $\rho = 1.225 \text{ kg/s}$)
- $V = 128 \text{ m/s}$
 - $V = 141 \text{ m/s}$
 - $V = 152 \text{ m/s}$
 - $V = 163 \text{ m/s}$
13. A small airplane of mass $m = 1000$ kg is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}) at an unknown altitude. We know its speed is $V = 2.5 V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the lift (L) if the plane is to keep its altitude and speed? (take $g \simeq 10 \text{ m/s}^2$)
- $L = 62500 \text{ N}$
 - $L = 43800 \text{ N}$
 - $L = 25000 \text{ N}$
 - It depends on the altitude.
14. The drag polar of an airplane changes from $C_D = 0.035 + 0.14C_L^2$ in clean configuration to $C_D = 0.05 + 0.08C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
- $C_L = 0.5$
 - $C_L = 0.638$
 - $C_L = 0.263$
 - $C_L = 0.159$
15. An airplane of mass $m = 900$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 24 = f_{max}$. Its velocity is $V = 50 \text{ m/s}$. What can we say about the total (P_R), induced (P_{R_i}) and parasitic (P_{R_p}) required power? (take $g \simeq 9.8 \text{ m/s}^2$)
- $P_R = 18.4 \text{ kW}$ and $P_{R_p} = P_{R_i} = 9.19 \text{ kW}$.
 - $P_R = 368 \text{ kW}$ and $P_{R_p} = P_{R_i} = 184 \text{ kW}$.
 - $P_R = 368 \text{ kW}$ and $P_{R_p} > P_{R_i}$.
 - $P_R = 18.4 \text{ kW}$ and $P_{R_p} \neq P_{R_i}$.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is required for dynamic stability around an axis?
- (a) Static stability, restoring force and damping force.
 - (b) Automatic control for high manoeuvrability.
 - (c) Ailerons and slats.
 - (d) A dynamic response in which all parameters grow without bounds.
17. Which of the following is not a Primary Control Surface?
- (a) Winglets.
 - (b) Ailerons.
 - (c) Elevator.
 - (d) Rudder.
18. Which of the following Secondary Control Surfaces is a High-Lift Device?
- (a) flaps.
 - (b) Spoilers.
 - (c) Ruddervators.
 - (d) Rudder trim tab.
19. Which of the following control surfaces is to balance pitching moment?
- (a) Elevator.
 - (b) Rudder.
 - (c) Ailerons.
 - (d) Slat.
20. What can be said about a canard airplane?
- (a) The foreplane must stall before the wing to be safe.
 - (b) The canard sees a perturbed flow.
 - (c) The canard downforce subtracts from the wing lift.
 - (d) The canard can by no means balance pitching moment.
21. What can be said about the pitch oscillation mode?
- (a) Angle of attack oscillates with a short period.
 - (b) Pitch does not oscillate.
 - (c) Angle of attack evolves aperiodically.
 - (d) It is damped by the vertical stabilizer.
22. What cannot be said about the phugoid mode?
- (a) It is strongly coupled to yaw.
 - (b) It has a long period.
 - (c) It is poorly damped.
 - (d) It corresponds to a potential-kinetic energy exchange.
23. Which of the following is not a hypothesis for the lateral equations?
- (a) No roll rotation velocity.
 - (b) Constant angle of attack.
 - (c) No pitch rotation velocity.
 - (d) Small sideslip.
24. How can directional stability be tuned?
- (a) Vertical stabilizer size.
 - (b) Horizontal stabilizer size.
 - (c) Ailerons trim tab size.
 - (d) Throttle.
25. What cannot be said about the roll subsidence mode?
- (a) It is unstable.
 - (b) It is damped through lift asymmetry between wings due to the increased angle of attack in the down-going wing and the reduced angle of attack in the up-going wing.
 - (c) It is aperiodic.
 - (d) The wing aspect ratio has a damping effect.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.24\bar{\rho}V^3 + 0.018W^2/\bar{\rho}V$. If the airplane has mass $m = 1400$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 1$, what is the optimal velocity for maximum range (V_R)? (take $g \simeq 9.8$ m/s²)
- $V_R = 61.3$ m/s
 - $V_R = 46.6$ m/s
 - $V_R = 80.7$ m/s
 - It depends on the available power P_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1200$ kg has a required power curve $P_R = 0.25\bar{\rho}V^3 + 0.04W^2/\bar{\rho}V$. What is the maximum rate of climb $((R/C)_{max})$ at sea level ($\bar{\rho} = 1$) if the available power is $P_A = 350 \bar{\rho} \delta_x$ kW? (take $g \simeq 9.8$ m/s²)
- $(R/C)_{max} = 17.7$ m/s
 - $(R/C)_{max} = 16$ m/s
 - $(R/C)_{max} = 13.9$ m/s
 - $(R/C)_{max} = 12.7$ m/s
28. An airplane with drag polar $C_D = 0.03 + 0.15C_L^2$, wing surface $S = 36$ m² and mass $m = 2000$ kg is flying at a speed $V = 110$ m/s. What is the engine power required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.7$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $P_R = 545$ kW
 - $P_R = 504$ kW
 - $P_R = 425$ kW
 - $P_R = 712$ kW
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 32$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 36$ km of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- $h = 1130$ m
 - $h = 2250$ m
 - $h = 1150$ m
 - $h = 889$ m
30. The required thrust of a jet-engined airplane of mass $m = 1800$ kg, wing area $S = 24$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 20000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 51$ m/s
 - $V \simeq 38.7$ m/s
 - $V \simeq 43.6$ m/s
 - $V \simeq 31.1$ m/s

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	b	a	b
2	a	b	b	b	c
3	a	a	b	d	c
4	a	b	c	c	a
5	a	c	d	c	d
6	a	c	a	a	c
7	a	d	a	a	b
8	a	b	a	c	b
9	a	a	d	c	b
10	a	d	b	d	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	c	b	b	a
12	a	c	b	a	d
13	a	a	b	c	a
14	a	c	a	a	b
15	a	b	a	c	a

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	b	b	c
17	a	c	c	c	b
18	a	d	c	b	a
19	a	a	c	d	a
20	a	c	b	d	c
21	a	a	b	c	c
22	a	a	b	a	c
23	a	b	d	b	c
24	a	d	b	d	b
25	a	b	b	a	a

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	b	d	c	d
27	a	d	a	d	a
28	a	c	d	c	d
29	a	c	a	c	b
30	a	d	d	a	b

BLO-1: Flight principles and aircraft performance

1. The angle of attack of an airfoil is defined as the angle between what and what?
 - (a) Airspeed and chord.
 - (b) Horizontal plane and airspeed.
 - (c) Chord and horizontal plane.
 - (d) Upper and lower surfaces.
2. What is true about lift generation in a wing?
 - (a) All options are true.
 - (b) The circulation required for lift is due to viscosity.
 - (c) Lift is the reaction force on the wing to the action of diverting air downwards.
 - (d) Lift is the result of a pressure difference between upper and lower surfaces.
3. In flight compressibility effects are governed by which nondimensional parameter?
 - (a) Mach number.
 - (b) Reynolds number.
 - (c) Peclet number.
 - (d) None of the others.
4. What is the effect on the zero-lift angle of attack (α_0) of an ideal high-lift device that only increases the chord?
 - (a) It has no effect.
 - (b) It increases.
 - (c) It decreases.
 - (d) It increases or decreases depending on the altitude.
5. In what sense does the air of a left wingtip vortex rotate as seen from behind a cruising aircraft?
 - (a) Clockwise.
 - (b) Anticlockwise.
 - (c) It does not rotate.
 - (d) In the same sense as the right wingtip vortex.
6. What happens to the minimum ($P_{R_{min}}$) of the power curve ($P_R(V)$) as the airplane loses altitude?
 - (a) It occurs at lower speeds (moves left) and decreases (moves down).
 - (b) It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. How is maximum climb angle affected by head wind?
 - (a) It increases.
 - (b) It decreases.
 - (c) It is not affected.
 - (d) It depends on air density.
8. Which of the following statements characterises the first flight regime (R1)?
 - (a) The airplane has a tendency to climb when pulling the yoke.
 - (b) The airplane has a tendency to descend when pulling the yoke.
 - (c) The flight velocity is lower than in the second regime (R2).
 - (d) Induced drag/power dominates over parasitic drag/power.
9. Which of the following conditions is about adequate for maximising cruise endurance of a propeller airplane?
 - (a) All options are correct.
 - (b) Minimum hourly fuel consumption (C, in lb/h).
 - (c) Minimum required power.
 - (d) Minimum flight altitude.
10. Why are runways of airports located at high elevation longer than at sea level?
 - (a) Because both landing and takeoff require longer ground runs.
 - (b) Only because landing requires longer ground runs, but not takeoff.
 - (c) Only because takeoff requires longer ground runs, but not landing.
 - (d) Conversely, high elevation airports require shorter runways.

BLO-2: Flight principles related problems

11. What is the maximum aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.025 + 0.12C_L^2$?
- (a) $f_{max} = 9.13$
 - (b) $f_{max} = 8.64$
 - (c) $f_{max} = 8.3$
 - (d) $f_{max} = 7.81$
12. An aerobatics airplane of mass $m = 1000 \text{ kg}$ and wing surface $S = 12 \text{ m}^2$ with drag polar $C_D = 0.07 + 0.15C_L^2$ engages in a perfect dive (vertical descent) with its engine at idle. What will the terminal dive speed (V) be? (take $g = 9.8 \text{ m/s}^2$ and $\rho = 1.225 \text{ kg/s}$)
- (a) $V = 138 \text{ m/s}$
 - (b) $V = 128 \text{ m/s}$
 - (c) $V = 115 \text{ m/s}$
 - (d) $V = 178 \text{ m/s}$
13. A small airplane of unknown mass is engaged in a coordinated banked turn with bank angle ($\phi = 45^\circ$) at a speed $V = 90 \text{ m/s}$. What is the angular speed of the turn (Ω)? (Take $g = 9.8 \text{ m/s}^2$)
- (a) $\Omega = 0.109 \text{ rad/s}$
 - (b) $\Omega = 0.103 \text{ rad/s}$
 - (c) $\Omega = 0.126 \text{ rad/s}$
 - (d) $\Omega = 0.0813 \text{ rad/s}$
14. The lift coefficient and drag polar of an airplane changes from $C_L = 0.15 + 0.12\alpha$ and $C_D = 0.05 + 0.13C_L^2$ in clean configuration to $C_L = 0.2 + 0.12\alpha$ and $C_D = 0.07 + 0.065C_L^2$ with full flaps. What is the value of the angle of attack (α) just before optimal flap deployment?
- (a) $\alpha = 3.37^\circ$
 - (b) $\alpha = 2.96^\circ$
 - (c) $\alpha = 3.24^\circ$
 - (d) $\alpha = 3.09^\circ$
15. An airplane of mass $m = 2000 \text{ kg}$ is cruising in straight and level flight at a speed that is lower than that for optimal aerodynamic efficiency. If the aerodynamic efficiency is $f = 20 < f_{max}$, what can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ m/s}^2$)
- (a) $D = 1000 \text{ N}$ and $D_p < D_i$.
 - (b) $D = 500 \text{ N}$ and $D_p = D_i = 250 \text{ N}$.
 - (c) $D = 1000 \text{ N}$ and $D_p > D_i$.
 - (d) $D = 500 \text{ N}$ and $D_p \neq D_i$, but there is not enough information to know which one is greater.

BLO-3: Aircraft stability & control

16. Which of the following is not required by dynamic stability around an axis?
- (a) High Manoeuvrability.
 - (b) Static Stability.
 - (c) Restoring Force.
 - (d) Damping Force.
17. Which of the following Primary Control Surfaces is commanded by Yoke?
- (a) Ailerons and elevator.
 - (b) Rudder.
 - (c) Spoilers.
 - (d) Vortex generators.
18. Which of the following is a high-lift secondary control surface...
- (a) flaps.
 - (b) Air Brakes.
 - (c) Winglets.
 - (d) Throttle.
19. Which of the following control surfaces is related to longitudinal balance?
- (a) Elevator.
 - (b) Rudder.
 - (c) Ailerons.
 - (d) All options are correct.
20. Which of the following relative positions of the Center of Gravity and Neutral Point of an airplane produces Neutral Stability?
- (a) Center of Gravity on Neutral Point.
 - (b) Center of Gravity ahead of Neutral Point.
 - (c) Center of Gravity after Neutral Point.
 - (d) All options are correct.
21. What can be said about the longitudinal modes?
- (a) All options are correct.
 - (b) Pitching oscillation is damped by wing and horizontal stabilizer.
 - (c) Phugoid has a long period.
 - (d) Straight and level flight recovery is stable and aperiodic.
22. Which of the following is not a hypothesis for lateral flight equations?
- (a) No roll or yaw rotation velocities.
 - (b) No pitch rotation velocity.
 - (c) Constant velocity and angle of attack.
 - (d) Small sideslip angle.
23. What can be said about Lateral Stability?
- (a) It is the tendency of the airplane to recover from a roll perturbation.
 - (b) It is decoupled from yaw.
 - (c) Recovery is mostly statical.
 - (d) All options are correct.
24. Which of the following is not a lateral mode?
- (a) Phugoid.
 - (b) Roll subsidence.
 - (c) Dutch roll.
 - (d) Spiral mode.
25. What can be said about Dutch roll mode?
- (a) all options are correct.
 - (b) It has a short period.
 - (c) Yaw dumpers are installed in some aircrafts to damp it.
 - (d) Size of vertical stabilizer affects damping.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.22\bar{\rho}V^3 + 0.025W^2/\bar{\rho}V$. If the airplane has mass $m = 1400$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.6$, what is the required power (P_R) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $P_R = 157$ kW
 - $P_R = 179$ kW
 - $P_R = 171$ kW
 - $P_R = 150$ kW
27. A propeller-powered airplane of mass $m = 1600$ kg has a required power curve $P_R = 0.28\bar{\rho}V^3 + 0.04W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s²)
- $V_{(R/C)_{max}} = 58.5$ m/s
 - $V_{(R/C)_{max}} = 77$ m/s
 - $V_{(R/C)_{max}} = 101$ m/s
 - $V_{(R/C)_{max}} = 44.4$ m/s
28. An airplane with drag polar $C_D = 0.025 + 0.16C_L^2$, wing surface $S = 35$ m² and mass $m = 2200$ kg is flying at a speed $V = 110$ m/s. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.33$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $T_R = 2810$ N
 - $T_R = 2930$ N
 - $T_R = 3040$ N
 - $T_R = 2510$ N
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 40$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 50$ km of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- $h = 1250$ m
 - $h = 2500$ m
 - $h = 2000$ m
 - $h = 800$ m
30. The required thrust of a jet-engined airplane of mass $m = 1800$ kg, wing area $S = 20$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.55\bar{\rho}V^2 + 0.015W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 10000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 1$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 34.6$ m/s
 - $V \simeq 21.9$ m/s
 - $V \simeq 39.8$ m/s
 - $V \simeq 28.9$ m/s

31. Which of the following is not a jet engine?
- (a) Piston engine.
 - (b) Turboprop.
 - (c) Ramjet.
 - (d) Turbofan.
32. What happens to turbojet thrust when altitude rises?
- (a) Decreases.
 - (b) Increases.
 - (c) Increases if the combustion chamber is annular.
 - (d) Increases in high bypass turbofan.
33. Which is the most common jet engine in current passenger transport airplanes?
- (a) Turbofan.
 - (b) Turbojet.
 - (c) Scramjet.
 - (d) Ramjet.
34. Which of the following statements related to turbojet engines is true?
- (a) All options are correct.
 - (b) Turbine extracts energy from the air flow to drive the compressor.
 - (c) Air flow speed decreases and pressure increases through inlet.
 - (d) Subsonic inlets are divergent.
35. What can be said about turbojet engines nozzles?
- (a) All options are correct.
 - (b) Air flow speed increases and pressure decreases through them.
 - (c) Afterburner devices can be installed in them.
 - (d) Supersonic nozzles are divergent.
36. Regarding the ground effect, which of the following statements is true?
- (a) It is effective only when the helicopter height is less than the main rotor diameter.
 - (b) It improves high speed flights.
 - (c) It permits the helicopter to perform an inverted flight.
 - (d) None of the other is true.
37. A helicopter in horizontal flight:
- (a) All of them are correct.
 - (b) The distribution of the main rotor airspeed induces a roll moment in the helicopter.
 - (c) The distribution of airspeeds along a particular blade depends on its position along the rotor disk.
 - (d) There exists a reverse flow zone somewhere in the rotor disk.
38. Which of the following statements is correct regarding stationary flight (hovering)?
- (a) All of the others are false.
 - (b) A fixed wing aircraft can perform a stationary flight.
 - (c) An autogyro can perform a stationary flight.
 - (d) A rocket can perform a stationary flight.
39. The suggested orbit for a navigation satellite mission is:
- (a) Medium Earth orbit.
 - (b) Geostationary orbit.
 - (c) Geosynchronous orbit
 - (d) Highly elliptical orbit.
40. When will helicopters fly faster than fixed wing aircraft?
- (a) Never.
 - (b) In less than 25 years.
 - (c) Once new powerful engines are developed.
 - (d) Once new powerful tail rotors are developed.

BLO-6: Navigation & altimetry

41. Which of the following sensors is the one used to measure airspeed of an aircraft?
- (a) Pitot tube.
 - (b) RAT turbine.
 - (c) Density sensor.
 - (d) None of others.
42. What is true about the Mach number?
- (a) All the others are true.
 - (b) At constant altitude, when TAS increases the Mach number increases.
 - (c) When climbing at constant TAS, the Mach number increases.
 - (d) When descending at constant Mach, the TAS increases.
43. When an aircraft is on a particular runway and the pilot has set the altimeter to the local QFE, the altimeter reading will be...
- (a) 0 ft.
 - (b) The altitude of the airport.
 - (c) The altitude of the runway.
 - (d) Further data is required to anticipate the altimeter reading.
44. When an aircraft is on a particular runway and the pilot has set the altimeter to the local QNH, the altimeter reading will be...
- (a) The reference altitude of the airport.
 - (b) 0 ft.
 - (c) The altitude of the highest surrounding obstacle.
 - (d) Further data is required to anticipate the altimeter reading.
45. What geographical area has been selected to establish zero reference of the geodetic longitude ($\lambda = 0^\circ$)?
- (a) It has been chosen arbitrarily.
 - (b) The Equator.
 - (c) The North Pole.
 - (d) The South Pole.
46. When travelling across the International Date Line in an easterly direction, the date...
- (a) decreases by one day.
 - (b) increases by one day.
 - (c) decreases 12 hours.
 - (d) increases 12 hours.
47. The Glide Path (GP) ILS subsystem...
- (a) defines the safe descent slope (normally 3°).
 - (b) is used to define airways (AWY).
 - (c) defines an extended centerline of the instrument runway.
 - (d) none of them are correct.
48. About the cone of silence of a VOR navaid...
- (a) the period during which the aircraft receives no signal increases with height.
 - (b) the period during which the aircraft receives no signal is independent of height.
 - (c) the period during which the aircraft receives no signal decreases with height.
 - (d) None of them are correct.
49. The QDM...
- (a) is a Q-Code that indicates the magnetic bearing to a station.
 - (b) is a Q-Code that indicates the longitude of the aircraft.
 - (c) stands for Quality Dynamics Measurement.
 - (d) none of the other is correct.
50. The DME navaid...
- (a) is an active system unlike VOR and ILS navaids.
 - (b) is a passive system like VOR and ILS navaids.
 - (c) is a passive system unlike VOR and ILS navaids.
 - (d) is an active system like VOR and ILS navaids.

BLO-7: Aircraft recognition and materials & structures

51. What aircraft model generates the greatest wake turbulence?

- (a) Airbus A380.
- (b) Boeing B747.
- (c) Airbus A340.
- (d) Boeing B737.

52. Regarding the separation distance between the preceding and following aircraft in landing procedures, which of the following statements is true?

- (a) The bigger the preceding aircraft is, the larger separation distance has to be.
- (b) The bigger the preceding aircraft is, the smaller the separation distance has to be.
- (c) The bigger the following aircraft is, the larger the separation distance has to be.
- (d) Separation distance is not related with aircraft size.

53. Identify the following aircraft:



- (a) Airbus A320.
- (b) Airbus A380.
- (c) Boeing B747.
- (d) Boeing B737.

54. Identify the following aircraft:



- (a) Boeing B737.
- (b) Airbus A320.
- (c) Airbus A380.
- (d) Boeing B747.

55. Which of the following aircraft has only two engines?

- (a) Boeing B777.
- (b) Airbus A380.
- (c) Airbus A340.
- (d) Boeing B747.

56. Young's Modulus...

- (a) is the stress-deformation ratio in the elastic regime.
- (b) is the stress-deformation ratio in the plastic regime.
- (c) is about crack propagation.
- (d) is the number of tension cycles before a catastrophic failure of a material.

57. What can be said about fatigue of materials?

- (a) It is the damage caused by cyclic stress in a material.
- (b) It is the amount of energy absorbed by a material during an impact.
- (c) It has none or small effect in aircraft structures.
- (d) All options are correct.

58. What can be said about Aluminium?

- (a) Its alloys are the most common metallic materials in airframe structures.
- (b) It has the highest melting point among the aerospace materials.
- (c) Its crystallographic structure is HCP.
- (d) It has been rarely used in airframe structures.

59. What can be said about Titanium?

- (a) All options are correct.
- (b) It has good resistance to corrosion.
- (c) It has good resistance to high temperature.
- (d) It has better specific properties than steel and aluminium alloys.

60. What can be said about composites?

- (a) All options are correct.
- (b) They have good resistance to fatigue.
- (c) They have higher production cost.
- (d) Their mechanical properties can be tailored through design.

BLO-8: Aircraft systems

61. What are the main advantages of the hydro-mechanical FCS with respect to the purely mechanical FCS?
- (a) Pilot reduces efforts on controls.
 - (b) Aircraft becomes lighter.
 - (c) The system does not depend on hydraulic power.
 - (d) Pilot real feel on controls is enhanced.
62. How many hydraulic systems are in an A320-type aircraft?
- (a) 3, redundant.
 - (b) 2, redundant.
 - (c) 1.
 - (d) 4, redundant.
63. Regarding the hydraulic system, which of the following devices is an alternative power source for some applications?
- (a) Hand-driven pump.
 - (b) Batteries.
 - (c) External hydraulic power supply.
 - (d) None of them.
64. Regarding the electrical system, which of the following devices is an alternative power source?
- (a) All of them.
 - (b) Emergency generator (blue hydraulic system).
 - (c) Batteries.
 - (d) APU.
65. Regarding the electrical system and considering nominal aircraft operation, how the DC buses are powered?
- (a) By means of the main generators through the transformer rectifiers.
 - (b) By means of the nominal batteries.
 - (c) By means of the hydraulic system.
 - (d) By means of the pneumatic generator.
66. Which of the following flight control options does NOT allow for computerised automatic control and flight augmentation?
- (a) An hydro-mechanical flight control system.
 - (b) An electro-hydro-mechanical flight control system.
 - (c) An electro-hydraulic flight control system.
 - (d) All of them allow for automatic control and flight augmentation.
67. What is an aim of the auxiliary power unit (APU) while the airplane is on ground?
- (a) Supply bleed air for engine start.
 - (b) The APU is always switched off on ground.
 - (c) Supply hydraulic power to be able to open the cargo door.
 - (d) Supply the passenger with food and beverages.
68. Which of the following elements cannot be found in an electrical system?
- (a) A cross-bleed duct.
 - (b) A transformer/rectifier.
 - (c) A static inverter.
 - (d) Batteries.
69. In a typical electrical system, where does the essential AC bus get the power from when all AC power sources are out of service?
- (a) From the batteries.
 - (b) The essential AC bus is not powered in this case.
 - (c) This unlikely case has not be considered by the aircraft systems designers.
 - (d) None of them are correct.
70. Where is the energy used to pressurise the hydraulic fluid usually taken from in normal operating conditions?
- (a) From the engines.
 - (b) From the APU.
 - (c) From the PTU.
 - (d) From the RAT.

Correct Answers

BLO-1: Flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	a	b	b	d
2	a	a	d	a	b
3	a	d	b	a	a
4	a	d	d	d	a
5	a	d	c	a	b
6	a	d	b	b	b
7	a	c	a	b	c
8	a	b	d	c	c
9	a	a	b	a	b
10	a	b	a	c	c

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	c	b	c
12	a	a	a	a	b
13	a	a	a	c	a
14	a	d	a	a	a
15	a	d	a	d	d

BLO-3: Aircraft stability & control

	Permutation Number				
Question	0	1	2	3	4
16	a	d	a	c	a
17	a	a	c	c	a
18	a	c	a	a	c
19	a	b	b	d	c
20	a	b	c	b	c
21	a	d	d	a	b
22	a	b	b	d	d
23	a	c	c	a	a
24	a	b	b	a	b
25	a	c	c	d	b

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	b	c	b
27	a	c	c	d	a
28	a	d	c	d	a
29	a	d	d	d	d
30	a	d	a	b	a

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	c	d	a
32	a	c	d	d	d
33	a	a	d	d	b
34	a	c	b	d	c
35	a	c	c	a	d
36	a	c	c	b	c
37	a	c	d	c	d
38	a	d	a	a	c
39	a	c	a	d	d
40	a	a	c	c	b

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
41	a	a	a	c	c
42	a	c	c	d	c
43	a	c	b	c	c
44	a	d	c	b	c
45	a	c	a	c	a
46	a	b	a	a	a
47	a	d	a	c	b
48	a	c	c	a	a
49	a	d	b	d	a
50	a	b	b	b	c

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
51	a	a	b	d	b
52	a	d	b	a	a
53	a	c	c	c	c
54	a	b	b	d	b
55	a	b	b	c	b
56	a	a	b	c	a
57	a	a	a	d	b
58	a	d	a	d	b
59	a	d	b	b	c
60	a	a	a	d	b

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	b	b	b	c
62	a	b	b	d	c
63	a	a	b	a	d
64	a	c	c	c	a
65	a	d	a	b	a
66	a	c	a	c	a
67	a	d	a	a	c
68	a	c	b	c	a
69	a	a	a	c	b
70	a	c	b	b	a

BLO-5: Aerospace propulsion, helicopters and space systems

1. What is the tail rotor used for?
 - (a) For compensating the torque from the main rotor.
 - (b) For generating lift.
 - (c) For compensating the drag force from the main rotor.
 - (d) For generating thrust.
2. Regarding the rotor controls, what is the result of a collective action?
 - (a) The helicopter increases or decreases its total lift.
 - (b) The rotor disk is tilted in a particular direction, resulting the helicopter moving in that direction.
 - (c) It causes the helicopter nose to yaw in a specified direction.
 - (d) The cyclic action is not defined in the helicopter rotor controls.
3. What statement is true regarding coaxial helicopters?
 - (a) Coaxial helicopters have two horizontal rotors mounted one above the other on concentric shafts.
 - (b) Coaxial helicopters have two large horizontal rotor assemblies mounted one in front of the other.
 - (c) Coaxial helicopters are based on the Coanda effect.
 - (d) The tail rotor of a coaxial helicopters reduces the whole helicopter noise.
4. What application is the low earth orbit (LEO) mainly used for?
 - (a) Remote sensing.
 - (b) Communications.
 - (c) Navigation.
 - (d) Micro-gravity experiments.
5. What pattern would a satellite do if it follows a geosynchronous earth orbit (GSO, $i \neq 0$) and it is viewed from an external, Earth fixed reference frame?
 - (a) An eight pattern.
 - (b) A circled pattern.
 - (c) As the satellite follows a GSO orbit, the satellite will remain still from this reference frame.
 - (d) None of them are correct..
6. In which type of engine part of the power generated by the turbine is used to drive a propeller?
 - (a) Turboprop.
 - (b) Turbofan.
 - (c) Ramjet.
 - (d) Alternative engine.
7. What happens to turbojet thrust when external air pressure rises?
 - (a) Increases.
 - (b) Decreases.
 - (c) External air pressure does not affect thrust.
 - (d) It depends on the bypass ratio.
8. Which of the following turbine engines has better performances at low speed?
 - (a) Turboprop.
 - (b) Turbojet.
 - (c) Scramjet.
 - (d) Turbofan.
9. Which of the following statements related to turbojet engines is true?
 - (a) Air flow speed decreases and pressure increases through inlet.
 - (b) Compressor extracts energy from the air flow to drive the turbine.
 - (c) Annular is the heaviest type of combustion chamber.
 - (d) Air flow speed decreases and pressure increases through nozzle.
10. Turbojet Thermodynamic cycle is described by?
 - (a) Brayton cycle.
 - (b) Otto cycle.
 - (c) Diesel cycle.
 - (d) Carnot cycle.

BLO-6: Navigation & altimetry

11. Regarding the navigation systems (e.g. GPS) what is the most used Earth model?
- (a) Elliptical Earth.
 - (b) Geoid.
 - (c) There is no need to use models in navigation systems.
 - (d) Flat Earth.
12. What is the range of latitude (ϕ) and longitude (λ) values?
- (a) $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ [rad] , $\lambda \in [-\pi, \pi]$ [rad].
 - (b) $\lambda \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ [rad] , $\pi \in [-\pi, \pi]$ [rad].
 - (c) $\phi \in [-\pi, \pi]$ [rad] , $\lambda \in [-\pi, \pi]$ [rad].
 - (d) $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ [rad] , $\lambda \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ [rad].
13. What is the Q-code used to specify the magnetic bearing to a navaid station?
- (a) QDM.
 - (b) QUJ.
 - (c) QDR.
 - (d) QDE.
14. Which of the following systems is NOT a navigation system?
- (a) RADAR.
 - (b) GPS.
 - (c) ILS.
 - (d) Locator.
15. What is the specific name of a high powered NDB?
- (a) NDB.
 - (b) Locator (L).
 - (c) Localizer (Loc).
 - (d) Localizer (L).
16. What subsystem responds the interrogation in a DME navaid?
- (a) The on-ground transponder.
 - (b) The on-board interrogator.
 - (c) The on-board transponder.
 - (d) The on-ground interrogator.
17. What is the accuracy of the VOR navaid?
- (a) 1° .
 - (b) 1 NM.
 - (c) 10 NM.
 - (d) 10° .
18. How many signals does a NDB navaid transmit?
- (a) One: An omni-directional signal.
 - (b) Two: Reference and variable signals.
 - (c) One: A variable signal.
 - (d) Three: Two reference signal and one variable.
19. What is the main difference between the TAS and G/S?
- (a) TAS is equal to G/S by correcting the wind effect to the latter one.
 - (b) G/S is equal to EAS by correcting the wind effect to the latter one.
 - (c) TAS is equal to G/S by correcting the density errors to the latter one.
 - (d) G/S is equal to TAS by correcting the density errors to the latter one.
20. What happens when an aircraft crosses the transition altitude (TA)?
- (a) When climbing, the pilot shall change the setting altitude from QNH to STD.
 - (b) When descending, the pilot shall change the setting altitude from STD to QNH.
 - (c) When descending, the pilot shall change the setting altitude from QNH to STD.
 - (d) When climbing, the pilot shall change the setting altitude from STD to QNH.

21. Identify the following aerospace vehicle:



- (a) Aerodyne
- (b) Rotorcraft
- (c) Tilt rotor
- (d) Lifting body

22. Identify the following aerospace vehicle:



- (a) Airship
- (b) Balloon
- (c) Aerodyne
- (d) Rotorcraft

23. Identify the following aerospace vehicle:



- (a) Aerostat
- (b) Balloons
- (c) Aerodyne
- (d) Rocket

24. Identify the following aerospace vehicle: C172 L1P

- (a) Cessna 172: Land Plane, single engine, piston type
- (b) Cessna 172: Land Plane, single wing, piston type
- (c) Cessna 172: Land Plane, single wing, pitot type
- (d) Cessna 172: Land Plane, single engine, pitot type

25. Identify the following aerospace vehicle:



- (a) A320
- (b) B737
- (c) A380
- (d) B777

26. elastic limit or yield point...

- (a) is the stress at which plastic deformation begins.
- (b) is the stress-deformation ratio.
- (c) is about crack propagation.
- (d) is the opposition shown by a material to any surface alteration.

27. What does 2024-T3 refer to?

- (a) To an Aluminium alloy.
- (b) To a Titanium alloy.
- (c) To a fighter aircraft.
- (d) To a glass fiber-epoxy composite.

28. What can be said about steel?

- (a) It is almost three times denser than Al.
- (b) It is never used in parts of airframe structures which require high mechanical performances.
- (c) Its crystallographic structure is HCP.
- (d) Around 80 per cent of modern airframe structures are made of steel.

29. Which of the following material are commonly used as reinforcement fibers in composites?

- (a) Carbon fiber.
- (b) Polyester resin.
- (c) Vynilester resin.
- (d) Wood.

30. What can be said about composites?

- (a) Their mechanical properties can be tailored through design.
- (b) They are too heavy for aerospace use.
- (c) They are easily corroded in dry environments.
- (d) They have higher assembly cost.

BLO-8: Aircraft systems

31. What system does the artificial feel subsystem belong to?
- (a) Flight control system.
 - (b) Hydraulic system.
 - (c) Pneumatic system.
 - (d) Air conditioning system.
32. Which of the following is NOT an energy source of the electrical system?
- (a) The pneumatic system.
 - (b) The main engines.
 - (c) Auxiliary power unit.
 - (d) Batteries.
33. Regarding the flight control laws, what law provides the aircraft with a flight envelope protection?
- (a) The nominal law.
 - (b) The direct law.
 - (c) The mechanical law.
 - (d) None of them.
34. In nominal operation, which of the following electrical buses are powered on?
- (a) All of them.
 - (b) DC BAT BUS.
 - (c) Essential buses.
 - (d) AC BUS.
35. Regarding the electrical system, what is the main reason to use 400 Hz alternate current?
- (a) The higher the frequency is the smaller the alternators and motors are.
 - (b) Large aircraft use 50 Hz, not 400 Hz.
 - (c) With 400 Hz, lower voltages are needed to generate the same power.
 - (d) None of them are correct.
36. In a typical electrical system during normal operation, where does DC power come from?
- (a) From the AC circuit, through the transformer rectifiers (TR)
 - (b) From nowhere. The DC circuits are only powered in case of an emergency.
 - (c) From batteries.
 - (d) From the hydraulic system.
37. What does the acronym FADEC, the system in charge of full engine management, stand for?
- (a) Full authority digital engine control.
 - (b) Fully automatic digital engine control.
 - (c) Full authority digital engine computer.
 - (d) Fully automatic digital engine computer.
38. What is the power transfer unit (PTU) for?
- (a) To transfer pressure among hydraulic circuits.
 - (b) To transfer pressurized fluid among hydraulic circuits.
 - (c) To transfer electrical power among electrical circuits.
 - (d) To transfer pneumatic power between the two packs of the pneumatic system.
39. What is the cross-bleed valve for?
- (a) To transfer pneumatic power between the two packs of the pneumatic system.
 - (b) To transfer pressure among hydraulic circuits.
 - (c) To transfer pressurized fluid among hydraulic circuits.
 - (d) To transfer electrical power among electrical circuits.
40. Which one of the following flight control systems options does NOT allow for computerized automatic control?
- (a) An hydro-mechanical flight control system.
 - (b) An electro-hydro-mechanical flight control system.
 - (c) An electro-hydraulic flight control system.
 - (d) All of them allow for computerized automatic control.

Correct Answers

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	a	a	c	d
2	a	c	b	c	c
3	a	b	d	d	a
4	a	d	b	c	b
5	a	c	d	c	c
6	a	d	c	c	c
7	a	a	b	b	d
8	a	d	c	d	c
9	a	d	c	a	a
10	a	a	a	a	c

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	b	b	d	d
12	a	c	a	a	a
13	a	b	a	a	b
14	a	c	d	d	d
15	a	c	d	c	b
16	a	b	b	a	a
17	a	b	a	d	b
18	a	b	a	d	d
19	a	a	a	b	b
20	a	b	c	c	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	c	b	c	b
22	a	a	b	d	c
23	a	d	c	a	d
24	a	c	c	a	a
25	a	c	c	a	d
26	a	d	b	d	b
27	a	b	d	b	c
28	a	c	a	b	a
29	a	a	a	d	c
30	a	c	b	d	a

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	b	a	a
32	a	b	a	c	d
33	a	a	b	c	d
34	a	c	c	c	b
35	a	d	c	a	b
36	a	a	a	c	b
37	a	a	d	a	c
38	a	c	b	c	b
39	a	c	b	d	b
40	a	a	b	c	c

BLO-5: Aerospace propulsion, helicopters and space systems

1. In helicopters, which of the following elements is NOT related to the generation of the lift force?
 - (a) The tail rotor.
 - (b) The main rotor.
 - (c) All of them are related.
 - (d) The main rotor blades.
2. Regarding the rotor controls, what is the result of a cyclic action?
 - (a) The rotor disk is tilted in a particular direction, resulting the helicopter moving in that direction.
 - (b) The helicopter increases or decreases its total lift.
 - (c) It causes the helicopter nose to yaw in a specified direction.
 - (d) The cyclic action is not defined within the helicopter rotor controls.
3. What statement is true regarding tandem helicopters?
 - (a) Tandem helicopters have two large horizontal rotor assemblies mounted one in front of the other.
 - (b) Tandem helicopters are based on the Coanda effect.
 - (c) The tail rotor of a tandem helicopters reduces the whole helicopter noise.
 - (d) Tandem helicopters have two horizontal rotors mounted one above the other on concentric shafts.
4. What application is the geostationary earth orbit (GEO) mainly used for?
 - (a) Communications.
 - (b) Remote sensing.
 - (c) Navigation.
 - (d) Micro-gravity experiments.
5. What application is the highly elliptical earth orbit (HEO) mainly used for?
 - (a) Communications for locations situated in extreme latitudes.
 - (b) Communications for locations situated in temperate latitudes.
 - (c) Communications for locations situated in extreme longitudes.
 - (d) Communications for locations situated in temperate longitudes.
6. In which type of jet engine part of the air flow is driven through a bypass?
 - (a) Turbofan.
 - (b) Turboprop.
 - (c) Ramjet.
 - (d) Alternative Engine.
7. What happens to turbojet thrust when external air temperature rises?
 - (a) Decreases.
 - (b) Increases.
 - (c) External air temperature does not affect thrust.
 - (d) It depends on the bypass ratio.
8. Which of the following turbine engines has better performances at high speed and altitude?
 - (a) Turbojet.
 - (b) High bypass ratio turbofan.
 - (c) Turboprop.
 - (d) Low bypass ratio turbofan.
9. Which of the following statements related to turbojet engines is true?
 - (a) Supersonic inlets are convergent.
 - (b) Compressors are driven by electric engines during the whole turbojet operation.
 - (c) Turbine blades operate at low temperatures.
 - (d) Supersonic nozzles are convergent.
10. Propulsive efficiency of turbojets is higher...?
 - (a) when exhaust air speed is close to aircraft airspeed.
 - (b) when a small airflow is accelerated to high speed.
 - (c) with centrifugal compressors.
 - (d) after turbine blades stall.

BLO-6: Navigation & altimetry

11. Regarding the shape of the Earth, which of the following models is the most accurate?
- (a) Geoid.
 - (b) Elliptical Earth.
 - (c) Spherical Earth.
 - (d) Flat Earth.
12. What is the equivalence between longitude (geodesic) and time?
- (a) $15^\circ \leftrightarrow 1$ hour.
 - (b) $1^\circ \leftrightarrow 15$ hours.
 - (c) $1^\circ \leftrightarrow 1$ hour.
 - (d) $15^\circ \leftrightarrow 15$ hours.
13. Complete the statement: A navaid is considered active when...
- (a) on-board transceivers transmit signals to compute the position solution (e.g. DME).
 - (b) on-board receivers never transmit signals to compute the position solution (e.g. DME).
 - (c) on-board receivers never transmit signals to compute the position solution (e.g. VOR).
 - (d) on-board transceivers transmit signals to compute the position solution (e.g. VOR).
14. Choose the correct statement about the relative bearing (RB), bearing (B) and heading (H):
- (a) $RB = B - H$.
 - (b) $B = RB - H$.
 - (c) $H = RB$.
 - (d) $RB = H - B$.
15. What is the specific name of a low powered NDB?
- (a) Locator (L).
 - (b) Localizer (Loc).
 - (c) Location (L).
 - (d) NDB.
16. What subsystem starts the communication in a DME navaid?
- (a) The on-board interrogator.
 - (b) The on-board transponder.
 - (c) The on-ground interrogator.
 - (d) The on-ground transponder.
17. What is the accuracy of the DME navaid?
- (a) 1 NM.
 - (b) 10 NM.
 - (c) 1° .
 - (d) 10° .
18. How many signals does a VOR navaid transmit?
- (a) Two: Reference and variable signals.
 - (b) One: An omni-directional signal.
 - (c) One: A variable signal.
 - (d) Three: Two reference signal and one variable.
19. What is the main difference between the IAS and RAS/CAS?
- (a) RAS/CAS is equal to IAS when correcting the pressure errors to the latter one.
 - (b) IAS is equal to RAS/CAS by correcting the pressure errors to the latter one.
 - (c) RAS/CAS is equal to IAS by correcting the density error to the latter one.
 - (d) IAS is equal to RAS/CAS by correcting the density error to the latter one.
20. What happens when an aircraft crosses the transition level (TL)?
- (a) When descending, the pilot shall change the setting altitude from STD to QNH.
 - (b) When climbing, the pilot shall change the setting altitude from QNH to STD.
 - (c) When descending, the pilot shall change the setting altitude from QNH to STD.
 - (d) When climbing, the pilot shall change the setting altitude from STD to QNH.

21. Identify the following aerospace vehicle:



- (a) Aerostat
- (b) Aerodyne
- (c) Rocket
- (d) Space Vehicle

22. Identify the following aerospace vehicle:



- (a) Aerodyne
- (b) Rotorcraft
- (c) Tilt rotor
- (d) Lifting body

23. Identify the following aerospace vehicle:



- (a) Airship
- (b) Balloon
- (c) Aerodyne
- (d) Rotorcraft

24. Identify the following aerospace vehicle: A346 L4J

- (a) Airbus 340-600: Land Plane, 4 engines, jet type
- (b) Airbus 346: Conventional Plane, 4 engines, jet type
- (c) Airbus 340: Conventional Plane, 4 pilots, jet type
- (d) Airbus 340-600: Conventional Plane, 4 engines, jet type

25. Identify the following aerospace vehicle:



- (a) B777
- (b) A320
- (c) B737
- (d) A380

26. Young's Modulus...

- (a) is about elastic deformation along an axis
- (b) is about plastic deformation along an axis
- (c) is the number of tension cycles before a catastrophic failure of a longeron
- (d) is bigger in rubber than steel

27. What can be said about composites?

- (a) They are made of two or more different materials
- (b) They are a kind of titanium alloy
- (c) Carbon fiber cannot be one of their constituents
- (d) They have lost importance as airframe structural material in the last 30 years

28. What can be said about Aluminium?

- (a) Its crystallographic structure is FCC.
- (b) It is the metal which has the highest density.
- (c) It is the metal which has the highest melting point.
- (d) It has been rarely used in airframe structures.

29. What can be said about Titanium?

- (a) It has good resistance to corrosion.
- (b) It has no allotropic transformation.
- (c) It is cheaper than Aluminium.
- (d) It has worst specific mechanical properties than Aluminium.

30. What can be said about fiber reinforced composites?

- (a) They have a very good resistance to fatigue.
- (b) They have lower production cost.
- (c) They cannot show delamination damage.
- (d) They have bad resistance to corrosion.

BLO-8: Aircraft systems

31. What are the main components of the flight control system?

- (a) Control surfaces, cockpit controls and transmission elements.
- (b) Cockpit surfaces, primary controls and transmission elements.
- (c) The green, yellow and blue control systems.
- (d) None of them are correct.

32. What is the name of the following device?



- (a) Ram air turbine.
- (b) Artificial feel system.
- (c) Auxiliary power unit.
- (d) Accessory gearbox.

33. In nominal operation, what system is in charge of pressurizing the hydraulic system reservoirs?

- (a) The pneumatic system.
- (b) The electrical system.
- (c) The power plant.
- (d) The hydraulic system.

34. Which of the following is not a source of the pneumatic system

- (a) Electrical system.
- (b) Auxiliary power unit.
- (c) Main engines.
- (d) High pressure ground connection.

35. In nominal operation, what is the energy source of the hydraulic system?

- (a) The main engines.
- (b) The auxiliary power unit.
- (c) The electrical system.
- (d) The flight control system.

36. What does *alternate law* mean for a flight control system?

- (a) It is a flight control reconfiguration law in case there are failures that hinder *normal control law*.
- (b) It is a flight control law that reverses the action of the commands.
- (c) It is a flight control law that power the electronics with alternate current AC.
- (d) None of them are correct.

37. Which of the following elements cannot be found in an electrical system?

- (a) A power transfer unit (PTU)
- (b) A transformer/rectifier.
- (c) Buses.
- (d) Batteries.

38. What is the aim of the auxiliary power unit (APU) in flight?

- (a) Backup the electrical system.
- (b) It is forbidden to switch on in flight.
- (c) Provide extra thrust.
- (d) Massage the flight crew.

39. In a electro-hydraulic flight control system, how the actuators are moved?

- (a) By means of hydraulic power.
- (b) By means of pneumatic power.
- (c) By means of electrical power.
- (d) The actuators remain still during the flight.

40. What is the main aim of the hydraulic system, apart from actuating control surfaces, in an Airbus A320 type aircraft?

- (a) To drive an emergency generator to produce electrical power if required.
- (b) To stabilize flight.
- (c) To power the pneumatic system.
- (d) To power the auxiliary power unit (APU).

Correct Answers

BLO-5: Aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	b	a	a	b
2	a	b	a	c	c
3	a	d	c	a	a
4	a	b	c	c	d
5	a	d	a	b	b
6	a	c	c	a	a
7	a	c	d	a	d
8	a	b	a	c	d
9	a	b	d	b	c
10	a	c	a	d	b

BLO-6: Navigation & altimetry

	Permutation Number				
Question	0	1	2	3	4
11	a	b	b	d	c
12	a	c	c	a	c
13	a	c	a	b	b
14	a	d	b	d	c
15	a	a	b	d	b
16	a	c	a	b	b
17	a	d	c	b	a
18	a	c	c	d	a
19	a	d	b	a	d
20	a	b	b	a	b

BLO-7: Aircraft recognition and materials & structures

	Permutation Number				
Question	0	1	2	3	4
21	a	d	a	a	a
22	a	b	c	d	d
23	a	c	b	c	b
24	a	d	a	a	d
25	a	a	b	b	d
26	a	c	b	d	a
27	a	b	b	c	c
28	a	a	d	c	b
29	a	b	d	b	a
30	a	a	c	c	c

BLO-8: Aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	d	b	c
32	a	d	b	a	d
33	a	c	d	d	a
34	a	c	d	b	d
35	a	a	d	c	c
36	a	d	a	a	a
37	a	b	c	d	b
38	a	b	a	d	a
39	a	a	b	c	b
40	a	d	a	c	d

BLO-1: Concepts on flight principles and aircraft performance

1. Which of the following types of vehicle does not rely on an atmosphere to fly?
 - (a) a rocket.
 - (b) an airplane.
 - (c) an helicopter.
 - (d) a hot air baloon.
2. Which of the following solutions does not reduce wing tip vortex formation?
 - (a) Increasing wing tip aerodynamic loading.
 - (b) Increasing the wing aspect ratio.
 - (c) Introducing taper.
 - (d) Introducing negative twist, either geometrical or aerodynamic.
3. Which of the following parameters is roughly independent of altitude if we neglect the effects of Reynolds and Mach numbers?
 - (a) Stall angle of attack.
 - (b) Stall speed.
 - (c) Maximum bank angle in a coordinated turn.
 - (d) Maximum available thrust.
4. Which of the following explanations of lift is plain false...
 - (a) The asymmetry of the upper and lower surfaces of the airfoil block the airflow through the extrados so that air must flow faster and lift is generated.
 - (b) The airfoil deflects air downwards, so that lift is the reaction force on the airfoil.
 - (c) Air flows faster along the upper surface than along the lower surface, so that a pressure difference is generated that accounts for lift.
 - (d) Viscosity generates vorticity in a thin layer around the airfoil, which produces a global circulation that generates lift.
5. How does maximum bank angle compare between an airplane flying high at an altitude h_h (ϕ_h) and the same airplane, at the same speed, flying at sea level at h_{SL} (ϕ_{SL})?
 - (a) $\phi_h < \phi_{SL}$
 - (b) $\phi_h > \phi_{SL}$
 - (c) $\phi_h = \phi_{SL}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
6. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the airplane burns fuel during cruise if altitude is kept constant?
 - (a) It occurs at lower speeds (moves left) and decreases (moves down).
 - (b) It occurs at lower speeds (moves left) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. Which of the following performance optima are not affected by wind?
 - (a) Endurance and maximum rate of climb.
 - (b) Range and maximum rate of climb.
 - (c) Endurance and maximum climb angle.
 - (d) Range and maximum climb angle.
8. Which of the following is not a limiting factor in constant speed and constant altitude turns?
 - (a) All are limiting factors.
 - (b) Maximum load factor.
 - (c) Maximum available power.
 - (d) Stall angle of attack.
9. Which of the following conditions is generally adequate for maximising cruise range of a propeller airplane?
 - (a) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (b) All options are correct.
 - (c) Minimum required power.
 - (d) Minimum flight altitude.
10. How are airport runways operated in wind conditions?
 - (a) Headwind for both takeoff and landing.
 - (b) Tailwind for both takeoff and landing.
 - (c) Headwind for takeoff, tailwind for landing.
 - (d) Tailwind for takeoff, headwind fot landing.

BLO-2: Flight principles related problems

11. What is the angle of attack ($\alpha_{f_{max}}$) that maximises aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.02 + 0.12C_L^2$ and lift characteristics $C_L = 0.1 + 1.1\alpha$ with α in $^\circ$?
- (a) $\alpha_{f_{max}} = 0.28^\circ$
 - (b) $\alpha_{f_{max}} = 0.311^\circ$
 - (c) $\alpha_{f_{max}} = 0.339^\circ$
 - (d) $\alpha_{f_{max}} = 0.37^\circ$
12. A fighter airplane of mass $m = 800$ kg and wing surface $S = 20$ m², with lift coefficient $C_L = 0.1 + 0.11\alpha$ (α in $^\circ$) and drag polar $C_D = 0.02 + 0.12C_L^2$, engages in a perfect vertical climb at a constant speed $V = 60$ m/s at sea level. How much thrust (T_A) must the engines produce (take $g \simeq 9.8$ m/s² and $\rho = 1.225$ kg/m³)?
- (a) $T_A = 872$ daN
 - (b) $T_A = 88.2$ daN
 - (c) $T_A = 784$ daN
 - (d) $T_A = 980$ daN
13. A small airplane is engaged in a coordinated banked turn at sea level with maximum angle $\phi_{max}^{SL} = 70^\circ$. What is the maximum bank angle ϕ_{max}^h for a turn with the same aircraft at the same speed but at an altitude such that $\rho^h = 0.5 \rho^{SL}$?
- (a) $\phi_{max}^h = 46.8^\circ$
 - (b) $\phi_{max}^h = 52.5^\circ$
 - (c) $\phi_{max}^h = 58^\circ$
 - (d) $\phi_{max}^h = 32^\circ$
14. The drag polar of an airplane changes from $C_D = 0.011 + 0.12C_L^2$ in clean configuration to $C_D = 0.02 + 0.09C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.047$
 - (b) $C_D = 0.0614$
 - (c) $C_D = 0.087$
 - (d) $C_D = -0.00774$
15. An airplane of mass $m = 900$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 24 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
- (a) $D = 368$ N and $D_p = D_i = 184$ N.
 - (b) $D = 368$ N and $D_p > D_i$.
 - (c) $D = 184$ N and $D_p = D_i = 91.9$ N.
 - (d) $D = 184$ N and $D_p > D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is not a stability condition?
- (a) Reduced.
 - (b) Unstable.
 - (c) Stable.
 - (d) Neutral.
17. Which of the following Primary Control Surfaces is commanded with pedals?
- (a) Rudder.
 - (b) Power plant setting.
 - (c) Ailerons.
 - (d) Elevator.
18. Flight equilibrium requires...
- (a) All options are correct.
 - (b) Summation of all forces over the aircraft equals zero.
 - (c) All moments over the aircraft are balanced.
 - (d) Zero linear and angular acceleration of the aircraft.
19. Which of the following relative positions of the Center of Gravity and Neutral Point of an airplane produces Positive Stability?
- (a) Center of Gravity ahead of Neutral Point.
 - (b) Center of Gravity on Neutral Point.
 - (c) Center of Gravity behind Neutral Point.
 - (d) It does not depend on the relative position.
20. Which of the following is not a longitudinal mode?
- (a) Dutch Roll.
 - (b) Pitch oscillation.
 - (c) Return to straight and level flight.
 - (d) Phugoid.
21. How does an aircraft damp roll?
- (a) Lift asymmetry between wings due to increased angle of attack in down-going wing and reduced angle of attack in up-going wing.
 - (b) Inertial forces due to fuel movement in wing integral fuel tank.
 - (c) Spoilers commanded by the Automatic Control System of the aircraft.
 - (d) Pilot acting on aileron trim tabs.
22. What can be said about Lateral Stability of an airplane?
- (a) Roll and Yaw are strongly coupled.
 - (b) A restoring moment appears, which depends on the size of the fin.
 - (c) A damping moment appears, which depends on the deployment angle of the flaps.
 - (d) A restoring moment appears, which depends on the size of the horizontal stabiliser.
23. Which of the following is not a stabilising contributor to Directional Stability in an airplane?
- (a) Forward wing sweep.
 - (b) Backward wing sweep.
 - (c) Vertical Stabiliser size.
 - (d) Center of Gravity to empennage distance.
24. What can be said about Dutch Roll?
- (a) Some aircraft install a Yaw Damper to damp it.
 - (b) It consists of a pitching oscillation.
 - (c) It has a long period.
 - (d) Horizontal Stabiliser affects damping.
25. Which of the following Secondary Control Surfaces are used to pilot roll in some airplanes?
- (a) Spoilers.
 - (b) Slats.
 - (c) Slots.
 - (d) Vortex Generators.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.24\bar{\rho}V^3 + 0.022W^2/\bar{\rho}V$. If the airplane has mass $m = 1500\text{ kg}$ and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8\text{ m/s}^2$)
- $V_E = 71.7\text{ m/s}$
 - $V_E = 94.3\text{ m/s}$
 - $V_E = 124\text{ m/s}$
 - $V_E = 80\text{ m/s}$
27. A jet-powered airplane of mass $m = 1400\text{ kg}$ has a required thrust curve $T_R = 0.28\bar{\rho}V^2 + 0.032W^2/\bar{\rho}V^2$. What is the velocity for best climb angle ($V_{\gamma_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8\text{ m/s}^2$)
- $V_{\gamma_{max}} = 68.1\text{ m/s}$
 - $V_{\gamma_{max}} = 51.7\text{ m/s}$
 - $V_{\gamma_{max}} = 89.6\text{ m/s}$
 - $V_{\gamma_{max}} = 39.3\text{ m/s}$
28. An airplane with drag polar $C_D = 0.02 + 0.18C_L^2$, wing surface $S = 35\text{ m}^2$ and mass $m = 2000\text{ kg}$ is flying at a speed $V = 120\text{ m/s}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.33\text{ kg/m}^3$? (take $g \simeq 9.8\text{ m/s}^2$)
- $T_R = 2490\text{ N}$
 - $T_R = 2580\text{ N}$
 - $T_R = 2680\text{ N}$
 - $T_R = 2230\text{ N}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 45$. What is the descent angle (γ)?
- $\gamma = -1.27^\circ$
 - $\gamma = -1.41^\circ$
 - $\gamma = -1.01^\circ$
 - $\gamma = -1.67^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 2000\text{ kg}$, wing area $S = 25\text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 20000\text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 9.8\text{ m/s}^2$ and $\rho_0 = 1.225\text{ kg/m}^3$)
- $V \simeq 52.6\text{ m/s}$
 - $V \simeq 43.2\text{ m/s}$
 - $V \simeq 47.5\text{ m/s}$
 - $V \simeq 46.9\text{ m/s}$

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	a	a	b	d
2	a	b	d	b	d
3	a	b	a	a	d
4	a	a	b	c	b
5	a	c	b	c	c
6	a	c	a	c	d
7	a	a	c	b	c
8	a	d	a	a	d
9	a	b	d	a	b
10	a	a	d	a	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	c	a	d
12	a	c	c	a	d
13	a	d	b	b	c
14	a	a	c	a	b
15	a	b	d	c	b

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	a	c	b	c
17	a	d	a	b	c
18	a	c	a	b	c
19	a	c	d	a	d
20	a	b	c	c	a
21	a	d	a	d	a
22	a	d	d	d	d
23	a	d	d	b	c
24	a	c	b	c	b
25	a	c	c	b	c

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	b	b	c
27	a	b	b	b	d
28	a	d	d	c	b
29	a	c	d	d	d
30	a	d	b	a	c

BLO-1: Concepts on flight principles and aircraft performance

1. The difference between autogyres and helicopters is...
 - (a) The autogyre's main rotor is not powered.
 - (b) The helicopter's main rotor is not powered.
 - (c) The autogyre is a rotorcraft while the helicopter isn't.
 - (d) They are synonyms.
2. What is the main effect on the drag coefficient of increasing the aspect ratio of a wing?
 - (a) It increases friction and reduces lift-induced drag.
 - (b) It reduces friction and increases lift-induced drag.
 - (c) It increases both friction and lift-induced drag.
 - (d) It reduces both friction and lift-induced drag.
3. Under which conditions are pitch angle (θ) and angle of attack (α) equal?
 - (a) In horizontal flight ($\gamma = 0$).
 - (b) In low subsonic flight ($M < 0.3$).
 - (c) In low Reynolds number flight.
 - (d) Always.
4. Which of the following statements is FALSE about lift of an airfoil?
 - (a) Lift is the result of air's friction on the airfoil surfaces.
 - (b) Lift is the result of deflecting air downwards.
 - (c) Lift is the result of generating a pressure difference between upper and lower surfaces.
 - (d) Lift is the result of generating circulation around the airfoil.
5. What happens to stall velocity V_s as an airplane engages into an altitude-preserving manoeuvre that increases its load factor (n)?
 - (a) V_s increases.
 - (b) V_s decreases.
 - (c) V_s stays the same.
 - (d) It depends on whether the airplane is flying in the first or second regime.
6. How does the required power curve ($P_R(V)$) roughly evolve as the airplane burns fuel in a constant altitude flight?
 - (a) Lift-induced power decreases while parasitic power remains the same.
 - (b) Parasitic power decreases while lift-induced power remains the same.
 - (c) Both parasitic and lift-induced power decrease.
 - (d) Both lift-induced and parasitic power remain the same.
7. Which of the following performance optimals critically depends on wind direction and intensity?
 - (a) Maximum range.
 - (b) Maximum endurance.
 - (c) Maximum rate of climb.
 - (d) None of the others.
8. What is the optimal way to pilot a glider?
 - (a) Best rate within thermals and best glide from thermal to thermal.
 - (b) Best glide within thermals and best rate from thermal to thermal.
 - (c) Best glide always.
 - (d) Best rate always.
9. Which of the following conditions is roughly adequate for maximising cruise endurance of a jet airplane?
 - (a) Minimum hourly fuel consumption (C , in lb/h).
 - (b) Maximum excess power.
 - (c) Minimum required power.
 - (d) All options are correct.
10. Of which two stages does the transition phase of a landing consist?
 - (a) Flare and float.
 - (b) Float and ground run.
 - (c) Final approach and ground run.
 - (d) Final approach and flare.

BLO-2: Flight principles related problems

11. What is the lift coefficient ($C_{L_{opt}}$) for the optimal condition providing maximum aerodynamic efficiency (f_{max}) for an airplane with drag polar $C_D = 0.05 + 0.2C_L^2$?
 - (a) $C_{L_{opt}} = 0.5$
 - (b) $C_{L_{opt}} = 0.478$
 - (c) $C_{L_{opt}} = 0.453$
 - (d) $C_{L_{opt}} = 0.426$

12. An aerobatics airplane with lift coefficient $C_L = 0.13 + 0.11\alpha$ with α in $^\circ$ is performing a balanced inverted flight. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.25$?
 - (a) $\alpha = -3.45^\circ$
 - (b) $\alpha = 1.09^\circ$
 - (c) $\alpha = -1.18^\circ$
 - (d) $\alpha = 3.45^\circ$

13. A small airplane is engaged in a coordinated banked turn at an unknown altitude. We know its speed is $V = 2.5 V_S|_{n=1}$, where $V_S|_{n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max}) that can be achieved if the plane is to keep its altitude and speed?
 - (a) $\phi_{max} = 80.8^\circ$
 - (b) $\phi_{max} = 66.4^\circ$
 - (c) $\phi_{max} = 14.1^\circ$
 - (d) It depends on the mass of the airplane.

14. The drag polar of an airplane changes from $C_D = 0.025 + 0.12C_L^2$ in clean configuration to $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
 - (a) $C_D = 0.07$
 - (b) $C_D = 0.0843$
 - (c) $C_D = 0.101$
 - (d) $C_D = 0.0113$

15. An airplane of mass $m = 1600$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 25 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
 - (a) $D = 627$ N and $D_p \neq D_i$.
 - (b) $D = 627$ N and $D_p = D_i = 314$ N.
 - (c) $D = 314$ N and $D_p = D_i = 157$ N.
 - (d) $D = 314$ N and $D_p \neq D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which of the following secondary flight surfaces is intended for reducing the aerodynamic efficiency of the wing?
- (a) Spoilers.
 - (b) Slats.
 - (c) Flaps.
 - (d) Trim tabs.
17. Which element (parameter) is in charge of damping the dutch roll mode?
- (a) Vertical stabiliser size (C_{n_r}).
 - (b) Wing dihedral angle (C_{l_β}).
 - (c) Horizontal stabilizer size (C_{m_q}).
 - (d) None of the others.
18. Static stability around an axis requires, at the very least...
- (a) The appearance of a restoring moment around this axis upon perturbation.
 - (b) All options are correct.
 - (c) Dynamic stability around this axis.
 - (d) The appearance of a damping moment around this axis during restoration after perturbation.
19. About manoeuvrability and natural stability of an airplane...
- (a) high manoeuvrability entails low natural stability.
 - (b) high manoeuvrability entails high natural stability.
 - (c) manoeuvrability is a necessary condition for natural stability.
 - (d) they are unrelated concepts.
20. For a stable airplane around its pitch axis, which of the following modes are usually aperiodic?
- (a) Return to straight and level flight.
 - (b) Pitch oscillation and phugoid.
 - (c) Phugoid.
 - (d) None of the longitudinal modes is aperiodic.
21. Why is it dangerous to pilot (damp manually) pitch oscillation?
- (a) The action of the pilot on the stick may amplify the oscillations.
 - (b) The action of the pilot on the pedals may induce dutch roll.
 - (c) The action of the pilot on stick and/or pedals may induce spiral instability.
 - (d) It is not dangerous, but rather recommended.
22. What can be said about the initial response of the airplane to a perturbation in yaw?
- (a) A restoring moment appears, which mainly depends on the size of the fin.
 - (b) No restoring moment appears. Recovery is purely dynamical.
 - (c) A restoring moment appears, which mainly depends on the size of the wings.
 - (d) A restoring moment appears, which mainly depends on the size of the horizontal stabiliser.
23. What is the main reason for considering straight and level flight an equilibrium while rectilinear ascent and descent are only considered pseudo-equilibria?
- (a) The air density dependence on altitude.
 - (b) The perturbation of rectilinear flight by the phugoid mode.
 - (c) The mass reduction due to fuel consumption.
 - (d) The increase of sun radiation with altitude.
24. How must the pilot induce sideslip to align the fuselage with the runway in a landing with side wind from the left?
- (a) Press the right pedal.
 - (b) Press the left pedal.
 - (c) Switch off one of the engines of the right wing.
 - (d) Press both pedals simultaneously.
25. Where are lifting canards usually placed in an airplane that mounts them?
- (a) Either side of the fuselage, ahead of the wings.
 - (b) Either side of the fuselage, behind the wings.
 - (c) On top of the fuselage, ahead of the wings.
 - (d) On top of the fuselage, behind the wings.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.22\bar{\rho}V^3 + 0.025W^2/\bar{\rho}V$. If the airplane has mass $m = 1500\text{ kg}$ and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.4$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8\text{ m/s}^2$)
- $V_E = 84.6\text{ m/s}$
 - $V_E = 111\text{ m/s}$
 - $V_E = 146\text{ m/s}$
 - It depends on the available power P_A , which is unknown.
27. A jet airplane of mass $m = 1200\text{ kg}$ has a drag curve $D = 0.28\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$. What is the minimum engine thrust needed at sea level ($\bar{\rho} = 1$) to achieve a climb angle of $\alpha = 6^\circ$? (take $g \simeq 9.8\text{ m/s}^2$)
- $T_A = 3200\text{ N}$
 - $T_A = 3460\text{ N}$
 - $T_A = 2640\text{ N}$
 - $T_A = 2300\text{ N}$
28. An airplane with drag polar $C_D = 0.03 + 0.15C_L^2$, wing surface $S = 36\text{ m}^2$ and mass $m = 1800\text{ kg}$ is flying at a speed $V = 100\text{ m/s}$. What is the minimum engine power required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.5\text{ kg/m}^3$? (take $g \simeq 9.8\text{ m/s}^2$)
- $P_R = 322\text{ kW}$
 - $P_R = 292\text{ kW}$
 - $P_R = 386\text{ kW}$
 - $P_R = 425\text{ kW}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 44$. What is the descent angle (γ)?
- $\gamma = -1.3^\circ$
 - $\gamma = -1.48^\circ$
 - $\gamma = -0.997^\circ$
 - $\gamma = -1.75^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 1500\text{ kg}$, wing area $S = 30\text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.5\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 12000\text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 9.8\text{ m/s}^2$ and $\rho_0 = 1.225\text{ kg/m}^3$)
- $V \simeq 62.9\text{ m/s}$
 - $V \simeq 44.9\text{ m/s}$
 - $V \simeq 142\text{ m/s}$
 - $V \simeq 94.4\text{ m/s}$

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

31. A two-stroke engine...
- (a) All options are false.
 - (b) may be based on a Diesel cycle but never on an Otto cycle.
 - (c) has lower specific power than a four-stroke engine.
 - (d) has higher efficiency than a four-stroke engine.
32. Which of the following statements are true?
- I A turboprop has the highest by-pass ratio.
 - II A ramjet has compressors and turbines.
 - III The optimal Mach number of a high by-pass ratio turbofan is usually higher than that of a turbojet.
- (a) I.
 - (b) I and II.
 - (c) I and III.
 - (d) II and III.
33. A helicopter is...
- (a) an aerodyne.
 - (b) an aerostat.
 - (c) a fixed wing.
 - (d) a variable wing.
34. In a helicopter the swashplate is responsible for...
- (a) All options are true.
 - (b) allowing the collective action.
 - (c) allowing the cyclic action.
 - (d) separating the control arms (fixed) from the rotor head (rotating).
35. In helicopters, the engine power derived to the tail rotor is...
- (a) about 10% of the total.
 - (b) about 50% of the total.
 - (c) constant whatever the regime is.
 - (d) nil. The tail rotor is not driven by the engine.
36. The speed limitation of a helicopter is due to...
- (a) All options are correct.
 - (b) the transonic flow and shock waves on the advancing side of the main rotor blades.
 - (c) the reverse flow region of the retreating side of the main rotor blades.
 - (d) blade tip noise and aeroelastic response.
37. In a helicopter, the typical main rotor speed is about...
- (a) 300 rpm.
 - (b) 300 m/s.
 - (c) 3000 m/s.
 - (d) 3000 rpm.
38. The specific impulse (Isp) of rockets is measured in seconds and corresponds to...
- (a) a measure of the engine ability to accelerate ejected gases.
 - (b) the time required to ignite the engine.
 - (c) the total burn time of the engine.
 - (d) Specific impulse is not measured in seconds but Newtons.
39. A Geosynchronous orbit (GSO)...
- (a) All options are correct.
 - (b) has the same period as the earth's rotation.
 - (c) leaves an 8-shaped ground track.
 - (d) returns over the same region after one sidereal day.
40. A Polar orbit...
- (a) has an inclination close to 90 degrees.
 - (b) covers only one of the poles and never crosses the Equator.
 - (c) is a cold orbit.
 - (d) is useless for Earth observation.

BLO-6: Concepts on avionics and navigation

41. What are the ISA Sea Level (SL) reference values?
- (a) All options are wrong.
 - (b) $p_0 = 1013.25 \text{ mbar}$, $T_0 = 15^\circ\text{C}$, $\rho_0 = 1225 \text{ kg/m}^3$.
 - (c) $p_0 = 101325 \text{ mPa}$, $T_0 = 15^\circ\text{C}$, $\rho_0 = 1.225 \text{ kg/m}^3$.
 - (d) $p_0 = 101325 \text{ Pa}$, $T_0 = 288.15 \text{ K}$, $\rho_0 = 1225 \text{ kg/m}^3$.
42. How does ISA temperature change for increasing altitude below 33000 ft?
- (a) It decreases 2°F per thousand feet.
 - (b) It increases 6.5°C per kilometer.
 - (c) It decreases 6.5°F per thousand feet.
 - (d) It increases 2°C per kilometer.
43. Right after take-off the barometric altimeter requires a calibration called...
- (a) QNH while below the transition altitude (TA).
 - (b) QNH while below the transition level (TL).
 - (c) QFE while climbing.
 - (d) STD while below the transition layer.
44. Which of the following is a possible navigation type and what is the related equipment?
- (a) All options are correct.
 - (b) Estimated navigation: IAS, clock and autonomous navigation systems like IRUs.
 - (c) Navigation by external references: systems defined in the ICAO Annex 10 like NDB, VOR, TACAN, DME, ILS, MLS, GNSS.
 - (d) Vector navigation: radar.
45. The Airborne Collision Avoidance System (ACAS) is based on:
- (a) A secondary surveillance radar.
 - (b) A primary surveillance radar.
 - (c) A datalink system.
 - (d) A voice control system.
46. An example of azimuthal guidance subsystem would be...
- (a) the Localizer (ILS/LLZ), which provides the horizontal angle.
 - (b) the Glide Slope (ILS/GS), which provides the vertical angle.
 - (c) the primary surveillance radar.
 - (d) the weather radar.
47. Why can't GPS be used as a standalone system for navigation?
- (a) Because the system cannot be trusted, as it does not issue a timely alarm in case of malfunction.
 - (b) Because of its low accuracy.
 - (c) Because the Selective Availability signal intentionally degrades the GPS accuracy.
 - (d) Because it is a military system.
48. The accuracy of a VOR is about...
- (a) $\pm 1.4^\circ$ (about 90 m crosstrack error in 2 nm).
 - (b) $\pm 185 \text{ m}$ for a limit of 100 aircrafts.
 - (c) in CAT-I, at the threshold of a 10000 ft -long runway, Azimuth = $\pm 9.1 \text{ m}$ and Elevation = $\pm 4.1 \text{ m}$.
 - (d) Horizontal = $\pm 13 \text{ m}$ and Vertical = $\pm 22 \text{ m}$.
49. Misleading information is...
- (a) an integrity risk because it does not provide a failure detection alarm upon position failure.
 - (b) a continuity risk because it does not provide a failure detection alarm upon position failure.
 - (c) an integrity risk because it provides a failure detection alarm without actual position failure.
 - (d) a continuity risk because it does not provide a failure detection alarm without actual position failure.
50. Integrity is...
- (a) the guarantee that the error will not be higher than some threshold and that, in case it is, an alarm will be received on time.
 - (b) the ability of the total system to perform its function without unscheduled interruptions during the intended operation.
 - (c) the ability of the total system to maintain the aircraft position within a given Total System Error (TSE) with a 95% probability.
 - (d) the ability of the total system to provide the required parameters at the start of the intended operation.

BLO-7: Concepts on materials and aeronautical regulations

51. Atoms join in molecules and cristal structures via...
- (a) ionic bonds, covalent bonds, metallic bonds, van der Waals and hydrogen bonds.
 - (b) yield stress, ultimate tensile stress and fracture toughness.
 - (c) cost, corrosion resistance, fatigue, easiness of transformation, etc.
 - (d) metals, ceramics, polymers, semi-conductors and composites.
52. Aluminium crystals are typically...
- (a) face-centered cubic (FCC).
 - (b) body-centered cubic (BCC).
 - (c) primitive cubic (PC).
 - (d) All options are false.
53. What is the main solute for a 2000 aluminum alloy series?
- (a) Copper.
 - (b) Zinc.
 - (c) Magnesium.
 - (d) Manganese.
54. Which statement is true about the properties of a 7000 aluminum alloy series when compared to 2000 series?
- (a) Higher hardness.
 - (b) Better behavior against crack propagation.
 - (c) Lower Young's modulus of elasticity.
 - (d) All options are true.
55. The best choice of material to be used for the wing box of a DC-10 type aircraft is...
- (a) 7075-T6 aluminum clad.
 - (b) 6Al-4V titanium.
 - (c) fiberglass reinforced plastic.
 - (d) 2024-T4 aluminum clad.
56. Which of the following ICAO annexes contains the standards and recommended practices regarding Aeronautical Charts?
- (a) 4
 - (b) 9
 - (c) 2
 - (d) 15
57. Which of the following ICAO annexes contains the standards and recommended practices regarding Environmental Protection?
- (a) 16
 - (b) 5
 - (c) 8
 - (d) 1
58. Which of the following organisations is a direct result of the Chicago convention of 1944?
- (a) ICAO
 - (b) EUROCONTROL
 - (c) EASA
 - (d) ICAN
59. Which of the following agencies is in charge of issuing the norms regulating the concession of the airworthiness certificate for European aircraft?
- (a) JAA
 - (b) FAA
 - (c) JAR
 - (d) FAR
60. The nationality of an aircraft is given by...
- (a) the registry country
 - (b) the manufacturing country
 - (c) the nationality of the airline that owns it
 - (d) Aircraft have no nationality

BLO-8: Concepts on aircraft systems

61. Which one of the following flight control system options helps relieve pilot effort on the commands but does not allow for automatic control?
- (a) An hydro-mechanical flight control system
 - (b) A purely mechanical flight control system
 - (c) An electro-hydraulic flight control system
 - (d) An electro-hydro-mechanical flight control system
62. Name a component that must be present in both hydro-mechanical and fly-by-wire electro-hydraulic flight control systems?
- (a) Artificial feel system.
 - (b) Flight Controls Computer.
 - (c) Sensors (gyros, accelerometers...)
 - (d) A full mechanical transmission.
63. Which options are there to power the hydraulic system of large aircraft?
- (a) All are possible.
 - (b) Engine driven pumps.
 - (c) Electric pumps.
 - (d) The Ram Air Turbine (RAT).
64. What is FALSE about the several hydraulic circuits of a large passenger aircraft?
- (a) Fluid may be transferred among circuits.
 - (b) Power may be transferred among circuits.
 - (c) Each circuit may be powered by different means.
 - (d) Each circuit may actuate different control surfaces.
65. How is AC electrical power usually generated in a large airplane while on ground with all engines off?
- (a) From the ground connection (external power).
 - (b) From the batteries, through DC/AC transformation.
 - (c) With solar panels.
 - (d) Engines must be started to have AC power.
66. Which of the following parts of the electrical system are disconnected in case of losing all normal AC power generation?
- (a) AC BUS 1.
 - (b) AC ESS BUS.
 - (c) DC ESS BUS.
 - (d) Emergency GEN.
67. Which of the following is a valid option for powering the pneumatic system?
- (a) All options are possible.
 - (b) From the engine bleed systems.
 - (c) From the APU load compressor.
 - (d) From a high pressure ground connection.
68. What is NOT a task of the engine in a large passenger aircraft?
- (a) All others are valid tasks.
 - (b) Provide pressurised air for the cabin.
 - (c) Provide electrical power.
 - (d) Provide thrust.
69. What does APU stand for?
- (a) Auxiliary Power Unit.
 - (b) Auxiliary Pressurising Unit.
 - (c) Automatic Power Unit.
 - (d) Automatic Pressurising Unit.
70. Where are the fuel system tanks usually placed in large aircraft?
- (a) In the wings.
 - (b) Within the engines casing.
 - (c) In the passengers cabin.
 - (d) In the rudder.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	a	d	d	b
2	a	b	c	d	d
3	a	d	d	c	b
4	a	d	b	c	b
5	a	b	b	a	c
6	a	a	d	d	d
7	a	a	d	b	c
8	a	b	d	a	d
9	a	a	b	d	b
10	a	c	d	b	a

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	c	c	d	d
12	a	a	d	b	c
13	a	b	b	c	c
14	a	c	a	c	a
15	a	b	d	c	b

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	d	a	c	c
17	a	c	d	d	a
18	a	a	a	a	b
19	a	c	b	c	d
20	a	d	c	a	a
21	a	d	a	c	c
22	a	b	d	c	a
23	a	c	b	c	b
24	a	a	a	a	b
25	a	d	a	b	a

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	d	b	c
27	a	b	a	c	c
28	a	c	b	c	d
29	a	d	c	c	d
30	a	b	d	c	d

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	b	d	a
32	a	c	d	a	b
33	a	c	b	a	c
34	a	c	d	d	b
35	a	c	b	a	b
36	a	b	b	b	d
37	a	b	d	d	a
38	a	a	a	c	b
39	a	a	a	d	a
40	a	d	d	c	b

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
41	a	c	c	a	d
42	a	b	b	a	d
43	a	a	a	b	c
44	a	b	b	c	d
45	a	a	a	b	a
46	a	a	a	c	b
47	a	c	a	c	c
48	a	c	d	a	c
49	a	c	b	d	b
50	a	b	a	c	c

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
51	a	b	c	d	a
52	a	b	d	b	b
53	a	c	a	a	a
54	a	a	a	c	a
55	a	a	d	a	d
56	a	d	b	c	c
57	a	b	b	d	d
58	a	b	d	a	d
59	a	c	b	d	a
60	a	d	b	a	b

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	d	d	b	d
62	a	c	d	b	d
63	a	d	b	d	d
64	a	a	d	b	b
65	a	c	c	b	b
66	a	c	b	b	a
67	a	d	d	d	b
68	a	b	a	b	d
69	a	b	b	a	c
70	a	b	b	b	d

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. For the same manufacturing parameters (engine displacement), which of the following statements is true about two- and four-stroke reciprocating engines?
 - (a) The two-stroke engine delivers more power than the four-stroke.
 - (b) The two-stroke engine consumes less fuel than the four-stroke.
 - (c) The two-stroke engine pollutes less than the four-stroke.
 - (d) Both engines have comparable characteristics.
2. Which of the following statements are true?
 - I Turboprops have high by-pass ratios.
 - II Ramjets have no compressor.
 - III High by-pass ratio turbofans are usually meant for higher speeds than turbojets.
 - (a) I and II.
 - (b) III.
 - (c) II and III.
 - (d) I and III.
3. A helicopter is...
 - (a) all answers are correct.
 - (b) an aircraft.
 - (c) an aerodyne.
 - (d) a rotorcraft.
4. A helicopter main rotor is responsible for...
 - (a) lateral and longitudinal manoeuvres.
 - (b) longitudinal manoeuvres.
 - (c) directional and lateral manoeuvres.
 - (d) directional manoeuvres.
5. Which fraction of engine power is destined to a helicopter tail rotor?
 - (a) around 10%.
 - (b) around 50%.
 - (c) around 1%.
 - (d) the tail rotor is usually driven by a separate engine.
6. The speed limit of a helicopter is a result of...
 - (a) All answers are correct.
 - (b) the advent of transonic flow and shock waves of the advancing blades of the main rotor.
 - (c) the advent of a reverse flow situation of the retreating blades of the main rotor.
 - (d) blade tip noise and the aerolastic response of the blade.
7. Typical main rotor speeds of average helicopters are:
 - (a) around 300 rpm.
 - (b) around 100 rpm.
 - (c) around 300 m/s.
 - (d) around 100 m/s.
8. The specific impulse of a rocket is related to...
 - (a) None of the others is correct.
 - (b) the time required for engine ignition.
 - (c) the total fuel burning time.
 - (d) the inertial mass of the rocket payload.
9. A sun-synchronous orbit is:
 - (a) all answers are correct.
 - (b) a geocentric orbit that always sweeps a given Earth latitude at the same local mean solar time.
 - (c) a geocentric orbit whose surface illumination angle of a given earth region is kept constant at every passage.
 - (d) useful for earth imaging and remote sensing that require consistent lighting conditions.
10. A launcher design has the following parameters: Total mass 15 Tn, Payload mass 1 Tn, structural mass 2 Tn and the exhaust gas velocity is 3048 m/s. The nozzle is adapted and the launching site is on the Equator. What is the minimum number of stages required to reach an eastward LEO orbit? (Tips: $V_e = I_{sp} g_0$, $\Delta V = V_e \ln(m_0/m_1)$ and for a LEO orbit $\Delta V > 7.6$ km/s)
 - (a) 3 stages.
 - (b) 1 stage.
 - (c) 2 stages.
 - (d) 5 stages.

BLO-6: Concepts on avionics and navigation

11. The ISA sea level reference values for pressure p_0 and temperature T_0 are...
 - (a) $p_0 = 1013.25\text{mbar}$, $T_0 = 288.15\text{ K}$.
 - (b) $p_0 = 1013.25\text{mbar}$, $T_0 = 288.15\text{ }^\circ\text{C}$.
 - (c) $p_0 = 1013.25\text{Pa}$, $T_0 = 288.15\text{ K}$.
 - (d) $p_0 = 1013.25\text{Pa}$, $T_0 = 288.15\text{ }^\circ\text{C}$.
12. The ISA vertical gradient of temperature T_h below 33000 ft is taken as:
 - (a) $-6.5\text{ }^\circ\text{C/km}$.
 - (b) $-6.5\text{ }^\circ\text{F/ft}$.
 - (c) $-6.5\text{ }^\circ\text{C/m}$.
 - (d) $-6.5\text{ }^\circ\text{C/ft}$.
13. How must the barometric altimeter be calibrated for landing?
 - (a) QNH below the Transition Level (TL).
 - (b) QNH below the Transition Altitude (TA).
 - (c) STD below the Transition Level (TL).
 - (d) STD below the Transition Altitude (TA).
14. Which of the following is an existing navigation option?
 - (a) All answers are correct.
 - (b) Esteemed navigation: IAS, Clock and autonomous navigation systems like IRUs.
 - (c) Navigation by external references: Systems defined in the ICAO Annex 10 like NDB, VOR, TACAN, DME, ILS, MLS, GNSS.
 - (d) Radar navigation.
15. For the ILS, the Decision Height (DH) is placed at:
 - (a) the Middle marker beacon.
 - (b) the Inner marker beacon.
 - (c) the Outer marker beacon.
 - (d) the Transition Altitude (TA).
16. Which of the following subsystems provides zenithal guidance:
 - (a) the Glide Slope (ILS/GS).
 - (b) the localizer (ILS/LLZ).
 - (c) the non-directional beacon (ADF/NDB).
 - (d) the tactical air navigation system (TACAN).
17. Since the Bernoulli equation is valid only at low speeds, how must Calibrated Air Speed (CAS) be corrected in cruise conditions?
 - (a) By true Mach and true density to estimate True Air Speed (TAS).
 - (b) By wind speed to estimate Ground Speed (GS).
 - (c) By true density only to estimate Equivalent Air Speed (EAS).
 - (d) By true Mach only to estimate True Air Speed (TAS).
18. The accuracy of a VOR is about:
 - (a) $\pm 1.4^\circ$ (about 90 m crosstrack error at 2 nm).
 - (b) $\pm 185\text{ m}$ for a limit of 100 aircraft.
 - (c) in CAT-I, a threshold of a 10000 ft-long runway, Azimuth= $\pm 9.1\text{ m}$ and Elevation= $\pm 4.1\text{ m}$.
 - (d) Horizontal= $\pm 13\text{ m}$ and Vertical= $\pm 22\text{ m}$.
19. Misleading information is...
 - (a) an integrity risk because it does not provide a failure detection alarm upon position failure.
 - (b) a continuity risk because it does not provide a failure detection alarm upon position failure.
 - (c) an integrity risk because it provides a failure detection alarm despite no position failure.
 - (d) a continuity risk because it provides a failure detection alarm despite no position failure.
20. Continuity is...
 - (a) the ability of the total system to perform its function without unscheduled interruptions during the intended operation.
 - (b) a measure of the trust that can be placed in the correctness of the information supplied by the total system, including the ability to provide timely warnings.
 - (c) the ability of the total system to maintain the aircraft position within a given Total System Error (TSE) with a 95% probability.
 - (d) the ability of the total system to provide the required parameters at the start of the intended operation.

BLO-7: Concepts on materials and aeronautical regulations

21. Which of the following types of material provides good mechanical properties for structural elements?
- (a) Metals.
 - (b) Ceramics.
 - (c) Polymers.
 - (d) Semi-conductors.
22. Which of the following characteristics of a material are usually required for structural aeronautic applications?
- (a) High yield stress.
 - (b) High brittleness.
 - (c) Low thermal conductivity.
 - (d) Low ductility.
23. What is the primary alloying element of the 7000 aluminium alloy series?
- (a) Zinc.
 - (b) Copper.
 - (c) Magnesium.
 - (d) Manganese.
24. The main advantage of the 2000 aluminium alloy series when compared with the 7000 series is...
- (a) better behaviour against crack propagation.
 - (b) higher toughness.
 - (c) higher Young modulus.
 - (d) All answers are correct.
25. The best choice of material for the fuselage frame and skin is:
- (a) Aluminum alloys.
 - (b) Hardened steel.
 - (c) Copper alloys.
 - (d) Kevlar.
26. Which of the following ICAO annexes contains the standards and recommended practices regarding Aerodromes?
- (a) 14
 - (b) 8
 - (c) 11
 - (d) 15
27. Which of the following ICAO annexes contains the standards and recommended practices regarding Aeronautical Telecommunications?
- (a) 10
 - (b) 7
 - (c) 15
 - (d) 5
28. The organization founded in 1960 whose original aim was to unify Control and Navigation air services in Europe is...
- (a) EUROCONTROL
 - (b) EASA
 - (c) ICAN
 - (d) ICAO
29. Which of the following agencies is the civil aviation authority in the United States?
- (a) FAA
 - (b) JAR
 - (c) JAA
 - (d) FAR
30. Which of the Spanish public bodies owns and operates the majority of airports in Spain?
- (a) AENA
 - (b) DGAC
 - (c) CASA
 - (d) EASA

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control system options does NOT require actuators?
- (a) A purely mechanical flight control system
 - (b) An electro-hydraulic flight control system
 - (c) An hydro-mechanical flight control system
 - (d) An electro-hydro-mechanical flight control system
32. What is the main disadvantage of a fly-by-wire electro-hydraulic flight control system with respect to an electro-hydro-mechanical fcs?
- (a) It is critically dependent on electronics.
 - (b) It does not allow for automatic control and flight augmentation capabilities.
 - (c) It is heavier.
 - (d) It only has advantages.
33. Which elements of the hydraulic system are responsible for undertaking intensive transient demands of hydraulic power?
- (a) Accumulators.
 - (b) Filters.
 - (c) Pumps.
 - (d) Valves.
34. What is true about the several hydraulic circuits of a large passenger aircraft?
- (a) Power may be transferred among circuits.
 - (b) Fluid may be transferred among circuits.
 - (c) The same pump is shared by all circuits.
 - (d) All control surfaces may be actuated by all circuits.
35. How is DC electrical power usually generated in a large aircraft during normal operation?
- (a) Through AC/DC transformation.
 - (b) Directly from batteries.
 - (c) From a DC generator attached to one of the engines.
 - (d) The electrical system does not provide DC power.
36. To which of the following elements of the electrical system must superfluous electrical equipment be connected?
- (a) BUS.
 - (b) ESS BUS.
 - (c) BATT.
 - (d) GEN.
37. Which of the following is a valid option for powering the pneumatic system?
- (a) From the engine bleed systems.
 - (b) From a compressor driven by the hydraulic system.
 - (c) From a compressor driven by the RAT.
 - (d) All options are possible.
38. What is NOT a task of the FADEC?
- (a) Provide an inertial reference for aircraft control.
 - (b) Control of gas generator.
 - (c) Protection against exceeding limits.
 - (d) Engine starting sequence.
39. What is an aim of the Auxiliary Power Unit (APU) while on ground?
- (a) Supply bleed air for engine start.
 - (b) It is forbidden to switch it on while on ground.
 - (c) Load luggage into the cargo area.
 - (d) Serve refreshments to passengers.
40. Which of the following components does NOT belong to the air conditioning system of a large aircraft?
- (a) Transformer/rectifier.
 - (b) Flow control valves.
 - (c) Mixing unit.
 - (d) Packs.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	c	b	d	c
2	a	c	b	c	c
3	a	b	a	c	a
4	a	a	d	d	c
5	a	d	d	b	b
6	a	c	c	b	d
7	a	d	d	a	b
8	a	b	c	a	b
9	a	a	d	c	a
10	a	c	c	a	a

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	b	c	a	b
12	a	b	a	a	d
13	a	c	a	c	c
14	a	b	c	c	a
15	a	b	a	a	a
16	a	a	b	b	c
17	a	d	c	d	c
18	a	a	b	a	d
19	a	d	b	a	c
20	a	a	d	c	a

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	a	c	b	d
22	a	b	c	a	b
23	a	a	d	a	b
24	a	a	b	b	b
25	a	a	a	c	a
26	a	b	b	c	d
27	a	c	a	a	b
28	a	b	c	c	c
29	a	b	c	d	c
30	a	b	c	a	d

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	c	a	d
32	a	c	c	a	c
33	a	d	a	a	d
34	a	a	c	a	a
35	a	c	d	d	d
36	a	b	d	d	a
37	a	c	c	a	a
38	a	a	a	a	b
39	a	b	b	d	c
40	a	c	a	c	c

BLO-1: Concepts on flight principles and aircraft performance

1. An archetypal example of aerostat is...
 - (a) a hot air balloon.
 - (b) a re-entry vehicle.
 - (c) an helicopter.
 - (d) a rotary-wing aerostat.
2. What is the main effect on drag of fitting a wing with a wingtip?
 - (a) It increases friction and reduces lift-induced drag.
 - (b) It reduces friction and increases lift-induced drag.
 - (c) It increases both friction and lift-induced drag.
 - (d) It reduces both friction and lift-induced drag.
3. Which of the following parameters is roughly independent of altitude if we neglect the effects of Reynolds and Mach numbers?
 - (a) Maximum lift coefficient.
 - (b) Stall speed.
 - (c) Maximum bank angle in a coordinated turn.
 - (d) Maximum lift force.
4. Air viscosity is responsible for...
 - (a) All answers are correct.
 - (b) Friction drag.
 - (c) Lift generation.
 - (d) Wake drag.
5. How does maximum bank angle compare between an airplane flying fast at a speed V_f (ϕ_f) and the same airplane, at the same altitude, flying slow at V_s (ϕ_s)?
 - (a) $\phi_f > \phi_s$
 - (b) $\phi_f < \phi_s$
 - (c) $\phi_f = \phi_s$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
6. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the altitude decreases?
 - (a) It occurs at lower speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (b) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at lower speeds (moves left) and decreases (moves down).
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. How does tail wind affect maximum range R_{max} and the corresponding aerodynamic speed $V_{R_{max}}$ of a given airplane?
 - (a) $V_{R_{max}}$ decreases and R_{max} improves.
 - (b) It doesn't affect whatsoever.
 - (c) $V_{R_{max}}$ increases and R_{max} improves.
 - (d) $V_{R_{max}}$ decreases and R_{max} worsens.
8. What is the operational ceiling of an airplane?
 - (a) The maximum altitude for which the airplane still keeps a given capability of ascending.
 - (b) The maximum altitude at which the airplane can fly.
 - (c) The interior part of the fuselage that is above the passengers heads.
 - (d) The maximum rate of climb at a given altitude.
9. Which of the following conditions is normally adequate for maximising cruise endurance of a propeller airplane?
 - (a) All options are correct.
 - (b) Minimum hourly fuel consumption (C, in lb/h).
 - (c) Minimum required power.
 - (d) Minimum flight altitude.
10. What must a four-engined airplane pilot do during takeoff if one of the engines fails before reaching the decision speed?
 - (a) Abort takeoff.
 - (b) Go ahead with takeoff and proceed to destination.
 - (c) Go ahead with takeoff and ask for immediate permission to land.
 - (d) It depends on whether stall speed has been reached or not.

BLO-2: Flight principles related problems

11. What is the drag coefficient ($C_{D_{opt}}$) for the optimal condition that maximises aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.02 + 0.12C_L^2$?
- (a) $C_{D_{opt}} = 0.04$
 - (b) $C_{D_{opt}} = 0.02$
 - (c) $C_{D_{opt}} = 0.408$
 - (d) $C_{D_{opt}} = 0.05$
12. An aerobatics airplane with lift coefficient $C_L = 0.1 + 0.11\alpha$ with α in $^\circ$ engages in a perfect dive (vertical descent). What should the angle of attack (α) be?
- (a) $\alpha = -0.909^\circ$
 - (b) $\alpha = 0.909^\circ$
 - (c) $\alpha = 1.1^\circ$
 - (d) $\alpha = -1.1^\circ$
13. A small airplane of unknown mass is engaged in a coordinated banked turn with bank angle ($\phi = 45^\circ$) at a speed $V = 90$ m/s. What is the turn radius (R_t)? (Take $g = 9.8$ m/s²)
- (a) $R_t = 827$ m
 - (b) $R_t = 783$ m
 - (c) $R_t = 695$ m
 - (d) It depends on the mass of the airplane.
14. The lift coefficient and drag polar of an airplane changes from $C_L = 0.2 + 0.12\alpha$ and $C_D = 0.03 + 0.16C_L^2$ in clean configuration to $C_L = 0.3 + 0.12\alpha$ and $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the angle of attack (α) just before optimal flap deployment?
- (a) $\alpha = 1.28^\circ$
 - (b) $\alpha = 0.446^\circ$
 - (c) $\alpha = 1.55^\circ$
 - (d) $\alpha = 0.171^\circ$
15. An airplane of mass $m = 800$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 14 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
- (a) $D = 560$ N and $D_p = D_i = 280$ N.
 - (b) $D = 560$ N and $D_p > D_i$.
 - (c) $D = 280$ N and $D_p = D_i = 140$ N.
 - (d) $D = 280$ N and $D_p > D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which of the following is considered a secondary control surface?
- (a) Slat.
 - (b) Rudder.
 - (c) Aileron.
 - (d) Elevator.
17. Which parameter is usually modified to tune the spiral mode?
- (a) Wing dihedral angle (C_{l_β}).
 - (b) Fin size (C_{n_r}).
 - (c) Horizontal stabilizer size (C_{m_q}).
 - (d) None of the others.
18. Dynamic stability around an axis requires, at the very least,...
- (a) All options are correct.
 - (b) Static stability around this axis.
 - (c) The appearance of a restoring moment around this axis upon perturbation.
 - (d) The appearance of a damping moment around this axis during restoration after perturbation.
19. What is favoured in the design stage of an aerobatics airplane?
- (a) High manoeuvrability.
 - (b) High stability.
 - (c) High wing aspect ratio.
 - (d) High cargo capability.
20. For a stable airplane around its pitch axis, in which order are the three axial modes usually damped?
- (a) Pitch oscillation, phugoid, return to straight and level flight.
 - (b) Phugoid, return to straight and level flight, pitch oscillation.
 - (c) Return to straight and level flight, pitch oscillation, phugoid.
 - (d) Phugoid, pitch oscillation, return to straight and level flight.
21. Upon which commands must a pilot act to perform a straight flight with sideslip, if the airplane is to preserve both altitude and speed?
- (a) Stick, pedals and thrust lever.
 - (b) Pedals and thrust lever.
 - (c) Pedals.
 - (d) Stick and thrust lever.
22. What can be said about the initial response of the airplane to a perturbation in roll?
- (a) No restoring moment appears. Recovery is purely dynamical.
 - (b) A restoring moment appears, which depends on the size of the fin.
 - (c) A restoring moment appears, which depends on the size of the wings.
 - (d) A restoring moment appears, which depends on the size of the horizontal stabiliser.
23. Why can the spiral mode be left unstable without critically compromising the aircraft integrity?
- (a) Because it is a very slow mode that can be easily controlled by the pilot.
 - (b) Because it couples rotation about both the yaw and roll axes.
 - (c) The spiral mode can never be unstable so the question makes no sense.
 - (d) The spiral mode can never be left unstable because it critically compromises the integrity of the aircraft.
24. How must the pilot of a four-engined airplane correct for excessive sideslip in case of failure of one of the engines on the right wing?
- (a) Press the left pedal.
 - (b) Press the right pedal.
 - (c) Switch off one of the engines of the left wing.
 - (d) Press both pedals simultaneously.
25. Which components does the empennage of an airplane generally consist of?
- (a) Fin and tailplane.
 - (b) Wing, slats and flaps.
 - (c) cockpit and fuselage.
 - (d) Ailerons, elevator and rudder.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.24\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $V_E = 97.4$ m/s
 - $V_E = 128$ m/s
 - $V_E = 74$ m/s
 - It depends on the available thrust T_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1200$ kg has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.03W^2/\bar{\rho}V$. What is the maximum rate of climb $((R/C)_{max})$ at sea level ($\bar{\rho} = 1$) if the available power is $P_A = 250 \bar{\rho} \delta_x$ kW? (take $g \simeq 9.8$ m/s²)
- $(R/C)_{max} = 11.1$ m/s
 - $(R/C)_{max} = 9.69$ m/s
 - $(R/C)_{max} = 7.9$ m/s
 - $(R/C)_{max} = 6.84$ m/s
28. An airplane with drag polar $C_D = 0.02 + 0.18C_L^2$, wing surface $S = 35$ m² and mass $m = 2000$ kg is flying at a speed $V = 100$ m/s. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4$ kg/m³? (take $g \simeq 9.8$ m/s²)
- $T_R = 2390$ N
 - $T_R = 2460$ N
 - $T_R = 2200$ N
 - $T_R = 2110$ N
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 45$. If its current height (above terrain) is $h = 800$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 36$ km
 - $d = 17.8$ km
 - $d = 0.8$ km
 - $d = 72$ km
30. The required thrust of a jet-engined airplane of mass $m = 1400$ kg, wing area $S = 20$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 12000$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 49.2$ m/s
 - $V \simeq 39.5$ m/s
 - $V \simeq 136$ m/s
 - $V \simeq 73.3$ m/s

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	b	d	b
2	a	d	c	d	b
3	a	d	d	d	d
4	a	b	a	b	a
5	a	c	d	d	c
6	a	b	c	b	c
7	a	a	b	b	b
8	a	c	b	b	b
9	a	b	d	a	c
10	a	b	b	a	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	b	c	b	d
12	a	d	b	c	d
13	a	c	b	d	c
14	a	d	a	c	b
15	a	d	d	c	a

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	b	d	c
17	a	a	a	c	a
18	a	a	b	b	d
19	a	d	b	b	b
20	a	c	b	d	b
21	a	a	b	c	b
22	a	b	b	c	b
23	a	a	a	b	d
24	a	c	b	c	d
25	a	d	b	d	d

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	d	b	a
27	a	a	c	c	c
28	a	b	b	d	c
29	a	c	d	b	c
30	a	c	a	b	d

BLO-1: Concepts on flight principles and aircraft performance

1. An example of an aerostat is...
 - (a) a hot air balloon.
 - (b) a Cessna.
 - (c) the International Space Station (ISS).
 - (d) an Exocet.
2. What is the main effect on an airplane drag polar ($C_D(C_L) = C_{D_0} + k C_L^2$) of deploying the flaps?
 - (a) C_{D_0} increases while k decreases.
 - (b) C_{D_0} decreases while k increases.
 - (c) Both C_{D_0} and k increase.
 - (d) Both C_{D_0} and k decrease.
3. Which of the following parameters is roughly independent of altitude if we neglect its effects through Reynolds and Mach numbers?
 - (a) Maximum aerodynamic efficiency.
 - (b) Full power cruising speed.
 - (c) Minimum angle of attack achievable in horizontal balanced flight.
 - (d) Best climb angle.
4. Which of the following drag sources is an immediate consequence of compressibility?
 - (a) Wave drag.
 - (b) Form or wake drag.
 - (c) Friction drag.
 - (d) Interference drag.
5. Which of the following equilibrium flights entails a higher load factor?
 - (a) Banked turn.
 - (b) Straight & level flight.
 - (c) Balanced ascent.
 - (d) Balanced descent.
6. What happens to the minimum ($P_{R_{min}}$) of the required power curve ($P_R(V)$) as the altitude increases?
 - (a) It occurs at higher speeds (moves right) and increases (moves up).
 - (b) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (c) It occurs at lower speeds (moves left) and decreases (moves down).
 - (d) It occurs at about the same speed and remains fairly constant.
7. Which of the following statements is generally true regarding the operation of airport runways?
 - (a) They can be used in either direction depending on wind conditions.
 - (b) They are usually designed to operate in only one direction.
 - (c) If the airport has two parallel runways they are operated in opposite directions, one for landing and the other for take-off operations.
 - (d) They can be rotated so that they instantaneously face the wind during operation.
8. The maximum rate of climb $(R/C)_{max}$ of an airplane...
 - (a) decreases with altitude.
 - (b) increases with altitude.
 - (c) decreases with altitude for a propeller but increases for a jet.
 - (d) is independent of the altitude.
9. Which of the following conditions maximises cruise range?
 - (a) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (b) Minimum hourly fuel consumption (C , in lb/h).
 - (c) Minimum drag.
 - (d) Minimum power.
10. How does tailwind affect maximum endurance E_{max} and the corresponding aerodynamic speed $V_{E_{max}}$ of a given airplane?
 - (a) It doesn't affect whatsoever.
 - (b) $V_{E_{max}}$ increases and E_{max} improves.
 - (c) $V_{E_{max}}$ decreases and E_{max} improves.
 - (d) $V_{E_{max}}$ stays the same but E_{max} worsens.

BLO-2: Flight principles related problems

11. What is the drag coefficient ($C_{D_{opt}}$) for the optimal condition that maximises aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.04 + 0.12C_L^2$?
 - (a) $C_{D_{opt}} = 0.08$
 - (b) $C_{D_{opt}} = 0.04$
 - (c) $C_{D_{opt}} = 0.577$
 - (d) $C_{D_{opt}} = 0.1$
12. An aerobatics airplane with lift coefficient $C_L = 0.05 + 0.1\alpha$ with α in $^\circ$ is engaged in an extreme coordinated turn with bank angle $\phi = 80^\circ$. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.1$?
 - (a) $\alpha = 5.26^\circ$
 - (b) $\alpha = 4.73^\circ$
 - (c) $\alpha = 6.31^\circ$
 - (d) $\alpha = 6.84^\circ$
13. What is the load factor of a balanced descent with a trajectory angle ($\gamma = -5^\circ$) with the engines off.
 - (a) $n = 0.996$
 - (b) $n = 0.976$
 - (c) $n = 0.956$
 - (d) It depends on the mass of the airplane.
14. The drag polar of an airplane changes from $C_D = 0.03 + 0.14C_L^2$ in clean configuration to $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.408$
 - (b) $C_L = 0.386$
 - (c) $C_L = 0.454$
 - (d) $C_L = 0.347$
15. An airplane of mass $m = 1500$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 10 < f_{max}$. If its speed is faster than the speed for maximum range, what can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $D = 1500$ N and $D_p > 750$ N $> D_i$.
 - (b) $D = 1500$ N and $D_i = D_p = 750$ N.
 - (c) $D = 1500$ N and $D_i > 750 > D_p$ N.
 - (d) $D = D_p = D_i = 1500$ N.

BLO-3: Concepts on aircraft stability and control

16. How is an airplane piloted around the yaw axis (z)?
- (a) With the pedals.
 - (b) With the stick or yoke.
 - (c) Airplanes need not be piloted around this axis.
 - (d) With a trim wheel.
17. Which parameter is mainly responsible for directional stability?
- (a) Fin size (C_{n_β}).
 - (b) Wing span and aspect ratio (C_{l_ϕ}).
 - (c) Horizontal stabilizer size (C_{m_α}).
 - (d) None of the others.
18. About static and dynamic stability...
- (a) the latter implies the former.
 - (b) the former implies the latter.
 - (c) they are synonyms.
 - (d) one is related to stability and the other to manoeuvrability.
19. An aerobatics airplane...
- (a) favors manoeuvrability over stability.
 - (b) favors stability over manoeuvrability.
 - (c) sacrifices both stability and manoeuvrability in favor of lower fuel consumption.
 - (d) is more stable than an average trainer airplane.
20. While in horizontal level flight, which of the following modes would you expect to see triggered by a longitudinal wind gust?
- (a) Pitch oscillation.
 - (b) Roll subsidence.
 - (c) Dutch roll.
 - (d) None of the others. The equilibrium would by no means be affected.
21. Which commands must a pilot use to perform a crosswind balanced flight, if the airplane is to preserve both altitude and speed?
- (a) Stick, pedals and thrust lever.
 - (b) Stick and thrust lever.
 - (c) Stick.
 - (d) Pedals and thrust lever.
22. What can be said about static lateral stability?
- (a) It doesn't really exist. Recovery from a perturbation in roll is mostly dynamical.
 - (b) A restoring moment appears that corrects any perturbation in roll.
 - (c) It is related to engine thrust.
 - (d) It is the ability of the airplane to recover from a perturbation in pitch.
23. What is true about pitch oscillation?
- (a) Trying to pilot it can put the airplane in jeopardy.
 - (b) It is a slow aperiodic mode.
 - (c) It corresponds to the periodic exchange between potential and kinetic energy.
 - (d) It couples pitch and yaw motion.
24. Given an airplane, which kind of turn requires a higher bank angle to achieve the same turning rate at the same speed and altitude?
- (a) Operational turn.
 - (b) Coordinated turn.
 - (c) Both Operational and coordinated turn require the same bank angle.
 - (d) It depends on engine type.
25. Which components does the tailplane of an airplane generally consist of?
- (a) Horizontal stabiliser and elevator.
 - (b) Vertical stabiliser and rudder.
 - (c) Vertical stabiliser and elevator.
 - (d) Horizontal stabiliser and rudder.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.23\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum range (V_R)? (take $g = 9.81 \text{ ms}^{-2}$)
- (a) $V_R = 130 \text{ ms}^{-1}$
 - (b) $V_R = 98.5 \text{ ms}^{-1}$
 - (c) $V_R = 74.8 \text{ ms}^{-1}$
 - (d) It depends on the available thrust T_A , which is unknown.
27. A jet-powered airplane of mass $m = 1400$ kg has a required thrust curve $T_R = 0.32\bar{\rho}V^2 + 0.03W^2/\bar{\rho}V^2$. What is the best climb angle (γ_{max}) at sea level ($\bar{\rho} = 1$) if the available thrust is $T_A = 4000 \bar{\rho} \delta_x \text{ N}$? (take $g = 9.81 \text{ ms}^{-2}$)
- (a) $\gamma_{max} = 5.47^\circ$
 - (b) $\gamma_{max} = 3.73^\circ$
 - (c) $\gamma_{max} = 6.07^\circ$
 - (d) $\gamma_{max} = 7.45^\circ$
28. What is the absolute ceiling of a jet-powered airplane of mass $m = 4000$ kg, required thrust $T_R = 0.306\bar{\rho}V^2 + 0.0057W^2/\bar{\rho}V^2$ and maximum available thrust $T_A = (T_A)_{SL} \bar{\rho} \delta_x$, with $(T_A)_{SL} = 8000 \text{ N}$? (take $g = 9.81 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
- (a) $h \simeq 8380 \text{ m}$ ($\rho \simeq 0.502 \text{ kg/m}^3$)
 - (b) $h \simeq 9240 \text{ m}$ ($\rho \simeq 0.453 \text{ kg/m}^3$)
 - (c) $h \simeq 7090 \text{ m}$ ($\rho \simeq 0.583 \text{ kg/m}^3$)
 - (d) $h \simeq 10800 \text{ m}$ ($\rho \simeq 0.375 \text{ kg/m}^3$)
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 50$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 30$ km of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- (a) $h = 600 \text{ m}$
 - (b) $h = 1200 \text{ m}$
 - (c) $h = 1500 \text{ m}$
 - (d) $h = 1670 \text{ m}$
30. The required thrust of a jet-engined airplane of mass $m = 2000$ kg is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 8000 \text{ N}$. What is the maximum speed this airplane can achieve in straight and level flight at an altitude such that $\bar{\rho} = \rho/\rho_0 = 0.4$? (take $g = 9.81 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
- (a) $V \simeq 93.5 \text{ ms}^{-1}$
 - (b) $V \simeq 99.6 \text{ ms}^{-1}$
 - (c) $V \simeq 78.6 \text{ ms}^{-1}$
 - (d) $V \simeq 120 \text{ ms}^{-1}$

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

31. Sort the following engines from lowest to highest bypass ratio.
- (a) Ramjet, low bypass turbofan, high bypass turbofan, turboprop.
 - (b) Turbojet, low bypass turbofan, turboprop, ramjet.
 - (c) Low bypass turbofan, ramjet, turboprop, high bypass turbofan.
 - (d) Turbojet, low bypass turbofan, ramjet, turboprop.
32. At equal engine size, which of the following statements is more accurate?
- (a) Diesel relies on a rich oxygen mixture while Otto is stoichiometric.
 - (b) Otto is always four stroke while Diesel is always two stroke.
 - (c) Otto has a diffusive flame while Diesel often has a premixed one.
 - (d) Diesel emits more CO₂ while Otto sends out more carbon particles.
33. What is the role of a diffuser in a jet-engine?
- (a) To reduce the velocity of the air that is fed to the compressor.
 - (b) To reduce the pressure of the air that is fed to the compressor.
 - (c) To reduce the density of the air that is fed to the compressor.
 - (d) All answers are correct.
34. Which one of the following statements is true?
- (a) Both helicopters and airplanes can be powered by gas turbine engines.
 - (b) Only helicopters can be powered by gas turbine engines.
 - (c) Only fixed wing aircraft can be powered by gas turbine engines.
 - (d) Neither helicopters nor airplanes can be powered by gas turbine engines.
35. Which of the following components CANNOT be found in a ramjet?
- (a) None can be found.
 - (b) A compressor.
 - (c) A turbine.
 - (d) A fan.
36. What is the difference between an helicopter and an autogyro?
- (a) Unlike helicopters, autogyros develop lift by a free spinning rotor.
 - (b) Unlike autogyros, helicopters develop lift by a free spinning rotor.
 - (c) Unlike autogyros, helicopters develop thrust by a conventional propeller.
 - (d) Unlike helicopters, autogyros do not have any rotor.
37. Which of the following missions will be better performed by an helicopter than an airplane?
- (a) All of them.
 - (b) Emergency medical service.
 - (c) Search and Rescue (SAR).
 - (d) Aerial surveillance.
38. How is the lift force distributed along a main rotor blade?
- (a) It increases rapidly from the central hub.
 - (b) It remains constant.
 - (c) It decreases rapidly from the central hub.
 - (d) Rotor blades do not generate lift.
39. Remote sensing satellites usually follow:
- (a) Low Earth Orbits (LEO).
 - (b) Medium Earth Orbits (MEO).
 - (c) Geostationary Orbits (GEO).
 - (d) Highly Elliptical Orbits (HEO).
40. Geostationary Orbits (GEO) are mainly used for:
- (a) Communications relay satellites for temperate latitude locations.
 - (b) Worldwide communications relay satellites.
 - (c) Communications relay satellites for extreme latitude locations.
 - (d) Global navigation services.

BLO-6: Concepts on avionics and navigation

41. If installation errors and compressibility effects are neglected, True Airspeed (TAS) can be inferred from Indicated Airspeed (IAS) only if...
- (a) air density is known.
 - (b) Mach number is known.
 - (c) viscosity is known.
 - (d) there's no way to infer TAS from IAS.
42. The Transition Level (TL) is situated:
- (a) Above the Transition Altitude (TA).
 - (b) Below the Transition Altitude (TA).
 - (c) Sometimes above the Transition Altitude (TA) and others below it, depending on the pressure gradient.
 - (d) Sometimes above the Transition Altitude (TA) and others below it, depending on the temperature gradient.
43. The Distance Measurement Equipment is:
- (a) 1 NM accurate.
 - (b) 1° accurate.
 - (c) 1 rad accurate.
 - (d) 2° accurate.
44. The statement "Civil aviation requires a pilot to be warned in a timely manner of a navigation system anomaly which may create a hazard" is related to:
- (a) integrity performance.
 - (b) accuracy performance.
 - (c) availability performance.
 - (d) None of the others.
45. Four satellites are needed to estimate the user's four dimensional position. Which are these dimensions?
- (a) Position (X, Y, Z) plus time.
 - (b) Position (X, Y, Z) plus speed.
 - (c) Position (X, Y, Z) plus acceleration.
 - (d) None of them.
46. Why is VHF Omnidirectional Range (VOR) used for en-route (ENR) navigation?
- (a) Because it provides accurate lateral guidance.
 - (b) Because it provides accurate vertical guidance.
 - (c) They are not used in ENR navigation.
 - (d) Because they are cheaper than ILS.
47. Which of these statements about Ground-Based Augmentation System (GBAS) is correct?
- (a) All statements are correct.
 - (b) GBAS is based on differential GPS (dGPS)
 - (c) GBAS provides higher levels of accuracy and integrity than bare GPS.
 - (d) GBAS provides lower coverage than Satellite-based Augmentation System (SBAS)
48. About geodesy concepts...
- (a) Latitude (ϕ) specifies the north-south position of a point with respect to the Equator.
 - (b) Longitude (λ) specifies the east-west position of a point with respect to the Equator.
 - (c) Latitude (ϕ) specifies the north-south position of a point with respect to the Meridian.
 - (d) Longitude (λ) specifies the east-west position of a point with respect to the Meridian.
49. About GPS Selective Availability (S/A)...
- I The GPS signals available for unrestricted use are purposely degraded under a policy called S/A.
 - II The aim of S/A is to deny adversary opportunity to generate and broadcast GPS signals with the aim of spoofing the receiver.
 - III S/A is done via encryption of the ranging code.
 - IV S/A degradation can be achieved by means of broadcasting erroneous values of satellite ephemeris parameters.
- (a) I, IV.
 - (b) I, III, IV.
 - (c) II, III.
 - (d) II, IV.
50. Which of the following navigation systems provides higher accuracy levels in the landing phase of flight?
- (a) ILS.
 - (b) VOR.
 - (c) None of them are accurate.
 - (d) DME.

BLO-7: Concepts on materials and aeronautical regulations

51. In which of the following situations is the wing upper skin subject to higher traction?
- (a) Fully loaded fuel tank while on ground.
 - (b) Empty fuel tank while on ground.
 - (c) Fully loaded fuel tank while cruising.
 - (d) Empty fuel tank while cruising.
52. The wingbox is...
- (a) the structural component of the fuselage from which the wings extend.
 - (b) an avionics receptacle that is located within the wing.
 - (c) a box-shaped wing.
 - (d) any wing-like structure that can be easily twisted.
53. Which of the following components is usually made of steel alloys?
- (a) Engine pylons.
 - (b) High pressure turbine nozzle guide vanes.
 - (c) Fuselage skin.
 - (d) Control surfaces.
54. Which of the following properties is NOT characteristic of ceramic materials?
- (a) Ductility.
 - (b) Chemical stability.
 - (c) Brittleness.
 - (d) Thermal and electrical insulation.
55. In which circumstances would polymers be usually chosen?
- (a) For applications that are not mechanically demanding and require light, flexible materials.
 - (b) For applications that are either thermally or mechanically demanding.
 - (c) For applications that need conductive materials.
 - (d) polymers are never used in aeronautical applications.
56. The ICAO annex 14 concerns...
- (a) Aerodromes.
 - (b) Aeronautical Information Services.
 - (c) Personnel Licensing.
 - (d) Rules of the Air.
57. The ICAO annex 15 concerns...
- (a) Aeronautical Information Services.
 - (b) Aerodromes.
 - (c) Personnel Licensing.
 - (d) Rules of the Air.
58. Which of the following is the organization created by European aircraft manufacturers in order to ease adoption of new products and coordinate airworthiness standards?
- (a) JAA.
 - (b) EUROCONTROL.
 - (c) DGAC.
 - (d) FAA.
59. The country in charge of issuing the airworthiness certificate of an aircraft is:
- (a) Registry country.
 - (b) Airline country.
 - (c) Manufacturing country.
 - (d) Captain country.
60. Which organisation was founded following the Chicago Convention of 1944?
- (a) OACI.
 - (b) EASA.
 - (c) JAR.
 - (d) AENA.

BLO-8: Concepts on aircraft systems

61. Which of the following components is part of the Flight Control System (FCS)?
- (a) All of them.
 - (b) Primary control surfaces.
 - (c) Secondary control surfaces.
 - (d) Transmission elements.
62. Which of the following is a primary objective of the Flight Control System (FCS)?
- (a) Render the aircraft pilotable.
 - (b) Pressurise the cabin.
 - (c) Drive an emergency generator.
 - (d) None of them.
63. Which is the main power generator, during an uneventful flight, of the hydraulic system?
- (a) Engine-driven pumps.
 - (b) Electric pumps.
 - (c) Batteries.
 - (d) Ram Air Turbine (RAT).
64. In uneventful cruise conditions, DC Buses are powered by means of:
- (a) Engine-driven generators producing AC power that is transformed by Transformer Rectifiers (TR).
 - (b) The batteries (BAT).
 - (c) The Auxiliary Power Unit (APU).
 - (d) None of them.
65. Which of the following are high pressure air sources for the pneumatic system?
- (a) Engine bleed systems.
 - (b) Batteries.
 - (c) The fly-by-wire control system.
 - (d) None of them.
66. Which of the following may be components of a purely mechanical Flight Control System (FCS)?
- (a) Pulleys & cables.
 - (b) An artificial feel system.
 - (c) Hydraulic actuators.
 - (d) None of them.
67. Why do some flight control systems need an artificial "feel"?
- (a) To give the pilot feedback on the effects of his actions on commands.
 - (b) To make the aircraft stable.
 - (c) To help the pilot reduce efforts on the commands.
 - (d) To make the passengers more comfortable.
68. How is redundancy achieved for the hydraulic system of a large passenger aircraft?
- (a) By having different hydraulic circuits that can exchange pressure but not hydraulic fluid.
 - (b) By having identical hydraulic circuits isolated from each other.
 - (c) By having identical hydraulic circuits that can exchange both hydraulic fluid and pressure.
 - (d) By having different hydraulic circuits that can exchange hydraulic fluid.
69. What does FADEC do?
- (a) Performs complete engine management.
 - (b) Control ailerons and elevators.
 - (c) Provides AC electrical power.
 - (d) It is the computer in charge of flight augmentation.
70. Which of the following is NOT an option to power the hydraulic system in an Airbus A320 aircraft?
- (a) The pneumatic system.
 - (b) Engine-driven pumps.
 - (c) All of them can power the hydraulic system.
 - (d) The Ram Air Turbine (RAT).

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	b	a	a
2	a	a	b	b	b
3	a	d	a	d	b
4	a	a	c	b	b
5	a	a	c	a	c
6	a	b	a	d	b
7	a	d	c	b	d
8	a	a	a	a	b
9	a	b	a	b	b
10	a	b	a	a	a

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	c	a	c	b
12	a	c	d	a	b
13	a	b	a	c	a
14	a	c	b	d	c
15	a	b	a	a	a

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	c	d	b	d
17	a	d	b	b	d
18	a	a	c	c	a
19	a	c	c	d	c
20	a	d	d	d	b
21	a	c	b	d	b
22	a	d	d	a	b
23	a	a	a	a	c
24	a	c	c	d	d
25	a	b	b	d	c

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	d	d	b
27	a	d	a	c	a
28	a	b	c	a	c
29	a	a	c	a	c
30	a	c	c	c	d

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	b	b	a	d
32	a	b	c	a	b
33	a	d	b	d	d
34	a	a	b	a	c
35	a	a	a	d	d
36	a	c	a	b	b
37	a	d	c	d	c
38	a	a	d	d	c
39	a	a	b	a	d
40	a	a	b	c	b

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
41	a	b	c	b	b
42	a	b	d	b	d
43	a	a	d	a	c
44	a	c	d	a	a
45	a	b	d	b	d
46	a	d	a	b	a
47	a	a	b	a	c
48	a	d	b	b	d
49	a	b	a	a	a
50	a	c	c	a	a

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
51	a	a	a	c	b
52	a	d	d	a	a
53	a	c	b	d	b
54	a	a	b	a	a
55	a	d	a	b	c
56	a	a	b	d	b
57	a	d	c	a	b
58	a	c	d	a	b
59	a	d	c	c	d
60	a	d	c	c	c

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	c	c	d	a
62	a	d	b	d	d
63	a	d	c	d	d
64	a	b	a	a	a
65	a	c	b	c	c
66	a	d	c	d	a
67	a	b	c	a	b
68	a	a	c	d	d
69	a	d	a	a	c
70	a	a	d	a	b

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. Which is the engine type having the lowest bypass ratio?
 - (a) Turbojet.
 - (b) Turboprop.
 - (c) High bypass ratio turbofan.
 - (d) Low bypass ratio turbofan.
2. Which of the following statements is TRUE?
 - (a) Otto engines are designed stoichiometric while Diesels aren't.
 - (b) Otto engines send out more carbon particles while Diesel engines emit more CO₂ at equal size.
 - (c) Otto engines are always two strokes while Diesel engines are always four stroke.
 - (d) Otto engines have a diffusive flame while Diesel engines have a premixed one.
3. To fly at 5 Mach, the most appropriate engine option is/are...
 - (a) Ramjet.
 - (b) Turbojet or low bypass turbofan.
 - (c) High bypass turbofan.
 - (d) Turboprop.
4. Which component drives the fan in a twin spool turbofan?
 - (a) The low pressure turbine.
 - (b) The high pressure turbine.
 - (c) The high pressure compressor.
 - (d) The exhaust nozzle.
5. How can the efficiency of some propellers be adjusted in-flight to varying cruising speeds?
 - (a) Modifying the blades pitch.
 - (b) Modifying the blades curvature with blade flaps.
 - (c) Modifying the blades length.
 - (d) Nothing can be done.
6. Which of the following airframe components of an helicopter generate thrust force?
 - (a) The main rotor.
 - (b) The tail cone.
 - (c) Both the tail rotor and the main rotor.
 - (d) Both the tail rotor and the tail cone.
7. Regarding the rotor controls, which of the following statements are correct?
 - I The cyclic behaves similar to the stick of an airplane.
 - II The pedals behave similar to the rudder pedals.
 - III The collective adjusts the pitch of the tail rotor.
 - (a) I and II.
 - (b) II and III.
 - (c) I and III.
 - (d) I.
8. In helicopters, the stall angle of attack limits:
 - (a) Its maximum horizontal speed.
 - (b) Its minimum horizontal speed.
 - (c) The stall angle of attack does not make sense in helicopters.
 - (d) Its inverted flight.
9. A sun-synchronous orbit (SSO):
 - (a) is a polar orbit ($i \approx 90^\circ$) in which the satellite flies a given point of the surface at the same local time.
 - (b) is a polar orbit ($i \approx 0^\circ$) in which the satellite flies a given point of the surface at the same local time.
 - (c) it is widely used in navigation applications.
 - (d) it is widely used in satellite-communications.
10. Regarding the Tsiolkovsky equation ($\Delta V = v_e \log \frac{m_0}{m_1}$)
 - (a) The greater ΔV , the more mass will be consumed, for a given effective exhaust velocity (specific impulse).
 - (b) The less ΔV , the more mass will be consumed, for a given specific effective exhaust velocity (specific impulse).
 - (c) The greater ΔV , the more distance will be traveled by the spacecraft
 - (d) The less ΔV , the less mass will be consumed, regardless of the effective exhaust velocity (specific impulse).

BLO-6: Concepts on avionics and navigation

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11. An anemometer measures:
- The Indicated Airspeed (IAS).
 - The Calibrated/Rectified Airspeed (CAS/RAS).
 - The True Airspeed (TAS).
 - The Ground Speed (GS).
12. To have the altimeter read true height of the aircraft above the Mean Sea Level (MSL), which is the altimeter setting?
- QNH.
 - QNE.
 - STD.
 - QFE.
13. The Transition Level (TL) is the altitude at which the pilot shifts the calibration of the altimeter:
- From STD to QNH.
 - From QNH to STD.
 - From QFE to QNH.
 - From QNH to QFE.
14. Which of the following statements are correct:
- In navigation, the *precision* of a system, also called repeatability, is the degree to which repeated measurements under unchanged conditions show the same results.
 - In navigation, the *accuracy* of a system, also called repeatability, is the degree to which repeated measurements under unchanged conditions show the same results.
 - In navigation, the *accuracy* of a system is the degree of closeness of measurements of a quantity to that quantity's actual (true) value.
 - In navigation, the *precision* of a system is the degree of closeness of measurements of a quantity to that quantity's actual (true) value.
- I, III
 - II, IV
 - I
 - IV
15. Regarding the Ground Based Augmentation System, which of the following statements are correct:
- The errors associated with the GPS measurements are similar for users located "not far" from each other, and they change "slowly" in time. In other words, the errors are correlated both spatially and temporally.
 - GBAS is similar to differential GPS (dGPS) but applied to air navigation.
 - The use of one or several relay communication satellites provides the system with wide coverage.
 - The accuracy of the whole system is enhanced but its integrity remains constant.
- I, II.
 - I, II, IV.
 - II, IV.
 - III, IV.
16. Which navigation (sub)system provides azimuthal guidance for a precision approach?
- Localizer.
 - ILS.
 - Glide Path.
 - Markers.
17. If we want to improve the quality of the guidance provided by an ILS we can replace:
- Markers by a DME.
 - The Glide Path by a Localizer.
 - DME by Markers.
 - The whole ILS by an NDB.
18. About latitude (ϕ) and longitude (λ) ranges:
- $\phi \in [-90, 90]$, $\lambda \in [-180, 180]$.
 - $\phi \in [-180, 180]$, $\lambda \in [-90, 90]$.
 - $\phi \in [-180, 180]$, $\lambda \in [-180, 180]$.
 - $\phi \in [-90, 90]$, $\lambda \in [-90, 90]$.
19. Which of the following technologies are critically important in GPS?
- Satellites stability and satellite clocks.
 - Satellites stability and both satellite and receiver clocks.
 - Transmission power and receiver clocks.
 - Transmission power and satellite clocks.
20. Which of the following navigation systems is used to define airways (AWY)?
- VOR.
 - None of the others.
 - ILS.
 - DME.

BLO-7: Concepts on materials and aeronautical regulations

21. In aeronautics, when a wing is structurally modeled as a beam, the wing root is considered..
- (a) a rigid union.
 - (b) a free end.
 - (c) an articulated union.
 - (d) an articulated gliding union.
22. Regarding stresses on the upper and lower surfaces of a cruising airplane's wing...
- (a) the upper surface is subject to compression while the lower surface is under traction.
 - (b) the upper surface is subject to traction while the lower surface is under compression.
 - (c) both the upper and lower surfaces are under traction.
 - (d) both the upper and lower surfaces are under compression.
23. How do the wing tips mainly flex during take-off and why?
- (a) Upwards, due to increasing lift.
 - (b) Rearwards, due to increasing drag.
 - (c) Downwards, due to increasing weight.
 - (d) It doesn't flex. It is a rigid structure.
24. An aeronautic engineer must choose the appropriate material being aware of...
- (a) all the other statements are correct.
 - (b) the environment in which it will be used.
 - (c) life time expected.
 - (d) its price.
25. The fuselage of a cruising airplane...
- (a) Is subject to traction stresses.
 - (b) Is subject to compression stresses.
 - (c) May be in risk of buckling.
 - (d) Supports no stresses whatsoever.
26. Which of the following ICAO annexes contains the standards and recommended practices regarding Personnel Licensing?
- (a) 1
 - (b) 14
 - (c) 13
 - (d) 15
27. Which of the following ICAO annexes contains the standards and recommended practices regarding Rules of the Air?
- (a) 2
 - (b) 8
 - (c) 10
 - (d) 5
28. The organization founded in 1960 whose original aim was to unify Control and Navigation air services in Europe is...
- (a) EUROCONTROL
 - (b) EASA
 - (c) FAA
 - (d) ICAO
29. To be given the certificate of airworthiness for an aircraft, the set of requirements that every American aircraft must fulfill is called...
- (a) FAR
 - (b) JAR
 - (c) JAA
 - (d) FAA
30. The civil aviation authority of Spain receives the name...?
- (a) DGAC
 - (b) ICAO
 - (c) EUROCONTROL
 - (d) EASA

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control system is the simplest but can lead to unaffordable pilot efforts?
- (a) A purely mechanical flight control system.
 - (b) An electro-hydraulic flight control system.
 - (c) An hydro-mechanical flight control system.
 - (d) An electro-hydro-mechanical flight control system.
32. Which of the following are the main advantages of fly-by-wire Flight Control System (FCS)?
- (a) It is light and provides flight augmentation capabilities.
 - (b) It is light although it provides no flight augmentation capabilities.
 - (c) It does not depend on electronics and is light.
 - (d) It does not depend on electronics and provides flight augmentation capabilities.
33. The Ram Air Turbine is used...
- (a) to provide power to the hydraulic system in case of a loss of engine power, among other failures.
 - (b) to provide power to the hydraulic system when on ground and the engines are stopped.
 - (c) to provide power to the pneumatic system in case of a loss of engine power, among other failures.
 - (d) to provide power to the pneumatic system when on ground and the engines are stopped.
34. Some of the components of an hydraulic system are:
- (a) The power transfer unit (PTU) and pressurized reservoirs.
 - (b) Pressurized reservoirs and the auxiliary Power Unit (APU).
 - (c) The auxiliary Power Unit (APU) and the power transfer unit (PTU)
 - (d) The power transfer unit (PTU) and the pitot tube.
35. What is the main provider of AC power while the aircraft is on ground with all main engines stopped?
- (a) The Auxiliary Power Unit (APU).
 - (b) The Ram Air Turbine (RAT).
 - (c) The batteries.
 - (d) The emergency generator.
36. How is the DC electric system usually powered on ground before starting any of the gas turbines (APU or engines)?
- (a) Either from the ground supply or the batteries.
 - (b) It cannot be done. The APU has to be started first.
 - (c) With the Ram Air Turbine (RAT).
 - (d) With the hydraulic system.
37. Which of the following components belongs in the pneumatic system of an Airbus A320 type aircraft?
- (a) Bleed Monitoring Computer (BMC).
 - (b) Auxiliary Power Unit (APU).
 - (c) Ram Air Turbine (RAT).
 - (d) Nozzle guide vanes.
38. The Power Plant aims at...
- (a) All of them.
 - (b) providing bleed air.
 - (c) providing AC electrical power.
 - (d) providing hydraulic power.
39. When is the Auxiliary Power Unit (APU) usually stopped?
- (a) During cruise and landing.
 - (b) On ground and during take-off.
 - (c) During cruise and on ground.
 - (d) On ground and during landing.
40. How are the brakes of an airplane usually actuated while taxiing?
- (a) From the rudder pedals.
 - (b) From the sidestick.
 - (c) From the elevator trim.
 - (d) None of the others.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	c	d	b	a
2	a	a	a	c	a
3	a	d	b	b	d
4	a	a	a	c	d
5	a	d	d	a	d
6	a	c	c	c	b
7	a	d	c	c	a
8	a	b	a	a	b
9	a	b	d	b	b
10	a	b	a	d	c

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	d	a	a	c
12	a	c	a	c	b
13	a	d	a	b	a
14	a	c	d	b	d
15	a	d	a	c	a
16	a	a	c	b	b
17	a	c	b	b	a
18	a	a	d	b	b
19	a	d	d	c	c
20	a	d	a	d	d

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	d	d	a	a
22	a	c	c	d	c
23	a	c	c	d	a
24	a	d	b	a	c
25	a	c	d	b	b
26	a	a	a	c	d
27	a	a	a	b	c
28	a	c	a	b	c
29	a	b	d	b	a
30	a	b	a	a	d

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	b	b	d	b
32	a	a	c	c	c
33	a	d	d	b	c
34	a	b	a	c	b
35	a	b	b	b	c
36	a	c	d	a	c
37	a	c	a	c	a
38	a	d	c	a	c
39	a	a	d	d	b
40	a	d	a	b	d

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. From the following types of engines, which is the one having the highest bypass ratio?
 - (a) Turboprop.
 - (b) Turbojet.
 - (c) Low bypass ratio turbofan.
 - (d) High bypass ratio turbofan.
2. Choose the correct statement.
 - (a) Otto engines emit more CO₂ while Diesel engines send out more carbon particles at equal engine size.
 - (b) Otto engines are always two stroke while Diesels are always four stroke.
 - (c) Otto engines have diffusive flames while Diesel engines have premixed flames.
 - (d) Otto engines use rich oxygen mixtures while Diesels are stoichiometric.
3. From the following engines, which are the most appropriate engine options to fly at 1.5 Mach?
 - (a) Turbojet and low bypass turbofan.
 - (b) Both low and high bypass turbofan.
 - (c) Turbojet and ramjet.
 - (d) Turboprop and high bypass turbofan.
4. Which of the following layouts, from intake to gas outlet, is representative of a 2-spool turbojet?
 - (a) diffuser, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (b) nozzle, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, diffuser
 - (c) diffuser, high pressure compressor, low pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (d) nozzle, high pressure compressor, low pressure compressor, combustion chamber, low pressure turbine, high pressure turbine, diffuser
5. To increase the thrust of a propeller engine without anticipating compression phenomena, what can be done?
 - (a) Increase the number of blades.
 - (b) Decrease the number of blades.
 - (c) Increase the length of the blades.
 - (d) Decrease the length of the blades.
6. Which of the following airframe components of an helicopter generates lift force?
 - (a) The main rotor.
 - (b) The tail cone.
 - (c) Both the tail rotor and the main rotor.
 - (d) Both the tail cone and the tail rotor.
7. Which of the following statements are correct?
 - I Most helicopter blades have a twist such that the angle of attack *decreases* from root to tip.
 - II Most helicopter blades have a twist such that the angle of attack *increases* from root to tip.
 - III If no blade twist is applied, the lift of the rotor increases rapidly from central hub to tip.
 - (a) I and III.
 - (b) II.
 - (c) II and III.
 - (d) I.
8. Which part of the retrieving blades of an helicopter will stop generating lift in forward flight?
 - (a) The inner part of the blade.
 - (b) The outer part of the blade.
 - (c) The mid part of the blade.
 - (d) The retrieving blades will never stop generating lift.
9. A geostationary orbit (GEO):
 - (a) is a particular geosynchronous orbit (GSO) with inclination: $i = 0^\circ$.
 - (b) is a particular geosynchronous orbit (GSO) with inclination: $i \approx 90^\circ$.
 - (c) it is widely used in navigation applications.
 - (d) has an orbital period: $T \ll 24$ hours.
10. The Highly Elliptical Orbits (HEO) are mostly used in countries situated:
 - (a) At extreme latitudes.
 - (b) At temperate latitudes.
 - (c) At extreme longitudes.
 - (d) At temperate longitudes.

BLO-6: Concepts on avionics and navigation

11. A Pitot-static tube measures
- (a) Both static and total pressure.
 - (b) Only static pressure.
 - (c) Both static pressure and temperature.
 - (d) True Airspeed (TAS).
12. Above the Transition Level (TL) the altitude setting must always be set to:
- (a) STD.
 - (b) QNH.
 - (c) QFE.
 - (d) Insufficient data provided.
13. Above the Transition Altitude (TA) the altitude setting must always be set to:
- (a) Insufficient data provided.
 - (b) STD.
 - (c) QNH.
 - (d) QFE.
14. Which of the following statements is correct:
- (a) GPS Selective Availability (S/A) can degrade system performance, mainly in terms of integrity and accuracy, which makes it unsuitable for civil aviation purposes.
 - (b) In civil aviation, accuracy and precision performances are the most important.
 - (c) In civil aviation, a Global Positioning System (GPS) complies with accuracy and precision performance requirements in all phases of flight. Hence, it can be used in Precision Approach (PA) procedures.
 - (d) None of the others is correct.
15. Regarding Satellite Based Augmentation Systems, which of the following statements is correct:
- (a) The use of one or several relay communication satellites provides the system with wide coverage.
 - (b) The position of the SBAS satellite(s) is critical in terms of accuracy improvement.
 - (c) SBAS has not been implemented yet. All SBAS systems are currently in development.
 - (d) The accuracy of the whole system is enhanced but its integrity remains constant.
16. Which navigation (sub)system provides the elevation (zenithal) guidance for the precision approach?
- (a) Glide Path.
 - (b) ILS.
 - (c) Localizer.
 - (d) Markers.
17. VOR is:
- (a) $\approx 1^\circ$ accurate.
 - (b) ≈ 1 NM accurate.
 - (c) ≈ 1 meter accurate.
 - (d) ≈ 1 km accurate.
18. What is the geographic position in terms of latitude (ϕ) and longitude (λ) of the geographic North Pole:
- (a) $\phi = 90^\circ, \forall \lambda$.
 - (b) $\lambda = 90^\circ, \forall \phi$.
 - (c) $\phi = 180^\circ, \forall \lambda$.
 - (d) $\lambda = 180^\circ, \forall \phi$.
19. How many GPS satellites, at least, must be in view in order to estimate the user's position (X, Y, Z)?
- (a) 4.
 - (b) 3.
 - (c) 5.
 - (d) 2.
20. About the DME working principle...
- (a) The aircraft interrogates and the ground installation responds.
 - (b) The ground installation interrogates and the aircraft responds.
 - (c) DME is a passive system, i.e. the signal only travels from the aircraft to the ground station.
 - (d) DME is a passive system, i.e. the signal only travels from the ground station to the aircraft.

BLO-7: Concepts on materials and aeronautical regulations

21. In aeronautics, the simplest scheme of a wing corresponds to...
- (a) a beam with a rigid union at one end while the opposite one is free.
 - (b) a beam with a rigid union at one end while the opposite one is articulated.
 - (c) an articulated beam at both ends.
 - (d) a rigid union at both the ends.
22. Which of the following statements are TRUE of biplane wings?
- I Wires were used by biplane aircraft to make their wings stronger.
 - II World War I biplane wings were mostly made with wood and cotton fabric.
 - III The usage of wires reduced the drag in comparison with the usage of fatter wings.
- (a) I and II
 - (b) II and III
 - (c) II
 - (d) III
23. Regarding aeronautics, in an optimal structure...
- I flying and stationary stresses are opposite but with the same modulus.
 - II there are no structural members loafing while others are overworking.
 - III the stresses are balanced in such a way that the stresses on all structural members are equal.
- (a) II and III
 - (b) I and II
 - (c) II
 - (d) III
24. Regarding stresses on the upper and lower surfaces of a parked airplane's wing...
- (a) the upper surface is subject to traction while the lower surface is under compression.
 - (b) the upper surface is subject to compression while the lower surface is under traction.
 - (c) both the upper and lower surfaces are under traction.
 - (d) both the upper and lower surfaces are under compression.
25. Which of the following ICAO annexes contains the standards and recommended practices regarding Aeronautical Information Services?
- (a) 15
 - (b) 14
 - (c) 1
 - (d) 3
26. Which of the following ICAO annexes contains the standards and recommended practices regarding Aeronautical Telecommunications?
- (a) 10
 - (b) 14
 - (c) 1
 - (d) 3
27. Which of the following ICAO annexes contains the standards and recommended practices regarding Aerodromes?
- (a) 14
 - (b) 18
 - (c) 1
 - (d) 3
28. To be given the certificate of airworthiness for an aircraft, the set of requirements that every European aircraft must fulfill is called...
- (a) JAR
 - (b) FAR
 - (c) JAA
 - (d) FAA
29. Which types of aircraft were considered in the agreements following the Chicago convention of 1944?
- (a) Civil
 - (b) Customs
 - (c) Police
 - (d) Military
30. The Chicago convention in 1944 started an international organization called...
- (a) ICAO
 - (b) EUROCONTROL
 - (c) EASA
 - (d) DGAC

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control systems is also known as "fly-by-wire"?
- (a) An electro-hydraulic flight control system.
 - (b) An hydro-mechanical flight control system.
 - (c) An electro-hydro-mechanical flight control system.
 - (d) A purely mechanical flight control system.
32. Which of the following is the main drawback of a fly-by-wire Flight Control System (FCS)?
- (a) It is too dependent on electronics.
 - (b) It is too light.
 - (c) Flight augmentation characteristics are deceptive.
 - (d) None of the others.
33. How many circuits does the hydraulic system of an Airbus A320-type aircraft consist of?
- (a) 3.
 - (b) 2.
 - (c) 1.
 - (d) 4.
34. Regarding the hydraulic system, which of the following statements are correct?
- I Fluid can be transferred among hydraulic circuits.
 - II Power can be transferred among hydraulic circuits.
 - III Pilot effort is increased in comparison with mechanical flight control.
 - IV Pilot effort is decreased in comparison with mechanical flight control.
- (a) II, IV
 - (b) I, IV
 - (c) II, III
 - (d) I, III
35. How is the main generator of the AC electric system driven while in normal uneventful flight?
- (a) The turbine(s).
 - (b) The Auxiliary Power Unit (APU).
 - (c) The Ram Air Turbine (RAT).
 - (d) The batteries.
36. How is DC electric power generated while the aircraft is in flight?
- (a) The turbine(s).
 - (b) The Auxiliary Power Unit (APU).
 - (c) The Ram Air Turbine (RAT).
 - (d) The batteries.
37. Which one of the following items is a component of the pneumatic system?
- (a) The crossbleed duct and valve.
 - (b) A static inverter.
 - (c) An accessory gearbox.
 - (d) Nozzle guide vanes.
38. The Full Authority Digital Engine Control (FADEC) aims at...
- (a) performing complete engine management.
 - (b) monitoring the engines.
 - (c) starting the engines.
 - (d) none of the others.
39. When is the Auxiliary Power Unit (APU) usually in operation?
- (a) On ground.
 - (b) Both on ground and during cruise.
 - (c) Both during turns and landing.
 - (d) During cruise and landing.
40. The brake system of large aircraft is powered by...
- (a) The hydraulic system.
 - (b) The pneumatic system.
 - (c) The electric system.
 - (d) None of the others.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	b	a	a	b
2	a	b	b	c	b
3	a	b	b	c	b
4	a	d	c	d	d
5	a	c	a	b	b
6	a	b	c	c	c
7	a	a	a	a	b
8	a	a	a	a	c
9	a	b	b	a	d
10	a	b	a	d	d

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	d	a	a	c
12	a	b	a	d	d
13	a	d	b	b	c
14	a	d	b	c	a
15	a	d	d	b	a
16	a	a	a	d	a
17	a	c	a	c	c
18	a	c	b	d	a
19	a	a	c	c	d
20	a	a	c	b	b

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	a	a	d	d
22	a	c	d	c	d
23	a	d	c	a	c
24	a	c	d	a	a
25	a	b	d	a	d
26	a	b	a	d	d
27	a	c	a	c	c
28	a	d	a	d	d
29	a	c	b	b	b
30	a	a	d	c	c

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	d	d	d	b
32	a	b	b	d	a
33	a	c	c	b	c
34	a	c	c	a	a
35	a	c	b	b	a
36	a	c	b	c	b
37	a	b	d	a	a
38	a	b	d	a	b
39	a	b	a	d	b
40	a	c	b	b	b

BLO-1: Concepts on flight principles and aircraft performance

1. An airplane is...
 - (a) a fixed-wing aerodyne.
 - (b) a rotary-wing aerodyne.
 - (c) a fixed-wing aerostat.
 - (d) a rotary-wing aerostat.
2. What is the main effect on a wing lift characteristic curve ($C_L(\alpha) = C_{L\alpha}(\alpha - \alpha_0)$ for $\alpha < \alpha_S$) of increasing its aspect ratio?
 - (a) It increases the slope $C_{L\alpha}$.
 - (b) It shifts the curve up by decreasing α_0 .
 - (c) It increases the stall angle α_S .
 - (d) It reduces $C_D(\alpha)$ but does nothing to $C_L(\alpha)$.
3. Which of the following parameters is roughly independent of velocity if we neglect its effects through Reynolds and Mach numbers?
 - (a) All of them.
 - (b) Stall angle of attack.
 - (c) Parasitic drag coefficient.
 - (d) Maximum aerodynamic efficiency.
4. What is true about airfoil drag?
 - (a) Friction drag is a consequence of air viscosity.
 - (b) A symmetric airfoil at zero angle of attack produces no drag.
 - (c) The wake of an airfoil contributes to lift but entails no drag.
 - (d) The drag coefficient is minimal close to stall.
5. How does stall speed compare between a plane in horizontal level flight ($V_{S_{cruise}}$) and the same plane, at the same altitude, engaged in a coordinated banked turn ($V_{S_{turn}}$)?
 - (a) $V_{S_{turn}} > V_{S_{cruise}}$
 - (b) $V_{S_{turn}} = V_{S_{cruise}}$
 - (c) $V_{S_{turn}} < V_{S_{cruise}}$
 - (d) Which one is bigger depends on the aerodynamic characteristics of the airplane.
6. What happens to the minimum (D_{min}) of the drag curve ($D(V)$) as the altitude increases?
 - (a) It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - (b) It occurs at lower speeds (moves left) and decreases (moves down).
 - (c) It occurs at about the same speed and remains fairly constant.
 - (d) It occurs at higher speeds (moves right) and increases (moves up).
7. How is a runway to be designed for an airport located high above sea level if it intends to be certified for large heavy aircraft?
 - (a) The runway needs be longer than for an equivalent airport at sea level.
 - (b) The runway can be shorter than for an equivalent airport at sea level.
 - (c) There is no particular restriction on runway length.
 - (d) The runway must include a slight slope at the very end to assist takeoff operations.
8. How do the operational and absolute ceiling compare?
 - (a) The absolute ceiling is above the operational ceiling.
 - (b) The operational ceiling is above the absolute ceiling.
 - (c) Which one is above depends on engine type.
 - (d) They are synonyms.
9. Which of the following conditions maximises cruise endurance?
 - (a) Minimum hourly fuel consumption (C , in lb/h).
 - (b) Minimum kilometric fuel consumption (C/V , in lb/km).
 - (c) Minimum drag.
 - (d) Minimum required power.
10. How does headwind affect the best rate of climb ($(R/C)_{max}$) and the corresponding aerodynamic speed $V_{(R/C)_{max}}$ of a given airplane?
 - (a) It doesn't affect whatsoever.
 - (b) $V_{(R/C)_{max}}$ increases and $(R/C)_{max}$ improves.
 - (c) $V_{(R/C)_{max}}$ decreases and $(R/C)_{max}$ improves.
 - (d) $V_{(R/C)_{max}}$ stays the same but $(R/C)_{max}$ worsens.

BLO-2: Flight principles related problems

11. What is the optimal angle of attack (α_{opt}) that maximises aerodynamic efficiency (f_{max}) for an airplane with lift coefficient $C_L = 0.18 + 0.09\alpha$ for $\alpha < \alpha_S = 12^\circ$ and drag polar $C_D = 0.02 + 0.06 C_L^2$?
- (a) $\alpha_{opt} = 4.42^\circ$
 - (b) $\alpha_{opt} = 12^\circ$
 - (c) $\alpha_{opt} = 4^\circ$
 - (d) $\alpha_{opt} = 3.43^\circ$
12. An aerobatics airplane with lift coefficient $C_L = 0.1 + 0.1\alpha$ with α in $^\circ$ is performing a balanced inverted flight. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires $\alpha = 2^\circ$?
- (a) $\alpha = -4^\circ$
 - (b) $\alpha = -3^\circ$
 - (c) $\alpha = -5^\circ$
 - (d) $\alpha = -6^\circ$
13. A small airplane of unknown mass is engaged in a coordinated banked turn with bank angle ($\phi = 45^\circ$). The lift coefficient in straight and level flight at the same altitude and speed is $C_{L_{n=1}} = 0.3$. What is the required lift coefficient for this turn?
- (a) $C_{L_\phi} = 0.424$
 - (b) $C_{L_\phi} = 0.467$
 - (c) $C_{L_\phi} = 0.339$
 - (d) It depends on the mass of the airplane.
14. The drag polar of an airplane changes from $C_D = 0.03 + 0.11C_L^2$ in clean configuration to $C_D = 0.04 + 0.09C_L^2$ with full flaps. What is the value of the drag coefficient (C_D) at optimal flap deployment?
- (a) $C_D = 0.085$
 - (b) $C_D = 0.102$
 - (c) $C_D = 0.0501$
 - (d) $C_D = 0.142$
15. An airplane of mass $m = 2000$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 20 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
- (a) $D = 1000$ N and $D_p = D_i = 500$ N.
 - (b) $D = 1000$ N and $D_p > D_i$.
 - (c) $D = 500$ N and $D_p = D_i = 250$ N.
 - (d) $D = 500$ N and $D_p > D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which primary control surfaces are actuated with the stick (or yoke)?
- (a) Ailerons and elevator.
 - (b) Rudder and ailerons.
 - (c) Flaps and ailerons.
 - (d) Elevator and rudder.
17. Which parameter is mainly responsible for the roll subsidence mode damping?
- (a) Wing span and aspect ratio (C_{l_p}).
 - (b) Fin size (C_{n_r}).
 - (c) Horizontal stabilizer size (C_{m_q}).
 - (d) None of the others.
18. If an airplane is dynamically stable around an axis, what can we say about its static stability around the same axis?
- (a) It is necessarily statically stable.
 - (b) It is necessarily statically unstable.
 - (c) It is necessarily statically neutral.
 - (d) It might or might not be statically stable.
19. What can be done to have an airplane that is at the same time highly stable and highly manoeuvrable?
- (a) Design it highly manoeuvrable and stabilise it with automatic control.
 - (b) Design it highly stable and make it manoeuvrable with automatic control.
 - (c) Design it highly stable, which will make it also highly manoeuvrable at no additional cost.
 - (d) Nothing. There is no way to meet both conditions.
20. Why is the stabilised climb *only* a *pseudoequilibrium*?
- (a) Because of the return to straight and level flight mode, due to density dependence on altitude.
 - (b) Because of the phugoid mode, which damps the climb slope.
 - (c) Because of the pitch oscillation, which renders the climb oscillatory.
 - (d) None of the others. It is a legitimate equilibrium in its own right.
21. Which commands must a pilot use to perform a coordinated banked turn, if the airplane is to preserve both altitude and speed?
- (a) Stick, pedals and thrust lever.
 - (b) Stick and thrust lever.
 - (c) Stick.
 - (d) Stick and pedals.
22. What can be said about the initial response of the airplane to a perturbation in pitch?
- (a) It depends on the relative position of the centre of gravity with respect to the aerodynamic centre.
 - (b) It depends on the size of the fin.
 - (c) It depends on the wings dihedral.
 - (d) It depends on the thrust of the engines.
23. What is true about the spiral mode?
- (a) It can be left unstable without critically compromising the aircraft integrity.
 - (b) It is fast and usually periodic.
 - (c) It corresponds to the periodic exchange between potential and kinetic energy.
 - (d) It couples pitch and yaw motion.
24. How must the pilot of a twin-engined airplane correct for excessive sideslip in case of left engine failure?
- (a) Press the right pedal.
 - (b) Press the left pedal.
 - (c) Switch off the right engine.
 - (d) Press both pedals simultaneously.
25. Which components does the fin of an airplane generally consist of?
- (a) Vertical stabiliser and rudder.
 - (b) Horizontal stabiliser and elevator.
 - (c) Vertical stabiliser and elevator.
 - (d) Horizontal stabiliser and rudder.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.22\bar{\rho}V^2 + 0.018W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1400\text{ kg}$ and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.4$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 10\text{ ms}^{-2}$)
- $D = 1760\text{ N}$
 - $D = 2030\text{ N}$
 - $D = 1850\text{ N}$
 - $D = 1940\text{ N}$
27. A propeller-powered airplane of mass $m = 1200\text{ kg}$ has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.02W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 10\text{ ms}^{-2}$)
- $V_{(R/C)_{max}} = 42.3\text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 55.7\text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 73.3\text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 96.4\text{ ms}^{-1}$
28. An airplane with drag polar $C_D = 0.03 + 0.16C_L^2$, wing surface $S = 35\text{ m}^2$ and mass $m = 1800\text{ kg}$ is flying at a speed $V = 100\text{ ms}^{-1}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.5\text{ kgm}^{-3}$? (take $g \simeq 10\text{ ms}^{-2}$)
- $T_R = 3220\text{ N}$
 - $T_R = 3280\text{ N}$
 - $T_R = 3090\text{ N}$
 - $T_R = 3410\text{ N}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 40$. What is the descent angle (γ)?
- $\gamma = -1.43^\circ$
 - $\gamma = -1.63^\circ$
 - $\gamma = -1.77^\circ$
 - $\gamma = -0.96^\circ$
30. The required thrust of a jet-engined airplane of mass $m = 1500\text{ kg}$, wing area $S = 20\text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.4$, is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 10000\text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10\text{ ms}^{-2}$ and $\rho_0 = 1.225\text{ kgm}^{-3}$)
- $V \simeq 51.5\text{ ms}^{-1}$
 - $V \simeq 49.2\text{ ms}^{-1}$
 - $V \simeq 119\text{ ms}^{-1}$
 - $V \simeq 76.6\text{ ms}^{-1}$

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	c	a	a	d
2	a	c	c	c	a
3	a	b	a	a	c
4	a	c	d	c	b
5	a	d	d	d	a
6	a	c	d	d	a
7	a	b	c	d	a
8	a	b	b	a	a
9	a	d	d	d	c
10	a	b	d	b	b

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	b	d	b	c
12	a	c	a	a	a
13	a	a	a	d	b
14	a	a	d	d	a
15	a	a	b	d	c

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	d	b	c	d
17	a	b	a	c	c
18	a	b	b	b	b
19	a	b	d	b	a
20	a	b	d	c	b
21	a	c	a	d	a
22	a	a	d	b	b
23	a	b	a	a	a
24	a	b	d	b	d
25	a	b	c	b	b

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	b	d	b
27	a	c	b	d	c
28	a	b	a	b	a
29	a	d	c	c	c
30	a	d	d	b	c

BLO-1: Concepts on flight principles and aircraft performance

1. Which nondimensional parameters affect lift and drag coefficients through the advent of compressibility¹ and turbulence² at high velocities?
 - (a) The Mach¹ and Reynolds² numbers, respectively.
 - (b) The Reynolds¹ and Mach² numbers, respectively.
 - (c) The Mach¹ and Prandtl² numbers, respectively.
 - (d) The Prandtl¹ and Reynolds² numbers, respectively.
2. The Airbus A320 and the Cessna 172 are aerospace vehicles that belong to the group of
 - (a) Fixed-wing aerodynes.
 - (b) Rotorcraft aerodynes.
 - (c) Aerostats.
 - (d) Fixed-wing and rotorcraft aerodynes, respectively.
3. Which of the following is a leading-edge high-lift device?
 - (a) Slat.
 - (b) Fowler flap.
 - (c) Hinged flap.
 - (d) Spoiler.
4. For a rectangular wing made of direct extrusion of a given aerodynamic airfoil, what is the main effect on its lift characteristic curve ($C_L(\alpha) = C_{L\alpha}(\alpha - \alpha_0)$ for $\alpha < \alpha_S$) of reducing the aspect ratio λ ?
 - (a) Slope reduction $C_{L\alpha} \searrow$.
 - (b) Slope increase $C_{L\alpha} \nearrow$.
 - (c) Upward shift $\alpha_0 \searrow$.
 - (d) Downward shift $\alpha_0 \nearrow$.
5. In wing aerodynamics, air viscosity is responsible for...
 - (a) both lift and drag.
 - (b) lift, but not drag.
 - (c) drag, but not lift.
 - (d) neither lift nor drag.
6. In a coordinated banked turn...
 - (a) lift outweighs weight.
 - (b) lift equals weight.
 - (c) weight outweighs lift.
 - (d) which one dominates between lift and weight depends on the sense of rotation (left or right).
7. Sort from lower to higher the following V-speeds during takeoff: takeoff configuration stall speed (V_s), takeoff/lift-off speed (V_{to}) and decision speed (V_d)?
 - (a) $V_d < V_s < V_{to}$.
 - (b) $V_s < V_{to} < V_d$.
 - (c) $V_{to} < V_d < V_s$.
 - (d) $V_s < V_d < V_{to}$.
8. What must a pilot do, regardless of how the aircraft is powered, to climb with maximum vertical velocity?
 - (a) Set full throttle and maximise excess power.
 - (b) Set full throttle and minimise required power.
 - (c) Set full throttle and minimise required thrust.
 - (d) Set full throttle and maximise excess thrust.
9. In which of the two possible straight and level flight regimes, stall may occur?
 - (a) R2.
 - (b) R1.
 - (c) In both R1 and R2.
 - (d) Neither in R1 nor in R2.
10. For a glider to gain altitude...
 - (a) it must fly in ascending air.
 - (b) it must fly in descending air.
 - (c) it must fly with headwind.
 - (d) it must fly with tailwind.

BLO-2: Flight principles related problems

11. What is the maximum aerodynamic efficiency (f_{max}) of a glider with drag polar $C_D = 0.016 + 0.008C_L^2$?
- (a) $f_{max} = 44.2$
 - (b) $f_{max} = 41.8$
 - (c) $f_{max} = 48.9$
 - (d) $f_{max} = 37.9$
12. An aerobatics airplane with lift coefficient $C_L = 0.2 + 0.1\alpha$ with α in $^\circ$ is performing a stabilised manoeuvre with load factor $n = 2$. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.3$?
- (a) $\alpha = 4^\circ$
 - (b) $\alpha = 1^\circ$
 - (c) $\alpha = -0.5^\circ$
 - (d) $\alpha = -2^\circ$
13. The sea-level maximum-weight stall velocity of an airplane is $V_s = 30$ m/s. What would the stall velocity V'_s be if the airplane's weight is 90% its maximum and the altitude is such that the density is 50% that of sea-level?
- (a) $V'_s = 40.2$ m/s
 - (b) $V'_s = 22.4$ m/s
 - (c) $V'_s = 54$ m/s
 - (d) $V'_s = 44.7$ m/s
14. The drag polar of an airplane changes from $C_D = 0.02 + 0.04C_L^2$ in clean configuration to $C_D = 0.03 + 0.02C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
- (a) $C_L = 0.707$
 - (b) $C_L = 0.731$
 - (c) $C_L = 0.757$
 - (d) $C_L = 0.646$
15. An airplane of mass $m = 2000$ kg is cruising in straight and level flight at a speed that is greater than that for optimal aerodynamic efficiency. If the aerodynamic efficiency is $f = 20 < f_{max}$, what can we say about induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
- (a) $D_p > 500 \text{ N} > D_i$.
 - (b) $D_p = D_i = 500 \text{ N}$.
 - (c) $D_p < D_i < 1000 \text{ N}$.
 - (d) $D_p + D_i = 1000 \text{ N}$ and $D_p \neq D_i$, but there is not enough information to know which one is greater.

BLO-3: Concepts on aircraft stability and control

16. Which command and primary control surface pilot the aircraft around its roll axis?
- (a) The stick (or yoke) and the ailerons.
 - (b) The pedals and the rudder.
 - (c) The stick (or yoke) and the elevator.
 - (d) The pedals and the elevator.
17. Why can the spiral mode be left unstable for some aircraft?
- (a) Because it is a very slow mode that results in no serious hazard.
 - (b) It can never be left unstable.
 - (c) Because it is a very fast mode and it is well damped even if it is unstable.
 - (d) Because it is extremely difficult to stabilise.
18. When we perturb a certain equilibrium of a system it starts to oscillate around the equilibrium with increasing amplitude. What can be said about static and dynamic stability?
- (a) The system is statically stable but dynamically unstable.
 - (b) The system is dynamically stable but statically unstable.
 - (c) The system is both statically and dynamically stable.
 - (d) The system is both statically and dynamically unstable.
19. What would the best choice be so as to have an airplane that is extremely manoeuvrable around its pitch axis and at the same time very stable?
- (a) Automatic control.
 - (b) A huge elevator.
 - (c) Place the center of gravity as forward as possible.
 - (d) A huge horizontal stabiliser.
20. Which one of the longitudinal modes acts as a successive exchange of potential and kinetic energy?
- (a) Phugoid.
 - (b) Pitch oscillation.
 - (c) Return to straight and level flight.
 - (d) Dutch roll.
21. For which of the following manoeuvres would the pedals be used?
- (a) Coordinated turn and crosswind straight flight (with sideslip).
 - (b) Coordinated and operational turns.
 - (c) Crosswind straight flight (with sideslip) and climb.
 - (d) Operational turn and climb.
22. What can be said about the response of the airplane to a perturbation in roll?
- (a) It depends on both directional and lateral stability, since there is no restoring moment and the return to straight and level flight is mostly dynamical.
 - (b) It depends on static directional stability, since there automatically appears a restoring yaw moment.
 - (c) It depends mainly on static lateral stability, since there automatically appears a restoring roll moment.
 - (d) It depends mainly on static longitudinal stability, since there automatically appears a restoring pitch moment.
23. What is true about dutch roll and pitch oscillation?
- (a) All options are correct.
 - (b) Both are oscillatory modes.
 - (c) While pitch oscillation only involves rotation around the pitch axis, dutch roll couples rotations around the yaw and roll axes.
 - (d) Both modes are relatively fast (in the order of seconds at most).
24. While flying a small stable airplane, how must pitch oscillation be piloted?
- (a) Leave it alone, trying to damp it actively can be counterproductive.
 - (b) With the stick, trying to damp it actively.
 - (c) With the pedals, trying to damp it actively.
 - (d) With the trim wheel, trying to damp it actively.
25. What is the main element responsible for directional stability?
- (a) The vertical stabiliser.
 - (b) The horizontal stabiliser.
 - (c) The stickers on the fuselage.
 - (d) The airliner chairman.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.25\bar{\rho}V^3 + 0.02W^2/\bar{\rho}V$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.3$, what is the optimal velocity for maximum range (V_R)? (take $g \simeq 10 \text{ ms}^{-2}$)
- $V_R = 119 \text{ ms}^{-1}$
 - $V_R = 90.4 \text{ ms}^{-1}$
 - $V_R = 157 \text{ ms}^{-1}$
 - It depends on the available power P_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1200$ kg has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.03W^2/\bar{\rho}V$. What is the maximum rate of climb $((R/C)_{max})$ at sea level ($\bar{\rho} = 1$) if the available power is $P_A = 300 \bar{\rho} \delta_x \text{ kW}$? (take $g \simeq 10 \text{ ms}^{-2}$)
- $(R/C)_{max} = 14.7 \text{ ms}^{-1}$
 - $(R/C)_{max} = 13.3 \text{ ms}^{-1}$
 - $(R/C)_{max} = 11.5 \text{ ms}^{-1}$
 - $(R/C)_{max} = 10.4 \text{ ms}^{-1}$
28. An airplane with drag polar $C_D = 0.02 + 0.05C_L^2$, wing surface $S = 40 \text{ m}^2$ and mass $m = 1500$ kg is flying at a speed $V = 120 \text{ ms}^{-1}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4 \text{ kgm}^{-3}$? (take $g \simeq 10 \text{ ms}^{-2}$)
- $T_R = 2400 \text{ N}$
 - $T_R = 2390 \text{ N}$
 - $T_R = 2380 \text{ N}$
 - $T_R = 2430 \text{ N}$
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 40$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 50 \text{ km}$ of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination?
- $h = 1250 \text{ m}$
 - $h = 2500 \text{ m}$
 - $h = 2000 \text{ m}$
 - $h = 800 \text{ m}$
30. The required thrust of a jet-engined airplane of mass $m = 1500$ kg, wing area $S = 20 \text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.5$, is given by $T_R = 0.6\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 7500 \text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
- $V \simeq 64 \text{ ms}^{-1}$
 - $V \simeq 49.7 \text{ ms}^{-1}$
 - $V \simeq 91.7 \text{ ms}^{-1}$
 - $V \simeq 76.6 \text{ ms}^{-1}$

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

31. An airplane is propelled by a single turbojet flies at 250m/s. The air mass flow at the intake is 10kg/s while the kerosene flow (with $h_{PR} = 42800\text{kJ/Kg}$) at the combustion chamber is 0.5kg/s. If the engine has an adapted nozzle and the exhaust gas velocity is 1100m/s, then the propulsive efficiency is
- 0.37
 - 0.141
 - 0.06
 - None of the other statements is true.
32. The Ministry of Industry plans to reduce CO_2 emissions. Considering parity of horsepower, they should know that...
- Diesel engines emit less CO_2 than Otto engine
 - Electric engines emit less CO_2 if the electricity comes from a coal power plant.
 - Hybrid engines based on Diesel and Otto engines emit less CO_2 than their normal counterparts.
- I and III
 - III
 - II and III
 - II
33. An airplane flies at $M = 1.5$, you can be sure it...
- has a convergent-divergent nozzle.
 - has a supersonic combustion chamber.
 - has a variable geometry diffuser.
 - is a twin engine powered aircraft.
34. An airplane is stopped at the head of the runway. You can be sure that the propulsive power ...
- is zero when it starts accelerating.
 - is constant during takeoff.
 - is such that thrust equals the forces that kept the airplane stopped when acceleration starts.
 - is maximum when it starts accelerating.
35. About turbojet turbines, the turbine...
- gets power from the gases to drive the compressor.
 - gets power from the gases to drive the diffuser.
 - does not need any refrigeration.
 - expands the gases providing engine thrust.
36. Which of the following options are part of the helicopter airframe¹?
- the cabin
 - tail boom
 - landing gear
- All of them
 - I
 - I and II
 - III
37. What is the limiting speed factor of a helicopter in horizontal flight with anti-clockwise main rotor?
- the compressibility effects at the tip of the advancing main rotor blades
 - the compressibility effects at the tip of the retreating main rotor blades
 - the vortex created at the tip of the advancing main rotor blades
 - the vortex created at the tip of the retreating main rotor blades
38. Ground effect
- increases the aircraft lift.
 - increases flight performances at high speed.
 - in helicopters is maintained while flight altitude is lower than main rotor diameter.
- I and III
 - III
 - II
 - I and II
39. The most appropriate orbit for Global Earth Observation mission is a
- polar orbit.
 - geostationary orbit.
 - highly elliptical orbit
 - geosynchronous orbit.
40. A 10kN rocket engine is ejecting a mass flow of 2.05kg/s, so it is a
- liquid cryogenic propellant rocket engine
 - liquid storable propellant rocket engine
 - solid propellant rocket engine
 - plasma ion electric rocket engine

¹helicopter airframe = célula del helicóptero

BLO-6: Concepts on avionics and navigation

41. An aneroid altimeter measures altitude using:
- (a) only static pressure from static ports.
 - (b) both stagnation pressure from a pitot tube and static pressure from static ports.
 - (c) only stagnation pressure from a pitot tube.
 - (d) static pressure from static ports and air temperature.
42. In which of the following situations does a typical anemometer read true airspeed?
- (a) Low speed flight at sea level in standard atmosphere conditions.
 - (b) High speed flight at sea level in standard atmosphere conditions.
 - (c) Low speed flight at high altitude in standard atmosphere conditions.
 - (d) High speed flight at sea level in real (non-standard) atmospheric conditions.
43. The Transition Level (TL) is the flight level at which the pilot shifts the calibration of the altimeter:
- (a) From STD to QNH
 - (b) From QNH to STD
 - (c) From QFE to QNH
 - (d) From QNH to QFE
44. The ability of a system to operate without unscheduled interruptions for the intended interval of operation is called...
- (a) continuity
 - (b) integrity
 - (c) accuracy
 - (d) availability
45. What is a Global Navigation Satellite System (GNSS)?
- (a) A navigation system based on a constellation of satellites such as GPS or GLONASS.
 - (b) A navigation system that enhances GPS / GLONASS systems performance.
 - (c) A communications system based on geostationary satellites.
 - (d) A surveillance system based on low earth orbit satellites.
46. Which navigation subsystem provides zenithal guidance for a precision approach?
- (a) Glideslope
 - (b) VOR
 - (c) ADF
 - (d) Localizer
47. An aircraft's heading indicator reads 120° while the VOR indicates FROM with the CDI needle perfectly centred when tuned with radial 300° . What is the airplane doing?
- (a) Flying towards the VOR station.
 - (b) Flying away from the VOR station.
 - (c) Flying around the VOR station.
 - (d) We need to couple this information with a DME reading to find out what the airplane is doing.
48. Which three coordinates are required to give the position of an aircraft in airspace?
- (a) Latitude, longitude and altitude.
 - (b) Altitude, latitude and heading.
 - (c) Heading, longitude and altitude.
 - (d) Latitude, longitude and heading.
49. What is a Primary Surveillance Radar (PSR)?
- (a) A ground based system that allows identification and tracking of aircraft by issuing an interrogation signal that passively bounces on the aircraft surface.
 - (b) A ground based system that allows identification and tracking of aircraft by issuing an interrogation signal that is actively responded by onboard equipment.
 - (c) An onboard system that allows identification of airspace obstacles by issuing an interrogation signal that passively bounces on these obstacles surface.
 - (d) None of the others.
50. On which of the following elements is the heading indicator (HI), or the horizontal situation indicator (HSI) in more advanced aircraft, based?
- (a) Gyros.
 - (b) Magnetic compass.
 - (c) VHF radio signals.
 - (d) Dice.

BLO-7: Concepts on materials and aeronautical regulations

51. The simplest scheme of a wing corresponds to
- (a) a beam with a rigid union in one end while the opposite one is free.
 - (b) a beam with a rigid union in one end while the opposite one is articulated.
 - (c) a articulated beam in both the ends.
 - (d) a rigid union in both the ends.
52. A composite material
- (a) none of the other statements is true.
 - (b) combines the compression resistance of its fibres with the traction properties of the matrix.
 - (c) is simple to recycle.
 - (d) must have the fibres aligned.
53. In an optimal structure...
- I the stresses are balanced in such a way that the stresses on all structural members are equal.
 - II there are no structural members loafing while others are overworking.
 - III flying and stationary stresses are opposite but with the same modulus.
- (a) I and II
 - (b) II and III
 - (c) II
 - (d) I
54. The wing of an airplane in horizontal flight
- (a) has its upper part under compression stresses
 - (b) has its upper part under traction stresses
 - (c) has its lower part under compression stresses
 - (d) supports less weight than the one supported on ground.
55. Which of the following properties discourages ceramic materials for structural applications?
- (a) their fragility
 - (b) their chemical stability
 - (c) their refractory properties
 - (d) their hardness
56. Which of the following ICAO annexes contains the standards and recommended practices regarding Aerodromes?
- (a) 14
 - (b) 12
 - (c) 2
 - (d) 7
57. The agency charged with coordinating and regulating international air travel that was established by the Chicago Convention is...
- (a) ICAO
 - (b) EUROCONTROL
 - (c) EASA
 - (d) FAA
58. The set of requirements that every american aircraft must fulfil to be given the certificate of airworthiness is called...
- (a) FAR
 - (b) JAR
 - (c) JAA
 - (d) FAA
59. The nationality of an aircraft is that of...
- (a) the registration.
 - (b) the pilot.
 - (c) the airline.
 - (d) the manufacturer.
60. Which organisation is the civil aviation authority in Spain?
- (a) DGAC.
 - (b) AENA.
 - (c) RENFE.
 - (d) EASA.

BLO-8: Concepts on aircraft systems

61. What is a fly-by-wire flight control system?
- (a) An electro-hydraulic flight control system.
 - (b) An hydro-mechanical flight control system.
 - (c) An electro-hydro-mechanical flight control system.
 - (d) A purely mechanical flight control system.
62. What does *alternate law* mean for a flight control system?
- (a) It is a flight control reconfiguration law in case there are failures that hinder the *normal control law*.
 - (b) It is a flight control law that reverses the action of the commands.
 - (c) It is a flight control law that powers the electronics with alternate current AC.
 - (d) None of the others.
63. How is redundancy implemented for the hydraulic system of large aircraft?
- (a) With several hydraulic circuits that are different and can exchange power but not fluid.
 - (b) With several hydraulic circuits that are identical and can exchange power but not fluid.
 - (c) With several hydraulic circuits that are different and can exchange both power and fluid.
 - (d) With several hydraulic circuits that are identical and can exchange both power and fluid.
64. How and why are the reservoirs pressurised in an hydraulic system?
- (a) With the pneumatic system, to avoid cavitation.
 - (b) With the pneumatic system, to discharge engine pumps from having to pressurise the hydraulic fluid all on their own.
 - (c) With pumps mounted on the engines gear box, to avoid ice formation.
 - (d) With pumps mounted on the engines gear box, to avoid hydraulic fluid leaks.
65. In a typical electrical system during normal operation, where does DC power come from?
- (a) From the AC circuit, via transformer rectifiers (TR).
 - (b) From nowhere. The DC circuit is only powered in case of an emergency.
 - (c) From synchronous DC generators.
 - (d) From disposable batteries.
66. Which of the following elements CANNOT power the electrical system of an Airbus A320?
- (a) Pneumatic-driven generator.
 - (b) Engine-driven generator.
 - (c) APU-driven generator.
 - (d) Hydraulic-driven generator.
67. Which of the following is NOT an application of the high-pressure air delivered by the pneumatic system?
- (a) Increase engine thrust.
 - (b) Pressurise hydraulic reservoirs.
 - (c) Air conditioning.
 - (d) Engine starting.
68. What does the acronym FADEC, the system in charge of full engine management, stand for?
- (a) Full Authority Digital Engine Control.
 - (b) Fully Automatic Digital Engine Control.
 - (c) Full Authority Digital Engine Computer.
 - (d) Fully Automatic Digital Engine Computer.
69. What is the use of thrust reversal?
- (a) Increase aircraft braking capabilities during landing.
 - (b) Recover energy when the aircraft reduces speed.
 - (c) In-flight efficient speed reduction.
 - (d) To take off backwards.
70. What is true about the engine anti-ice system of a big airplane?
- (a) Its main aim is to avoid the formation of ice rocks on the nacelle that can damage the fan as they are released and swallowed by the engine.
 - (b) Its main aim is to avoid the formation of ice that would make the airplane too heavy to sustain flight.
 - (c) It is powered by the hydraulic system.
 - (d) Together with the wing anti-ice system they work as a whole in what is known as the integrated anti-ice system.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	a	b	c	c
2	a	b	d	d	b
3	a	c	a	c	a
4	a	c	a	a	c
5	a	c	c	c	d
6	a	d	c	b	a
7	a	c	c	a	d
8	a	d	a	a	d
9	a	c	d	c	d
10	a	b	b	c	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	b	d	a
12	a	c	a	a	c
13	a	a	b	b	b
14	a	d	a	d	b
15	a	b	b	b	c

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	a	b	d
17	a	c	d	a	a
18	a	b	d	a	b
19	a	b	d	a	a
20	a	b	b	c	c
21	a	b	d	c	b
22	a	a	c	c	b
23	a	c	a	b	b
24	a	d	b	b	d
25	a	c	c	b	c

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	b	a	c	a
27	a	d	c	c	c
28	a	a	d	a	d
29	a	d	b	b	d
30	a	b	d	a	c

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	c	c	c	a
32	a	d	a	c	c
33	a	d	d	c	a
34	a	b	c	d	c
35	a	a	a	c	d
36	a	d	d	a	d
37	a	d	b	d	a
38	a	b	c	a	d
39	a	d	c	b	c
40	a	d	c	b	c

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
41	a	d	c	d	c
42	a	a	b	b	b
43	a	c	a	a	a
44	a	a	a	a	c
45	a	c	c	b	c
46	a	a	c	c	d
47	a	b	b	a	a
48	a	c	b	b	d
49	a	a	b	a	c
50	a	d	a	d	d

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
51	a	d	b	c	b
52	a	d	a	d	d
53	a	b	a	c	d
54	a	d	a	b	a
55	a	d	c	a	c
56	a	a	a	a	d
57	a	a	b	b	a
58	a	a	b	c	c
59	a	d	a	d	d
60	a	d	c	b	d

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	b	a	d	d
62	a	a	c	d	a
63	a	b	b	c	c
64	a	d	c	b	b
65	a	b	b	c	a
66	a	c	c	c	b
67	a	b	c	a	d
68	a	d	b	d	b
69	a	d	b	a	a
70	a	c	d	c	d

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. A lucky student is planning to go to ski and is thinking between taking his father car driven by a Diesel engine or his mother one, driven by an Otto engine. Assuming that both cars are identical and that the only difference is the engine, he must know that:
 - (a) the Otto engine emits more CO₂ than the Diesel engine but the Diesel engine sends out more carbon particles than the Otto one.
 - (b) will get the destination sooner riding a donkey.
 - (c) Otto engine has a diffusive flame while Diesel engine has a premixed one.
 - (d) Otto engine uses a rich oxygen mixture while the Diesel one is stoichiometric.
2. Which of the following engines has a highest bypass ratio?
 - (a) Turboprop
 - (b) Turbojet
 - (c) Low bypass ratio turbofan
 - (d) High bypass ratio turbofan
3. The most appropriate engine options to fly at M=1.5 are:
 - (a) Turbojet or low bypass turbofan
 - (b) Turbofan (all kind)
 - (c) Ramjet or turbojet
 - (d) High bypass turbofan or turboprops
4. From the entrance to the exit of the gases, a 2 spool turbojet is composed by
 - (a) diffuser, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (b) nozzle, low pressure compressor, high pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, diffuser
 - (c) diffuser, high pressure compressor, low pressure compressor, combustion chamber, high pressure turbine, low pressure turbine, nozzle
 - (d) nozzle, high pressure compressor, low pressure compressor, combustion chamber, low pressure turbine, high pressure turbine, diffuser
5. To increase the thrust of a propeller engine without anticipating compression phenomena, you should...
 - (a) increase the number of blades.
 - (b) decrease the number of blades.
 - (c) increase the length of the blades.
 - (d) decrease the length of the blades.
6. The Coanda effect...
 - (a) is used to create thrust along the tail boom to compensate the torque created by the main rotor in NOTAR helicopters.
 - (b) is used to create thrust along the tail boom to compensate the torque created by the main rotor in coaxial main rotor helicopters.
 - (c) is the air flux expelled at the end of the tail of a NOTAR helicopter
 - (d) is the flux interaction created between the two rotors of a coaxial main rotor helicopter.
7. Which of the following statements are true?
 - I An autogiro can perform stationary flight.
 - II The rotors of tandem helicopter are counter-rotating.
 - III The rotors of a coaxial helicopter are contra-rotating.
 - (a) II and III
 - (b) I
 - (c) II
 - (d) I and III
8. When will helicopters fly faster than fixed wing aircrafts?
 - (a) Never.
 - (b) In less than 20 years.
 - (c) Once new powerful engines are developed.
 - (d) Once new high specific power engines are developed.
9. Ion electric rocket engines are appropriate for
 - I deep space missions.
 - II orbit maintenance mission.
 - III launch mission.
 - (a) I and II
 - (b) I
 - (c) II
 - (d) I and III
10. To launch a satellite into geostationary orbit, the launcher flies to the
 - (a) East.
 - (b) West.
 - (c) North-East.
 - (d) North-West

BLO-6: Concepts on avionics and navigation

11. An anemometer measures indicated airspeed using:
- (a) both stagnation pressure from a pitot tube and static pressure from static ports.
 - (b) only static pressure from static ports.
 - (c) only stagnation pressure from a pitot tube.
 - (d) static pressure from static ports and air temperature.
12. To have the altimeter read true height of the aircraft above a nearby airport, which is the altimeter setting of choice?
- (a) QFE.
 - (b) QNE.
 - (c) QNH.
 - (d) None of the others.
13. The Transition Altitude (TA) is the altitude at which the pilot shifts the calibration of the altimeter:
- (a) From QNH to STD
 - (b) From STD to QNH
 - (c) From QFE to QNH
 - (d) From QNH to QFE
14. The ability of a system to guarantee correctness of operation and to issue due warning in case of impairment to provide such guarantee is called
- (a) integrity
 - (b) accuracy
 - (c) continuity
 - (d) availability
15. What is a Satellite Based Augmentation System (SBAS)?
- (a) A navigation system that enhances GPS / GLONASS systems performance.
 - (b) None of the others.
 - (c) A communications system based on geostationary satellites.
 - (d) A surveillance system based on low earth orbit satellites.
16. Which navigation system provides both azimuthal and vertical guidance for a precision approach?
- (a) ILS
 - (b) Glide Path
 - (c) Markers
 - (d) A localizer
17. An aircraft's heading indicator reads 150° while the VOR indicates TO with the CDI needle perfectly centred when tuned with radial 330° . What is the airplane doing?
- (a) Flying away from the VOR station.
 - (b) Flying towards the VOR station.
 - (c) Flying around the VOR station.
 - (d) We need to couple this information with a DME reading to find out what the airplane is doing.
18. What name receives the angle between the magnetic and the geographic north of the earth?
- (a) Magnetic declination.
 - (b) Magnetic deviation.
 - (c) Geographic drift.
 - (d) Geographic setting.
19. What is a Secondary Surveillance Radar (SSR)?
- (a) A ground based system that allows identification and tracking of aircraft by issuing an interrogation signal that is actively responded by onboard equipment.
 - (b) A ground based system that allows identification and tracking of aircraft by issuing an interrogation signal that passively bounces on the aircraft surface.
 - (c) An onboard system that allows identification of airspace obstacles by issuing an interrogation signal that passively bounces on these obstacles surface.
 - (d) None of the others.
20. Which of the following navigation systems accumulates error with time?
- (a) INS
 - (b) ILS
 - (c) DME
 - (d) VOR

BLO-7: Concepts on materials and aeronautical regulations

21. The simplest scheme of a wing corresponds to
- (a) a beam with a rigid union in one end while the opposite one is free.
 - (b) a beam with a rigid union in one end while the opposite one is articulated.
 - (c) an articulated beam in both the ends.
 - (d) a rigid union in both the ends.
22. About biplane wings, which of the following statements are true?
- I Wires were used by biplane aircraft to make their wings stronger.
 - II The usage of wires reduced the drag with respect to the usage of fatter wings.
 - III World War I biplane wings were mostly made with wood and cotton fabric.
- (a) I and III
 - (b) II and III
 - (c) II
 - (d) III
23. In an optimal structure...
- I the stresses are balanced in such a way that the stresses on all structural members are equal.
 - II there are no structural members loafing while others are overworking.
 - III flying and stationary stresses are opposite but with the same modulus.
- (a) I and II
 - (b) II and III
 - (c) II
 - (d) I
24. The wing of an airplane in horizontal flight
- (a) has its upper part under compression stresses
 - (b) has its upper part under traction stresses
 - (c) has its lower part under compression stresses
 - (d) supports less weight than the one supported on ground.
25. To choose the appropriate material you must be aware of
- (a) all the other statements are correct.
 - (b) its price.
 - (c) the environment in which it will be used.
 - (d) life time expected.
26. Which of the following ICAO annexes contains the standards and recommended practices regarding Aeronautical Information Services?
- (a) 15
 - (b) 14
 - (c) 1
 - (d) 3
27. The organisation founded in 1960 whose original aim was to unify Control and Navigation air services in Europe is...
- (a) EUROCONTROL
 - (b) EASA
 - (c) FAA
 - (d) ICAO
28. The set of requirements that every european aircraft must fulfil to be given the certificate of airworthiness is called...
- (a) JAR
 - (b) FAR
 - (c) JAA
 - (d) FAA
29. Which of the following types of aircraft are considered in the agreements following the Chicago convention of 1944?
- (a) Civil
 - (b) Customs
 - (c) Police
 - (d) Military
30. Which one of the following organisations is a direct result of the Chicago convention of 1944?
- (a) ICAO
 - (b) EUROCONTROL
 - (c) EASA
 - (d) DGAC

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control system options offers a clear advantage in terms of weight for large airplanes?
- (a) An electro-hydraulic flight control system
 - (b) An hydro-mechanical flight control system
 - (c) An electro-hydro-mechanical flight control system
 - (d) A purely mechanical flight control system
32. Which one of the following elements of a flight control system are essential to the implementation of flight augmentation capabilities and automatic control?
- (a) All of them.
 - (b) Hydraulic and/or electromechanic actuators.
 - (c) Sensors (gyros, accelerometers, transducers...)
 - (d) Computers (FCC, SEC, ELAC, FAC)
33. How is the hydraulic system powered in an Airbus A320 type aircraft during normal operation?
- (a) With engine-driven pumps.
 - (b) With the pneumatic system.
 - (c) With the auxiliary power unit (APU).
 - (d) With the ram air turbine (RAT).
34. What are the reservoirs for in an hydraulic system?
- (a) To store hydraulic fluid that is not being used by the system at any particular time.
 - (b) To transfer pressurised fluid among hydraulic circuits.
 - (c) To transfer power among hydraulic circuits.
 - (d) To absorb the transient demand of pressurised fluid during intensive hydraulic circuit operation.
35. In a typical electrical system, where does the AC BUS 1 get the power from if all AC generation is lost during flight?
- (a) From nowhere. Both AC BUS 1 and AC BUS 2 would be disconnected.
 - (b) From an electrical emergency generator powered by the hydraulic system.
 - (c) From the RAT via the AC ESS BUS.
 - (d) From the batteries and the DC ESS bus through DC/AC transformation.
36. Which of the following elements CANNOT be found in an electrical system?
- (a) A power transfer unit (PTU).
 - (b) A transformer/rectifier.
 - (c) Buses.
 - (d) Batteries.
37. Which of the following is a component of the pneumatic system of an Airbus A320 type aircraft?
- (a) The crossbleed duct and valve.
 - (b) A static inverter.
 - (c) An accessory gearbox.
 - (d) Nozzle guide vanes.
38. What does the acronym FADEC, the system in charge of full engine management, stand for?
- (a) Full Authority Digital Engine Control.
 - (b) Fully Automatic Digital Engine Control.
 - (c) Full Authority Digital Engine Computer.
 - (d) Fully Automatic Digital Engine Computer.
39. What is usually NOT a direct aim of the Auxiliary Power Unit (APU) while on ground?
- (a) Drive the fuel pumps to feed the engines.
 - (b) Supply bleed air for air conditioning.
 - (c) Supply bleed air for engine start.
 - (d) Provide electrical power.
40. What is true about the wing anti-ice system of a big airplane?
- (a) It is usually based on hot air circulation, and sometimes on electric power dissipation.
 - (b) Its main aim is to avoid the formation of ice that would make the airplane too heavy to sustain flight.
 - (c) It is powered by the hydraulic system.
 - (d) It includes an anti-skid system so that blue-collars do not slip during maintenance walks on the wing.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	b	c	b	c
2	a	c	a	b	a
3	a	d	b	b	d
4	a	d	d	d	d
5	a	c	c	a	b
6	a	b	c	b	b
7	a	b	c	a	a
8	a	d	d	c	d
9	a	b	c	d	b
10	a	a	b	c	b

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	d	b	b	b
12	a	d	d	a	d
13	a	b	b	d	c
14	a	c	d	d	d
15	a	c	b	d	b
16	a	c	b	b	b
17	a	d	b	a	b
18	a	c	a	b	b
19	a	c	d	b	c
20	a	d	c	c	c

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	b	b	b	c
22	a	b	b	b	d
23	a	c	d	a	a
24	a	b	c	a	a
25	a	d	b	b	a
26	a	b	a	c	b
27	a	a	a	a	d
28	a	b	c	c	a
29	a	b	b	d	b
30	a	b	d	d	c

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	c	d	a	d
32	a	a	c	a	c
33	a	c	d	b	c
34	a	d	a	a	c
35	a	b	b	a	c
36	a	a	b	c	c
37	a	d	a	d	b
38	a	b	c	c	d
39	a	b	b	b	a
40	a	a	c	d	d

BLO-1: Concepts on flight principles and aircraft performance

1. What is the main effect on an aerofoil lift characteristic curve ($C_L(\alpha) = C_{L\alpha}(\alpha - \alpha_0)$ for $\alpha < \alpha_S$) of a high-lift device that only increases the chord of the airfoil?
 - (a) It increases the slope $C_{L\alpha}$.
 - (b) It shifts the curve up by decreasing α_0 .
 - (c) It increases the stall angle α_S .
 - (d) It reduces $C_D(\alpha)$ but does nothing to $C_L(\alpha)$.
2. Which of the following parameters is roughly independent of velocity?
 - (a) Maximum aerodynamic efficiency.
 - (b) Maximum available thrust and power.
 - (c) Maximum bank angle for a coordinated turn.
 - (d) Maximum achievable lift.
3. How does an airfoil generate lift?
 - (a) All options are correct.
 - (b) By deflecting air downwards.
 - (c) By creating a pressure difference between upper and lower surfaces.
 - (d) Through the effects of viscosity, by introducing a circulation in the airflow.
4. Which of the following manoeuvres entails a higher load factor...
 - (a) Coordinated banked turn.
 - (b) Straight and level flight.
 - (c) Balanced ascent.
 - (d) Glide.
5. Which of the following drag sources is exclusive to 3D wings?
 - (a) Lift-induced drag.
 - (b) Friction drag.
 - (c) Form drag.
 - (d) Wave drag.
6. Which velocity sets the dividing line between the first (R1) and second (R2) flight regimes?
 - (a) Minimum drag velocity for jet-powered aircraft and minimum required power velocity for propeller aircraft.
 - (b) Minimum drag velocity.
 - (c) Minimum required power velocity.
 - (d) Stall velocity.
7. Which of the following factors needs NOT be considered upon airport runway design?
 - (a) All factors mentioned require consideration.
 - (b) Wind intensity and preferred direction at airport location.
 - (c) Type of aircraft that will operate in the airport.
 - (d) Airport altitude.
8. Which of the following statements is true about the velocities for best angle ($V_{\gamma_{max}}$) and for maximum rate of climb ($V_{(R/C)_{max}}$)?
 - (a) $V_{\gamma_{max}} < V_{(R/C)_{max}}$
 - (b) $V_{\gamma_{max}} = V_{(R/C)_{max}}$
 - (c) $V_{\gamma_{max}} > V_{(R/C)_{max}}$
 - (d) There is no way to know which one of $V_{\gamma_{max}}$ and $V_{(R/C)_{max}}$ is greater because it depends on the type of engine.
9. Why do air passenger carriers usually fly at speeds faster than that for optimum range?
 - (a) Because time-to-destination may be considerably reduced at a small fuel consumption penalty.
 - (b) Because optimum endurance is a better choice.
 - (c) Because fuel consumption per distance unit is smaller at higher velocities than that for optimum range.
 - (d) Airliners never do that.
10. Which of the following factors is responsible for the absolute ceiling of both jet-engined and propeller-driven aircraft?
 - (a) Maximum available thrust/power decreases with altitude.
 - (b) Solar radiation increases with altitude.
 - (c) Minimum required thrust/power increases with altitude.
 - (d) All of them.

BLO-2: Flight principles related problems

11. What is the drag coefficient ($C_{D_{opt}}$) for the optimal condition that maximises aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.025 + 0.125C_L^2$?
 - (a) $C_{D_{opt}} = 0.05$
 - (b) $C_{D_{opt}} = 0.025$
 - (c) $C_{D_{opt}} = 0.447$
 - (d) $C_{D_{opt}} = 0.0375$

12. An aerobatics airplane with lift coefficient $C_L = 0.15 + 0.12\alpha$ with α in degrees ($^\circ$) engages in a perfect dive (vertical descent). What should the angle of attack (α) be?
 - (a) $\alpha = -1.25^\circ$
 - (b) $\alpha = 1.25^\circ$
 - (c) $\alpha = 0.8^\circ$
 - (d) $\alpha = -0.8^\circ$

13. A small airplane of unknown mass is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}). We know its speed is $V = 2V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max})?
 - (a) $\phi_{max} = 75.5^\circ$
 - (b) $\phi_{max} = 48.2^\circ$
 - (c) $\phi_{max} = 60^\circ$
 - (d) It depends on the mass of the airplane.

14. The drag polar of an airplane changes from $C_D = 0.02 + 0.12C_L^2$ in clean configuration to $C_D = 0.04 + 0.08C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.707$
 - (b) $C_L = 0.681$
 - (c) $C_L = 0.752$
 - (d) $C_L = 0.64$

15. An airplane of mass $m = 1000$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 40 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $D = 250$ N and $D_p = D_i = 125$ N.
 - (b) $D = 250$ N and $D_p > D_i$.
 - (c) $D = 125$ N and $D_p = D_i = 62.5$ N.
 - (d) $D = 125$ N and $D_p > D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which command and primary control surface pilot the aircraft around its yaw axis?
- The pedals and the rudder.
 - The stick (or yoke) and the elevator.
 - The stick (or yoke) and the ailerons.
 - The throttle and the flaps.
17. Which parameters have a disengaging (stabilising) effect on the spiral mode?
- Positive dihedral (wings up, $C_{l_\beta} < 0$).
 - High directional stability (big fin, $C_{n_\beta} > 0$).
 - High roll damping (big wing span, $C_{l_p} < 0$).
 - None of the others.
18. About static and dynamic stability...
- To have static stability a restoring force is required but not necessarily a damping force.
 - To have dynamic stability a damping force is required but not necessarily a restoring force.
 - To have dynamic stability a restoring force is required but not necessarily a damping force.
 - To have static stability a damping force is required but not necessarily a restoring force.
19. What are the risks of having an extremely stable airplane?
- It will be hard to manoeuvre.
 - None. Stabilising airplanes is a painstaking task already because they tend to be intrinsically unstable.
 - It is more expensive to design and produce.
 - None. It only has advantages.
20. Sort the longitudinal modes from faster to slower.
- Pitch oscillation, phugoid and return to straight and level flight.
 - Pitch oscillation, return to straight and level flight and phugoid.
 - Phugoid, pitch oscillation and return to straight and level flight.
 - Phugoid, return to straight and level flight and pitch oscillation.
21. What can be said about the bank angle of two level turns with equal angular velocity, one coordinated (ϕ_{CT}) and the other operational (ϕ_{OT} , skidding)?
- $\phi_{CT} < \phi_{OT}$.
 - $\phi_{CT} > \phi_{OT}$.
 - $\phi_{CT} = \phi_{OT}$.
 - Which bank angle is larger depends on many factors.
22. What can be said about the response of the airplane to a perturbation in yaw?
- It depends on static directional stability, since there automatically appears a restoring yaw moment.
 - It depends on both directional and lateral stability, since there is no restoring moment and the return to straight and level flight is mostly dynamical.
 - It depends mainly on static lateral stability, since there automatically appears a restoring roll moment.
 - It depends mainly on static longitudinal stability, since there automatically appears a restoring pitch moment.
23. What is true about the pitch oscillation mode?
- It is fast and usually periodic.
 - It corresponds to the periodic exchange between potential and kinetic energy.
 - It couples pitch and yaw motion.
 - It can be left unstable without critically compromising the aircraft integrity.
24. How is crosswind straight flight (with sideslip) piloted?
- Mostly with the pedals, using the stick to compensate for the induced roll.
 - Mostly with the stick, using the pedals to compensate for the induced yaw.
 - Only with the pedals.
 - Only with the stick.
25. What is true about trim tabs?
- They are meant to assist the pilot by easing the continuous effort on the commands that may be required during certain phases of flight.
 - Rudder trim tabs are as common as elevator trim tabs in small aircraft.
 - They are tablets that must be carried onboard the aircraft for the crew to keep trim after a long flight.
 - They are meant to increase aircraft manoeuvrability.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.2\bar{\rho}V^3 + 0.04W^2/\bar{\rho}V$. If the airplane has mass $m = 1500\text{ kg}$ and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the required power (P_R) at the optimal velocity for maximum endurance (V_E)?
- $P_R = 273\text{ kW}$
 - $P_R = 311\text{ kW}$
 - $P_R = 323\text{ kW}$
 - $P_R = 285\text{ kW}$
27. A jet airplane of mass $m = 1600\text{ kg}$ has a drag curve $D = 0.25\bar{\rho}V^2 + 0.05W^2/\bar{\rho}V^2$. What is the minimum engine thrust needed at sea level ($\bar{\rho} = 1$) to achieve a climb angle of $\alpha = 4^\circ$? (take $g \simeq 10\text{ ms}^{-2}$)
- $T_A = 4690\text{ N}$
 - $T_A = 5250\text{ N}$
 - $T_A = 5540\text{ N}$
 - $T_A = 6030\text{ N}$
28. An airplane with drag polar $C_D = 0.03 + 0.12C_L^2$, wing surface $S = 40\text{ m}^2$ and mass $m = 2500\text{ kg}$ is flying at a speed $V = 120\text{ ms}^{-1}$. What is the minimum engine power required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4\text{ kgm}^{-3}$? (take $g \simeq 10\text{ ms}^{-2}$)
- $P_R = 493\text{ kW}$
 - $P_R = 437\text{ kW}$
 - $P_R = 387\text{ kW}$
 - $P_R = 338\text{ kW}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 50$. If its current altitude is $h = 3000\text{ m}$, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 150\text{ km}$
 - $d = 60\text{ km}$
 - $d = 3\text{ km}$
 - $d = 300\text{ km}$
30. The required thrust of a jet-engined airplane of mass $m = 2000\text{ kg}$, wing area $S = 20\text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.64\bar{\rho}V^2 + 0.008W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 9200\text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10\text{ ms}^{-2}$ and $\rho_0 = 1.225\text{ kgm}^{-3}$)
- $V \simeq 69.2\text{ ms}^{-1}$
 - $V \simeq 64.2\text{ ms}^{-1}$
 - $V \simeq 97.9\text{ ms}^{-1}$
 - $V \simeq 82.3\text{ ms}^{-1}$

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	b	a	a	d
2	a	c	b	a	b
3	a	a	a	b	a
4	a	b	a	b	c
5	a	a	d	d	c
6	a	d	b	d	b
7	a	a	d	a	d
8	a	b	d	d	a
9	a	b	c	b	c
10	a	a	a	a	b

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	d	d	a
12	a	a	d	d	a
13	a	d	c	c	b
14	a	a	a	b	d
15	a	a	a	c	b

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	c	d	b	c
17	a	c	d	b	d
18	a	d	b	d	d
19	a	d	a	b	c
20	a	b	a	d	b
21	a	b	d	d	d
22	a	b	b	d	d
23	a	c	a	b	a
24	a	d	a	c	c
25	a	b	b	c	a

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	a	d	c
27	a	b	c	a	b
28	a	a	c	a	d
29	a	a	b	a	d
30	a	b	a	d	b

BLO-1: Concepts on flight principles and aircraft performance

1. What is the usual effect of flap deployment on an aircraft drag polar ($C_D(C_L) = C_{D_0} + k C_L^2$)?
 - (a) Increase C_{D_0} and reduce k .
 - (b) Reduce C_{D_0} and increase k .
 - (c) Increase both C_{D_0} and k .
 - (d) Reduce both C_{D_0} and k .
2. Which of the following lift-related aerodynamic parameters is roughly independent of flight altitude?
 - (a) All of them can be considered independent of altitude.
 - (b) Lift characteristics slope C_{L_α} .
 - (c) 0-aoa lift coefficient C_{L_0} .
 - (d) Stall aoa α_S .
3. Which non-dimensional parameter determines the flow regime with regard to the effects of compressibility (incompressible, subsonic compressible, transonic, supersonic, hypersonic)?
 - (a) The Mach number.
 - (b) The Reynolds number.
 - (c) The Prandtl number.
 - (d) The Bernoulli number.
4. How do winglets increase the aerodynamic efficiency of some aircraft?
 - (a) By reducing lift-induced drag.
 - (b) By reducing parasitic drag.
 - (c) They do not increase efficiency, they are just used because they are trendy.
 - (d) By increasing lift.
5. What is the definition of load factor n ?
 - (a) It is the lift-to-weight ratio $n = L/W$.
 - (b) It is the drag-to-thrust ratio $n = D/T$.
 - (c) It is the lift-to-drag ratio $n = L/D$.
 - (d) It is the thrust-to-weight ratio $n = T/W$.
6. Which of the following statements is true about the first (R1) flight regime?
 - (a) Parasitic drag/power outweighs lift-induced drag/power.
 - (b) Airspeeds are slower than for the second regime (R2).
 - (c) Drag/power decreases with increasing airspeed.
 - (d) All options are true.
7. Which of the following parameters has an influence on takeoff and landing distances?
 - (a) All of them.
 - (b) Airport altitude.
 - (c) Wind.
 - (d) Airplane weight, thrust and braking capabilities.
8. Which of the following statements is true about climb?
 - (a) All statements are true.
 - (b) The best climb angle corresponds to maximum excess thrust.
 - (c) The maximum rate of climb corresponds to maximum excess power.
 - (d) For a jet powered airplane, the best climb angle roughly corresponds to minimum drag conditions.
9. Why do aircraft often cruise at speeds slightly above maximum range conditions?
 - (a) Because the benefit in time-to-destination outweighs the small penalty in fuel consumption.
 - (b) Because this reduces total fuel consumption.
 - (c) Because the usual choice is maximum endurance, which corresponds to higher speed.
 - (d) Typical aircraft never cruise at speeds above maximum range.
10. An airplane in straight and level flight engages in a coordinated banked turn at constant altitude. What can be said about its stall speed?
 - (a) It increases.
 - (b) It diminishes.
 - (c) It stays the same.
 - (d) It increases or diminishes depending on the engine type.

BLO-2: Flight principles related problems

11. What is the optimal drag coefficient ($C_{D_{f_{max}}}$) that maximises aerodynamic efficiency (f_{max}) for an airplane with drag polar $C_D = 0.035 + 0.15C_L^2$?
- (a) $C_{D_{f_{max}}} = 0.07$
 - (b) $C_{D_{f_{max}}} = 0.035$
 - (c) $C_{D_{f_{max}}} = 0.483$
 - (d) $C_{D_{f_{max}}} = 0.0525$
12. An aerobatics airplane with lift coefficient $C_L = 0.05 + 0.1\alpha$ with α in $^\circ$ is performing a balanced inverted flight. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.3$?
- (a) $\alpha = -3.5^\circ$
 - (b) $\alpha = 2.5^\circ$
 - (c) $\alpha = -0.5^\circ$
 - (d) $\alpha = 3.5^\circ$
13. A small airplane with mass $m = 1200$ kg is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}) at an unknown altitude. We know its speed is $V = 2 V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the lift (L) if the plane is to keep its altitude and speed? (take $g \simeq 10 \text{ ms}^{-2}$)
- (a) $L = 48000$ N
 - (b) $L = 45600$ N
 - (c) $L = 24000$ N
 - (d) It depends on the altitude.
14. The drag polar of an airplane changes from $C_D = 0.03 + 0.14C_L^2$ in clean configuration to $C_D = 0.04 + 0.07C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
- (a) $C_L = 0.378$
 - (b) $C_L = 0.399$
 - (c) $C_L = 0.333$
 - (d) $C_L = 0.443$
15. An airplane of mass $m = 1200$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 25 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
- (a) $D = 480$ N and $D_p = D_i = 240$ N.
 - (b) $D = 480$ N and $D_p > D_i$.
 - (c) $D = 240$ N and $D_p = D_i = 120$ N.
 - (d) $D = 240$ N and $D_p > D_i$.

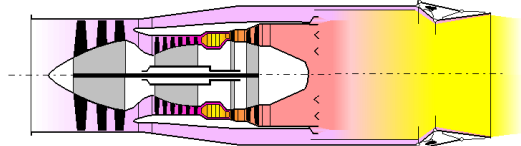
BLO-3: Concepts on aircraft stability and control

16. Which command and primary control surface pilot the aircraft around its roll axis?
- (a) The stick (or yoke) and the ailerons.
 - (b) The pedals and the rudder.
 - (c) The stick (or yoke) and the elevator.
 - (d) The throttle and the flaps.
17. Which aerodynamic element of the airplane design is usually set to tune the spiral mode?
- (a) The wing dihedral, through C_{l_β} .
 - (b) The vertical stabiliser, through C_{n_β} .
 - (c) The horizontal stabiliser, through C_{m_α} .
 - (d) The fuselage, through C_{n_r} .
18. About static and dynamic stability...
- (a) To have dynamic stability we need both a restoring force/moment and a damping force/moment.
 - (b) They refer to stability around the pitch axis only.
 - (c) Static stability implies dynamic stability.
 - (d) To have static stability we need a damping force/moment but not necessarily a restoring force/moment.
19. About stability and manoeuvrability...
- (a) To have high manoeuvrability a bit of stability must be sacrificed.
 - (b) They are totally unrelated concepts.
 - (c) To have high manoeuvrability, stability must also be high.
 - (d) They are synonyms.
20. What is true about the roll subsidence mode?
- (a) Its damping strongly depends on the wings aspect ratio (C_{l_p} , the roll moment due to roll rotational velocity).
 - (b) It can be left unstable.
 - (c) It couples rotation about the yaw and roll axes.
 - (d) It is strongly dependent on the size and distance of the vertical stabiliser.
21. Which of the following equilibria involve action on both pedals and stick?
- (a) Coordinated banked turn and crosswind straight flight.
 - (b) Operational turn and straight and level flight.
 - (c) Coordinated banked turn and best angle climb.
 - (d) Glide and operational turn.
22. How can (and should) the pilot partially compensate for the sideslip induced by an engine failure in a twin-engined airplane?
- (a) Deflecting the rudder towards the working engine.
 - (b) Deflecting the rudder towards the failed engine.
 - (c) There is nothing that can be done.
 - (d) Switching off the other engine.
23. Which of the following longitudinal modes is usually fastest?
- (a) Pitch oscillation.
 - (b) Phugoide.
 - (c) Return to straight and level flight.
 - (d) Dutch Roll.
24. Which parameter (and coefficient) are crucial to longitudinal stability?
- (a) The relative position of the aerodynamic centre and the centre of gravity (C_{m_α}).
 - (b) The size and position of the vertical stabiliser (C_{n_β}).
 - (c) The wing dihedral (C_{l_β}).
 - (d) None of the others.
25. What are trim tabs for?
- (a) None of the other options is true.
 - (b) Increase stability.
 - (c) Tune manoeuvrability.
 - (d) Prevent stall.

BLO-4: Aircraft performance related problems

26. The required power of a propeller-powered airplane is given by $P_R = 0.24\bar{\rho}V^3 + 0.018W^2/\bar{\rho}V$. If the airplane has mass $m = 1400$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)?
- $V_E = 66.5 \text{ ms}^{-1}$
 - $V_E = 87.6 \text{ ms}^{-1}$
 - $V_E = 115 \text{ ms}^{-1}$
 - It depends on the available power P_A , which is unknown.
27. A jet-powered airplane of mass $m = 1300$ kg has a required thrust curve $T_R = 0.3\bar{\rho}V^2 + 0.03W^2/\bar{\rho}V^2$. What is the velocity for best angle ($V_{\gamma_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 10 \text{ ms}^{-2}$)
- $V_{\gamma_{max}} = 64.1 \text{ ms}^{-1}$
 - $V_{\gamma_{max}} = 48.7 \text{ ms}^{-1}$
 - $V_{\gamma_{max}} = 84.4 \text{ ms}^{-1}$
 - $V_{\gamma_{max}} = 37 \text{ ms}^{-1}$
28. An airplane with drag polar $C_D = 0.04 + 0.15C_L^2$, wing surface $S = 40 \text{ m}^2$ and mass $m = 1800$ kg is flying at a speed $V = 100 \text{ ms}^{-1}$. What is the minimum engine power required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4 \text{ kgm}^{-3}$? (take $g \simeq 10 \text{ ms}^{-2}$)
- $P_R = 381 \text{ kW}$
 - $P_R = 334 \text{ kW}$
 - $P_R = 307 \text{ kW}$
 - $P_R = 504 \text{ kW}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 50$. If its current altitude is $h = 500$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 25 \text{ km}$
 - $d = 10 \text{ km}$
 - $d = 0.5 \text{ km}$
 - $d = 50 \text{ km}$
30. The required thrust of a jet-engined airplane of mass $m = 1900$ kg, wing area $S = 25 \text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.5\bar{\rho}V^2 + 0.01W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 20000 \text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
- $V \simeq 56 \text{ ms}^{-1}$
 - $V \simeq 41.6 \text{ ms}^{-1}$
 - $V \simeq 196 \text{ ms}^{-1}$
 - $V \simeq 90.2 \text{ ms}^{-1}$

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

31. Which of the following engines has better thermal efficiency?
- 4 stroke Diesel engine.
 - 4 stroke Otto engine.
 - 2 stroke Otto engine.
 - None of the other statements is true.
32. Which of the following engines has better propulsive efficiency if it is used on an aircraft?
- The propulsive efficiency depends on the propeller.
 - 4 stroke Diesel engine.
 - 4 stroke Otto engine.
 - 2 stroke Otto engine.
33. A helicopter in horizontal flight,
- the distribution of the main rotor speeds induces a roll momentum in the helicopter.
 - the distribution of the main rotor speeds is proportional to the radius.
 - the retreating blade moves up to compensate the reverse flow zone.
 - there is no reverse flow in the main rotor.
34. About a propeller,
- the angle of attack is positive when the aircraft is taking off.
 - the advance ratio decreases if the aircraft speed is increased.
 - the longer it is, the faster it will fly.
 - None of the other statements is true.
35. Which of the following statements is true?
- None of the other statements is true.
 - The efficiency of the propeller is proportional to the number of blades it has.
 - The more blades the propeller has, the minor are the air perturbations created
 - A turbofan uses an external propeller.
36. An airplane is flying at a speed of 100 m/s at an altitude for which the atmospheric pressure is 0.265 bar. At the exit of the nozzle, with a diameter of 0.5 m, the pressure of the gases is 2 bar and their speed is 1275 m/s. If the fuel consumption is 2.65 kg/s and the air massflow into the engine is 45 kg/s, how much is its thrust-specific fuel consumption (TSFC)?
- 0.029 g/N
 - 0.032 g/N
 - 0.047 g/N
 - None of the others.
37. The orbit suggested for a GPS satellite mission is a
- medium Earth orbit.
 - geostationary orbit.
 - polar orbit.
 - geosynchronous orbit.
38. Which of the following aircrafts has the highest flight speed?
- A turbojet powered airplane.
 - A turbofan powered airplane.
 - A helicopter.
 - An autogyro.
39. What kind of aircraft engine scheme can you distinguish in the figure?
- 
- Low by-pass ratio turbofan with post-combustion.
 - High by-pass ratio turbofan with post-combustion.
 - Double spool turbojet with post-combustion.
 - Double spool turbojet without post-combustion.
40. Which of the following statements is true?
- An autogyro can perform stationary flight.
 - The main rotors of a tandem helicopter are counter-rotating.
 - A convertible aircraft can perform stationary flight.
- II and III are true.
 - I is true.
 - II is true.
 - All the statements are false.

BLO-6: Concepts on avionics and navigation

41. The Transition Altitude (TA) is the altitude at which the pilot shifts the calibration of the altimeter:
- (a) From QNH to STD
 - (b) From STD to QNH
 - (c) From QNH to QFE
 - (d) None of the other answers is correct
42. An aircraft flies in an atmosphere that is colder than the ISA. Thus, for a given variation in altitude, the corresponding variation of pressure is:
- (a) Larger than in the ISA
 - (b) Smaller than in the ISA
 - (c) Equal to that in the ISA
 - (d) We do NOT have enough information to choose which of the other answers is correct
43. Which of the following statements is true?
- (a) The Transition Altitude (TA) is 6000ft in most of the Spanish airports
 - (b) The conditions of the International Standard Atmosphere at Sea Level are the annual average values measured in Greenwich, UK
 - (c) In general, when an aircraft flies with QNH calibration over the arrival airport, the altimeter reading will be similar to the real height
 - (d) None of the other answers is correct
44. In defining the International Standard Atmosphere, ICAO assumed that temperature decreases with altitude:
- (a) 6.5°C every km , in the troposphere
 - (b) 6.5°C every km , in the stratosphere
 - (c) 6.5°C every km , above FL360
 - (d) None of the other answers is correct
45. If the pilot shifts from QNH to STD, and the QNH calibration pressure was 1025 mb, the new altitude indicated by the altimeter with STD calibration is:
- (a) Smaller than the altitude indicated with QNH calibration
 - (b) Larger than the altitude indicated with QNH calibration
 - (c) Equal to the altitude indicated with QNH calibration
 - (d) We do NOT have enough information to choose which of the other answers is correct
46. Which is the main disadvantage of navigation systems based on external references?
- (a) These navigation systems are NOT autonomous
 - (b) The position error is accumulative with time
 - (c) These navigation systems need an initial position
 - (d) None of the other answers is correct
47. For a precision approach:
- (a) It is necessary to provide azimuthal and zenithal guidance, and distance information
 - (b) It is only necessary to provide azimuthal guidance
 - (c) It is only necessary to provide azimuthal and zenithal guidance
 - (d) None of the other answers is correct
48. Which of the following statements is true?
- (a) None of the other answers is correct
 - (b) The NDB-ADF provides altitude information
 - (c) The NDB-ADF provides zenithal guidance
 - (d) The NDB-ADF cannot be used for a non-precision approach
49. The navigation with VOR is based in the emission of:
- (a) An omnidirectional reference signal plus a highly directional variable signal emitted by a set of antennae
 - (b) A series of highly directional signals, emitted by a set of antennae placed orthogonally to the runway axis beyond the runway end
 - (c) A series of signals by a set of antennae in a tower placed 300 m beyond the runway threshold and at 150 m from the runway axis
 - (d) None of the other answers is correct
50. What does accuracy stand for when talking about aircraft navigation systems?
- (a) The degree of conformance of the position measured by the system with the true position of the aircraft at a given time
 - (b) The warranty that the position error will not be higher than some threshold and that, if the threshold is exceeded, an alarm will be received on time
 - (c) The average length of the time window between two alarms issued due to malfunction of the system
 - (d) The percentage of times that the navigation system is available when an aircraft needs it service

51. The wing of an airplane in horizontal leveled flight
- (a) is supporting traction stresses in the lower part.
 - (b) is supporting compression stresses in the lower part.
 - (c) has the same stresses on both parts.
 - (d) None of the other statements is right.
52. Which of the following statements is true?
- (a) A fatigue cycle begins at take off and ends after landing.
 - (b) Aircraft fatigue cycles are counted as the number of bendings the structure suffers.
 - (c) Fatigue does not affect aircraft structure controls.
 - (d) None of the other statements is true.
53. A structural stress
- (a) is the load per area applied to a structure.
 - (b) is equivalent to the load applied to the structure.
 - (c) is the load applied per volume of the structure.
 - (d) is the force applied to the structure when Jupiter and Mars are aligned.
54. Which of the following statements is true?
- (a) An alloy is a solid - solid homogeneous solution of two or more metals or even non metallic elements.
 - (b) Pure metals are very resistant to chemical attacks.
 - (c) A pure metal has better properties than its derived alloys.
 - (d) None of the other statements is true.
55. Which of the following statements is right?
- I Composite structures were used during World War II.
 - II Composite structures are used on secondary structures.
 - III Composite structures are used on primary structures.
- (a) All the statements are right.
 - (b) Only II is right.
 - (c) Only III is right.
 - (d) Only II and III are right.
56. The nationality of an aircraft is given by:
- (a) Registry country.
 - (b) Airline country.
 - (c) Manufacturing country.
 - (d) Captain country.
57. Which of the following organization is in charge of managing and exploiting the Air Traffic Services in Spain?
- (a) AENA
 - (b) JAA
 - (c) EASA
 - (d) EUROCONTROL
58. Which of the following ICAO annexes contains the standards and recommended practices for the Meteorological Service For International Air Navigation?
- (a) 3
 - (b) 2
 - (c) 4
 - (d) 1
59. Which of the following ICAO annexes contains the standards and recommended practices for Personnel Licensing?
- (a) 1
 - (b) 10
 - (c) 3
 - (d) 16
60. Which of the following ICAO annexes contains the standards and recommended practices for Aeronautical Charts?
- (a) 4
 - (b) 5
 - (c) 11
 - (d) 15

BLO-8: Concepts on aircraft systems

61. Which one of the following flight control system options does NOT require an artificial feel system?
- (a) A purely mechanical flight control system
 - (b) An electro-hydraulic flight control system
 - (c) An hydro-mechanical flight control system
 - (d) An electro-hydro-mechanical flight control system
62. What is the main advantage of a fly-by-wire electro-hydraulic flight control system over an electro-hydro-mechanical fcs?
- (a) It is lighter
 - (b) It allows for automatic control and flight augmentation capabilities.
 - (c) It is less dependent on electronics.
 - (d) It has no advantages whatsoever.
63. How are the hydraulic system reservoirs (tanks) pressurised in an Airbus A320 type aircraft?
- (a) With the pneumatic system.
 - (b) With engine-driven pumps.
 - (c) With electrical motors.
 - (d) With the ram air turbine (RAT).
64. How is redundancy achieved for the hydraulic system of a large passenger aircraft?
- (a) by having different hydraulic circuits that can exchange pressure but not hydraulic fluid.
 - (b) by having identical hydraulic circuits isolated from one another.
 - (c) by having identical hydraulic circuits that can exchange both hydraulic fluid and pressure.
 - (d) by having different hydraulic circuits that can exchange hydraulic fluid.
65. Typical electrical systems of large aircraft provide...
- (a) both AC (Alternate Current) power and DC (Direct Current) power.
 - (b) either AC (Alternate Current) power or DC (Direct Current) power, depending on the airplane.
 - (c) only AC (Alternate Current) power.
 - (d) only DC (Direct Current) power.
66. Which of the following elements CANNOT be found in an electrical system?
- (a) A crossfeed valve.
 - (b) A transformer/rectifier.
 - (c) Switches.
 - (d) Batteries.
67. Where can the pneumatic system get the pressurised air from?
- (a) All options are possible.
 - (b) From the engine bleed systems.
 - (c) From the APU load compressor.
 - (d) From a high pressure ground connection.
68. What does the FADEC do?
- (a) Performs complete engine management.
 - (b) Controls ailerons and elevators.
 - (c) Provides AC electrical power.
 - (d) It is the computer in charge of flight augmentation.
69. What is the aim of the Auxiliary Power Unit (APU) in flight?
- (a) Backup the electrical system.
 - (b) It is forbidden to switch it on during flight.
 - (c) Provide extra thrust.
 - (d) Massage the flight crew to relieve stress.
70. What is the main aim of the air conditioning system in large aircraft during normal cruise?
- (a) Heat up the cabin to provide a comfortable temperature.
 - (b) Cool down the cabin to provide a comfortable temperature.
 - (c) Depressurise the cabin.
 - (d) Remove moisture from the air for better cabin comfort.

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	d	d	c
2	a	a	a	c	c
3	a	a	b	b	c
4	a	d	c	b	d
5	a	a	d	d	d
6	a	d	c	c	d
7	a	c	b	c	a
8	a	a	b	d	c
9	a	d	b	d	d
10	a	b	b	c	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	b	a	d	d
12	a	b	b	a	b
13	a	a	c	d	c
14	a	c	c	a	c
15	a	c	d	c	b

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	d	d	a
17	a	a	d	d	a
18	a	c	d	b	b
19	a	d	b	a	b
20	a	c	c	a	a
21	a	d	a	d	c
22	a	d	b	c	c
23	a	b	d	c	c
24	a	d	d	b	a
25	a	b	b	c	d

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	d	c	b	a
27	a	b	a	a	c
28	a	c	b	b	a
29	a	d	c	b	d
30	a	c	a	b	c

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	b	a	d
32	a	b	a	a	a
33	a	d	c	a	b
34	a	c	d	a	c
35	a	b	b	c	c
36	a	b	c	c	d
37	a	b	b	c	d
38	a	b	c	b	c
39	a	c	c	d	b
40	a	a	c	c	c

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
41	a	d	b	d	a
42	a	c	a	c	b
43	a	c	b	d	d
44	a	c	b	a	d
45	a	a	a	d	c
46	a	d	a	a	c
47	a	a	d	c	c
48	a	a	a	b	c
49	a	d	c	a	b
50	a	a	a	b	c

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
51	a	a	b	b	d
52	a	c	b	d	a
53	a	a	a	d	a
54	a	b	d	d	c
55	a	a	d	a	b
56	a	d	b	a	c
57	a	a	b	c	a
58	a	c	c	b	b
59	a	d	b	c	d
60	a	c	b	b	a

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
61	a	b	a	d	b
62	a	b	c	c	d
63	a	c	a	b	c
64	a	d	a	b	c
65	a	b	c	c	b
66	a	b	c	c	c
67	a	c	a	c	b
68	a	d	a	c	b
69	a	b	d	c	c
70	a	b	b	d	c

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. To increase the thrust of a rocket engine
 - (a) both statements are correct.
 - (b) both statement are false.
 - (c) the mass flow may be increased.
 - (d) exhaust gases velocity may be increased.
2. Turboprop engines, which of the following statements is true?
 - (a) a turboprop uses the propeller as thrust reverser.
 - (b) a turboprop needs a gear box to increase the speed of the propeller.
 - (c) a turboprop is less efficient than a turbofan engine.
 - (d) None of the other statements is true.
3. A helicopter in horizontal flight,
 - (a) the distribution of the main rotor speeds induced a roll momentum in the helicopter.
 - (b) the distribution of the main rotor speeds is proportional to the radius.
 - (c) the retreating blade moves up to compensate the reverse flow zone.
 - (d) there is no reverse flow in the main rotor.
4. The propulsive power is
 - (a) the thrust of the engines times the aircraft speed.
 - (b) constant with the aircraft speed.
 - (c) the mass flow times the exhaust gases speed.
 - (d) None of the other statements is true.
5. Which of the following statements is true?
 - (a) None of the other statements is true.
 - (b) The efficiency of the propeller is proportional to the number of blades it has.
 - (c) The more blades the propeller has, the minor are the air perturbations created
 - (d) A turbofan uses an external propeller.
6. The main function of the combustion chamber of a turbojet engine is
 - (a) to increase the temperature of the fluid.
 - (b) to increase the pressure of the fluid.
 - (c) to increase the pressure and the temperature of the fluid.
 - (d) None of the other statement is true.
7. The orbit suggested for a GPS satellite mission is a
 - (a) medium Earth orbit.
 - (b) geostationary orbit.
 - (c) polar orbit.
 - (d) geosynchronous orbit.
8. Which of the following engines emits more uncombusted hydrocarbons?
 - (a) Two strokes Otto engine.
 - (b) Two strokes Diesel naval engine.
 - (c) Four strokes Diesel engine.
 - (d) Four strokes Otto engine.
9. What kind of flame has an Otto engine?
 - (a) Premixed flame
 - (b) Diffusive flame
 - (c) Detonative flame
 - (d) Barbecue flame
10. The by-pass ratio of
 - (a) a turbofan is higher than a turbojet.
 - (b) a turboprop is lower than a turbofan.
 - (c) a ramjet is higher than a turbofan.
 - (d) a turbofan is lower than a rocket engine.

BLO-6: Concepts on avionics and navigation

11. An aircraft flies in an atmosphere that is warmer than the ISA. Thus, for a given variation in pressure, the corresponding variation of altitude is:
 - (a) larger than in the ISA
 - (b) smaller than in the ISA
 - (c) equal to that in the ISA
 - (d) We do NOT have enough information to choose which of the other answers is correct
12. A pilot is making a voice report to ATC. If he says that he is flying at FL110, it means that:
 - (a) The altimeter calibration is STD and the altimeter reading is 11000 ft
 - (b) The altimeter calibration is STD and the altimeter reading is 11000 m
 - (c) The altimeter calibration is QNH and the altimeter reading is 11000 ft
 - (d) The altimeter calibration is QNH and the altimeter reading is 11000 m
13. Which of the following statements is true?
 - (a) None of the other answers is correct
 - (b) The Transition Level (TL) is fixed for a given airport
 - (c) The transition Altitude (TA) varies depending on the local QNH pressure
 - (d) Commercial civil aviation aircraft no longer have analogic altitude indicators in the instrument panel
14. Which of the following statements is true?
 - (a) When an aircraft flies with the QFE calibration over the departure airport, the altimeter reading will be similar to the real height
 - (b) The altitude is the vertical distance between the aircraft and the ground
 - (c) The height is the vertical distance between the aircraft and the Sea Level
 - (d) All the other answers are correct
15. Which of the following statements is true?
 - (a) Aircraft fly using STD calibration always in the cruise phase
 - (b) Aircraft fly using STD calibration only in transoceanic flights
 - (c) Aircraft fly using STD calibration only in continental flights
 - (d) All the other answers are correct
16. What does integrity stand for when talking about aircraft navigation systems?
 - (a) The warranty that the position error will not be higher than some threshold and that, if the threshold is exceeded, an alarm will be received on time
 - (b) The degree of conformance of the position measured by the system with the true position of the aircraft at a given time
 - (c) The average length of the time window between two alarms issued due to malfunction of the system
 - (d) The percentage of times that the navigation system is available when an aircraft needs it service
17. What equipment can provide guidance for a precision approach?
 - (a) ILS
 - (b) VOR+DME
 - (c) VOR+NDB
 - (d) VOR+NDB+DME
18. Why is it necessary that the GNSS receptor be visible by at least 4 satellites to be able to compute its position?
 - (a) To correct the time error due to the time drift between the clock in the receptor and the atomic clocks in the satellites
 - (b) Because the 4th satellite transfers housekeeping data from the satellites that are servicing a receptor at a given time
 - (c) To have a redundant signal such that the integrity of the information provided by the GNSS is enhanced
 - (d) None of the other answers is correct
19. Which is the main advantage of the dead reckoning navigation systems ("navegación a estima")?
 - (a) These navigation systems are autonomous
 - (b) For these navigation systems the error is NOT accumulative with time
 - (c) These navigation systems do NOT need an initial position
 - (d) None of the other answers is correct
20. Which of these navigation systems provides zenithal guidance?
 - (a) ILS/GP
 - (b) INS
 - (c) VOR
 - (d) DME

BLO-7: Concepts on materials and aeronautical regulations

21. The wing of an airplane parked in the airport
- (a) is supporting compression stresses in the lower part.
 - (b) is supporting traction stresses in the lower part.
 - (c) has the same stresses on both parts.
 - (d) None of the other statements is true.
22. Which of the following materials has the highest thermal resistance?
- (a) Ceramic material.
 - (b) Polymer.
 - (c) Aluminum alloy.
 - (d) None of the other statements is true.
23. What type of material is/are used for aircraft structural purposes?
- (a) Metallic and composite materials.
 - (b) Composite materials
 - (c) Metallic materials
 - (d) Ceramic materials
24. Which of the following statement about pure aluminum is true?
- (a) The mechanical properties are poor.
 - (b) The melting temperature is so high.
 - (c) It's chemical stability is so high.
 - (d) None of the other statements is true.
25. What kind of alloys have been mostly used in aircraft structures?
- (a) Aluminum, titanium and steel alloys.
 - (b) Aluminum and steel alloys.
 - (c) Only aluminum alloys.
 - (d) None of the other statements is true.
26. El Anexo Técnico de la OACI encargado de las Cartas Aeronáuticas es el:
- (a) 4
 - (b) 14
 - (c) 1
 - (d) 3
27. El Anexo Técnico de la OACI encargado de las Licencias al Personal es el:
- (a) 1
 - (b) 2
 - (c) 7
 - (d) 14
28. El Anexo Técnico de la OACI encargado de las Telecomunicaciones Aeronáuticas es el:
- (a) 10
 - (b) 12
 - (c) 7
 - (d) 4
29. En el Convenio de Chicago de 1944 se formó la organización (u organizaciones):
- (a) OPACI y OACI
 - (b) CINA y EASA
 - (c) EASA y OPACI
 - (d) JAA y CINA
30. La organización que nació en 1960 con el objetivo de lograr la unificación de los servicios de Control y de navegación aérea europeos es:
- (a) EUROCONTROL
 - (b) La FAA
 - (c) La OACI
 - (d) La DGAC

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control system options does NOT entitle the pilot to a full control recovery in case of a complete electrical blackout?
- (a) An electro-hydraulic flight control system
 - (b) An hydro-mechanical flight control system
 - (c) An electro-hydro-mechanical flight control system
 - (d) A purely mechanical flight control system
32. Which one of the following elements of a flight control system provides the pilot with feedback on what his actions on the commands are producing on the control surfaces?
- (a) Artificial feel system
 - (b) Hydraulic actuators
 - (c) Sensors (gyros, accelerometers...)
 - (d) The flight controls computer (FCC)
33. Which of the following is NOT an option to power the hydraulic system in an Airbus A320 type aircraft?
- (a) The pneumatic system.
 - (b) Engine-driven pumps.
 - (c) Electrical motors.
 - (d) The ram air turbine (RAT).
34. What is the power transfer unit (PTU) for?
- (a) To transfer pressure among hydraulic circuits.
 - (b) To transfer pressurised fluid among hydraulic circuits.
 - (c) To transfer electrical power among electrical circuits.
 - (d) To transfer pneumatic power between the two packs of the pneumatic system.
35. In a typical electrical system, where does the AC ESS bus get the power from if normal AC generation is lost during flight?
- (a) From an electrical emergency generator powered by the hydraulic system.
 - (b) From the engine generators via the normal AC buses.
 - (c) From the engine generators directly.
 - (d) From the batteries and the DC ESS bus through DC/AC transformation.
36. Which of the following elements CANNOT be found in an electrical system?
- (a) An accumulator.
 - (b) A transformer/rectifier.
 - (c) Switches.
 - (d) Batteries.
37. Which of the following is NOT a component of the pneumatic system of an Airbus A320 type aircraft?
- (a) A static inverter.
 - (b) The crossbleed duct and valve.
 - (c) Leak detection system.
 - (d) Bleed monitoring computer.
38. Which of the following tasks is not performed by the engine in large aircraft?
- (a) All options are correct.
 - (b) Provide hydraulical power.
 - (c) Provide AC electrical power.
 - (d) Provide thrust.
39. What is an aim of the Auxiliary Power Unit (APU) during takeoff?
- (a) Supply bleed air to reduce stress on the engines.
 - (b) The APU is always switched off during takeoff.
 - (c) Provide extra thrust.
 - (d) Supply the passengers with food and drinks.
40. What is FALSE about the brake system acting on the wheels of large aircraft?
- (a) It controls thrust reversal.
 - (b) It is controlled from the rudder pedals.
 - (c) It is powered by the hydraulic system.
 - (d) It includes an anti-skid system.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	a	d	c	b
2	a	a	c	d	a
3	a	a	a	a	c
4	a	a	d	a	b
5	a	b	d	b	b
6	a	b	c	b	c
7	a	a	b	d	d
8	a	c	d	b	c
9	a	d	b	b	a
10	a	a	c	c	b

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	a	c	b	a
12	a	d	b	c	b
13	a	b	a	a	a
14	a	d	a	a	d
15	a	c	a	d	a
16	a	b	a	b	b
17	a	b	c	b	d
18	a	b	d	b	c
19	a	b	a	b	a
20	a	b	a	a	a

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	b	b	b	d
22	a	d	b	b	a
23	a	d	c	a	a
24	a	a	b	b	b
25	a	b	a	d	b
26	a	a	c	d	c
27	a	b	d	b	c
28	a	b	b	a	c
29	a	a	d	c	c
30	a	c	b	b	a

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	a	c	c	d
32	a	c	a	b	b
33	a	b	a	b	a
34	a	a	d	d	b
35	a	b	b	a	a
36	a	d	b	b	d
37	a	b	a	d	c
38	a	b	b	a	a
39	a	b	d	b	a
40	a	d	a	c	b

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. Which of the following statements is true?
 - (a) Thrust reversing is used on landing to slow down the airplane.
 - (b) Thrust vectoring increases the power of the aircraft.
 - (c) An afterburner increases the efficiency of the engine.
 - (d) The other statements are wrong.
2. To obtain the maximum propulsive efficiency
 - (a) the mass flow must be maximized with the minimum possible acceleration.
 - (b) the thermal efficiency must be maximized.
 - (c) the exhaust gases must be accelerated as much as possible and the mass flow reduced.
 - (d) it is needed to get a compromise between what is written in the other statements.
3. If a compressor is working properly, this compressor
 - (a) all the other statements are true.
 - (b) increases the density of the air.
 - (c) increases the pressure of the air.
 - (d) avoids the blow-out phenomena.
4. About turbojet combustion chambers,
 - (a) the temperature of the flame is higher than the temperature of the combustion chamber walls.
 - (b) the combustion chamber is usually cooled injecting water from the walls.
 - (c) there is almost no oxygen at the exit of the combustion chamber.
 - (d) the other statements are wrong.
5. An airplane is stopped at the head of the runway. You can be sure that
 - (a) when it starts accelerating, the propulsive power is zero.
 - (b) the propulsive power will be constant during the take off.
 - (c) when it starts accelerating, the thrust will be equal to the forces that were keeping the airplane stopped.
 - (d) when it starts accelerating, the propulsive power will be maximum.
6. Piston engines, which of the following statements is true?
 - (a) The mixture of a Diesel engine is rich in oxygen.
 - (b) High compression ratios in Diesel engines risk a self-detonation in the combustion chamber.
 - (c) Otto compression ratios are higher than Diesel compression ratios.
 - (d) None of the other statements are true.
7. Piston engines, which of the following statements is true?
 - (a) the specific power of a two strokes piston engine is higher than a 4 strokes piston engine.
 - (b) Two strokes Otto piston engine is more efficient than 4 strokes piston engine.
 - (c) It is not possible to build a two strokes Diesel engine.
 - (d) None of the other statements are true.
8. The orbit suggested for an Earth observation mission is
 - (a) polar orbit
 - (b) geostationary orbit
 - (c) Molniya orbit
 - (d) Medium Earth orbit
9. The maximum tip speed of the main rotor of an helicopter
 - (a) is limited by the noise produced and compressibility effects.
 - (b) is limited by the compressibility effects and the risk of flow separation at the blades.
 - (c) is limited by the stored kinetic energy of the main rotor and the risk of flow separation at the blades.
 - (d) The other statements are wrong.
10. Rocket engine, which of the following statements is true?
 - (a) The ratio thrust vs weight of a solid propellant engine is higher than the ratio T/W of a liquid propellant.
 - (b) The specific impulse of a solid propellant rocket is higher than the specific impulse of a liquid propellant.
 - (c) The required thrust/weight ratio for an interplanetary mission is higher than for a launching mission.
 - (d) The other statements are wrong.

BLO-6: Concepts on avionics and navigation

11. The aneroid altimeter measures the altitude:
 - (a) Using static pressure readings from several static ports placed strategically in the fuselage of the aircraft
 - (b) Thanks to the Doppler effect observed in electromagnetic signals that the aircraft sends towards the ground surface
 - (c) Using the echo that a DME station broadcasts after receiving the interrogation from the transponder on board the aircraft
 - (d) None of the other answers is correct
12. If the pilot of an aircraft in an airport wants the altitude reading in the altimeter to be equal to the elevation of the airport, which calibration must he select?
 - (a) None of the other answers is correct
 - (b) He must select the real pressure in the airport as the reference pressure for altitude zero
 - (c) He must select 1013 mb as the reference pressure for altitude zero
 - (d) He must select a QFE calibration
13. The Transition Level (TL) is the FL at which the pilot shifts the calibration of the altimeter:
 - (a) From STD to QNH
 - (b) From QNH to STD
 - (c) From QFE to QNH
 - (d) From QNH to QFE
14. Which of the following statements is true?
 - (a) None of the other answers is correct
 - (b) Aircraft are generally allowed to fly their cruise phase inside the transition layer
 - (c) The Transition Altitude (TA) is always above the Transition Level (TL)
 - (d) The QFE calibration is typically used for take-off and final approach phases in commercial civil aviation flights
15. To define the International Standard Atmosphere, ICAO assumed that:
 - (a) The air behaves as a perfect gas
 - (b) The gravity varies with the altitude
 - (c) The fluidostatic equation is not valid
 - (d) None of the other answers is correct
16. What ILS subsystem provides azimuthal guidance in the final approach phase?
 - (a) The Localizer
 - (b) The Glide Path
 - (c) The markers
 - (d) All the other answers are correct
17. Which of the following systems does NOT belong to the group of radionavigation systems?
 - (a) INS
 - (b) NDB-ADF
 - (c) GNSS
 - (d) DME
18. The navigation with VOR is based in the emission of:
 - (a) An omnidirectional reference signal plus a highly directional variable signal emitted by a set of antennae
 - (b) A series of highly directional signals, emitted by a set of antennae placed orthogonally to the runway axis beyond the runway end
 - (c) A series of signals by a set of antennae in a tower placed 300 m beyond the runway threshold and at 150 m from the runway axis
 - (d) None of the other answers is correct
19. The DME provides the following information:
 - (a) The distance between the DME station and the aircraft
 - (b) The azimuthal position of the aircraft respect to the DME station
 - (c) The latitude and longitude of the aircraft
 - (d) None of the other answers is correct
20. For which of the following systems the error is NOT accumulative with time?
 - (a) All the other answers are correct
 - (b) ILS
 - (c) DME
 - (d) VOR

BLO-7: Concepts on materials and aeronautical regulations

21. The wing of an airplane in horizontal leveled flight
- (a) is supporting traction stresses in the lower part.
 - (b) is supporting compression stresses in the lower part.
 - (c) has the same stresses on both parts.
 - (d) None of the other statements is right.
22. Which of the following materials has the weakest dynamic resistance?
- (a) Ceramic material.
 - (b) Polymer.
 - (c) Aluminum alloy.
 - (d) None of the other statements is true.
23. Which of the following properties is appropriate for a ceramic material as an structural component?
- (a) None of the other statements is true.
 - (b) Its hardness.
 - (c) Its resistance to dynamic forces.
 - (d) Its thermal properties.
24. Which of the following statements is true?
- (a) An alloy is a solid - solid homogeneous solution of two or more metals or even non metallic elements.
 - (b) Pure metals are so resistant to chemical attacks.
 - (c) A pure metal has better properties than it's derived alloys.
 - (d) None of the other statements is true.
25. What kind of alloys have been mostly used in aircraft structures?
- (a) Aluminum, titanium and steel alloys.
 - (b) Aluminum and steel alloys.
 - (c) Only aluminum alloys.
 - (d) None of the other statements is true.
26. El Anexo Técnico de la OACI encargado de los Aeródromos es el:
- (a) 14
 - (b) 1
 - (c) 7
 - (d) 3
27. El Anexo Técnico de la OACI encargado de los Servicios de Tránsito Aéreo es el:
- (a) 11
 - (b) 2
 - (c) 3
 - (d) 7
28. El Anexo Técnico de la OACI encargado de la Meteorología es el:
- (a) 3
 - (b) 4
 - (c) 10
 - (d) 14
29. La nacionalidad de una aeronave es la de:
- (a) El país de matriculación
 - (b) El país de construcción
 - (c) El país de su línea aérea
 - (d) El país de origen del capitán
30. Cuál o cuáles tipos de aeronaves son cubiertas por las resoluciones del Convenio de Chicago de 1944?
- (a) Civiles
 - (b) Policía
 - (c) Aduanas
 - (d) Militares

BLO-8: Concepts on aircraft systems

31. Which one of the following flight control system options does NOT allow for computerised automatic control and flight augmentation?
- (a) An hydro-mechanical flight control system
 - (b) An electro-hydro-mechanical flight control system
 - (c) An electro-hydraulic flight control system
 - (d) All of them allow for automatic control and flight augmentation
32. Which one of the following elements of a flight control system reduces the effort the pilot has to put into actuating control surfaces?
- (a) Hydraulic actuators
 - (b) Artificial feel system
 - (c) Sensors (gyros, accelerometers...)
 - (d) The flight controls computer (FCC)
33. What is the main aim of the hydraulic system, apart from actuating control surfaces, in an Airbus A320 type aircraft?
- (a) To drive an emergency generator to produce electrical power if required.
 - (b) To refrigerate air for air conditioning.
 - (c) To stabilise flight.
 - (d) To power the pneumatic system.
34. What can be said, in terms of power generation (how hydraulic power is produced) and distribution (how and where hydraulic power is used), of the several hydraulic circuits that comprise a redundant hydraulic system?
- (a) They are different in both generation and distribution.
 - (b) They are different in generation but equal in distribution.
 - (c) They are equal in both generation and distribution.
 - (d) They are equal in generation but different in distribution.
35. In a typical electrical system, where does the AC ESS bus get the power from in normal operation during flight?
- (a) From the engine generators via the normal AC buses.
 - (b) From an electrical generator powered by the hydraulic system.
 - (c) From the engine generators directly.
 - (d) From the batteries and the DC ESS bus through DC/AC transformation.
36. Which of the following elements cannot be found in an electrical system?
- (a) A crossbleed duct.
 - (b) A transformer/rectifier.
 - (c) A Static Inverter.
 - (d) Batteries.
37. Which of the following high pressure sources in an aircraft cannot supply the pneumatic system?
- (a) The ram air turbine.
 - (b) Engine bleed systems.
 - (c) The APU load compressor.
 - (d) The High Pressure ground connection.
38. Which of the following tasks is not performed by the engine in large aircraft?
- (a) All options are correct.
 - (b) Provide bleed air.
 - (c) Provide AC electrical power.
 - (d) Provide thrust.
39. What is an aim of the Auxiliary Power Unit (APU) while the airplane is on ground?
- (a) Supply bleed air for engine start.
 - (b) The APU is always switched off on ground.
 - (c) Supply hydraulic power to be able to open the cargo door.
 - (d) Supply the passengers with food and drinks.
40. Where are the tanks of the fuel system usually placed in large airplanes?
- (a) In the wings and in the fuselage.
 - (b) In the empennage.
 - (c) They are external.
 - (d) In the engines.

Correct Answers

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Permutation Number				
Question	0	1	2	3	4
1	a	d	d	d	a
2	a	d	d	d	c
3	a	c	c	b	a
4	a	d	d	d	a
5	a	b	d	a	c
6	a	c	a	c	d
7	a	c	c	d	a
8	a	c	a	c	b
9	a	d	d	a	d
10	a	b	a	c	a

BLO-6: Concepts on avionics and navigation

	Permutation Number				
Question	0	1	2	3	4
11	a	b	d	a	b
12	a	c	b	c	a
13	a	c	c	c	b
14	a	b	a	c	c
15	a	d	c	c	c
16	a	a	c	a	d
17	a	a	b	d	a
18	a	a	a	c	a
19	a	b	b	a	c
20	a	b	d	c	d

BLO-7: Concepts on materials and aeronautical regulations

	Permutation Number				
Question	0	1	2	3	4
21	a	a	c	c	b
22	a	c	a	d	c
23	a	b	c	b	d
24	a	d	b	c	b
25	a	d	d	d	c
26	a	d	a	d	d
27	a	c	a	c	d
28	a	a	b	c	d
29	a	d	d	b	b
30	a	a	c	a	b

BLO-8: Concepts on aircraft systems

	Permutation Number				
Question	0	1	2	3	4
31	a	c	c	c	c
32	a	a	b	c	b
33	a	c	b	b	b
34	a	a	c	a	a
35	a	c	b	d	a
36	a	c	d	d	a
37	a	d	b	b	d
38	a	c	d	d	a
39	a	c	b	b	d
40	a	b	c	d	d

BLO-1: Concepts on flight principles and aircraft performance

1. What is the effect on an aerofoil lift characteristic curve ($C_L(\alpha) = C_{L_0} + C_{L_\alpha} \alpha$ for $\alpha < \alpha_S$) of a high-lift device that only increases the camber of the aerofoil?
 - (a) It shifts the curve up by increasing C_{L_0} .
 - (b) It increases the slope C_{L_α} .
 - (c) It increases the stall angle α_S .
 - (d) It reduces $C_D(\alpha)$ but does nothing to $C_L(\alpha)$.
2. Which of the following stall parameters is roughly independent of altitude?
 - (a) Stall angle of attack.
 - (b) Stall velocity.
 - (c) Stall bank angle for a coordinated turn.
 - (d) Stall thrust.
3. Which non-dimensional parameter determines the flow regime with regard to turbulence and friction (laminar, transitional or turbulent)?
 - (a) The Reynolds number.
 - (b) The Mach number.
 - (c) The Prandtl number.
 - (d) The Bernoulli number.
4. The maximum aerodynamic efficiency of a wing with elliptical lift distribution can be increased by...
 - (a) Reducing friction.
 - (b) Reducing span.
 - (c) Transferring lift production from wing root to wing tip.
 - (d) Increasing the chord.
5. What is the International Standard Atmosphere (ISA)?
 - (a) An atmospheric model providing a common reference for temperature and pressure over a wide range of altitudes.
 - (b) The earth average atmospheric conditions at sea level.
 - (c) The atmospheric conditions that allow international flight.
 - (d) The layer of air that surrounds the earth.
6. Which of the following statements is true about the first (R1) and second (R2) flight regimes?
 - (a) Pulling the stick has opposite effects on the ascent/descent slope.
 - (b) Pushing the thrust lever has opposite effects on the ascent/descent slope.
 - (c) R1 corresponds to high subsonic flight (compressible regime) while R2 corresponds to low subsonic flight (incompressible regime).
 - (d) None of the others is true.
7. Which of the following statements is true about takeoff and landing in case of wind?
 - (a) Both takeoff and landing must be performed with head wind.
 - (b) Both takeoff and landing must be performed with tail wind.
 - (c) Takeoff must be performed with head wind while landing must be performed with tail wind.
 - (d) Takeoff must be performed with tail wind while landing must be performed with head wind.
8. Which of the following statements is FALSE about climb?
 - (a) The velocities for best angle and for maximum rate of climb are the same.
 - (b) The best climb angle corresponds to maximum excess thrust.
 - (c) The maximum rate of climb corresponds to maximum excess power.
 - (d) For a jet powered airplane, the best climb angle roughly corresponds to minimum drag conditions.
9. Which of the following conditions corresponds to maximum cruise range, independently of how the airplane is powered (jet or propeller)?
 - (a) Minimum hourly fuel consumption per speed unit (C/V).
 - (b) Minimum hourly fuel consumption (C).
 - (c) Minimum required power.
 - (d) Minimum drag.
10. Which of the following is a limiting factor for the maximum achievable bank angle (ϕ_{max}) in turning performance?
 - (a) Stall speed.
 - (b) Structural strength.
 - (c) Available power.
 - (d) All of them.

BLO-2: Flight principles related problems

11. What is the optimal angle of attack (α_{opt}) that maximises aerodynamic efficiency (f_{max}) for an airplane with lift coefficient $C_L = 0.2 + 0.1\alpha$ for $\alpha < \alpha_S = 10^\circ$ and drag polar $C_D = 0.03 + 0.15C_L^2$?
 - (a) $\alpha_{opt} = 2.47^\circ$
 - (b) $\alpha_{opt} = 10^\circ$
 - (c) $\alpha_{opt} = 2.21^\circ$
 - (d) $\alpha_{opt} = 1.92^\circ$

12. An aerobatics airplane with lift coefficient $C_L = 0.16 + 0.11\alpha$ with α in degrees ($^\circ$) engages in a perfect dive (vertical descent). What should the angle of attack (α) be?
 - (a) $\alpha = -1.45^\circ$
 - (b) $\alpha = 1.45^\circ$
 - (c) $\alpha = 0.688^\circ$
 - (d) $\alpha = -0.688^\circ$

13. A small airplane is engaged in a coordinated banked turn at an unknown altitude. We know its speed is $V = 1.25 V_{S|n=1}$, where $V_{S|n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max}) that can be achieved if the plane is to keep its altitude and speed?
 - (a) $\phi_{max} = 50.2^\circ$
 - (b) $\phi_{max} = 36.9^\circ$
 - (c) $\phi_{max} = 8.76^\circ$
 - (d) It depends on the mass of the airplane.

14. The lift coefficient and drag polar of an airplane changes from $C_L = 0.2 + 0.1\alpha$ and $C_D = 0.05 + 0.075C_L^2$ with full flaps to $C_L = 0.1 + 0.1\alpha$ and $C_D = 0.04 + 0.15C_L^2$ in clean configuration. What is the value of the angle of attack (α) just before optimal flap removal?
 - (a) $\alpha = 1.65^\circ$
 - (b) $\alpha = 2.65^\circ$
 - (c) $\alpha = 2.32^\circ$
 - (d) $\alpha = 1.98^\circ$

15. An airplane of mass $m = 2000$ kg is cruising in straight and level flight with suboptimal aerodynamic efficiency $f = 20 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $D = 1000$ N and $D_p \neq D_i$.
 - (b) $D = 1000$ N and $D_p = D_i = 500$ N.
 - (c) $D = 500$ N and $D_p = D_i = 250$ N.
 - (d) $D = 500$ N and $D_p \neq D_i$.

BLO-3: Concepts on aircraft stability and control

16. Which command and primary control surface pilot the aircraft around its pitch axis?
- (a) The stick (or yoke) and the elevator.
 - (b) The pedals and the rudder.
 - (c) The stick (or yoke) and the ailerons.
 - (d) The throttle and the flaps.
17. Which aerodynamic element is mainly responsible for directional stability?
- (a) The vertical stabiliser.
 - (b) The horizontal stabiliser.
 - (c) The wing dihedral.
 - (d) The fuselage.
18. About static and dynamic stability...
- (a) Dynamic stability implies static stability.
 - (b) Static stability implies dynamic stability.
 - (c) To have dynamic stability we need a restoring force but not necessarily a damping force.
 - (d) To have static stability we need a damping force but not necessarily a restoring force.
19. We are at the design stage of an aerobatics airplane and we want it to be highly manoeuvrable. Which of the following options should we consider if our preliminary design is still hard to manoeuvre in all three axes?
- (a) Sacrifice a bit of stability.
 - (b) Move the centre of gravity forward to increase the static margin.
 - (c) Increase wing span to enhance roll damping.
 - (d) Increase the size of the stabilisers to damp oscillatory modes.
20. What is true about the dutch roll mode?
- (a) Its damping strongly depends on the coefficient C_{n_r} , the yaw moment due to yaw rotational velocity.
 - (b) It is a relatively fast, highly damped, aperiodic mode.
 - (c) It couples rotation about the yaw and pitch axes.
 - (d) It is strongly dependent on the size and distance of the horizontal stabiliser.
21. What is an operational (skidding) turn?
- (a) It is a turn with stick only (no pedals).
 - (b) It is a turn that applies pedals to correct for the sideslip induced by stick operation.
 - (c) It is a turn with pedals only (no stick).
 - (d) It is an aerobatic manoeuvre.
22. What can we say about the response of the airplane to a perturbation in roll?
- (a) It depends on both directional and lateral stability, since there is no restoring moment and the return to level flight is mostly dynamical.
 - (b) It depends mainly on static directional stability, since there automatically appears a restoring yaw moment.
 - (c) It depends mainly on static longitudinal stability, since there automatically appears a restoring pitch moment.
 - (d) It depends mainly on static lateral stability, since there automatically appears a restoring roll moment.
23. What is true about the phugoid motion?
- (a) It corresponds to the periodic exchange between potential and kinetic energy.
 - (b) It is a stable and fast periodic mode.
 - (c) It couples roll and yaw motion.
 - (d) It is responsible for the return to straight flight after a turn.
24. When is crosswind straight flight (with sideslip) useful?
- (a) For crosswind landing, to align the fuselage with the runway when there is transversal wind.
 - (b) For crosswind cruise, to align the fuselage to point towards destination when wind is transversal to the trajectory.
 - (c) For crosswind glide, to enhance aerodynamic efficiency when wind comes from one side.
 - (d) None of the others.
25. What is the main aim of canards?
- (a) Either lift production or longitudinal stabilisation, depending on the type of canard.
 - (b) Lay eggs.
 - (c) Directional stability.
 - (d) Prevent wing stall.

BLO-4: Aircraft performance related problems

26. The required thrust of a jet airplane is given by $T_R = 0.25\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1200$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.7$, what is the optimal velocity for maximum range (V_R)?
- $V_R = 91.6 \text{ ms}^{-1}$
 - $V_R = 69.6 \text{ ms}^{-1}$
 - $V_R = 52.9 \text{ ms}^{-1}$
 - It depends on the available thrust T_A , which is unknown.
27. A propeller-powered airplane of mass $m = 1500$ kg has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.03W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 10 \text{ ms}^{-2}$)
- $V_{(R/C)_{max}} = 52.3 \text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 68.9 \text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 90.6 \text{ ms}^{-1}$
 - $V_{(R/C)_{max}} = 119 \text{ ms}^{-1}$
28. An airplane with drag polar $C_D = 0.03 + 0.15C_L^2$, wing surface $S = 40 \text{ m}^2$ and mass $m = 2500$ kg is flying at a speed $V = 120 \text{ ms}^{-1}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.5 \text{ kgm}^{-3}$? (take $g \simeq 10 \text{ ms}^{-2}$)
- $T_A = 4970 \text{ N}$
 - $T_A = 5050 \text{ N}$
 - $T_A = 4830 \text{ N}$
 - $T_A = 5170 \text{ N}$
29. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 60$. If its current altitude is $h = 1000$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
- $d = 60 \text{ km}$
 - $d = 16.7 \text{ km}$
 - $d = 1 \text{ km}$
 - $d = 120 \text{ km}$
30. The required thrust of a jet-engined airplane of mass $m = 1900$ kg, wing area $S = 25 \text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.81\bar{\rho}V^2 + 0.009W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 15000 \text{ N}$. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
- $V \simeq 56 \text{ ms}^{-1}$
 - $V \simeq 47.6 \text{ ms}^{-1}$
 - $V \simeq 127 \text{ ms}^{-1}$
 - $V \simeq 77.9 \text{ ms}^{-1}$

Correct Answers

BLO-1: Concepts on flight principles and aircraft performance

	Permutation Number				
Question	0	1	2	3	4
1	a	d	c	a	c
2	a	c	b	c	c
3	a	a	c	d	c
4	a	c	b	a	a
5	a	d	d	c	d
6	a	d	b	b	c
7	a	c	a	c	b
8	a	c	b	a	b
9	a	d	c	b	c
10	d	c	c	a	d

BLO-2: Flight principles related problems

	Permutation Number				
Question	0	1	2	3	4
11	a	d	c	c	a
12	a	d	a	d	b
13	a	d	a	c	a
14	a	b	b	c	b
15	a	a	a	a	c

BLO-3: Concepts on aircraft stability and control

	Permutation Number				
Question	0	1	2	3	4
16	a	b	c	d	c
17	a	d	a	c	d
18	a	d	a	b	b
19	a	c	a	a	b
20	a	c	a	a	c
21	a	c	b	c	a
22	a	d	d	a	a
23	a	d	d	c	a
24	a	d	b	c	b
25	a	c	c	d	d

BLO-4: Aircraft performance related problems

	Permutation Number				
Question	0	1	2	3	4
26	a	c	a	b	b
27	a	d	a	a	d
28	a	c	b	b	c
29	a	c	d	a	d
30	a	d	b	b	c

BLO-1: Concepts on flight principles and aircraft performance

1. Which non-dimensional parameter determines the compressibility regime of flight (low subsonic, subsonic compressible, transonic, supersonic or hypersonic)?
 - (a) The Mach number.
 - (b) The Reynolds number.
 - (c) The Prandtl number.
 - (d) The Bernoulli number.
2. How should the pilot modify the engine setting in order to keep a straight and level flight if the speed is to be increased?
 - (a) It depends on whether the airplane is being flown in the first or second regime.
 - (b) It has to be increased.
 - (c) It has to be lowered.
 - (d) It must be kept as is.
3. What is the main effect on the lift coefficient (C_L) of a high-lift device that only retards boundary layer detachment.
 - (a) Increase the angle of attack at which stall occurs (α_s).
 - (b) Increase the lift coefficient at 0-angle of attack (C_{L_0}).
 - (c) Increase the slope of the lift coefficient curve (C_{L_α}).
 - (d) Decrease the drag coefficient (C_D) without modifying C_L .
4. Which of the following parameters of an airplane is, approximately, independent of the altitude and speed at which it is flying?
 - (a) The angle of attack at which stall occurs.
 - (b) The speed at which stall occurs.
 - (c) The pitch angle at which stall occurs.
 - (d) The bank angle at which stall occurs.
5. What happens to parasitic and induced drag when an airplane deploys its flaps at constant speed and altitude?
 - (a) Parasitic drag is increased and induced drag is reduced.
 - (b) Parasitic drag is reduced and induced drag is increased.
 - (c) Both parasitic and induced drag are increased.
 - (d) Both parasitic and induced drag are reduced.
6. Which is the velocity that separates the first and second regime for a jet engine in straight and level flight?
 - (a) The speed for minimum drag.
 - (b) The speed for minimum required power.
 - (c) The speed for maximum range.
 - (d) The stall speed.
7. A glider flying with maximum aerodynamic efficiency in still air (no wind) will...
 - (a) fly further (maximum distance).
 - (b) fly longer (maximum time).
 - (c) fly forever.
 - (d) fly in circles.
8. Which of the following conditions corresponds to a best angle climb of an airplane, independently of how it is powered (jet or propeller)?
 - (a) Maximum excess thrust.
 - (b) Maximum excess power.
 - (c) Minimum power.
 - (d) Minimum drag.
9. Which of the following statements is always true about the takeoff speed V_{to} ?
 - (a) It is faster than the stall speed: $V_{to} > V_s$.
 - (b) It is slower than the stall speed: $V_{to} < V_s$.
 - (c) It is equal to the stall speed: $V_{to} = V_s$.
 - (d) It can be faster, slower or equal to the stall speed, depending on the airport and the airplane.
10. Why do airplanes step-climb during cruise rather than climbing gradually?
 - (a) Due to aerospace regulations, to avoid collision with other aircraft, even if step-climb is less optimal than gradual climb.
 - (b) Because it produces more optimal cruise conditions than gradual climb.
 - (c) To avoid engine overheating due to constant climb demands.
 - (d) The statement is wrong: airplanes usually climb gradually and not in steps.

BLO-2: Flight principles related problems

1. A small airplane with mass $m = 1600$ kg is engaged in a coordinated banked turn of maximum bank angle (ϕ_{max}) at an unknown altitude. We know its speed is $V = 2 V_S|_{n=1}$, where $V_S|_{n=1}$ is its stall speed in straight and level flight at the same altitude. What is the lift (L) if the plane is to keep its altitude and speed? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $L = 64000$ N
 - (b) $L = 80000$ N
 - (c) $L = 32000$ N
 - (d) It depends on the altitude.

2. The lift coefficient and drag polar of an airplane changes from $C_L = 0.1 + 0.1\alpha$ and $C_D = 0.03 + 0.15C_L^2$ in clean configuration to $C_L = 0.2 + 0.1\alpha$ and $C_D = 0.05 + 0.075C_L^2$ with full flaps. What is the value of the angle of attack (α) just before optimal flap deployment?
 - (a) $\alpha = 4.16^\circ$
 - (b) $\alpha = 3.16^\circ$
 - (c) $\alpha = 3.83^\circ$
 - (d) $\alpha = 2.83^\circ$

3. An aerobatics airplane with lift coefficient $C_L = 0.15 + 0.11\alpha$ with α in $^\circ$ is performing a balanced inverted flight. What should the angle of attack (α) be if a balanced straight and level flight at the same altitude and speed requires a $C_L = 0.21$?
 - (a) $\alpha = -3.27^\circ$
 - (b) $\alpha = 0.545^\circ$
 - (c) $\alpha = -1.36^\circ$
 - (d) $\alpha = 3.27^\circ$

4. What is the optimal drag coefficient ($C_{D_{opt}}$) that maximises aerodynamic efficiency (f_{max}) for an airplane with drag polar $C_D = 0.03 + 0.15C_L^2$?
 - (a) $C_{D_{opt}} = 0.06$
 - (b) $C_{D_{opt}} = 0.03$
 - (c) $C_{D_{opt}} = 0.447$
 - (d) $C_{D_{opt}} = 0.045$

5. An airplane of mass $m = 2000$ kg is cruising in straight and level flight with optimal aerodynamic efficiency $f = 25 = f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $D = 800$ N and $D_p = D_i = 400$ N.
 - (b) $D = 800$ N and $D_p > D_i$.
 - (c) $D = 400$ N and $D_p = D_i = 200$ N.
 - (d) $D = 400$ N and $D_p > D_i$.

BLO-3: Concepts on aircraft stability and control

1. Which command and primary control surface pilot the aircraft around its yaw axis?
 - (a) The pedals and the rudder.
 - (b) The stick (or yoke) and the ailerons.
 - (c) The stick (or yoke) and the elevator.
 - (d) The throttle and the flaps.
2. The sign of which aerodynamic coefficient is responsible for directional stability?
 - (a) $C_{n\beta}$, the change in yaw moment due to a change of sideslip angle.
 - (b) $C_{m\alpha}$, the change in pitching moment due to a change of angle of attack.
 - (c) $C_{Y\beta}$, the change in lateral force due to a change of sideslip angle.
 - (d) $C_{L\alpha}$, the change in lift due to a change of angle of attack.
3. An aircraft is dynamically stable around one of its axis. What can we say about static stability around that axis?
 - (a) It must be stable.
 - (b) It may or may not be stable.
 - (c) It must be unstable.
 - (d) It depends on the axis we refer to.
4. What is usually true about an airplane with no automatic control system?
 - (a) More stability implies less manoeuvrability.
 - (b) More manoeuvrability implies more stability.
 - (c) More stability implies more manoeuvrability.
 - (d) Stability and manoeuvrability are two completely unrelated concepts.
5. What is FALSE about the dutch roll mode?
 - (a) It is usually left unstable.
 - (b) It is a relatively fast, poorly damped, periodic mode.
 - (c) It couples rotation about the yaw and roll axes.
 - (d) It is strongly dependent on the size and distance of the vertical stabiliser.
6. What is a coordinated banked turn?
 - (a) It is a turn that applies pedals to correct for the sideslip induced by stick operation.
 - (b) It is a turn with stick only (no pedals).
 - (c) It is a turn with pedals only (no stick).
 - (d) It is an aerobatic manoeuvre.
7. The lateral modes of an airplane are:
 - (a) the roll subsidence, the dutch roll and the spiral mode.
 - (b) the pitch oscillation, the phugoid and the return to straight and level flight.
 - (c) the roll subsidence, the phugoid and the spiral mode.
 - (d) the pitch oscillation, the dutch roll and the spiral mode.
8. What is true about the return to straight and level flight mode?
 - (a) It is a stable and slow aperiodic mode.
 - (b) It is an unstable and slow aperiodic mode.
 - (c) It is a stable and fast periodic mode.
 - (d) It is an unstable and fast periodic mode.
9. How is crosswind straight flight (with sideslip) piloted?
 - (a) Mostly with the pedals, using the stick to compensate for the induced roll.
 - (b) Mostly with the stick, using the pedals to compensate for the induced yaw.
 - (c) Only with the pedals.
 - (d) Only with the stick.
10. What is the aim of trim tabs?
 - (a) Ease the continuous effort the pilot would have to apply on the commands.
 - (b) Allow for dynamic mode damping.
 - (c) Achieve an extra lift when needed.
 - (d) Increase aircraft manoeuvrability.

BLO-4: Aircraft performance related problems

1. The required thrust of a jet airplane is given by $T_R = 0.25\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$. If the airplane has mass $m = 2000$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)?
 - (a) $D = 2830$ N
 - (b) $D = 3270$ N
 - (c) $D = 2680$ N
 - (d) $D = 3410$ N
2. An airplane with drag polar $C_D = 0.04 + 0.15C_L^2$, wing surface $S = 40$ m² and mass $m = 1500$ kg is flying at a speed $V = 120$ ms⁻¹. What is the minimum engine power required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4$ kgm⁻³? (take $g \simeq 10$ ms⁻²)
 - (a) $P_R = 588$ kW
 - (b) $P_R = 521$ kW
 - (c) $P_R = 469$ kW
 - (d) $P_R = 420$ kW
3. A jet airplane of mass $m = 1500$ kg has a drag curve $D = 0.3\bar{\rho}V^2 + 0.03W^2/\bar{\rho}V^2$. What is the minimum engine thrust needed at sea level ($\bar{\rho} = 1$) to achieve a climb angle of $\alpha = 5^\circ$? (take $g \simeq 10$ ms⁻²)
 - (a) $T_A = 4150$ N
 - (b) $T_A = 4510$ N
 - (c) $T_A = 4880$ N
 - (d) $T_A = 2930$ N
4. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 40$. If its current altitude is $h = 2500$ m, what is the maximum horizontal distance (d) it can fly before hitting the ground?
 - (a) $d = 100$ km
 - (b) $d = 62.5$ km
 - (c) $d = 2.5$ km
 - (d) $d = 200$ km
5. The required thrust of a jet-engined airplane of mass $m = 1600$ kg, wing area $S = 25$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.86\bar{\rho}V^2 + 0.007W^2/\bar{\rho}V^2$, and the maximum available thrust can be considered independent of V : $T_A \leq 2500$ N. Which is the lowest velocity at which this airplane can be flown at straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10$ ms⁻² and $\rho_0 = 1.225$ kgm⁻³)
 - (a) $V \simeq 62.4$ ms⁻¹
 - (b) $V \simeq 51.4$ ms⁻¹
 - (c) $V \simeq 70.1$ ms⁻¹
 - (d) $V \simeq 66.1$ ms⁻¹

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. Quin del següents motors té millor eficiència tèrmica?
 - (a) Motor Diesel 4T
 - (b) Motor Otto 2T
 - (c) Motor Otto 4T
 - (d) Tots els motors a pistons tenen la mateixa eficiència.
2. Sobre una hèlix,
 - (a) el pas de l'hèlix augmenta a mesura que augmenta la velocitat.
 - (b) l'angle d'atac és positiu durant la carrera d'aterratge.
 - (c) l'angle d'atac és negatiu durant la carrera d'enlairament.
 - (d) Cap de les anteriors és certa.
3. Un avió vola a 300m/s i pressió atmosfèrica de 0.265bar. A la sortida de la tovera de 0.5m de diàmetre, la pressió dels gasos de sortida és de 2bar i la velocitat de sortida d'aquests gasos és de 1275m/s. Sabent que el consum de combustible és de 2.65kg/s i el d'aire que entra al motor és de 45kg/s, calculeu-ne el Thrust Specific Fuel Consumption (TSFC).
 - (a) 0.032 g/N
 - (b) 0.056 g/N
 - (c) 0.01 g/N
 - (d) Cap de les anteriors.
4. Quina de les següents aeronaus té la menor velocitat màxima de vol?
 - (a) un helicòpter.
 - (b) un avió propulsat per un turboreactor.
 - (c) un avió propulsat per un turbohèlix.
 - (d) un avió propulsat per hèlix i motor a pistons.
5. Les diferents etapes d'un turbofan en el sentit de circulació de l'aire són:
 - (a) fan, difusor, compressors, cambra de combustió, turbines, tovera.
 - (b) difusor, fan, compressors, cambra de combustió, turbines, tovera.
 - (c) fan, difusor, turbines, cambra de combustió, compressors, tovera.
 - (d) difusor, fan, compressors, turbines, cambra de combustió, tovera.
6. La funció de la cambra de combustió d'un turbojet és:
 - (a) proporcionar temperatura al fluid
 - (b) proporcionar pressió i temperatura al fluid
 - (c) proporcionar pressió al fluid
 - (d) cap de les anteriors
7. Un turbojet vola a $M=2.2$. Podem assegurar que:
 - (a) la tovera és convergent - divergent.
 - (b) la tovera és convergent.
 - (c) el difusor és subsònic.
 - (d) el motor no té parts mòvils.
8. En un helicòpter en vol horitzontal
 - I - la distribució de la velocitat a les pales del rotor principal és proporcional a la distància respecte l'eix del rotor.
 - II - la distribució de velocitats al rotor principal induïx un moment de balanceig a l'helicòpter.
 - III - la pala que es retira tendeix a moure's amunt per tal de compensar la zona de velocitat invertida.
 - (a) només la 2 és certa.
 - (b) només la 3 és certa.
 - (c) només la 2 i la 3 són certes.
 - (d) només a 1 i la 3 són certes.
9. Sigui un helicòpter en una condició de vol en la que el pas de les pales del rotor és el mateix independentment de la posició azimuthal que ocupen en el seu moviment de rotació. El pilot, mitjançant els comandaments de vol, dona una ordre que provoca que el pas de les pales sigui més gran quan passen sobre la cabina que quan passen sobre el con de cua. Què ha fet el pilot?
 - (a) Ha estirat cap enrere la palanca de pas del cíclic.
 - (b) Ha empès cap endavant la palanca de pas del cíclic.
 - (c) Ha estirat cap amunt la palanca de pas del col·lectiu.
 - (d) Ha empès avall la palanca de pas del col·lectiu.
10. Per al llançament d'un coet que porta una càrrega destinada a Mart, quin dels següents punts de la Terra és el més adequat per a instal·lar-hi una base de llançament?
 - (a) un punt proper a l'equador.
 - (b) un punt proper al pol.
 - (c) un punt amb 20 de latitud.
 - (d) un punt amb 45 de latitud.

BLO-6: Concepts on avionics and navigation

1. Radionavigation aid system that provides information about the vertical angle to approach a runway:
 - (a) Glide Slope.
 - (b) Localizer.
 - (c) Automatic Direction Finder.
 - (d) VHF Omnidirectional Range.
2. 'The ability of the total system to perform its function without unscheduled interruptions during the intended operation' is the definition of:
 - (a) Continuity.
 - (b) Accuracy.
 - (c) Integrity.
 - (d) Availability.
3. Radionavigation aid system that is often installed in order to signal the decision height when approaching a runway:
 - (a) Middle Marker.
 - (b) Localizer.
 - (c) Glide Slope.
 - (d) VHF Omnidirectional Range.
4. An example of a 'Fix point direction' radionavigation aid system is:
 - (a) Automatic Direction Finder.
 - (b) Glide Slope.
 - (c) Localizer.
 - (d) VHF Omnidirectional Range.
5. 'The ability of the total system to maintain the aircraft position within a given Total System Error (TSE) with a 95 percent of probability' corresponds to the definition of:
 - (a) Accuracy.
 - (b) Integrity.
 - (c) Continuity.
 - (d) Availability.
6. The concept of 'Alert Limit' is related to the 'outer tunnel', and therefore in an indirect way it is related to:
 - (a) Integrity.
 - (b) Accuracy.
 - (c) Continuity.
 - (d) Availability.
7. Which is the Operational System Mode when a position failure happened but a failure wasn't detected?
 - (a) Misleading information.
 - (b) False alarm.
 - (c) Nominal operation.
 - (d) Justified alarm.
8. The imaginary line used to change the altimeter setting from STD to QNH is called:
 - (a) Transition Level.
 - (b) Transition Altitude.
 - (c) Flight Level.
 - (d) QFE.
9. The altimeter setting needed for the altimeter to indicate 0 feet of height when the aircraft is parked at the airport is called:
 - (a) QFE.
 - (b) QNH.
 - (c) STD.
 - (d) TA.
10. Which of the following Operational System Modes represents NO 'continuity risk' and NO 'integrity risk'?
 - (a) Nominal operation.
 - (b) Misleading information.
 - (c) False alarm.
 - (d) Justified alarm.

BLO-7: Concepts on materials and aeronautical regulations

1. La necessitat de fer estructures lleugeres ve donada per
 - I - la necessitat d'augmentar la càrrega útil
 - II - la necessitat de disminuir la potència necessària per volar
 - III - la necessitat de donar rigidesa estructural a l'avió
 - (a) només la I i la II són certes.
 - (b) només la I és certa.
 - (c) només la III és certa.
 - (d) només la II i la III són certes.
2. Quina de les següents afirmacions és certa?
 - I - Una estructura ideal estaria dissenyada de tal manera que totes les parts fallessin per al mateix límit de càrrega i fatiga.
 - II - Es pot dissenyar una estructura alar que suporti una càrrega aerodinàmica major del que requereixi la normativa reguladora.
 - III - L'objectiu principal d'una estructura alar és suportar les càrregues aerodinàmiques amb el major pes estructural possible.
 - (a) només la I i la II són certes.
 - (b) només la I és certa.
 - (c) només la III és certa.
 - (d) només la II i la III són certes.
3. En un material compost
 - (a) és difícil fer-ne un bon reciclatge.
 - (b) es combina la resistència a compressió de les fibres amb la resistència a tracció de la matriu.
 - (c) cap de les anteriors.
 - (d) les fibres han d'estar aliniades.
4. Quina de les següents afirmacions és certa?
 - I - Els materials metàl·lics tenen enllaços metàl·lics i iònics.
 - II - Els materials ceràmics tenen enllaços covalents i iònics.
 - III - Els materials polimèrics tenen enllaços covalents i de Van der Waals.
 - (a) només la II i la III són certes.
 - (b) només la I i la II són certes.
 - (c) totes són certes.
 - (d) només la III és certa.
5. En un diagrama tensió vs. deformació,
 - (a) la tensió de la proveta segueix augmentant a partir del punt d'inici de l'estracció.
 - (b) la força que suporta la proveta segueix augmentant a partir del punt d'inici de l'estracció.
 - (c) en una deformació plàstica el material torna a la seva longitud original quan se li deixa d'aplicar tensió.
 - (d) cap de les anteriors
6. Which of the following is the organization created just after the Second World War with the objective of rule the development of the Civil Aviation?
 - (a) OACI.
 - (b) EUROCONTROL.
 - (c) FAA.
 - (d) EASA.
7. The OACI technical annex dealing with the Investigation of Aircraft Accidents is Annex number?
 - (a) 13.
 - (b) 14.
 - (c) 4.
 - (d) 3.
8. Which of the following is the organization created by European aircraft manufacturers in order to ease adoption of new products and coordinate airworthiness standards?
 - (a) JAA.
 - (b) EUROCONTROL.
 - (c) DGAC.
 - (d) EASA.
9. The country in charge of issuing the airworthiness certificate of an aircraft is the:
 - (a) Registry country.
 - (b) Airline country.
 - (c) Manufacturing country.
 - (d) Captain country.
10. Which of the following is the organization created by the Commission of the European Union in order to substitute the JAA and, in the future, become the sole European Aeronautical Authority?
 - (a) EASA.
 - (b) JAR.
 - (c) EUROCONTROL.
 - (d) DGAC.

BLO-8: Concepts on aircraft systems

1. What is the main aim of the hydraulic system?
 - (a) To actuate flight control surfaces.
 - (b) To provide the cabin with pressurised water.
 - (c) To stabilise flight.
 - (d) To power the pneumatic system.
2. Which of the following elements cannot be found in an hydraulic system?
 - (a) A cross-bleed valve.
 - (b) A power transfer unit (PTU).
 - (c) Reservoirs.
 - (d) Accumulators.
3. What is a Ram Air Turbine (RAT)?
 - (a) An emergency wind-driven generator to provide hydraulic or electric power.
 - (b) An auxiliary gas turbine to provide hydraulic or electric power.
 - (c) The computer that controls engine operation.
 - (d) A device used to measure airspeed.
4. What is the advantage of an hydro-mechanical flight control system as opposed to a purely mechanical flight control system?
 - (a) The pilot has to apply less effort on the controls.
 - (b) Automatic control is more easily implemented.
 - (c) It is much lighter.
 - (d) It has no advantages.
5. In which sense are the computers of a fly-by-wire system such as that of the A320 redundant?
 - (a) There are different types of computer and several computers of each type.
 - (b) All computers are different but their tasks overlap.
 - (c) All computers are identical and capable of performing the same tasks.
 - (d) There is a unique computer that computes everything several times to avoid mistakes.
6. Why do aircraft need an anti-ice system?
 - (a) To prevent deformation of the wing profile's leading edge due to ice accumulation, which can deteriorate aerodynamic performance.
 - (b) To prevent the airplane from getting heavier due to ice formation.
 - (c) To prevent the fuel in the tanks from freezing.
 - (d) To preheat the air into the engine to improve its performance.
7. Where is the energy used to pressurise the hydraulic fluid usually taken from in normal operating conditions?
 - (a) From the engine(s).
 - (b) From the APU.
 - (c) From the PTU.
 - (d) From the RAT.
8. What is the ESSENTIAL BUS?
 - (a) It is an electrical distribution bar from which all essential equipment needed to sustain flight obtain electrical power.
 - (b) It is a data bus that carries all data from the ADIRU and FMGC to the flight computers.
 - (c) It is the electrical supply system that have some airports to provide on-ground electrical power to parked aircraft.
 - (d) It is the component of the pneumatic system that pressurises the cabin so that the airplane can carry people.
9. Which of the following statements is FALSE about the fuel system of a large passenger airplane?
 - (a) Fuel cannot be circulated among tanks.
 - (b) Engines can be fed with fuel from different tanks.
 - (c) Fuel is often used to cool other fluids such as engine oil or hydraulic fluid.
 - (d) Fuel is usually stored in the wings to compensate for the bending moment caused by lift distribution.
10. What is true about an electro-hydraulic flight control system (fly-by-wire) when compared with the classical mechanical control system?
 - (a) All options are correct.
 - (b) The fly-by-wire system allows for aircraft protection against flying out of its flight envelope.
 - (c) A mechanical system is typically much heavier than the electro-hydraulic system.
 - (d) The fly-by-wire system allows for flight augmentation capabilities, enhancing stability and tuning manoeuvrability.

Correct Answers:

BLO-1: Concepts on flight principles and aircraft performance

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	d	d	b	c
2	a	b	c	c	d
3	a	c	c	c	c
4	a	b	c	c	c
5	a	c	c	a	b
6	a	a	a	c	a
7	a	d	d	c	a
8	a	c	a	d	a
9	a	b	d	b	a
10	a	b	a	a	d

BLO-2: Flight principles related problems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	a	c	c
2	a	c	b	a	d
3	a	b	b	d	b
4	a	c	d	c	d
5	a	d	b	c	d

BLO-3: Concepts on aircraft stability and control

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	c	d	d
2	a	b	c	b	c
3	a	a	d	a	c
4	a	c	b	b	d
5	a	c	a	d	c
6	a	d	d	c	a
7	a	b	b	c	b
8	a	d	a	b	d
9	a	b	c	a	a
10	a	d	a	a	b

BLO-4: Aircraft performance related problems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	d	b	b	b
2	a	a	c	d	b
3	a	b	b	d	c
4	a	c	d	c	d
5	a	c	d	c	c

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	d	a	b	c
2	a	a	d	b	c
3	a	a	c	a	d
4	a	c	a	d	c
5	b	d	a	d	a
6	a	a	b	d	c
7	a	b	c	a	d
8	a	a	c	a	b
9	a	d	c	b	c
10	a	b	d	b	c

BLO-6: Concepts on avionics and navigation

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	d	a	a
2	a	b	a	c	d
3	a	c	c	b	d
4	a	a	a	c	d
5	a	b	d	b	b
6	a	c	a	c	c
7	a	c	a	c	a
8	a	a	b	a	a
9	a	b	a	c	a
10	a	b	d	c	c

BLO-7: Concepts on materials and aeronautical regulations

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	d	c	c	a
2	a	d	d	c	c
3	a	a	b	c	c
4	a	d	b	b	d
5	a	c	a	b	d
6	a	d	c	c	b
7	a	b	c	b	d
8	a	d	d	a	b
9	a	b	a	d	c
10	a	c	a	d	c

BLO-8: Concepts on aircraft systems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	c	c	a
2	a	b	c	a	c
3	a	b	d	d	d
4	a	a	a	a	d
5	a	c	c	c	d
6	a	d	c	b	b
7	a	c	a	a	c
8	a	a	a	a	a
9	a	b	c	b	c
10	a	c	d	c	b

BLO-1: Concepts on flight principles and aircraft performance

1. Which non-dimensional parameter determines the compressibility regime of flight (low subsonic, subsonic compressible, transonic, supersonic or hypersonic)?
 - (a) The Mach number.
 - (b) The Reynolds number.
 - (c) The Prandtl number.
 - (d) The Bernoulli number.
2. What is the main effect on the lift coefficient (C_L) of a high-lift device that increases the camber?
 - (a) Increase the lift coefficient at 0-angle of attack (C_{L_0}).
 - (b) Increase the slope of the lift coefficient curve (C_{L_α}).
 - (c) Increase the angle of attack at which stall occurs (α_s).
 - (d) Decrease the drag coefficient (C_D) without modifying C_L .
3. Which of the following mechanisms is responsible for induced drag in a finite-span wing?
 - (a) The wing tip vortices due to the pressure difference between the wing's lower and upper surfaces.
 - (b) The skin friction due to the contact of the air stream with the wing surface.
 - (c) The pressure drop at the trailing edge due to the wake.
 - (d) The impact of the air molecules on the wing's leading edge.
4. During a coordinated banked turn, what happens to the load factor n ?
 - (a) $n > 1$
 - (b) $n = 1$
 - (c) $n < 1$
 - (d) The sign of n depends on whether it's a left or right turn.
5. Two identical airplanes that weigh the same are in balanced straight and level flight at different altitudes. Which of the following statements is decidedly FALSE?
 - (a) The stall true airspeed is the same for both.
 - (b) The stall angle of attack is the same for both.
 - (c) The lift is the same for both.
 - (d) The stall equivalent airspeed is the same for both.
6. Which of the following conditions does definitely NOT correspond to a best angle climb of a jet airplane?
 - (a) Maximum excess power.
 - (b) Minimum drag.
 - (c) Maximum excess thrust.
 - (d) Maximum aerodynamic efficiency.
7. Why do airplanes usually climb during cruise?
 - (a) Due to loss of weight, to keep cruise conditions optimal.
 - (b) Due to engine overheating, to keep cruise conditions optimal.
 - (c) To avoid collision with other aircraft, even if cruise conditions deteriorate.
 - (d) So the pilot doesn't get bored.
8. For a given airplane in straight and level flight at a given altitude, what can we say about the relative value of the velocities for maximum endurance V_E , maximum range V_R , best aerodynamic efficiency $V_{f_{max}}$ and minimum drag $V_{D_{min}}$?
 - (a) $V_R > V_E$ and $V_{f_{max}} = V_{D_{min}}$
 - (b) $V_R > V_E = V_{f_{max}} = V_{D_{min}}$
 - (c) $V_{D_{min}} = V_{f_{max}} = V_R > V_E$
 - (d) $V_{f_{max}} = V_R > V_E = V_{D_{min}}$
9. Which of the following statements concerning takeoff and landing is true?
 - (a) Both takeoff and landing distances are influenced by wind and airport altitude.
 - (b) Takeoff distance is influenced by wind and airport altitude, but not landing.
 - (c) Landing distance is influenced by wind and airport altitude, but not takeoff.
 - (d) Both takeoff and landing distances are influenced by wind but not by airport altitude.
10. Which of the following performance optimals is NOT affected by wind?
 - (a) Endurance.
 - (b) Best angle of climb.
 - (c) Range.
 - (d) Maximum gliding distance.

BLO-2: Flight principles related problems

1. A small airplane is engaged in a coordinated banked turn at an unknown altitude. We know its speed is $V = 2 V_S|_{n=1}$, where $V_S|_{n=1}$ is its stall speed in straight and level flight at the same altitude. What is the maximum bank angle (ϕ_{max}) that can be achieved if the plane is to keep its altitude and speed?
 - (a) $\phi_{max} = 75.5^\circ$
 - (b) $\phi_{max} = 60^\circ$
 - (c) $\phi_{max} = 13.2^\circ$
 - (d) It depends on the mass of the airplane.

2. The drag polar of an airplane changes from $C_D = 0.04 + 0.15C_L^2$ in clean configuration to $C_D = 0.05 + 0.075C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.365$
 - (b) $C_L = 0.341$
 - (c) $C_L = 0.408$
 - (d) $C_L = 0.432$

3. An aerobatics airplane with lift coefficient $C_L = 0.2 + 0.1\alpha$ with α in $^\circ$ engages in a perfect dive (vertical descent). What should the angle of attack (α) be?
 - (a) $\alpha = -2^\circ$
 - (b) $\alpha = 2^\circ$
 - (c) $\alpha = 0.5^\circ$
 - (d) $\alpha = -0.5^\circ$

4. What is the maximum aerodynamic efficiency (f_{max}) of an airplane with drag polar $C_D = 0.03 + 0.15C_L^2$?
 - (a) $f_{max} = 7.45$
 - (b) $f_{max} = 15.5$
 - (c) $f_{max} = 16.4$
 - (d) $f_{max} = 12.7$

5. An airplane of mass $m = 2000$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 20 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $D = 1000$ N and $D_p \neq D_i$.
 - (b) $D = 1000$ N and $D_p = D_i = 500$ N.
 - (c) $D = 500$ N and $D_p = D_i = 250$ N.
 - (d) $D = 500$ N and $D_p \neq D_i$.

BLO-3: Concepts on aircraft stability and control

1. Which command and primary control surface pilot the aircraft around its roll axis?
 - (a) The stick (or yoke) and the ailerons.
 - (b) The stick (or yoke) and the elevator.
 - (c) The pedals and the rudder.
 - (d) The throttle and the flaps.
2. The sign of which aerodynamic coefficient is responsible for longitudinal stability?
 - (a) C_{m_α} , the change in pitching moment due to a change of angle of attack.
 - (b) C_{Y_β} , the change in lateral force due to a change of sideslip angle.
 - (c) C_{L_α} , the change in lift due to a change of angle of attack.
 - (d) C_{n_β} , the change in yaw moment due to a change of sideslip angle.
3. An aircraft is statically stable around one of its axis. What can we say about dynamic stability around that axis?
 - (a) It may or may not be stable.
 - (b) It must be unstable.
 - (c) It must be stable.
 - (d) It depends on the axis we refer to.
4. What is the aim of trim tabs?
 - (a) Ease the continuous effort the pilot would have to apply on the commands.
 - (b) Allow for dynamic mode damping.
 - (c) Achieve an extra lift when needed.
 - (d) Increase aircraft manoeuvrability.
5. The stability of which mode is critically compromised by the relative position of the aerodynamic centre and the centre of gravity?
 - (a) Pitch oscillation.
 - (b) Phugoid.
 - (c) Dutch roll.
 - (d) Spiral.
6. Which of the following features are NOT characteristic of the pitch oscillation mode?
 - (a) The angle of attack remains constant along the oscillation.
 - (b) It is fast and usually periodic.
 - (c) It can be left unstable if appropriately stabilised with automatic control.
 - (d) It can be dangerous if it is unstable.
7. What is FALSE about the phugoid motion?
 - (a) The vertical stabiliser plays an important role in making it stable.
 - (b) It corresponds to a periodic exchange of potential and kinetic energy.
 - (c) It is a slow, periodic, poorly damped mode.
 - (d) It is related to longitudinal flight in the vertical plane.
8. What is a skidding (operational) turn?
 - (a) It is a turn with stick only (no pedals).
 - (b) It is a turn with pedals only (no stick).
 - (c) It is a turn that applies pedals to correct for the sideslip induced by stick operation.
 - (d) It is an aerobatic manoeuvre.
9. When is crosswind straight flight (with sideslip) desirable?
 - (a) During a crosswind landing to align the aircraft with the runway.
 - (b) For usual cruise flight to optimise fuel consumption when the wind is transversal to the trajectory.
 - (c) When it rains, to avoid engine water ingestion.
 - (d) When one of the hydraulic circuits fails.
10. What should the pilot of a twin-engined airplane do if one engine fails to correct for yaw moment imbalance?
 - (a) Deflect the rudder toward the operating engine.
 - (b) Deflect the ailerons to lift the wing on which the working engine is.
 - (c) Deploy the ram air turbine.
 - (d) Switch off the working engine.

BLO-4: Aircraft performance related problems

1. The required thrust of a jet airplane is given by $T_R = 0.25\bar{\rho}V^2 + 0.02W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.5$, what is the optimal velocity for maximum endurance (V_E)?
 - (a) $V_E = 92.1 \text{ ms}^{-1}$
 - (b) $V_E = 121 \text{ ms}^{-1}$
 - (c) $V_E = 70 \text{ ms}^{-1}$
 - (d) It depends on the available thrust T_A , which is unknown.
2. An airplane with drag polar $C_D = 0.04 + 0.15C_L^2$, wing surface $S = 40 \text{ m}^2$ and mass $m = 2000$ kg is flying at a speed $V = 120 \text{ ms}^{-1}$. What is the minimum engine thrust required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.4 \text{ kgm}^{-3}$? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $T_R = 5130 \text{ N}$
 - (b) $T_R = 5180 \text{ N}$
 - (c) $T_R = 5240 \text{ N}$
 - (d) $T_R = 4960 \text{ N}$
3. A propeller-powered airplane of mass $m = 1500$ kg has a required power curve $P_R = 0.3\bar{\rho}V^3 + 0.03W^2/\bar{\rho}V$. What is the minimum engine power needed at sea level ($\bar{\rho} = 1$) to achieve a rate of climb $\dot{h} = 5 \text{ ms}^{-1}$? (take $g \simeq 10 \text{ ms}^{-2}$)
 - (a) $P_A = 247 \text{ kW}$
 - (b) $P_A = 269 \text{ kW}$
 - (c) $P_A = 293 \text{ kW}$
 - (d) $P_A = 315 \text{ kW}$
4. A glider is flying in still air at its maximum aerodynamic efficiency $f_{max} = 50$. If its current altitude is $h = 1000 \text{ m}$, what is the maximum horizontal distance (d) it can fly before hitting the ground?
 - (a) $d = 50 \text{ km}$
 - (b) $d = 20 \text{ km}$
 - (c) $d = 1 \text{ km}$
 - (d) $d = 100 \text{ km}$
5. The required thrust of a jet-engined airplane of mass $m = 1500$ kg, wing area $S = 25 \text{ m}^2$, and maximum lift coefficient before stall $C_{L_{max}} = 1.2$, is given by $T_R = 0.86\bar{\rho}V^2 + 0.007W^2/\bar{\rho}V^2$, and the maximum available thrust can be considered independent of V : $T_A \leq 3000 \text{ N}$. Which is the lowest velocity at which this airplane can be flown at straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.33$)? (take $g \simeq 10 \text{ ms}^{-2}$ and $\rho_0 = 1.225 \text{ kgm}^{-3}$)
 - (a) $V \simeq 49.7 \text{ ms}^{-1}$
 - (b) $V \simeq 44.2 \text{ ms}^{-1}$
 - (c) $V \simeq 92.8 \text{ ms}^{-1}$
 - (d) $V \simeq 64 \text{ ms}^{-1}$

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

1. Quina de les següents afirmacions és certa?
 - (a) Els motor a coet i els turbojet són motors a jet.
 - (b) El motor a coet pot funcionar si la pressió a la cambra de combustió és inferior a la pressió exterior.
 - (c) Un motor coet no pot funcionar sota l'aigua.
 - (d) Cap de les anteriors és certa.
2. Sobre una hèlix,
 - (a) el pas de l'hèlix augmenta a mesura que augmenta la velocitat.
 - (b) l'angle d'atac és positiu durant la carrera d'aterratge.
 - (c) l'angle d'atac és negatiu durant la carrera d'enlairament.
 - (d) Cap de les anteriors és certa.
3. Motors a pistons,
 - (a) la relació de compressió d'un motor Otto és menor que un motor Diesel.
 - (b) a paritat de potència, un motor Diesel emet més diòxid de carboni que un motor Otto.
 - (c) els motors Otto de 2 temps tenen major rendiment que els de 4 temps.
 - (d) Cap de les anteriors.
4. Quin dels següents motors ens permetrà assolir velocitats més elevades?
 - (a) un estatoreactor (ramjet)
 - (b) un turboreactor
 - (c) un turbofan
 - (d) un turbopropulsor (turboprop)
5. Un difusor
 - I - permet disminuir la velocitat d'entrada de l'aire al compressors
 - II - augmenta la pressió de l'aire
 - III - augmenta la densitat de l'aire
 - (a) totes són certes
 - (b) només la I i la II són certes
 - (c) només la II i la III són certes
 - (d) només la I i la III són certes
6. La funció de la cambra de combustió d'un turbojet és:
 - (a) proporcionar temperatura al fluid
 - (b) proporcionar pressió i temperatura al fluid
 - (c) proporcionar pressió al fluid
 - (d) cap de les anteriors
7. En un helicòpter, l'efecte terra
 - I - augmenta la sustentació de l'helicòpter
 - II - millora el vol a altes velocitats
 - III - es manté mentre l'alçada de vol és inferior al diàmetre del rotor principal
 - (a) només la 1 i la 3 són certes
 - (b) només la 3 és certa
 - (c) només la 2 és certa
 - (d) només la 2 i la 3 són certes
8. Per un helicòpter en vol horitzontal
 - I - la distribució de la velocitat a les pales del rotor principal és proporcional a la distància respecte l'eix del rotor.
 - II - la distribució de velocitats al rotor principal induïx un moment de balanceig a l'helicòpter.
 - III - la pala que es retira tendeix a moure's amunt per tal de compensar la zona de velocitat invertida.
 - (a) només la 2 és certa.
 - (b) només la 3 és certa.
 - (c) només la 2 i la 3 són certes.
 - (d) només la 1 i la 3 són certes.
9. La velocitat mínima a la punta dels àleps del rotor principal d'un helicòpter,
 - (a) està limitada per l'energia cinètica del rotor principal i el risc d'entrada en pèrdua dels àleps
 - (b) està limitada pel soroll que aquests produeixen i els efectes de compressibilitat
 - (c) està limitada pels efectes de compressibilitat i el risc d'entrada en pèrdua de l'àlep
 - (d) cap de les anteriors
10. Un motor coet proporciona 10kN d'empenta amb un flux màssic de 3.5kg/s. Quin tipus de propulsió equipa el coet sonda?
 - (a) Propulsor sòlid
 - (b) Propulsor híbrid
 - (c) Propulsor líquid bipropelent criogènic
 - (d) Propulsor líquid bipropelent emmagatzemable a temperatura ambient

BLO-6: Concepts on avionics and navigation

1. 'The warranty that the error will not be higher than some threshold and, if it happens, an alarm will be received on time' is the definition of:
 - (a) Integrity.
 - (b) Accuracy.
 - (c) Continuity.
 - (d) Availability.
2. 'The ability of the total system to perform its function without unscheduled interruptions during the intended operation' is the definition of:
 - (a) Continuity.
 - (b) Accuracy.
 - (c) Integrity.
 - (d) Availability.
3. If we say that DME accuracy is 185 meters, we are saying that:
 - (a) 95 percent of the time the navigation error is equal to or smaller than 185 m.
 - (b) The DME accuracy is better than GPS accuracy.
 - (c) The standard deviation of the error is almost 200 m.
 - (d) The system provides no integrity and must not be used.
4. Radionavigation aid system that provides information about the horizontal angle to approach a runway:
 - (a) Localizer.
 - (b) Automatic Direction Finder.
 - (c) Glide Slope.
 - (d) VHF Omnidirectional Range.
5. Radionavigation aid system that is often installed in order to signal the decision height when approaching a runway:
 - (a) Middle Marker.
 - (b) Localizer.
 - (c) Glide Slope.
 - (d) VHF Omnidirectional Range.
6. Which is the Operational System Mode when a position failure didn't happen but a failure is detected?
 - (a) False alarm.
 - (b) Nominal operation.
 - (c) Justified alarm.
 - (d) Misleading information.
7. In a typical radionavigation system, there is a monitor that shutdowns the system if the following happens:
 - (a) The measured error is smaller than the error threshold.
 - (b) The measured error is bigger than zero.
 - (c) The measured error is bigger than the error threshold.
 - (d) The error threshold generates too many false alarms.
8. The altimeter setting needed to guarantee the vertical separation of aircraft during cruise is called:
 - (a) STD.
 - (b) QNH.
 - (c) FL.
 - (d) QNE.
9. The imaginary line used to change the altimeter setting from QNH to STD is called:
 - (a) Transition Altitude.
 - (b) QNE.
 - (c) Flight Level.
 - (d) Transition Level.
10. The altimeter setting needed for the altimeter to indicate the official airport altitude is called:
 - (a) QNH.
 - (b) STD.
 - (c) FL.
 - (d) QNE.

BLO-7: Concepts on materials and aeronautical regulations

1. L'ala d'un avió en vol horitzontal té

I - la part superior sotmesa a esforços de tracció.

II - la part inferior sotmesa a esforços de tracció.

III - la part inferior sotmesa a esforços de compressió.

(a) la II és certa

(b) la I és certa

(c) la II i la III són certes

(d) la I i la III són certes

2. En una estructura alar "*wing box*",

(a) La superfície alar forma part de la biga.

(b) la superfície alar és fina i transmet els esforços a la biga.

(c) les costelles transmeten els esforços de la la superfície alar a la biga.

(d) cap de les anteriors

3. En un material compost

(a) cap de les anteriors

(b) es combina la resistència a compressió de les fibres amb la resistència a tracció de la matriu.

(c) És senzill fer-ne un bon reciclatge.

(d) les fibres han d'estar aliniades.

4. Quina de les opcions és correcta respecte a la valoració de les propietats de l'alumini en estat pur, si el comparem amb altres materials?

(a) Les propietats mecàniques de l'alumini pur són en general molt pobres.

(b) La temperatura d'ebullició de l'alumini és molt elevada.

(c) La resistència a la corrosió és molt baixa.

(d) Cap de les anteriors

5. Quina propietat dels materials ceràmics no els fa aptes per a aplicacions estructurals?

(a) la seva fragilitat

(b) la seva estabilitat química

(c) el seu caràcter refractari

(d) la seva duresa

6. Which of the following is the organization created on 1960 with the objective of unifying European Air Traffic Control and Air Navigation services?

(a) EUROCONTROL.

(b) DGAC.

(c) JAA.

(d) FAA.

7. The nationality of an aircraft is given by:

(a) Registry country.

(b) Airline country.

(c) Manufacturing country.

(d) Captain country.

8. The OACI technical annex dealing with Aeronautical Telecommunications is Annex number?

(a) 10.

(b) 4.

(c) 3.

(d) 14.

9. Which of the following is the organization created by European aircraft manufacturers in order to ease adoption of new products and coordinate airworthiness standards?

(a) JAA.

(b) EUROCONTROL.

(c) DGAC.

(d) FAA.

10. The country in charge of issuing the airworthiness certificate of an aircraft is the:

(a) Registry country.

(b) Airline country.

(c) Manufacturing country.

(d) Captain country.

BLO-8: Concepts on aircraft systems

1. What is a fly-by wire system?
 - (a) An electro-hydraulic flight control system
 - (b) A mechanical flight control system
 - (c) An hydro-mechanical flight control system
 - (d) None of the others
2. Why do some flight control systems need an artificial “feel”?
 - (a) To give the pilot feedback on the effects of his actions on the commands.
 - (b) To make the aircraft stable.
 - (c) To help the pilot reduce the effort on the commands.
 - (d) To make the passengers more comfortable.
3. What is the main aim of the hydraulic system?
 - (a) To actuate flight control surfaces.
 - (b) To provide the cabin with pressurised water.
 - (c) To stabilise flight.
 - (d) To power the pneumatic system.
4. How is redundancy achieved for the hydraulic system of a large passenger aircraft?
 - (a) by having different hydraulic circuits that can exchange pressure but not hydraulic fluid.
 - (b) by having identical hydraulic circuits isolated from one another.
 - (c) by having identical hydraulic circuits that can exchange both hydraulic fluid and pressure.
 - (d) by having different hydraulic circuits that can exchange hydraulic fluid.
5. What is a Ram Air Turbine (RAT)?
 - (a) An emergency wind-driven generator to provide hydraulic or electric power.
 - (b) An auxiliary gas turbine to provide hydraulic or electric power.
 - (c) The computer that controls engine operation.
 - (d) A device used to measure airspeed.
6. A typical electrical system of a large aircraft provides...
 - (a) both AC (Alternate Current) power and DC (Direct Current) power.
 - (b) either AC (Alternate Current) power or DC (Direct Current) power, depending on the airplane.
 - (c) only AC (Alternate Current) power.
 - (d) only DC (Direct Current) power.
7. Which of the following elements cannot be found in an electrical system?
 - (a) A power transfer unit (PTU).
 - (b) A transformer/rectifier.
 - (c) A Static Inverter.
 - (d) Batteries.
8. Which of the following tasks does not involve the pneumatic system?
 - (a) control surface actuation.
 - (b) engine starting.
 - (c) hydraulic reservoir pressurisation.
 - (d) air conditioning.
9. What is the Auxiliary Power Unit (APU)?
 - (a) A gas turbine that can deliver shaft power for electrical generation and produce bleed air.
 - (b) A valve that allows transferring power from an hydraulic circuit to another.
 - (c) An emergency wind-driven generator to produce power in case of power loss in the electrical system.
 - (d) A set of batteries to deliver electrical power in case of an emergency.
10. Which parts of an aircraft are usually reached by the anti-ice system?
 - (a) The wing leading edge and the front section of the engine nacelle.
 - (b) The trailing edge of the wing.
 - (c) The winglets and the landing gear.
 - (d) The engine bleed valves.

Correct Answers:

BLO-1: Concepts on flight principles and aircraft performance

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	c	a	a	d
2	a	d	a	a	b
3	a	b	d	a	c
4	a	a	c	a	d
5	a	a	a	d	a
6	a	a	a	c	b
7	a	a	c	c	b
8	a	b	c	a	d
9	a	c	a	a	d
10	a	b	a	c	d

BLO-2: Flight principles related problems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	b	d	c	d
2	a	d	d	d	b
3	a	b	b	b	c
4	a	b	a	c	b
5	a	d	c	d	b

BLO-3: Concepts on aircraft stability and control

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	c	a	b
2	a	b	a	d	d
3	a	d	b	a	d
4	a	c	c	a	b
5	a	a	c	c	a
6	a	b	c	a	b
7	a	b	d	b	b
8	a	c	d	b	c
9	a	c	a	c	c
10	a	a	d	d	b

BLO-4: Aircraft performance related problems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	a	a	d
2	a	a	a	b	a
3	a	a	c	c	c
4	a	b	c	a	a
5	a	b	a	c	c

BLO-5: Concepts on aerospace propulsion, helicopters and space systems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	b	b	a	b
2	a	c	b	b	a
3	a	a	b	c	a
4	a	d	d	a	c
5	a	b	a	b	c
6	a	c	d	a	d
7	a	b	d	a	a
8	a	d	b	a	c
9	a	d	a	b	d
10	a	d	d	d	d

BLO-6: Concepts on avionics and navigation

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	c	d	d	a
2	a	b	a	d	d
3	a	a	b	b	d
4	a	b	a	c	c
5	a	a	c	c	d
6	a	c	b	a	a
7	c	a	d	d	b
8	a	b	b	c	c
9	a	a	c	c	b
10	a	c	b	b	b

BLO-7: Concepts on materials and aeronautical regulations

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	c	a	d
2	a	d	d	b	b
3	a	b	b	c	b
4	a	a	b	b	d
5	a	c	b	d	a
6	a	c	a	c	b
7	a	b	c	b	c
8	a	c	d	a	a
9	a	b	c	d	d
10	a	a	c	c	c

BLO-8: Concepts on aircraft systems

	Correct Answers				
Question	P00	P01	P02	P03	P04
1	a	a	a	a	a
2	a	b	b	c	d
3	a	c	c	c	c
4	a	d	c	b	d
5	a	a	c	c	a
6	a	a	c	a	a
7	a	a	b	c	d
8	a	b	c	b	a
9	a	b	c	d	c
10	a	d	a	d	d

BLO-1: Flight Principles

1. How is the wing tip placed/oriented with respect to wing root for a plane with positive dihedral?
 - (a) Higher up.
 - (b) Further back.
 - (c) Lower down.
 - (d) Further front.
2. Which of the following statements about ISA is FALSE?
 - (a) The speed of sound increases with altitude.
 - (b) Sea level pressure is considered the same in the Equator and the North Pole.
 - (c) Air density decreases with altitude.
 - (d) Temperature is considered constant for some altitude above the Tropopause.
3. Consider a plane with positive dihedral. What is the position of the wing root with respect to the wing tip?
 - (a) Lower down.
 - (b) Further ahead.
 - (c) Further back.
 - (d) None of the others.
4. Sweep helps an aircraft ...
 - (a) to recover stability about the z-axis.
 - (b) to recover stability about the y-axis.
 - (c) to recover stability about the x-axis.
 - (d) to achieve higher speeds.
5. Which one of the following manoeuvres entails a higher load factor?
 - (a) Coordinated banked turn.
 - (b) Rectilinear ascent.
 - (c) Rectilinear decent.
 - (d) Straight and level flight.
6. An aerobatics airplane with lift coefficient $C_L = 0.013 + 0.21\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 145$ kts ($1 \text{ m/s} = 1.944$ kts) with $\alpha = 0.7^\circ$. What should the speed (V_{if}) be to perform a balanced inverted flight at the same altitude with an angle of attack $\alpha = -0.8^\circ$?
 - (a) $V_{if} = 147$ kts.
 - (b) $V_{if} = 159$ kts.
 - (c) $V_{if} = 124$ kts.
 - (d) $V_{if} = 105$ kts.
7. Calculate the stall speed V_S of an airplane of mass $m = 820$ kg and wing surface $S = 10.5 \text{ m}^2$ with $C_L = 0.16 + 0.27\alpha$ with $\alpha_S = 14.5^\circ$ at an altitude such that $\bar{\rho} = 0.51$ (take $g = 9.8 \text{ m/s}^2$ and $\rho_0 = 1.225 \text{ kg/m}^3$).
 - (a) $V_S = 24.5 \text{ m/s}^2$.
 - (b) $V_S = 23.3 \text{ m/s}^2$.
 - (c) $V_S = 32.9 \text{ m/s}^2$.
 - (d) $V_S = 30.4 \text{ m/s}^2$.
8. A small airplane of mass $m = 1150$ kg is engaged in a coordinated banked turn with bank angle $\Phi = 45^\circ$ at a speed $V = 260$ km/h. What is the turn radius R_t for this manoeuvre? (consider $g \simeq 9.8 \text{ m/s}^2$).
 - (a) $R_t = 532$ m.
 - (b) $R_t = 658$ m.
 - (c) $R_t = 241$ m.
 - (d) $R_t = 464$ m.
9. An aircraft with a drag polar $C_D = 0.022 + 0.05C_L^2$. Which of the following conditions corresponds to maximum aerodynamic efficiency?
 - (a) All options are correct.
 - (b) $C_L = 0.663$.
 - (c) $C_D = 0.044$.
 - (d) $f = 15.1$.
10. An airplane of mass $m = 1450$ kg is cruising in straight and level flight with a suboptimal aerodynamic efficiency $f = 17 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8 \text{ m/s}^2$)
 - (a) $D = 836$ N and $D_p \neq D_i$.
 - (b) $D = 836$ N and $D_p = D_i = 418$ N.
 - (c) $D = 1670$ N and $D_p = D_i = 836$ N.
 - (d) $D = 1670$ N and $D_p \neq D_i$.

BLO-2: Aircraft Stability & Control

11. Which of the following statements is FALSE?
- (a) For $C_{l_r} < 0$ one observes Roll-Yaw Coupling.
 - (b) For $C_{l_\beta} < 0$ one observes Dihedral effect.
 - (c) For $C_{l_p} < 0$ one observes Roll Damping.
 - (d) For $C_{n_r} < 0$ one observes Yaw Damping.
12. Which of the following manoeuvres can be considered a lateral equilibrium?
- (a) All are lateral equilibria.
 - (b) Crosswind straight flight.
 - (c) Coordinated banked turn.
 - (d) Operational banked turn.
13. Which of the two possible straight and level flight regimes is easier to fly stably?
- (a) Regime 1 (R1).
 - (b) Regime 2 (R2).
 - (c) Both Regime 1 (R1) and Regime 2 (R2).
 - (d) Regime 0 (R0).
14. Which statement is FALSE for a positive static stability?
- (a) All are incorrect.
 - (b) Implies good maneuverability
 - (c) Generates instability
 - (d) Is achieved when Neutral point is ahead of center of gravity
15. Dutch roll couples oscillations in ...
- (a) Yaw and roll.
 - (b) Yaw and pitch.
 - (c) Pitch and roll.
 - (d) Spiral and roll.
16. What is the effect of increasing the aircraft stability?
- (a) Less manoeuvrability.
 - (b) More manoeuvrability.
 - (c) Manoeuvrability is not affected by stability.
 - (d) Increase endurance.
17. How can directional stability be tuned?
- (a) Vertical stabilizer size.
 - (b) Horizontal stabilizer size.
 - (c) Throttle.
 - (d) Ailerons trim tab size.
18. Which of the following hypotheses is NOT a requirement for longitudinal flight?
- (a) The angle of attack must be small.
 - (b) Propulsion moments must be orthogonal to the symmetry plane of the aircraft.
 - (c) Aerodynamics forces must be parallel to the symmetry plane of the aircraft.
 - (d) Aircraft reflexion-symmetry plane must be vertical.
19. Which of the following statements is TRUE regarding a negative static stability?
- (a) All options are correct.
 - (b) Implies good maneuverability.
 - (c) Generates instability.
 - (d) Is achieved when the neutral point is ahead of center of gravity.
20. What does Dynamic Stability around an axis requires, at the very least?
- (a) All options are correct.
 - (b) The appearance of a restoring moment around this axis upon perturbation.
 - (c) The appearance of a damping moment around this axis during restoration after perturbation.
 - (d) Static stability around this axis.

BLO-3: Aircraft Performances

21. What happens to the minimum drag D_{min} as a constant mass airplane loses altitude?
- It occurs at lower speeds (moves left) but remains fairly constant (neither goes up nor down).
 - It occurs at higher speeds (moves right) and decreases (moves down).
 - It occurs at higher speeds (moves right) but remains fairly constant (neither goes up nor down).
 - It occurs at lower speeds (moves left) and increases (moves up).
22. What is TRUE for operational and absolute ceiling?
- Operational is lower than absolute.
 - Operational is higher than absolute.
 - The type of engine defines the higher one.
 - They are synonyms.
23. What is essentially TRUE about maximum endurance conditions at cruise?
- For a propeller-powered airplane, it roughly corresponds to minimum required power.
 - Tail wind increases endurance.
 - High altitude is better.
 - None of the others.
24. How does tail wind affect maximum range R_{max} and the corresponding aerodynamic speed $V_{R_{max}}$ of a given airplane?
- $V_{R_{max}}$ decreases and R_{max} improves.
 - $V_{R_{max}}$ increases and R_{max} improves.
 - $V_{R_{max}}$ decreases and R_{max} worsens.
 - It doesn't affect whatsoever.
25. The required thrust of a jet airplane is given by $T_R = 0.24\bar{\rho}V^2 + 0.025W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1500$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = \rho/\rho_0 = 0.5$, what is velocity (V_E) for maximum endurance? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V_E = 97.4$ m/s.
 - $V_E = 61.4$ m/s.
 - $V_E = 129$ m/s.
 - $V_E = 81.9$ m/s.
26. The required thrust of a jet airplane is given by $T_R = 0.24\bar{\rho}V^2 + 0.018W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1350$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.52$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D = 1740$ N.
 - $D = 2010$ N.
 - $D = 1650$ N.
 - $D = 1920$ N.
27. A propeller-powered airplane of mass $m = 1440$ kg has a required power curve $P_R = 0.24\bar{\rho}V^3 + 0.016W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s²)
- $V_{(R/C)_{max}} = 45.9$ m/s.
 - $V_{(R/C)_{max}} = 60.4$ m/s.
 - $V_{(R/C)_{max}} = 79.4$ m/s.
 - $V_{(R/C)_{max}} = 105$ m/s.
28. A propeller-powered airplane with drag polar $C_D = 0.017 + 0.131C_L^2$, wing surface $S = 27$ m² and mass $m = 1650$ kg is flying at a speed $V = 105$ m/s. The available engine power at sea level is $(P_A)_{SL} = 850$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.52$ kg/m³? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $\delta_x = 0.512$.
 - $\delta_x = 0.57$.
 - $\delta_x = 0.423$.
 - $\delta_x = 0.217$.
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 43$ finds an isolated thermal column amidst an otherwise still atmosphere. If the glider flies at $h = 5700$ ft, what is the maximum distance he can reach? (take 1 m = 3.28 ft)
- $d = 74.7$ km.
 - $d = 106$ km.
 - $d = 86$ km.
 - $d = 41.5$ km.
30. The required thrust of a jet-engined airplane of mass $m = 1500$ kg, wing area $S = 21$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.05$, is given by $T_R = 0.25\bar{\rho}V^2 + 0.037W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 14500$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.71$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 39.2$ m/s.
 - $V \simeq 33.4$ m/s.
 - $V \simeq 42.8$ m/s.
 - $V \simeq 36.9$ m/s.

BLO-4: Aircraft Classification & Recognition / Materials & Structures

31. The S-N curve is related to ...
- fatigue.
 - plasticity.
 - elasticity.
 - hardness.
32. What can be said about composite materials?
- They are made of at least two or more different materials.
 - They are made of at least three or more different materials.
 - They can't be discontinuously reinforced.
 - They can't be arranged in different layers on top of each-other.
33. Which material is more suitable for turbine blades?
- Titanium.
 - Steel.
 - Aluminium.
 - Copper.
34. A structural composite material ...
- combines the better compression properties of the matrix with the better tension properties of the fibers.
 - combines the better tension properties of the matrix with the better compression properties of the fibers.
 - is highly anisotropic if layers of fibers are aligned in alternated directions.
 - None of the others.
35. Aluminium has, in general, good qualities for aircraft construction. What is its melting point?
- 660° C.
 - 1400° C.
 - 882,5° C.
 - 911° C.
36. To which airplane corresponds ICAO designation AT43?
- ATR 42-300.
 - Cessna 40-300.
 - Boeing 747-300.
 - Airbus 400-300.

37. According to ICAO, below which mass is defined an aircraft as "light" L in wake turbulence?

- 7000 kg.
- 3500 kg.
- 19000 kg.
- 1500 kg.

38. Identify the following aircraft



- Airbus A350-1000.
- Boeing 777-200.
- Boeing 777-300.
- Airbus A350-900.

39. Which aircraft manufacturer built the following airplane?



- Embraer.
- Boeing.
- McDonald Douglas.
- Airbus.

40. Which of the following designation identifies a wide body aircraft?

- A310.
- A320.
- A220.
- A321.

BLO-5: Propulsion systems

41. Propulsive efficiency of turbojets ...

- (a) depends on aircraft speed and exhaust jet velocity.
- (b) depends on compressor type.
- (c) is higher with tubular combustion chambers.
- (d) reaches its maximum at very low speed.

42. Subsonic nozzles are ...

- (a) convergent.
- (b) divergent.
- (c) obsolete in modern aircraft.
- (d) equipped with an afterburner only in turbofans.

43. A turbofan engine has a typical fuel mass flow $m_f = 5$ kg/s and a thrust $T = 250$ kN. Calculate its TSFC.

- (a) TSFC=20 (g/s)/kN.
- (b) TSFC=0.02 (g/s)/kN.
- (c) TSFC=50 (g/s)/kN.
- (d) TSFC=0.05 (g/s)/kN.

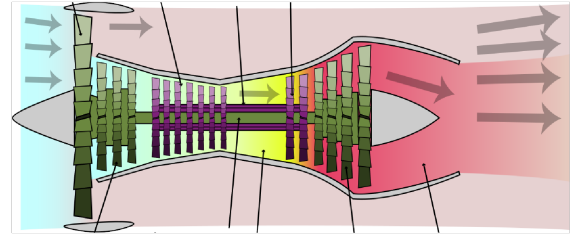
44. What is the by-pass ratio in a turbofan engine?

- (a) The ratio of air massflow through the *cold* duct to that entering the core engine (*hot*).
- (b) The ratio of fan blade count to compressor blade count.
- (c) The volumetric compression ratio of the flow.
- (d) The ratio of maximum to minimum nozzle cross-section areas.

45. What happens to the the thrust of a turbojet when ambient pressure rises?

- (a) It increases.
- (b) It decreases.
- (c) It remains approximately constant.
- (d) It increases for centrifugal-compressor turbojets and decreases for axial-compressor turbojets.

46. What kind of aircraft engine scheme can you distinguish in the figure?



- (a) High by-pass ratio turbofan with post-combustion.
- (b) Low by-pass ratio turbofan with post-combustion.
- (c) Double spool turbojet with post-combustion.
- (d) Double spool turbojet without post-combustion.

47. Which thermodynamic cycle does gas turbines follow ideally?

- (a) Brayton.
- (b) Otto or Diesel.
- (c) Sabathé.
- (d) Combustion.

48. Which of the following statements is true regarding low bypass ratio turbofans?

- (a) Their optimal speed is $M = 1 \sim 2$.
- (b) All are correct.
- (c) They are mainly used for low range airliners.
- (d) They generate a compression ratio of around 10-15.

49. Regarding turbojet engines, adapted nozzle means...?

- (a) Such a final section that makes $P_e = P_0$
- (b) Temperature adaptation after the combustion chamber
- (c) Speed adaptation before the compressor
- (d) Such a final section that makes $V_e = V_0$

50. Sort the following engine types by maximum achievable speed.

- (a) Ramjet, turboprop, turboshaft for helicopter application.
- (b) Turbofan, turboshaft for helicopter application, piston-engine-driven propeller.
- (c) Rocket, turboprop, turbofan.
- (d) Turbofan, turbojet, ramjet.

BLO-6: Aircraft Systems & Helicopters

51. A typical aircraft AC electrical system is based on:
- (a) AC 115/200V 400 Hz - DC 28V.
 - (b) DC 115/200V 400 Hz - AC 28V.
 - (c) AC 1 230V 50Hz - DC 12V.
 - (d) AC 3 380/50 Hz - DC 24V.
52. What is the function of the PTU?
- (a) It transfers pressure among hydraulic circuits.
 - (b) It transfers fluid among hydraulic circuits.
 - (c) It transfers electrical power among electrical circuits.
 - (d) It transfers pneumatic power between the two packs of the pneumatic system.
53. Which of the following tasks does NOT involve the pneumatic system?
- (a) Control surface actuation.
 - (b) Wing anti-icing.
 - (c) Hydraulic reservoir pressurisation.
 - (d) Engine starting.
54. Which one of the following is NOT an input for the flight control system (FCS)?
- (a) FMCS.
 - (b) ADIRU.
 - (c) LGCIU.
 - (d) SFCC.
55. What does the FADEC do?
- (a) Performs complete engine management.
 - (b) It is the computer in charge of flight augmentation.
 - (c) Provides AC electrical power.
 - (d) Control ailerons and elevators.
56. Which of the following components is NOT part of the electrical system?
- (a) LP compressor.
 - (b) ESS Bus.
 - (c) Battery.
 - (d) Transformer.
57. What is the function of the swash-plate of an a helicopter?
- (a) All options are true.
 - (b) Allowing the cyclic action.
 - (c) Separating the control arms (fixed) from the rotor head (rotating).
 - (d) Allowing the collective action.
58. What is TRUE about tail rotor compensation for a helicopter?
- (a) It uses around 10% of the engine power.
 - (b) It requires the installation of a second dedicated engine.
 - (c) It is not possible with gas turbine engine helicopters.
 - (d) It is not possible on piston-based engine helicopters.
59. Performing a longitudinal flight a helicopter ...
- (a) has not the same lift distribution along all its blades.
 - (b) has a zero specific fuel consumption.
 - (c) requires no tail compensation.
 - (d) cannot start moving laterally.
60. What is the maximum allowable blade tip speed in a typical helicopter?
- (a) Close to $M = 1$.
 - (b) $M = 0.7$.
 - (c) 300 m/s.
 - (d) $0.3 \Omega R$.

BLO-7: Space Systems

61. A rocket with initial mass of 20 t and final mass of 3.5 t has a specific impulse of 400 s. What is its ΔV ?
- (a) $\Delta V = 6.8$ km/s.
 - (b) $\Delta V = 7.6$ km/s.
 - (c) $\Delta V = 4.2$ km/s.
 - (d) None of the others.
62. What is the advantage of launching space missions from sites on the Equator?
- (a) ΔV is already large when launching eastwards.
 - (b) ΔV is already small when launching westwards.
 - (c) ΔV is already large when launching northwards to intercept Polar Orbits.
 - (d) All options are correct.
63. What is necessary in order to obtain the maximum propulsive efficiency?
- (a) Maximize the mass flow with the minimum possible acceleration.
 - (b) Maximize thermal efficiency.
 - (c) Maximize the acceleration of the exhaust gases and reduce mass flow.
 - (d) Minimize the mass flow with the maximum possible acceleration.
64. What is TRUE according the Tsiolkovsky equation?
- (a) Multistage rockets maximise the velocity increase.
 - (b) The larger the propellant-to-total mass ratio, the lower the velocity increase.
 - (c) The total mass of the rocket can be considered constant throughout the flight.
 - (d) single stage rockets maximise the velocity increase.
65. Which of the following propulsive systems is suitable for Earth launches?
- (a) Chemical rockets.
 - (b) Antimatter rockets.
 - (c) Continuous Fusion rockets.
 - (d) Electrostatic rockets.
66. In which of the following orbits the satellite is always in the vertical of a same fixed point on the Earth surface?
- (a) GEO.
 - (b) LEO.
 - (c) GSO.
 - (d) HEO.
67. What is the specific impulse of a rocket engine?
- (a) The thrust per unit mass-flow of ejected gases.
 - (b) The time it takes to burn all of the fuel.
 - (c) The energy contained in the fuel divided by the power.
 - (d) The maximum time it takes to reach full speed.
68. What of the following orbital elements are classical orbital elements:
- (a) The inclination of the orbit, the eccentricity and the argument of the periapsis.
 - (b) The periapsis and the apoapsis.
 - (c) The vernal equinox direction and the ecliptic.
 - (d) The inclination of the orbit and the ecliptic.
69. When comparing rocket propulsion systems, which of the following statements is TRUE?
- (a) All answers are true.
 - (b) Cold-gas thrusters provide the smallest specific impulse.
 - (c) Chemical propulsion provides higher thrust-to-weight ratios.
 - (d) Electrostatic propulsion provides higher specific impulse than chemical propulsion.
70. The Global Positioning System (GPS) deploys a constellation of ...
- (a) Medium Earth Orbit satellites.
 - (b) Low Earth Orbit satellites.
 - (c) Polar Orbit satellites.
 - (d) Geostationary Orbit satellites.

BLO-8: Navigation & Altimetry

71. Which of the following is NOT an ILS (Instrument landing System)?
- (a) DME.
 - (b) Localizer.
 - (c) Marker beacons.
 - (d) Glide Slope.
72. What does accuracy refer to in air navigation?
- (a) To the conformance between calculated and actual position of the aircraft.
 - (b) To the issuing of an alarm whenever the aircraft cannot be guaranteed as being inside the outer tunnel.
 - (c) To the guarantee that the outer tunnel will not disappear during operation.
 - (d) To the guarantee that the tunnel will be in place when the operation starts.
73. Which component is NOT necessary for INS?
- (a) Magnetometers.
 - (b) Gyroscopes.
 - (c) Accelerometers.
 - (d) Computers.
74. What is TRUE of using the QNH calibration setting in the altimeter?
- (a) Indicated altitude is referred to an estimated sea level's pressure.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter indicates flight levels.
 - (d) All options are correct.
75. If the airplane locates to the North East direction about VOR station, which is the phase difference of received VOR signals?
- (a) 45°.
 - (b) 90°.
 - (c) 135°.
 - (d) 180°.
76. What is Dead Reckoning?
- (a) The process of calculating the current position from a previously known position and an estimation of subsequent speeds, directions and elapsed time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) An airship-based radio navigation aid.
 - (d) A generic name for ground stations.
77. Which of the following navigation systems is used to define airways (AWY)?
- (a) VOR.
 - (b) DME.
 - (c) ILS.
 - (d) None of the others.
78. What is TRUE for CAS (Calibrated Air Speed)?
- (a) It is the IAS corrected for instrument, position and installation errors based on tabulated data.
 - (b) It is the IAS corrected for compressibility but not for density.
 - (c) It is the actual speed of the aircraft relative to the surrounding air.
 - (d) It is the raw estimation from the pitot-static tube reading.
79. What is TRUE when using the QNE calibration setting in the altimeter?
- (a) All of the other.
 - (b) It is always referred to 1013 hPa.
 - (c) It is only used above the transition altitude.
 - (d) Must switch to QFE/QNH below the transition level.
80. How many GNSS satellites are at least needed to estimate the position on the Earth surface?
- (a) 4.
 - (b) 3.
 - (c) 6.
 - (d) 5.

BLO-1: Flight Principles

1. Which of the following statements is TRUE regarding load factor?
 - (a) Ascent and descent: $n < 1$.
 - (b) Banked turn: $n < 1$.
 - (c) Slope increase: $n < 1$.
 - (d) Slope decrease: $n > 1$.
2. What is TRUE regarding Lift-to-Drag ratio?
 - (a) All other options are correct.
 - (b) It depends on wing planform and configuration.
 - (c) Mach number M .
 - (d) Reynolds number Re .
3. What is FALSE regarding flap deployment?
 - (a) k increases.
 - (b) C_{D0} increases.
 - (c) Deploy flaps when high C_L is required.
 - (d) Camber of the airfoil increases.
4. Within the ISA Troposphere and decreasing altitude.
 - (a) The speed of sound increases.
 - (b) Air density decreases.
 - (c) Temperature is considered constant for some altitude above the Tropopause.
 - (d) Air pressure decreases.
5. Two identical airplanes flying at the same altitude. One airplane flying at slow speed $V_s(\phi_s)$ and the other one flying at fast speed $V_f(\Phi_f)$. Which statement regarding the maximum bank angle is FALSE?
 - (a) $\Phi_f < \Phi_s$.
 - (b) $\Phi_f = \Phi_s$.
 - (c) $\Phi_f > \Phi_s$.
 - (d) It is independent of the aerodynamic characteristics of the airplane.
6. A small airplane of mass $m = 980$ kg is engaged in a coordinated banked turn with bank angle $\Phi = 41^\circ$ at a speed $V = 240$ km/h. What is the turn radius R_t for this manoeuvre? (consider $g \simeq 9.8$ m/s², 1mi = 1.609 km.)
 - (a) $R_t = 0.324$ mi.
 - (b) $R_t = 0.652$ mi.
 - (c) $R_t = 0.237$ mi.
 - (d) $R_t = 0.453$ mi.
7. An aerobatics airplane with lift coefficient $C_L = 0.03 + 0.18\alpha$ (α in $^\circ$) is flying in balanced straight and level flight at $V = 60$ m/s with $\alpha = 1.3^\circ$. What should the speed (V_{if}) be to perform a balanced inverted flight at the same altitude and angle of attack $\alpha = -0.7^\circ$?
 - (a) $V_{if} = 99.5$ m/s.
 - (b) $V_{if} = 106$ m/s.
 - (c) $V_{if} = 115$ m/s.
 - (d) $V_{if} = 73.1$ m/s.
8. An airplane of mass $m = 1180$ kg is cruising in straight and level flight with an optimal aerodynamic efficiency $f = 23 < f_{max}$. What can we say about the total (D), induced (D_i) and parasitic (D_p) drag? (take $g \simeq 9.8$ m/s²)
 - (a) $D = 503$ N and $D_p = 251$ N = D_i .
 - (b) $D = 503$ N and $D_p \neq D_i$.
 - (c) $D = 1010$ N and $D_p = 503$ N = D_i .
 - (d) $D = 1010$ N and $D_p \neq D_i$.
9. The drag polar of an airplane changes from $C_D = 0.021 + 0.131C_L^2$ in clean configuration to $C_D = 0.029 + 0.081C_L^2$ with full flaps. What is the value of the lift coefficient (C_L) at optimal flap deployment?
 - (a) $C_L = 0.4$.
 - (b) $C_L = 0.283$.
 - (c) $C_L = 0.586$.
 - (d) $C_L = 0.134$.
10. An Aircraft has the following clean-configuration aerodynamic characteristics: $C_L = 0.19 + 0.33\alpha$ with $\alpha_S = 11^\circ$ and $C_D = 0.015 + 0.052C_L^2$. With full flaps, it changes to: $C_L^f = 0.27 + 0.38\alpha$ with $\alpha_S = 12^\circ$ and $C_D^f = 0.032 + 0.041C_L^2$. What is the value of the angle of attack for optimal flap deployment?
 - (a) $\alpha = 3.19^\circ$.
 - (b) $\alpha = 2.88^\circ$.
 - (c) $\alpha = 0.989^\circ$.
 - (d) $\alpha = 5.12^\circ$.

BLO-2: Aircraft Stability & Control

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11. What is TRUE when the centre of lift is located behind the center of gravity?
- Both canard and tailplane have to generate an equilibrating pitching moment.
 - A canard must generate a downforce.
 - A tailplane may generate an unstable moment.
 - Stability does not depend on the location of the centre of gravity.
12. Lateral stability is influenced by which of the following wing parameters?
- Dihedral and sweep.
 - Aspect ratio and camber.
 - Taper and twist.
 - Span and chord.
13. Order the longitudinal modes from slower to faster.
- Return to straight and level flight, phugoid and pitch oscillation.
 - Return to straight and level flight, pitch oscillation and phugoid.
 - Phugoid, return to straight and level flight and pitch oscillation.
 - Pitch oscillation, return to straight and level flight and phugoid.
14. Which parameters have a disengaging (stabilizing) effect on the spiral mode?
- Positive dihedral (wings up, $C_{l\beta} < 0$).
 - High roll damping (big wing span, $C_{lp} < 0$).
 - High directional stability (big fin, $C_{n\beta} < 0$).
 - None of the others.
15. What of the following assertions is TRUE about specific points on an aircraft?
- Around the centre of lift, X_L , no pitching aerodynamic moment exists.
 - The centre of gravity, X_{cg} , is a function of α .
 - The aerodynamic centre, X_{ac} , strongly depends on α .
 - None of the others is correct.
16. Which of the following statements is TRUE about the response of an airplane to a perturbation in yaw?
- It depends on static directional stability, since there automatically appears a restoring yaw moment.
 - It depends on both directional and lateral stability, since there is no restoring moment and the return to straight and level flight is mostly dynamical.
 - It depends mainly on static longitudinal stability, since there automatically appears a restoring pitch moment.
 - It depends mainly on static lateral stability, since there automatically appears a restoring roll moment.
17. Which statement is TRUE for the roll subsidence mode?
- Large aspect ratio wings result in better damping.
 - The damping is related to the Dutch Roll.
 - It is a longitudinal mode.
 - It increases maneuverability.
18. What is TRUE for dynamic stability?
- All other options are true.
 - Roll subsidence is stable.
 - Dutch roll is stable.
 - Spiral mode is unstable.
19. Which of the following statements is TRUE regarding static stability?
- $C_{m,\alpha} < 0$ has to be negative.
 - $C_{m,\alpha} = 0$ has to be negative
 - $C_{m,\alpha} > 0$ has to be negative.
 - Center of gravity must be behind of the neutral point.
20. What is TRUE for the elevator trim angle?
- All other options are true.
 - Depends on α_e .
 - Depends on the centering $x_{cg} - x_{np}$.
 - It is independent of V , h , W , γ .

BLO-3: Aircraft Performances

21. What is TRUE for glide?
- The velocity for maximum glide distance is larger than the velocity for maximum glide time.
 - Maximum glide distance is achieved close to stall speed.
 - The minimum descent rate is obtained at minimum D .
 - The minimum descent angle is obtained at minimum DV .
22. What is TRUE for operational or absolute ceiling?
- All other options are true.
 - Operational ceiling, $ROC = 100$ ft/s.
 - Operational is lower than absolute.
 - Absolute ceiling, $ROC = 0$.
23. Which statement is FALSE about performance?
- $(P_R)_{min}$ increases with decreasing W .
 - Available thrust and available power decrease with altitude.
 - Maximum endurance corresponds to the minimum of the drag or power curve.
 - Maximum range corresponds to the tangent from the origin to the drag or power curve.
24. What is essentially TRUE in a constant-altitude coordinated banked turn?
- Weight is outweighed by lift.
 - Weight equals lift.
 - Lift is outweighs by weight.
 - Whether lift or weight is higher is determined by the sense of rotation (left or right).
25. How is maximum distance achieved in gliding flight?
- For minimum drag conditions.
 - For minimum descent rate conditions.
 - For minimum required power conditions.
 - For minimum aerodynamic efficiency.
26. The required thrust of a jet airplane is given by $T_R = 0.19\bar{\rho}V^2 + 0.033W^2/\bar{\rho}V^2$. If the airplane has mass $m = 1890$ kg and is cruising at an altitude such that the relative density is $\bar{\rho} = 0.57$, what is the drag (D) at the optimal velocity for maximum endurance (V_E)? (take $g \simeq 9.8$ m/s²)
- $D = 2930$ N.
 - $D = 3390$ N.
 - $D = 3080$ N.
 - $D = 3240$ N.
27. A propeller-powered airplane of mass $m = 1490$ kg has a required power curve $P_R = 0.26\bar{\rho}V^3 + 0.029W^2/\bar{\rho}V$. What is the velocity for maximum rate of climb ($V_{(R/C)_{max}}$) at sea level ($\bar{\rho} = 1$)? (take $g \simeq 9.8$ m/s², 1 ft = 0.305 m)
- $V_{(R/C)_{max}} = 6.27 \times 10^5$ ft/h.
 - $V_{(R/C)_{max}} = 8.25 \times 10^5$ ft/h.
 - $V_{(R/C)_{max}} = 10.9 \times 10^5$ ft/h.
 - $V_{(R/C)_{max}} = 14.3 \times 10^5$ ft/h.
28. A propeller-powered airplane with drag polar $C_D = 0.021 + 0.151C_L^2$, wing surface $S = 24$ m² and mass $m = 1490$ kg is flying at a speed $V = 98.8$ m/s. The available engine power at sea level is $(P_A)_{SL} = 880$ kW. What is the engine power setting (δ_x) required to sustain straight and level flight if the altitude is such that the density is $\rho = 0.61$ kg/m³? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $\delta_x = 0.44$.
 - $\delta_x = 0.395$.
 - $\delta_x = 0.531$.
 - $\delta_x = 0.219$.
29. A glider whose maximum aerodynamic efficiency is $f_{max} = 36$ finds an isolated thermal column amidst an otherwise still atmosphere. If the target airfield is within a distance $d = 36$ mi of the thermal, how high (height over terrain h) does it need to climb in order to reach its destination? (take 1 ft = 0.305 m and 1 km = 0.621 mi)
- $h = 5270$ ft.
 - $h = 1610$ ft.
 - $h = 6830$ ft.
 - $h = 3280$ ft.
30. The required thrust of a jet-engined airplane of mass $m = 1350$ kg, wing area $S = 17.5$ m², and maximum lift coefficient before stall $C_{L_{max}} = 1.14$, is given by $T_R = 0.27\bar{\rho}V^2 + 0.023W^2/\bar{\rho}V^2$, and the maximum available sea level thrust can be considered independent of V : $(T_A)_{SL} \leq 14200$ N. What is the lowest velocity at which this airplane can be flown in straight and level flight at high altitude ($\bar{\rho} = \rho/\rho_0 = 0.71$)? (take $g \simeq 9.8$ m/s² and $\rho_0 = 1.225$ kg/m³)
- $V \simeq 39.1$ m/s.
 - $V \simeq 23.8$ m/s.
 - $V \simeq 32.8$ m/s.
 - $V \simeq 19$ m/s.

31. The ability of a body to resist a distorting influence and to return to its original size and shape when that influence or force is called:

- (a) Elasticity.
- (b) Plasticity.
- (c) Toughness.
- (d) Resistance.

32. Which of the following is a selection criteria for materials?

- (a) All other options are correct.
- (b) Fatigue.
- (c) Compatibility with other materials.
- (d) Thermal and electric properties.

33. Which statement is TRUE regarding steel?

- (a) It is an allotrope.
- (b) Ferrite (α -Fe) has a face-centered cubic structure.
- (c) Austenite (γ -Fe) has a body-centered cubic structure.
- (d) Martensite is formed by diffusion.

34. Which of the following materials is NOT used as reinforcement fiber/filament in composites?

- (a) Metallic.
- (b) Spectra.
- (c) Glass.
- (d) Carbon.

35. What is FALSE about Aluminium (Al) in aeronautical applications?

- (a) Bad thermal and electric conductivity.
- (b) Good corrosion resistance.
- (c) Its crystal structure is face-centred cubic (FCC).
- (d) The 2 and 7 series are the most used, both with thermal treatment.

36. Identify the following aircraft: A223 L2J.

- (a) Airbus 220-300: Land plane, 2 engines, jet type.
- (b) Antonov 223: Land plane, 2 engines, jet type.
- (c) Antonov 223: Large plane, 2 engines, jet type.
- (d) Antonov 200-23: Land plane, 2 engines, jet type.

37. According to ICAO, how is an A320-200 (78 tons MTOW) classified in terms of wake turbulence?

- (a) M: Medium.
- (b) S: Super.
- (c) L: Light.
- (d) H: Heavy.

38. Identify the following aircraft.



- (a) Boeing 757.
- (b) Sukhoi Superjet 100.
- (c) Comac C919.
- (d) Tupolev Tu-204.

39. Which company constructs the following aircraft?



- (a) McDonnell Douglas.
- (b) Tupolev.
- (c) Lockheed Martin.
- (d) Embraer.

40. Which of the following is NOT a wide body aircraft?

- (a) Comac C919.
- (b) Airbus A310.
- (c) Airbus A300.
- (d) Iljushin Il-96.

BLO-5: Aerospace Propulsion

41. Sort the following engine types from minimum to maximum achievable speed.

- (a) Turboshift for helicopter application, turboprop, scramjet.
- (b) Turbofan, turboshift for helicopter application, piston-engine-driven propeller, rocket.
- (c) Piston-engine-driven propeller, rocket, turboprop, turbofan.
- (d) Turbofan, turbojet, Rocket, ramjet.

42. Which of the following statements is TRUE regarding high bypass ratio turbofans.

- (a) All are correct.
- (b) Optimal speed $M = 0.8 \sim 0.9$.
- (c) Are mainly used for medium and long range airliners.
- (d) Generate a compression ratio around 30 – 40.

43. A turbofan engine has a typical fuel mass flow $m_f = 22$ t/h and a thrust $T = 310$ kN. Calculate its TSFC.

- (a) TSFC = 19.7 (g/s)/kN.
- (b) TSFC = 11.1 (g/s)/kN.
- (c) TSFC = 44.4 (g/s)/kN.
- (d) TSFC = 28.4 (g/s)/kN.

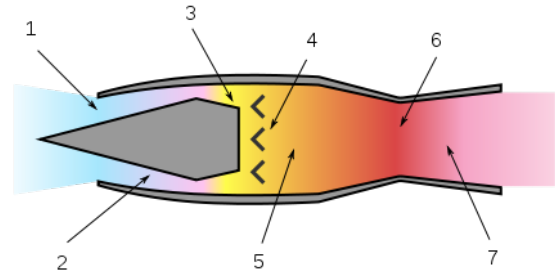
44. Which statement is TRUE for turboprops?

- (a) All other options are true.
- (b) Better power-to-weight ratio and less frontal area than piston engines.
- (c) Limitations of the propeller: Small velocity and low altitude.
- (d) Advantages of the propeller: Excellent performance during take-off, low-speed low-altitude flight.

45. What is FALSE for engine-dependent exponent λ_T ?

- (a) $\lambda_T \simeq 3$ for propeller / turbofan.
- (b) $\lambda_T \simeq 0$ for turbojet.
- (c) $\lambda_T \simeq 1$ for turbojet + afterburner.
- (d) $\lambda_T \simeq 2$ for ram-jet.

46. What kind of aircraft engine scheme can you distinguish in the figure?



- (a) Ramjet.
- (b) High by-pass ratio turbofan.
- (c) Low by-pass ratio turbofan.
- (d) Scramjet.

47. Which of the following statements regarding propeller is TRUE?

- (a) None of the other statements is true.
- (b) A turbofan uses an external propeller.
- (c) The more blades the propeller has, the minor are the air perturbations created.
- (d) The efficiency of the propeller is proportional to the number of blades it has.

48. Which of the following is a component of the gas generator?

- (a) All are components of a gas generator.
- (b) Compressor.
- (c) Combustor.
- (d) Turbine.

49. Which thermodynamic cycle does alternative engines follow ideally?

- (a) Otto or Diesel.
- (b) Brayton.
- (c) Sabathé.
- (d) Combustion.

50. Which statement is FALSE for ramjets?

- (a) Allows for $M \gtrsim 2$.
- (b) Subsonic combustion.
- (c) Needs another propulsion system for the vehicle to reach supersonic speeds.
- (d) Allows high temperature operation, high thrust-per-weight and high specific thrust.

BLO-6: Aircraft Systems & Helicopters

51. Which one of the following elements of a flight control system (FCS) are essential to the implementation of flight augmentation capabilities and automatic control?
- (a) All of them.
 - (b) Computers (FCC, SEC, ELAC, FAC).
 - (c) Hydraulic and/or electro-mechanic actuators.
 - (d) Sensors (gyros, accelerometers, transducers, ...).
52. What can be transferred between the 3 hydraulic systems of an A320?
- (a) Power but not fluid.
 - (b) Power and fluid.
 - (c) Fluid but not power.
 - (d) Water in case of breakdown.
53. Which one of the following flight control system is the simplest but can lead to unaffordable pilot efforts?
- (a) A purely mechanical flight control system.
 - (b) An electro-hydraulic flight control system.
 - (c) An electro-mechanical flight control system.
 - (d) An hydro-mechanical flight control system.
54. Which of the following is NOT an operation mode of the A320 break system?
- (a) Emergency braking.
 - (b) Normal braking.
 - (c) Alternate braking.
 - (d) Parking brakes.
55. What is TRUE for an electro-hydraulic FCS?
- (a) All are correct.
 - (b) It is lighter than a hydro-mechanical FCS.
 - (c) It allows Flight Augmentation Capabilities.
 - (d) It is also known as Fly-by-wire system.
56. What does the FADEC do?
- (a) Performs complete engine management.
 - (b) It is the computer in charge of flight augmentation.
 - (c) Provides AC electrical power.
 - (d) Control ailerons and elevators.
57. What is the main restriction regarding minimum allowable blade tip speed for a typical helicopter?
- (a) Kinetic energy and stall effects.
 - (b) Compressibility and noise effects.
 - (c) Compressibility and stall effects.
 - (d) None of the others.
58. Which of the following statements is TRUE for a helicopter in horizontal flight?
- (a) The distribution of the main rotor speeds induces a roll momentum in the helicopter.
 - (b) There is no reverse flow in the main rotor.
 - (c) The distribution of the main rotor speeds is proportional to the radius.
 - (d) The retreating blade moves up to compensate the reverse flow zone.
59. What is TRUE for a helicopter performing a lateral flight?
- (a) The helicopter has different lift distribution along different blades.
 - (b) The helicopter has a zero specific fuel consumption.
 - (c) The helicopter requires no tail compensation.
 - (d) The helicopter cannot stop moving laterally.
60. What is the typical main rotor speed for a helicopter?
- (a) 300 rpm.
 - (b) 300 m/s.
 - (c) 300 kts.
 - (d) 300 ft/s.

BLO-7: Space Systems

61. Which direction does the rocket launch take advantage?
- (a) East.
 - (b) West.
 - (c) South.
 - (d) North.
62. What of the following orbital elements are classical orbital elements:
- (a) The inclination of the orbit, the eccentricity and the argument of the periapsis.
 - (b) The periapsis and the apoapsis.
 - (c) The vernal equinox direction and the ecliptic.
 - (d) The inclination of the orbit and the ecliptic.
63. When comparing rocket propulsion systems, which of the following statements is TRUE?
- (a) All answers are true.
 - (b) Cold-gas thrusters provide the smallest specific impulse.
 - (c) Chemical propulsion provides higher thrust-to-weight ratios.
 - (d) Electrostatic propulsion provides higher specific impulse than chemical propulsion.
64. Which of the following is NOT a possible fluid propellant for a rocket propulsion system
- (a) Parapropellant.
 - (b) Bipropellant.
 - (c) Monopropellant.
 - (d) Cold-gas propellant.
65. Which of the following statements is TRUE for a rocket engine?
- (a) The ratio thrust vs weight (T/W) of a solid propellant engine is higher than the ratio T/W of a liquid propellant.
 - (b) The specific impulse of a solid propellant rocket is higher than the specific impulse of a liquid propellant.
 - (c) The required thrust/weight ratio (T/W) for an interplanetary mission is higher than for a launching mission.
 - (d) All statements are wrong.
66. Where are countries situated that mostly use Highly Elliptical Orbits (HEO)?
- (a) At extreme latitudes.
 - (b) At extreme longitudes.
 - (c) At temperate latitudes.
 - (d) At temperate longitudes.
67. What is FALSE for Solid-Propellant Rocket Motors?
- (a) They have a smaller thrust-to-volume ratios.
 - (b) Engine is simpler, safer, and more reliable.
 - (c) Solid propellants are more storable and stable.
 - (d) One ignited, it cannot be turned off.
68. What is TRUE for the specific impulse I_{SP} ?
- (a) All other options are correct.
 - (b) It increases with increasing adiabatic flame temperature.
 - (c) It decreases with increasing molecular weight of the propellant.
 - (d) Higher specific impulse means higher efficiency.
69. Which of the following orbits is the most suitable to cover all the Earth's surface?
- (a) Polar orbit.
 - (b) LEO.
 - (c) MEO.
 - (d) GEO.
70. Which orbit does give the same local time when the satellite crosses?
- (a) Sun-Synchronous Orbit.
 - (b) Geo Stationary Orbit.
 - (c) Medium Earth Orbit.
 - (d) Molniyar Orbit.

BLO-8: Navigation & Altimetry

71. When the airplane fly, the static pressure is measured as 850 mb. What is the altitude of the airplane approximately?
- (a) 1.4 km.
 - (b) 0.9 km.
 - (c) 2.5 km.
 - (d) We cannot measure it.
72. What does the following definition stands for? "The ability of the total system to perform its function without unscheduled interruptions during the intended operation."
- (a) Continuity.
 - (b) Accuracy.
 - (c) Integrity.
 - (d) Availability.
73. Which of the following radionavigation aids does NOT provide distance information?
- (a) VOR.
 - (b) DME.
 - (c) Radioaltimeter.
 - (d) GNSS.
74. What is TRUE of using the QNH calibration setting in the altimeter?
- (a) Indicated altitude is referred to an estimated sea level's pressure.
 - (b) Vertical separation is not guaranteed.
 - (c) The altimeter indicates flight levels.
 - (d) All options are correct.
75. Which satellite navigation system is different concept to the others?
- (a) QZSS.
 - (b) GPS.
 - (c) Galileo.
 - (d) Glonass.
76. What is Dead Reckoning?
- (a) The process of calculating the current position from a previously known position and an estimation of subsequent speeds, directions and elapsed time.
 - (b) The process of confirming the current position from satellite navigation aids.
 - (c) An airship-based radio navigation aid.
 - (d) A generic name for ground stations.
77. Which equipment can provide guidance for a precision approach to runway?
- (a) ILS.
 - (b) VOR+DME.
 - (c) VOR+NDB.
 - (d) VOR+NDB+DME.
78. Which of the following statements is TRUE if the DME accuracy is 185 meters?
- (a) 95% of the time the navigation error is equal or smaller than 185 m.
 - (b) The standard deviation of the error is almost 200 m.
 - (c) Its accuracy is lost when the AC is at 185 m or less.
 - (d) Its integrity is lost when the AC is at 185 m or less.
79. If the pilot of an aircraft in an airport wants the altitude reading in the altimeter to be equal to the elevation of the airport, which calibration must he select?
- (a) None of the other answers is correct.
 - (b) He must select a QFE calibration.
 - (c) He must select 1013 mb as the reference pressure for altitude zero.
 - (d) He must select the real pressure in the airport as the reference pressure for altitude zero.
80. Which is NOT an ILS (Instrument Landing System)?
- (a) DME.
 - (b) Localizer.
 - (c) Mark Beacons.
 - (d) Glide Slope.